

Current Electricity

Question1

A battery with EMF E and internal resistance r is connected across a resistance R . The power consumption in R will be maximum when:

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Options:

A.

$$R = \sqrt{2}r$$

B.

$$R = 2r$$

C.

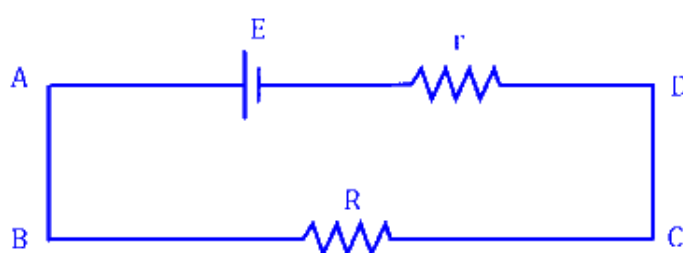
$$R = \frac{r}{2}$$

D.

$$R = r$$

Answer: D

Solution:



The total resistance of the circuit is $(R + r)$.

Using Ohm's Law, the current I flowing through the circuit is:

$$I = \frac{E}{R+r}$$

The power consumption P in the external resistor R is given by the formula:

$$P = I^2 R$$

Substituting the expression for I :

$$P = \left(\frac{E}{R+r} \right)^2 R = \frac{E^2 R}{(R+r)^2}$$

To find the value of R for which P is maximum, we differentiate P with respect to R and set the derivative to zero ($\frac{dP}{dR} = 0$).

$$\frac{dP}{dR} = E^2 \left[\frac{(R+r)^2 \cdot 1 - R \cdot 2(R+r)}{(R+r)^4} \right] = 0$$

$$\Rightarrow (R+r)^2 - 2R(R+r) = 0$$

$$\Rightarrow (R+r) - 2R = 0$$

$$\Rightarrow r - R = 0$$

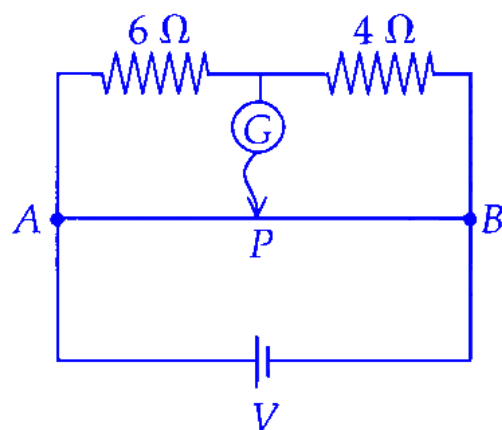
$$\Rightarrow R = r$$

Therefore, the power consumption in the external load R is at its maximum when the external resistance is equal to the internal resistance of the battery, $r = R$.

Hence, the correct option is (D).

Question2

The total length of potentiometer wire AB is 50 cm in the arrangement as shown in the figure. If P is the point where the galvanometer shows zero reading then the length AP is _____ cm.



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Options:

A.

25

B.

30

C.

15

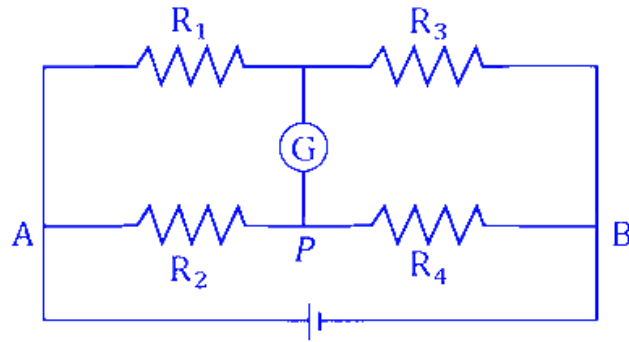
D.

20

Answer: B

Solution:

When the galvanometer (G) shows a zero reading, it means no current is flowing through the galvanometer branch. This occurs when the circuit is in a balanced state.



Resistance $R_1 = 6\Omega$.

Resistance $R_2 = 4\Omega$.

Let the length be l . Its resistance is R_{AP} .

Since the total length AB is 50 cm, the length of PB is $(50 - l)$. Its resistance is R_{PB} .

The resistance R of a uniform wire is given by the formula :

$$R = \frac{\rho L}{A}$$

Where ρ is resistivity, L is length, and A is the cross-sectional area. Since the wire is uniform, the resistance is directly proportional to the length ($R \propto L$).

$$R_{AP} = k \cdot l$$

$$\Rightarrow R_{PB} = k \cdot (50 - l)$$

where $k = \frac{\rho}{A}$ is a constant.

For a balanced bridge:

$$\frac{R_1}{R_2} = \frac{R_{AP}}{R_{PB}}$$

Substituting the known values :

$$\frac{6}{4} = \frac{k \cdot l}{k \cdot (50 - l)}$$

$$\Rightarrow \frac{3}{2} = \frac{l}{50 - l}$$

$$\Rightarrow 3(50 - l) = 2l$$

$$\Rightarrow 150 - 3l = 2l$$

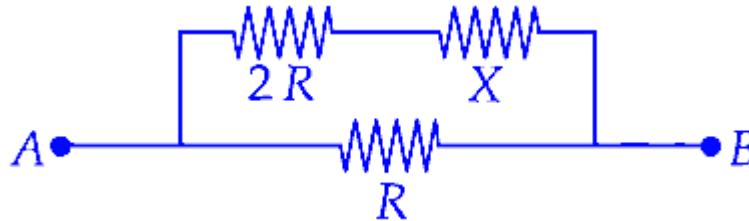
$$\Rightarrow 150 = 5l$$

$$\Rightarrow l = \frac{150}{5} = 30 \text{ cm}$$

Therefore, the length of AP is 30 cm. Hence, the correct option is (B).

Question3

Two known resistances of $R \Omega$ and $2R \Omega$ and one unknown resistance $X \Omega$ are connected in a circuit as shown in the figure. If the equivalent resistance between points A and B in the circuit is $X \Omega$, then the value of X is _____ Ω .



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Options:

A.

$$2(\sqrt{3} - 1)R$$

B.

$$R$$

C.

$$(\sqrt{3} - 1)R$$

D.

$$(\sqrt{3} + 1)R$$

Answer: C

Solution:

Resistors $2R$ and X are connected in series.

$$R_1 = 2R + X$$

And the combination of $(2R + X)$ is connected across R in parallel.

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R} = \frac{1}{2R+X} + \frac{1}{R}$$

$$\Rightarrow \frac{1}{X} = \frac{R+(2R+X)}{R(2R+X)}$$

$$\Rightarrow X(3R + X) = R(2R + X)$$

$$\Rightarrow 3RX + X^2 = 2R^2 + RX$$

$$\Rightarrow X^2 + 2RX - 2R^2 = 0$$

Using the quadratic equation formula,

$$X = \frac{-2R \pm \sqrt{(2R)^2 - 4(1)(-2R^2)}}{2 \times 1}$$

$$\Rightarrow X = \frac{-2R \pm \sqrt{4R^2 + 8R^2}}{2}$$

$$\Rightarrow X = \frac{-2R \pm \sqrt{12R^2}}{2}$$

$$\Rightarrow X = \frac{-2R \pm 2\sqrt{3}R}{2}$$

$$\Rightarrow X = -R \pm \sqrt{3}R$$

$$\Rightarrow X = (-1 \pm \sqrt{3})R$$

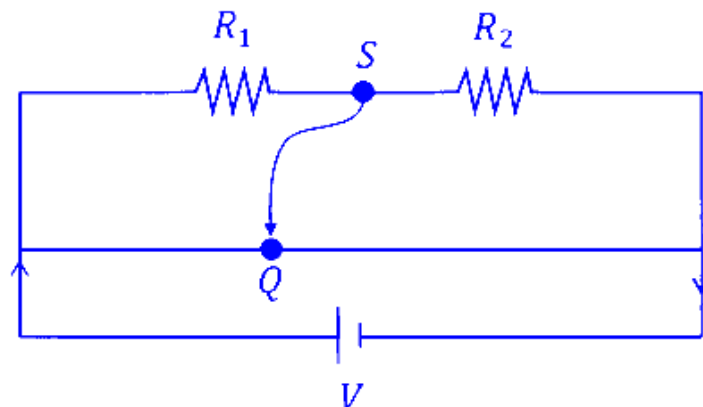
Since resistance cannot be negative, we take the positive root:

$$X = (\sqrt{3} - 1)R$$

Hence, the correct option is (C).

Question4

A meter bridge with two resistances R_1 and R_2 as shown in figure was balanced (null point) at 40 cm from the point P . The null point changed to 50 cm from the point P , when 16Ω resistance is connected in parallel to R_2 . The values of resistances R_1 and R_2 are _____ .



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Options:

A.

$$R_2 = 8\Omega, R_1 = \frac{16}{3}\Omega$$

B.

$$R_2 = 12\Omega, R_1 = \frac{12}{3}\Omega$$

C.

$$R_2 = 4\Omega, R_1 = \frac{4}{3}\Omega$$

D.

$$R_2 = 16\Omega, R_1 = \frac{16}{3}\Omega$$

Answer: A

Solution:

A meter bridge operates on the principle of a Wheatstone bridge.

Let the total length of the wire be $L = 100$ cm.

If the null point is found at a distance l from the left end, the length of the right segment is $(100 - l)$.

The balancing condition is given by:

$$\frac{R_{\text{left gap}}}{R_{\text{right gap}}} = \frac{l}{100-l}$$

The resistance in left gap is $R_{\text{left gap}} = R_1$ and the resistance in the right gap is $R_{\text{right gap}} = R_2$

Case 1

We are given that initially; the bridge is balanced at a distance of $l_1 = 40$ cm from point P .

The remaining length is $100 - 40 = 60$ cm.

Applying the balancing condition :

$$\frac{R_1}{R_2} = \frac{40}{60}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{2}{3}$$

$$\Rightarrow R_1 = \frac{2}{3}R_2 \dots (i)$$

Case 2

When a 16Ω resistance is connected in parallel to R_2 .

So, the new resistance in the right gap be R'_2 :

$$\frac{1}{R'_2} = \frac{1}{R_2} + \frac{1}{16} \Rightarrow R'_2 = \frac{16R_2}{16+R_2}$$

With this new resistance, the null point shifts to $l_2 = 50$ cm from point P .

The remaining length is now $100 - 50 = 50$ cm.

Applying the balancing condition again,

$$\frac{R_1}{R'_2} = \frac{50}{50} = 1$$

$$R_1 = R'_2 \dots (ii)$$

Substituting the expression for R'_2 into equation (ii):

$$R_1 = \frac{16R_2}{R_2+16} \dots (iii)$$

Now, substituting the expression for R_1 from equation into equation (iii) :

$$\frac{2}{3}R_2 = \frac{16R_2}{R_2+16}$$

$$\Rightarrow \frac{1}{3} = \frac{8}{R_2+16} \Rightarrow R_2 + 16 = 24$$

$$\Rightarrow R_2 = 8\Omega$$

$$\text{And, } R_1 = \frac{2}{3} \times 8\Omega = \frac{16}{3}\Omega$$

Therefore, the values are $R_2 = 8\Omega$ and $R_1 = \frac{16}{3}\Omega$.

Hence, the correct option is (A).

Question5

An electric power line having total resistance of 2Ω , delivers 1 kW of power at 250 V . The percentage efficiency of transmission line is _____ .

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Options:

A.

96.9

B.

92.5

C.

86.5

D.

100

Answer: A

Solution:

The resistance of the transmission line, $R = 2\Omega$

The power delivered (Output Power), $P_{\text{out}} = 1 \text{ kW} = 1000 \text{ W}$

The given power is delivered at voltage, $V = 250 \text{ V}$

The power delivered to the load is the product of the voltage across the load and the current flowing through it.

$$P_{\text{out}} = V \cdot I \Rightarrow I = \frac{P_{\text{out}}}{V}$$

$$\Rightarrow I = \frac{1000}{250} \text{ A} = 4 \text{ A}$$

So, a current of 4 A flows through the transmission line.

As the current flows through the transmission wires, some electrical energy is lost as heat due to the line's resistance. This power loss (P_{loss}) is calculated using Joule's heating formula :

$$P_{\text{loss}} = I^2 \cdot R$$

$$\Rightarrow P_{\text{loss}} = (4)^2 \times 2 \text{ W} = 32 \text{ W}$$

The total power supplied at the source (Input Power, P_{in}) must equal the power successfully delivered to the load plus the power lost in the transmission line.

$$P_{\text{in}} = P_{\text{out}} + P_{\text{loss}}$$

$$\Rightarrow P_{\text{in}} = 1000 + 32$$

$$\Rightarrow P_{\text{in}} = 1032 \text{ W}$$

The efficiency (η) of the transmission line is the ratio of the useful power delivered to the total power supplied, expressed as a percentage :

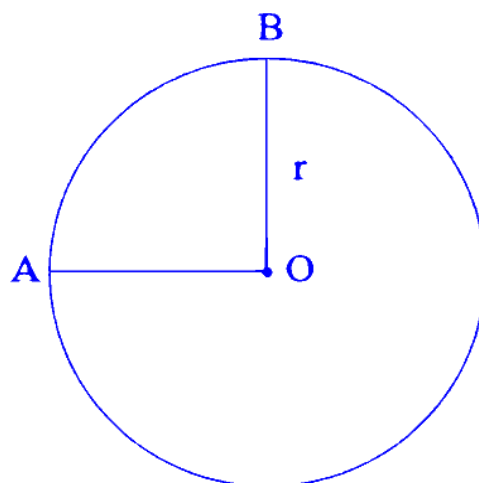
$$\eta = \left(\frac{P_{\text{out}}}{P_{\text{in}}} \right) \times 100\%$$

$$\Rightarrow \eta = \left(\frac{1000}{1032} \right) \times 100\% \approx 96.9\%$$

Therefore, the percentage efficiency of the transmission line is 96.9%. Hence, the correct option is (A).

Question6

A wire of uniform resistance $\lambda \Omega/\text{m}$ is bent into a circle of radius r and another piece of wire with length $2r$ is connected between points A and B (AOB) as shown in figure. The equivalent resistance between points A and B is _____ Ω .



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Options:

A.

$$(\pi + 1)2r\lambda$$

B.

$$\frac{6\pi\lambda r}{3\pi+16}$$

C.

$$\frac{3\pi\lambda r}{8}$$

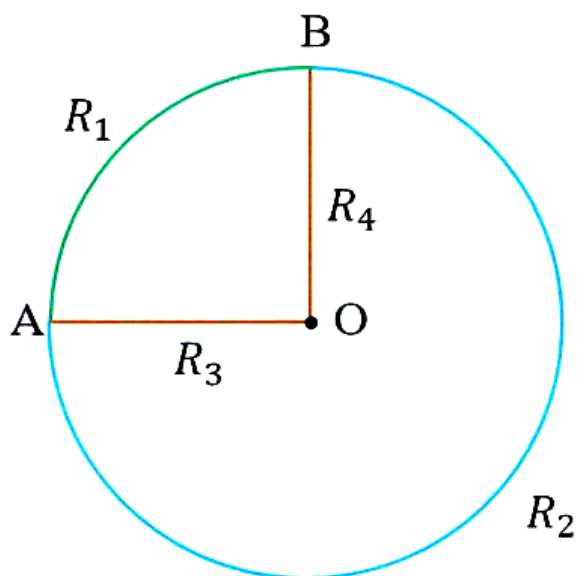
D.

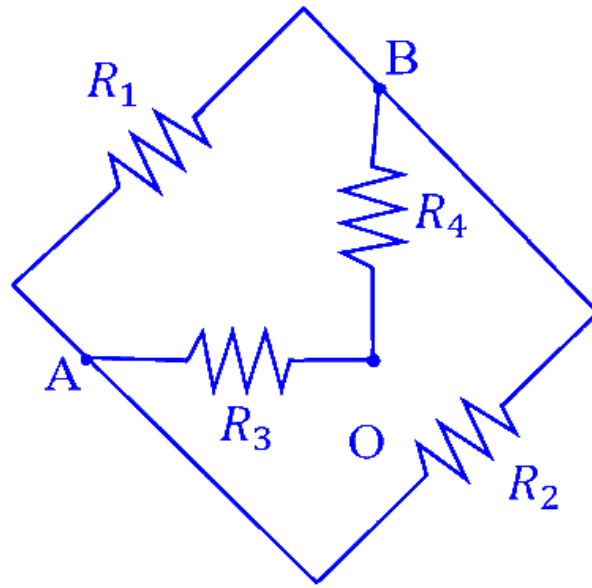
$$2\pi\lambda r$$

Answer: B

Solution:

For the equivalent resistance between points A and B we can identify three distinct paths for the current to flow from node A to node B. These paths are in parallel.





The resistance per unit for the wire is $\lambda \Omega/\text{m}$

The resistance is given by $R = \text{Resistance per unit length} \times \text{Length} = \lambda \times L$.

Path 1 (Minor Arc AB):

The shorter curved wire along the circle's circumference. Since $\angle AOB = 90^\circ$ (a quarter circle), the length is $\frac{1}{4}$ of the circumference.

$$L_1 = \frac{1}{4} \times 2\pi r = \frac{\pi r}{2}$$

So, the resistance of path 1 is,

$$R_1 = \lambda \left(\frac{\pi r}{2} \right)$$

Path 2 (Major Arc AOB):

The longer curved wire along the remaining part of the circle. This is $\frac{3}{4}$ of the circumference.

$$L_2 = \frac{3}{4} \times 2\pi r = \frac{3\pi r}{2}$$

So, the resistance of path 2 is,

$$R_2 = \lambda \left(\frac{3\pi r}{2} \right)$$

Path 3 (Wire AOB):

The straight wire sections connecting A to the centre O and then O to B. Thus, the two segments AO and OB are in series. It is given that the length of this wire is $2r$.

$$L_3 = r + r = 2r$$

So, the resistance of path 3 is,

$$R_{34} = R_3 + R_4 = \lambda(2r)$$

Since the three paths are in parallel, the equivalent resistance (R_{eq}) is,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_{34}}$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{\lambda\pi r/2} + \frac{1}{3\lambda\pi r/2} + \frac{1}{2\lambda r}$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{2}{\lambda\pi r} + \frac{2}{3\lambda\pi r} + \frac{1}{2\lambda r}$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{\lambda r} \left[\frac{2}{\pi} + \frac{2}{3\pi} + \frac{1}{2} \right]$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{\lambda r} \left[\frac{12+4+3\pi}{6\pi} \right]$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{\lambda r} \left[\frac{16+3\pi}{6\pi} \right]$$

$$\Rightarrow R_{eq} = \lambda r \left[\frac{6\pi}{16+3\pi} \right] = \frac{6\lambda\pi r}{3\pi+16}$$

Therefore, the equivalent resistance between points A and B is $\frac{6\lambda\pi r}{3\pi+16}$.

Hence, the correct option is (B).

Question7

To compare EMF of two cells using potentiometer the balancing lengths obtained are 200 cm and 150 cm . The least count of scale is 1 cm . The percentage error in the ratio of EMFs is _____

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Options:

A.

1.75

B.

1.55

C.

1.17

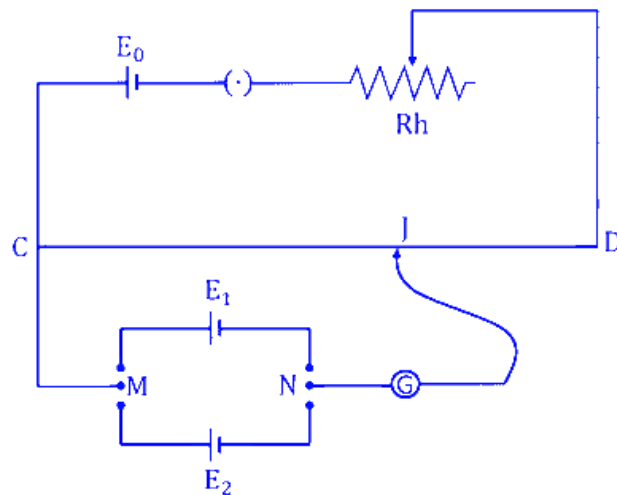
D.

1.65

Answer: C

Solution:

The EMF (E) of a cell in a potentiometer is directly proportional to its balancing length (l).



Therefore, the ratio of the EMFs of two cells is given by :

$$R = \frac{E_1}{E_2} = \frac{l_1}{l_2}$$

Balancing length for first cell $l_1 = 200$ cm

Balancing length for second cell $l_2 = 150$ cm

Absolute error in measurement $\Delta l_1 = \Delta l_2 = 1$ cm (the least count of the scale)

For a ratio $R = \frac{l_1}{l_2}$, the relative error $\left(\frac{\Delta R}{R}\right)$ is the sum of the relative errors of the individual components :

$$\frac{\Delta R}{R} = \frac{\Delta l_1}{l_1} + \frac{\Delta l_2}{l_2}$$

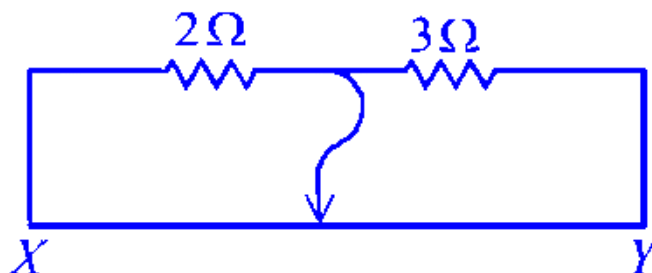
$$\text{Percentage Error} = \left(\frac{\Delta l_1}{l_1} + \frac{\Delta l_2}{l_2} \right) \times 100 = \left(\frac{1}{200} + \frac{1}{150} \right) \times 100 \approx 1.17\%$$

The calculated percentage error is approximately 1.17%.

Question8

Two resistors 2Ω and 3Ω are connected in the gaps of bridge as shown in figure. The null point is obtained with the contact of jockey

at some point on wire XY . When an unknown resistor is connected in parallel with 3Ω resistor, the null point is shifted by 22.5 cm toward Y . The resistance of unknown resistor is _____ Ω .



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Options:

A.

3

B.

1

C.

4

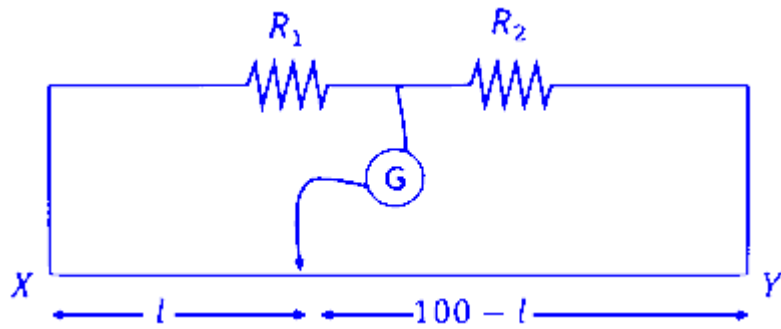
D.

2

Answer: D

Solution:

A meter bridge works on the principle of a Wheatstone bridge. When the bridge is balanced (null point), no current flows through the galvanometer.



The relationship between the resistors in the gaps (R_1, R_2) and the balancing lengths ($l, 100 - l$) is:

$$\frac{R_1}{R_2} = \frac{l}{100-l}$$

Here, $R_1 = 2\Omega, R_2 = 3\Omega$

Length of the wire from the left end (X) to the null point is l_1 then remaining length of the wire (assuming standard 100 cm wire) is $(100 - l_1)$.

$$\frac{2}{3} = \frac{l_1}{100-l_1}$$

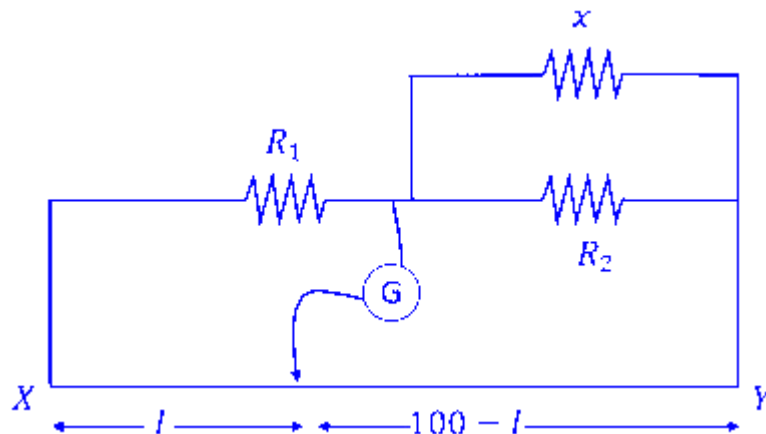
$$\Rightarrow 2(100 - l_1) = 3l_1$$

$$\Rightarrow 200 - 2l_1 = 3l_1$$

$$\Rightarrow 200 = 5l_1 \Rightarrow l_1 = 40 \text{ cm}$$

So, the initial null point is at 40 cm from X.

When an unknown resistor X is connected in parallel with the 3Ω resistor in the right gap.



The new right gap resistance is R'_2 .

For two resistors in parallel, the equivalent resistance is:

$$R'_2 = \frac{3 \times X}{3+X}$$

The null point shifts by 22.5 cm towards Y (the right end).

$$l_2 = l_1 + 22.5 \text{ cm}$$

$$\Rightarrow l_2 = 40 + 22.5 = 62.5 \text{ cm}$$

Using balanced condition,

$$\frac{R_1}{R'_2} = \frac{l_2}{100-l_2}$$

$$\Rightarrow \frac{2}{R'_2} = \frac{62.5}{100-62.5}$$

$$\Rightarrow \frac{2}{R'_2} = \frac{62.5}{37.5} = \frac{5}{3}$$

$$R'_2 = \frac{6}{5} = 1.2\Omega$$

We know that R'_2 is the parallel combination of 3Ω and X .

$$1.2 = \frac{3X}{3+X} \Rightarrow 1.2(3+X) = 3X$$

$$\Rightarrow 3.6 + 1.2X = 3X$$

$$\Rightarrow 3.6 = 3X - 1.2X$$

$$\Rightarrow 3.6 = 1.8X$$

$$\Rightarrow X = 2\Omega$$

Therefore, the value of X is 2Ω . Hence, the correct option is (D).

Question9

Two resistors of 100Ω each are connected in series with a 9 V battery. A voltmeter of 400Ω resistance is connected to measure the voltage drop across one of the resistors. The voltmeter reading is _____ V.

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Options:

A.

3

B.

2

C.

4

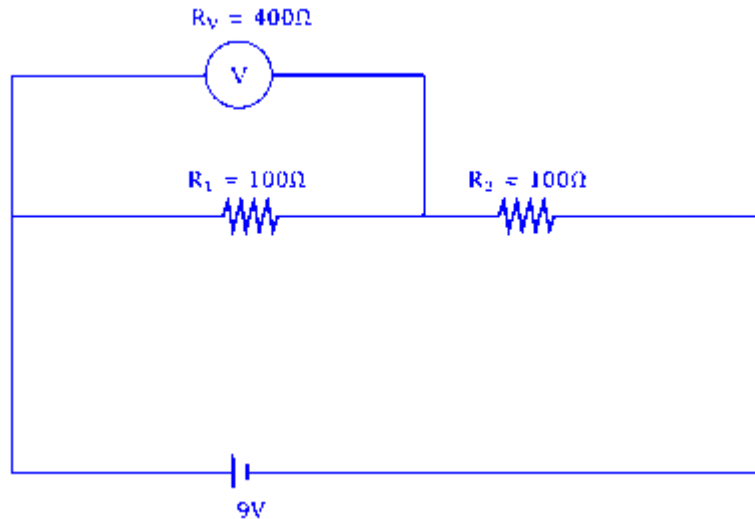
D.

4.5

Answer: C

Solution:

The initial circuit is,



When the voltmeter is connected to measure the voltage across one resistor, it is connected in parallel with that resistor.

Resistance of Voltmeter $R_V = 400\Omega$

The voltmeter (R_V) and the resistor (R_1) are in parallel.

$$\frac{1}{R_{1,V}} = \frac{1}{R_1} + \frac{1}{R_V}$$

$$\Rightarrow \frac{1}{R_{1,V}} = \frac{1}{100} + \frac{1}{400} = \frac{5}{400}$$

$$R_{1,V} = \frac{400}{5} = 80\Omega$$

Now, R_2 is in series with the equivalent resistance ($R_{1,V}$). So, the total resistance of the circuit is,

$$R_{\text{total}} = R_{1,V} + R_2$$

$$\Rightarrow R_{\text{total}} = 100\Omega + 80\Omega = 180\Omega$$

Using Ohm's Law ($V = I \cdot R$) for the entire circuit :

$$I = \frac{V}{R_{\text{total}}}$$

$$\Rightarrow I = \frac{9}{180} = \frac{1}{20} = 0.05 \text{ A}$$

The voltmeter measures the potential difference across the parallel section ($R_{1,V}$).

Using Ohm's Law for that specific section :

$$V_{\text{reading}} = I \times R_{1,V}$$

$$\Rightarrow V_{\text{reading}} = 0.05 \text{ A} \times 80\Omega$$

$$\Rightarrow V_{\text{reading}} = 4 \text{ V}$$

Therefore, the voltmeter reading is 4 V. Hence, the correct option is (C).

Question10

A regular hexagon is formed by six wires each of resistance $r\Omega$ and the corners are joined to the centre by wires of same resistance. If the current enters at one corner and leaves at the opposite corner, the equivalent resistance of the hexagon between the two opposite corners will be

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Options:

A.

$$\frac{3}{5}r$$

B.

$$\frac{4}{5}r$$

C.

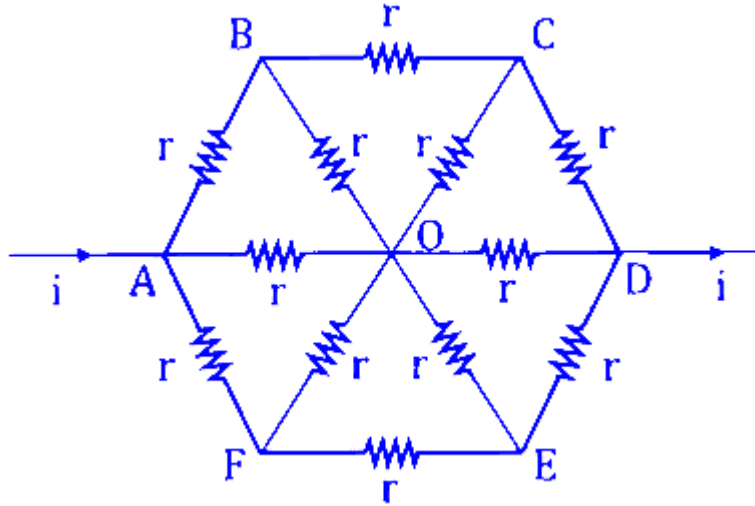
$$\frac{5}{8}r$$

D.

$$\frac{3}{4}r$$

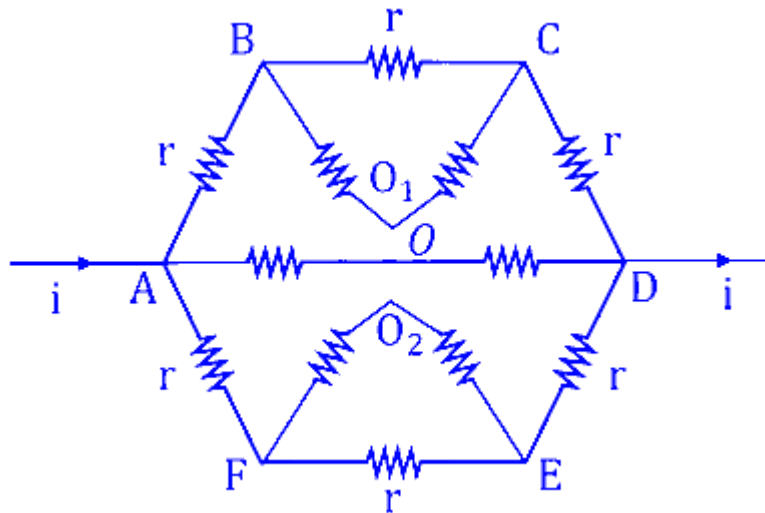
Answer: B

Solution:



Because the hexagon is regular and the entry/exit points (A and D) are symmetrical, we can identify points with equal potential relative to the axis AD :

- 1. Points B and F are at the same potential due to symmetry relative to the axis AD $\Rightarrow V_B = V_F$.
- 2. Points C and E are also at the same potential $\Rightarrow V_C = V_E$.
- 3. The centre point O lies exactly halfway between the entry and exit points on the direct path AOD.



$R_{BO_1C} = r + r = 2r$ and this is in parallel with point B and C,

$$R_{BC} = \frac{r \times 2r}{r + 2r} = \frac{2r}{3}$$

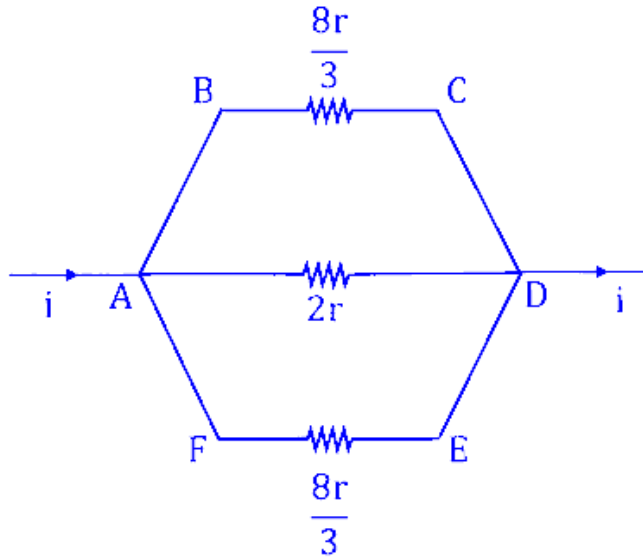
Now, R_{AB} , R_{BC} and R_{CD} are in series,

$$R_{ABCD} = r + \frac{2r}{3} + r = \frac{8r}{3}$$

Similarly, due to symmetry, $R_{AFED} = \frac{8r}{3}$

R_{AO} and R_{OD} are in series, $R_{OD} = r + r = 2r$

The resulting circuit is,



Now, all are in parallel, so the equivalent resistance between A and D is,

$$\frac{1}{R} = \frac{1}{R_{ABCD}} + \frac{1}{R_{AOB}} + \frac{1}{R_{AFED}} = \frac{1}{(8r/3)} + \frac{1}{(2r)} + \frac{1}{(8r/3)} = \frac{3}{8r} + \frac{1}{2r} + \frac{3}{8r} = \frac{3+4+3}{8r}$$

$$\Rightarrow \frac{1}{R} = \frac{10}{8r} \Rightarrow R = \frac{4r}{5}$$

Therefore, the equivalent resistance between any two corners opposite to each other is $\frac{4r}{5}$.

Hence, the correct option is (B).

Question11

A moving coil galvanometer of resistance 100Ω shows a full scale deflection for a current of 1 mA . The value of resistance required to convert this galvanometer into an ammeter, showing full scale deflection for a current of 5 mA , is _____ Ω

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Options:

A.

25

B.

2.5

C.

0.5

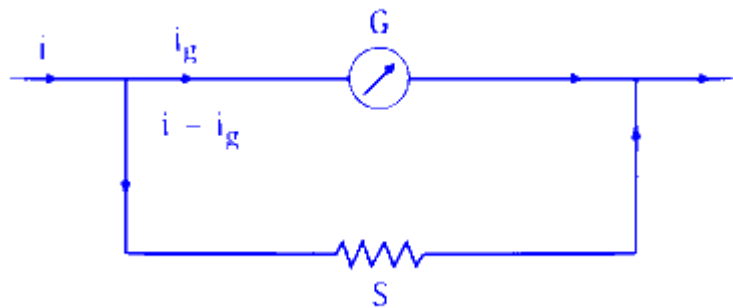
D.

10

Answer: A

Solution:

To convert a galvanometer into an ammeter of a higher range, a small shunt resistance (S) must be connected in parallel with the galvanometer.



Galvanometer resistance = $G = 100\Omega$

Full-scale deflection current = $I_g = 1 \text{ mA} = 1 \times 10^{-3} \text{ A}$

The required ammeter range = $I = 5 \text{ mA} = 5 \times 10^{-3} \text{ A}$

In a parallel circuit, the potential difference across the galvanometer is equal to the potential difference across the shunt:

$$V_g = V_s$$

$$\Rightarrow I_g \times G = (I - I_g) \times S$$

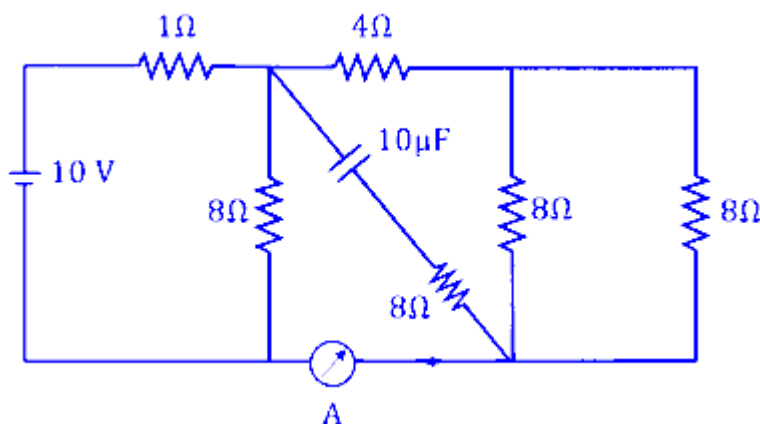
$$\Rightarrow S = \frac{I_g \times G}{I - I_g}$$

$$\Rightarrow S = \frac{1 \times (100)}{5 - 1} = 25\Omega$$

To measure a current of 5 mA with a 100Ω galvanometer that deflects at 1 mA, a shunt resistance of 25Ω is required. Hence, the correct option is (A).

Question12

The reading of the ammeter (A) in steady state in the following circuit (assuming negligible internal resistance of the ammeter) is ____ A.



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Options:

A.

$\frac{1}{2}$

B.

1

C.

2

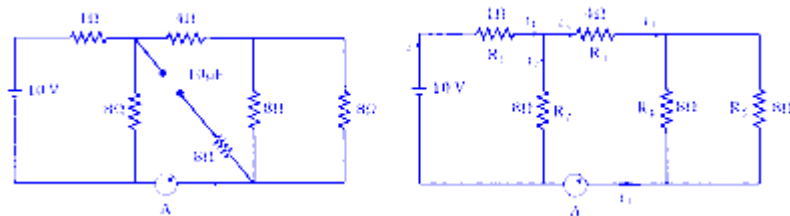
D.

0

Answer: B

Solution:

In a DC circuit, a capacitor reaches a steady state after being connected for a long time. In this state the capacitor is fully charged and it acts as an infinite resistance (an open circuit). This means no current flows through the branch containing the $10\mu\text{F}$ capacitor and the 8Ω resistor connected in series with it.



The two 8Ω resistors R_4 and R_5 are in parallel.

$$R_{4,5} = \frac{8 \times 8}{8+8} = \frac{64}{16} = 4\Omega$$

This $R_{4,5} = 4\Omega$ equivalent is in series with the $R_3 = 4\Omega$ resistor.

$$R_{3,4,5} = 4\Omega + 4\Omega = 8\Omega$$

This $R_{3,4,5} = 8\Omega$ combination is now in parallel with the $R_2 = 8\Omega$ vertical resistor in the middle.

$$R_{2,3,4,5} = \frac{8 \times 8}{8+8} = 4\Omega$$

Finally, this $R_{2,3,4,5} = 4\Omega$ is in series with the $R_1 = 1\Omega$.

$$R_{\text{total}} = 1\Omega + 4\Omega = 5\Omega$$

Using Ohm's Law for the whole circuit:

$$i = \frac{V}{R_{\text{total}}} = \frac{10}{5} \text{ A} = 2 \text{ A} = i_1$$

The reading of the ammeter is equal to the value of i_3 .

Since the resistances R_2 and $R_{3,4,5}$ are equal to 8Ω , the current splits equally, $i_2 = i_3$

Using Kirchhoff's current law at junction, $i_1 = i_2 + i_3 = 2i_2 = 2i_3 \Rightarrow i_2 = i_3 = \frac{i_1}{2}$

So, the reading of the ammeter is $i_3 = \frac{i_1}{2} = \frac{2}{2} = 1 \text{ A}$

Therefore, the ammeter reading is 1A in steady state. Hence, the correct option is (B).

Question13

In the potentiometer, when the cell in the secondary circuit is shunted with 4Ω resistance, the balance is obtained at the length 120 cm of wire. Now when the same cell is shunted with 12Ω resistance, the balance is shifted to a length of 180 cm . The internal resistance of cell is _____ Ω

JEE Main 2026 (Online) 28th January Morning Shift

Options:

A.

4

B.

12

C.

6

D.

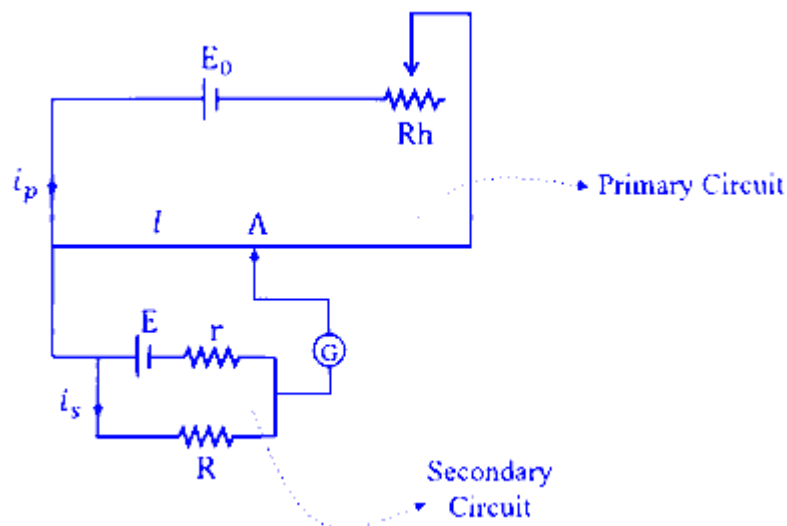
3

Answer: A

Solution:

Let the current in the primary circuit is i_p . If the resistance of potentiometer wire of length L is R , then the resistance of the balancing length l will be $R_l = \left(\frac{R}{L}\right)l$

The voltage drop at point A is. $V = i_p R_l = \left(\frac{i_p R}{L}\right)l = k \cdot l$, where k is constant.



the balancing length l is proportional to the terminal potential difference V :

$$V = k \cdot l \dots (i)$$

When a cell of EMF E and internal resistance r is shunted (connected in parallel) with an external resistance R ,

$$\text{The current in the secondary circuit is, } i_s = \frac{E}{R+r}$$

$$\text{Then terminal voltage is, } V = iR = \frac{ER}{R+r} \dots (ii)$$

From (i) and (ii),

$$k \cdot l = \frac{E \cdot R}{R+r}$$

$$\Rightarrow l = \frac{1}{k} \left(\frac{R}{R+r} \right)$$

Case 1 when the external resistance is $R_1 = 4\Omega$ and the balancing length $l_1 = 120$ cm

$$120 = \frac{1}{k} \left[\frac{4}{4+r} \right] \dots$$

Case 2 when the external resistance is $R_2 = 12\Omega$ and the balancing length $l_2 = 180$ cm

$$180 = \frac{1}{k} \left[\frac{12}{12+r} \right] \dots$$

Dividing equation (iv) by equation (iii) :

$$\frac{180}{120} = \frac{\frac{12}{12+r}}{\frac{4}{4+r}}$$

$$\Rightarrow \frac{3}{2} = \frac{3 \cdot (4+r)}{12+r}$$

$$\Rightarrow 12 + r = 2(4 + r)$$

$$\Rightarrow 12 + r = 2(4 + r)$$

$$\Rightarrow r = 4\Omega$$

Therefore, the internal resistance of the cell is 4Ω . Hence, the correct option is (A).

Question14

For the two cells having same EMF E and internal resistance r , the current passing through the external resistor 6Ω is same when both the cells are connected either in parallel or in series. The value of internal resistance r is _____ Ω .

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Options:

A.

3

B.

6

C.

4

D.

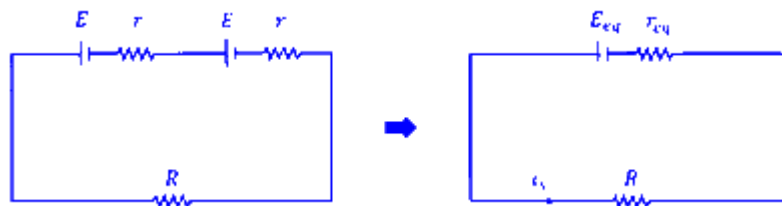
9

Answer: B

Solution:

Case 1 : when cells are in series :

When two identical cells (E, r) are connected in series with an external resistor $R = 6\Omega$:



Total EMF $E_{eq} = E + E = 2E$

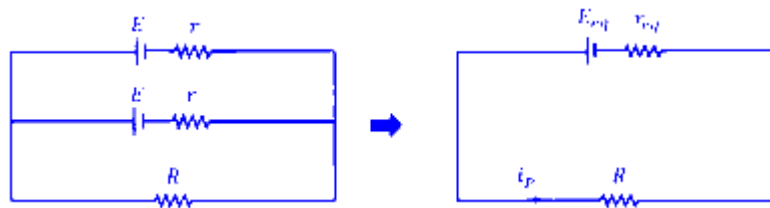
Total internal resistance $r_{eq} = r + r = 2r$

So, the current in the circuit is,

$$i_s = \frac{E_{eq}}{R + r_{eq}} = \frac{2E}{6 + 2r}$$

2. Case 2 : Cells in Parallel

When two identical cells (E, r) are connected in parallel with an external resistor $R = 6\Omega$:



Equivalent EMF (E_{eq}) in parallel is given as $\frac{\frac{E_1}{r_1} + \frac{E_2}{r_2} + \dots + \frac{E_n}{r_n}}{\frac{1}{r_1} + \frac{1}{r_2} + \dots + \frac{1}{r_n}}$

Here, $E_1 = E_2 = E$ and $r_1 = r_2 = r$,

$$E_{eq} = \frac{\frac{E}{r} + \frac{E}{r}}{\frac{1}{r} + \frac{1}{r}} = \frac{\frac{2E}{r}}{\frac{2}{r}} = E$$

As the internal resistances are in parallel, equivalent internal resistance (r_{eq}) :

$$\frac{1}{r_{eq}} = \frac{1}{r} + \frac{1}{r} = \frac{2}{r} \Rightarrow r_{eq} = \frac{r}{2}$$

Hence, the current through external resistor is

$$i_p = \frac{E_{eq}}{R+r_{eq}} = \frac{E}{6+\frac{r}{2}}$$

According to the question, $i_s = i_p$:

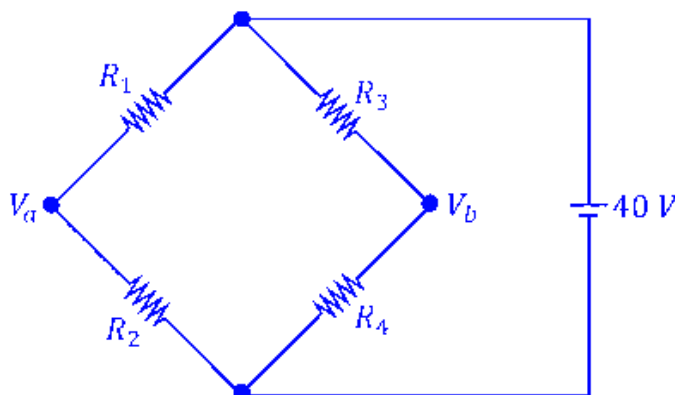
$$\begin{aligned} \frac{2E}{6+2r} &= \frac{E}{6+\frac{r}{2}} \Rightarrow \frac{1}{6+2r} = \frac{1}{12+r} \\ \Rightarrow 12+r &= 6+2r \Rightarrow 12-6 = 2r-r \\ &\Rightarrow r = 6\Omega \end{aligned}$$

So, the value of internal resistance r is 6Ω .

Hence, the correct option is (B).

Question15

A Wheatstone bridge is initially at room temperature and all arms of the bridge have same value of resistances ($R_1 = R_2 = R_3 = R_4$). When R_3 resistance is heated to some temperature, its resistance value has gone up by 10%. The potential difference ($V_a - V_b$) (after R_3 is heated) is _____ V.



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Options:

A.

1.05

B.

0.95

C.

0

D.

2

Answer: B

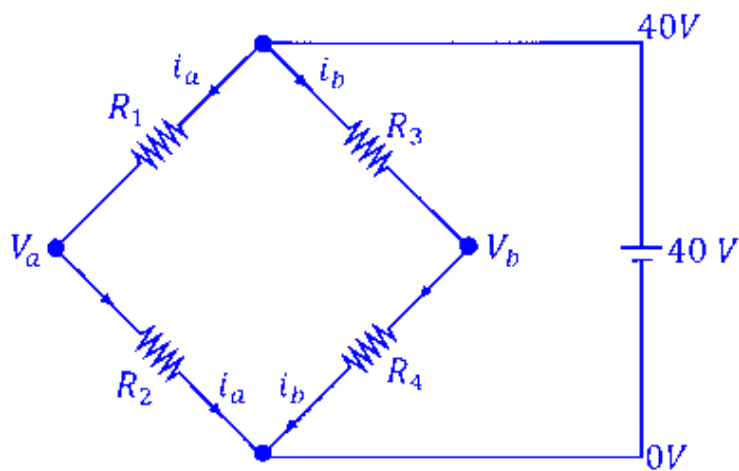
Solution:

Initially : $R_1 = R_2 = R_3 = R_4 = R$.

After heating R_3 :

- 1. $R_1 = R$
- 2. $R_2 = R$
- 3. $R_3 = R + 10\% \text{ of } R = 1.1R$
- 4. $R_4 = R$

Supply voltage $V = 40 \text{ V}$.



The voltage across R_1 and R_2 is 40 V . So the current in this branch is $i_a = \frac{40 \text{ V}}{R_1 + R_2}$.

The voltage across $R_1 = V_1 = i_a R_1 = \left(\frac{40 \text{ V}}{R_1 + R_2} \right) R_1 = \left(\frac{40 \text{ V}}{R + R} \right) R = 20 \text{ V}$

The voltage across $R_2 = V_2 = i_a R_2 = \left(\frac{40 \text{ V}}{R_1 + R_2} \right) R_2 = \left(\frac{40 \text{ V}}{R + R} \right) R = 20 \text{ V}$

Assuming the potential of negative terminal of battery as 0 V then $v_a = V_2 = 20 \text{ V}$

Similarly for the branch containing R_3 and R_4 ,

The voltage across R_3 and R_4 is 40 V . So the current in this branch is $i_b = \frac{40 \text{ V}}{R_3 + R_4}$.

The voltage across $R_3 = V_3 = i_b R_3 = \left(\frac{40 \text{ V}}{R_3 + R_4} \right) R_3 = \left(\frac{40 \text{ V}}{1.1R + R} \right) (1.1R) = 20.95 \text{ V}$

The voltage across $R_4 = V_4 = i_b R_4 = \left(\frac{40 \text{ V}}{R_3 + R_4} \right) R_4 = \left(\frac{40 \text{ V}}{1.1R + R} \right) R = 19.05 \text{ V}$

Assuming the potential of negative terminal of battery as 0 V then $v_b = V_4 = 19.05 \text{ V}$

So, the potential difference between points a and b is,

$$(V_a - V_b) = 20 \text{ V} - 19.05 \text{ V} = 0.95 \text{ V}$$

Therefore, the potential difference is 0.95 V . Hence, the correct option is (B).

Question16

Which one of the following is not a measurable quantity?

JEE Main 2026 (Online) 28th January Evening Shift

Options:

A.

Resistance

B.

Voltage difference

C.

Displacement current

D.

Voltage

Answer: D

Solution:

Resistance is a physical property of a conductor that opposes current. It is easily measured using an Ohmmeter or by calculating the ratio of voltage to current ($R = V/I$). Hence, option (A) is not the correct option.

The voltage of a single point is representing the electric potential of that point which is defined as the work done to move a unit charge from infinity to a specific point.

Since infinity is an arbitrary reference point, the absolute value of voltage at a single point cannot be measured directly. It is a mathematical construct. So, voltage cannot be measured, i.e., not a measurable quantity. Hence, option (D) is the correct option.

Voltage difference (Potential difference) is the difference in electric potential between two specific points. This is measured by a voltmeter. So, it is a measurable quantity. Hence, option (B) is not the correct option.

Displacement current is a quantity appearing in Maxwell's equations, representing a changing electric field (like between the plates of a capacitor). While it isn't a flow of charges like normal current, its magnetic effects are real and can be measured using magnetic sensors. Hence, it is a measurable quantity. So option (C) is not the correct option.

Therefore, the correct option is (D).

Question 17

Two resistors of $200\ \Omega$ and $400\ \Omega$ are connected in series with a battery of 100 V . A bulb rated at 200 V , 100 W is connected across the $400\ \Omega$ resistance. The potential drop across the bulb is _____ V .

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

25

B.

50

C.

66.6

D.

100

Answer: B

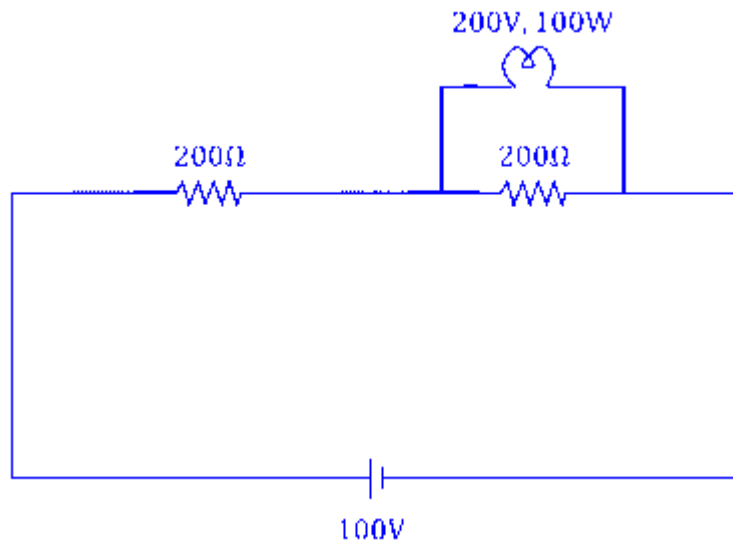
Solution:

The bulb is rated at power $P = 100 \text{ W}$ and the voltage rating is $V_{\text{rated}} = 200 \text{ V}$.

Using the power formula $P = \frac{V^2}{R}$, the resistance of the bulb is,

$$R_b = \frac{V_{\text{rated}}^2}{P}$$

$$\Rightarrow R_b = \frac{(200)^2}{100} = \frac{40000}{100} = 400\Omega$$



The bulb (400Ω) is connected in parallel with the 400Ω resistor. The equivalent resistance (R_p) of two equal resistances in parallel is :

$$\frac{1}{R_p} = \frac{1}{400} + \frac{1}{400} = \frac{2}{400}$$

$$\Rightarrow R_p = 200\Omega$$

This parallel combination is in series with the 200Ω resistor.

$$R_{\text{total}} = 200 + R_p = 200 + 200 = 400\Omega$$

$$\text{So, the total current in the circuit is } i = \frac{V}{R_{\text{total}}} = \frac{100}{400} = \frac{1}{4} \text{ A}$$

As the bulb is in parallel with 400Ω resistance, so the voltage drop across the bulb is same as across the parallel combination;

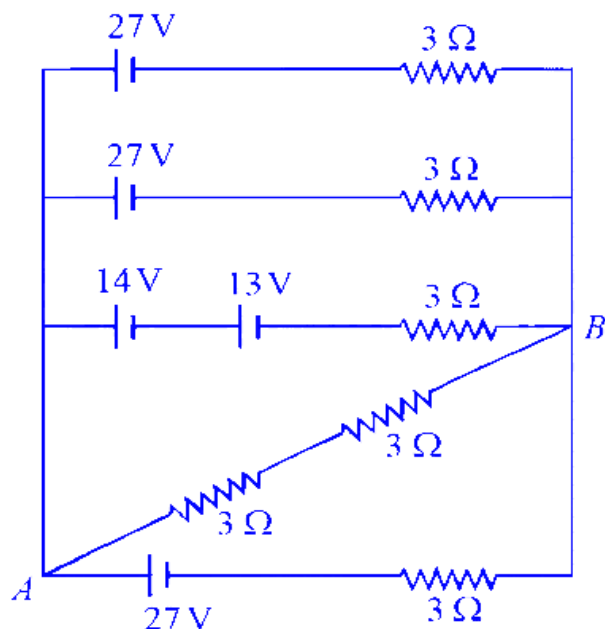
$$V_{\text{bulb}} = iR_p = \frac{1}{4} \times 200 \text{ V} = 50 \text{ V}$$

Therefore, the potential drop across the bulb is 50 V .

Hence, the correct option is (B).

Question18

The voltage and the current between A and B points shown in the circuit are ____ .



JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

$24\text{ V}, 12\text{ A}$

B.

$24\text{ V}, 4\text{ A}$

C. $18\text{ V}, 12\text{ A}$

D.

$27\text{ V}, 4\text{ A}$

Answer: B

Solution:

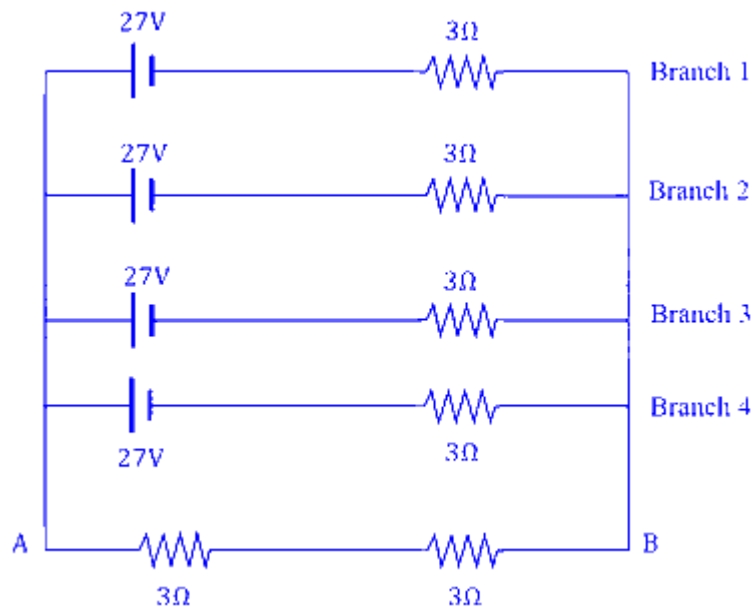
The circuit consists of four parallel branches between points A and B :

Branch 1: $E_1 = 27 \text{ V}$, $r_1 = 3\Omega$

Branch 2: $E_2 = 27 \text{ V}$, $r_2 = 3\Omega$

Branch 3: Two cells in series (14 V and 13 V) and a resistor. The total EMF is $E_3 = 14 + 13 = 27 \text{ V}$, $r_3 = 3\Omega$

Branch 4: $E_4 = 27 \text{ V}$, $r_4 = 3\Omega$



Since all branches have the same EMF (27 V) and are connected in parallel with the same polarity, the equivalent EMF of the combination is

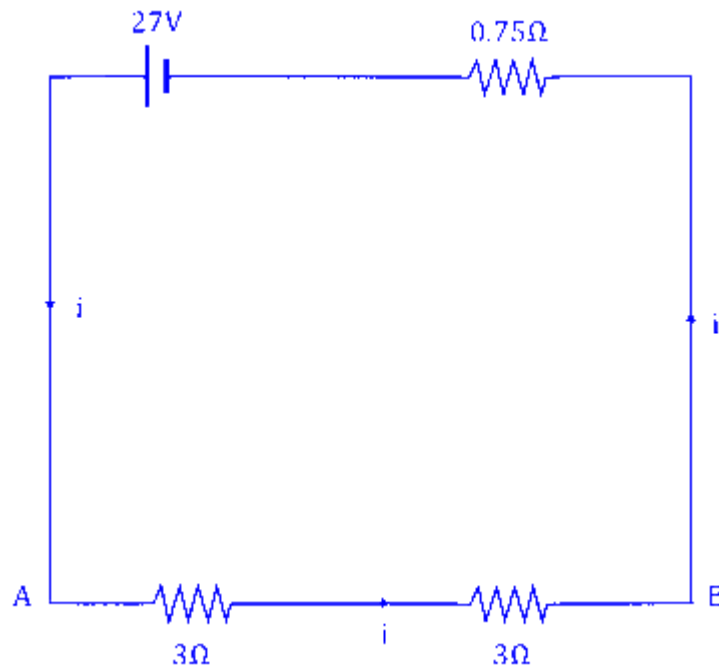
$$E_{\text{eq}} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2} + \frac{E_3}{r_3} + \frac{E_4}{r_4}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4}} = \frac{\frac{27}{3} + \frac{27}{3} + \frac{27}{3} + \frac{27}{3}}{\frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3}} = 27 \text{ V}$$

The four 3Ω resistors are in parallel. The equivalent internal resistance is :

$$\frac{1}{r_{\text{eq}}} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{4}{3}$$

$$\Rightarrow r_{\text{eq}} = \frac{3}{4}\Omega = 0.75\Omega$$

So, the equivalent circuit is,



Between points A and B , the load consisting of two 3Ω resistors in series.

So, total external resistance $R = 3 + 3 = 6\Omega$.

The total current (i) in the circuit is :

$$i = \frac{E_{eq}}{R + r_{eq}} = \frac{27}{6 + 0.75} = \frac{27}{6.75} = 4 \text{ A}$$

The terminal voltage (V_{AB}) across points A and B is :

$$V_{AB} = E_{eq} - ir_{eq} \text{ or } V_{AB} = I \times R$$

$$\Rightarrow V_{AB} = 4 \times 6 = 24 \text{ V}$$

Therefore, the voltage is 24 V and the current is 4 A .

Thus, the correct option is (B).

Question19

A voltmeter with internal resistance of $x\Omega$ can be used to measure upto 20 V . In order to increase its measuring range to 30 V , the required modification is to _____ .

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

connect resistor of $\frac{x}{2}\Omega$, in series with voltmeter.

B.

connect resistor of $\frac{x}{2}\Omega$, in parallel to voltmeter.

C.

connect a resistor of $x\Omega$ in series with voltmeter.

D.

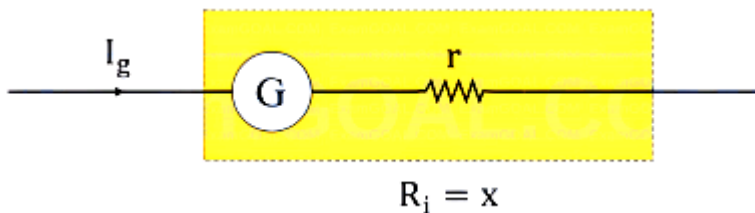
connect resistor of $2x\Omega$ in parallel to voltmeter.

Answer: A

Solution:

A voltmeter is a galvanometer with a high resistance connected in series. To increase the range of an existing voltmeter, we must add more resistance in series.

Let the full-scale deflection current of the voltmeter be I_g .

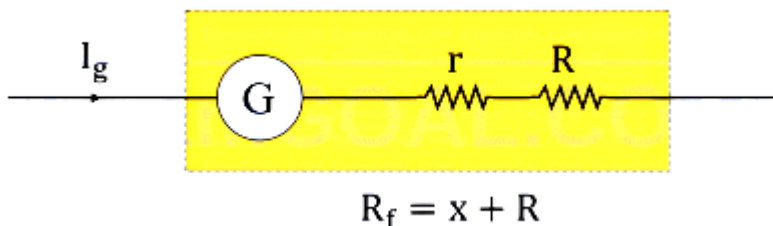


Initially, the voltmeter has a resistance x and measures up to $V_1 = 20\text{ V}$.

According to Ohm's Law ($V = IR$):

$$20 = I_g \cdot x \Rightarrow I_g = \frac{20}{x}$$

To measure a new maximum voltage $V_2 = 30\text{ V}$, we connect an additional resistance R in series with the voltmeter.



The total resistance becomes $(x + R)$. The current required for full-scale deflection remains I_g :

$$V_2 = I_g(x + R)$$

Substituting $V_2 = 30$ and $I_g = \frac{20}{x}$ into the equation :

$$30 = \left(\frac{20}{x}\right)(x + R)$$

$$\Rightarrow 3 = \frac{2}{x}(x + R)$$

$$\Rightarrow 3x = 2(x + R)$$

$$\Rightarrow 3x = 2x + 2R$$

$$\Rightarrow x = 2R \Rightarrow R = \frac{x}{2} \Omega$$

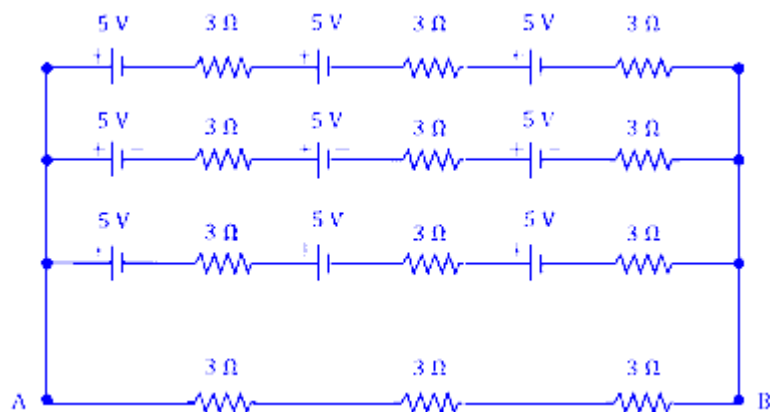
Thus, to increase the range to 30 V , a resistor of value $\frac{x}{2} \Omega$ must be connected in series with the voltmeter.

Hence, the correct option is (A).

Question20

Refer to the figure given below, current between terminals A and B is

_____ A.



JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

12.5

B.

1.25

C.

7.5

D.

5

Answer: B

Solution:

We have three identical branches in parallel. Each branch has an emf $E = 5 \text{ V}$ and an internal resistance $r = 3\Omega$.

When cells are connected in series with their polarity in order, the equivalent emf is given as,

$$E_{\text{eq(s)}} = E_1 + E_2 + \dots + E_n$$

The equivalent internal resistance is given as $r_{\text{eq(s)}} = r_1 + r_2 + \dots + r_n$.

When cells are connected in parallel with their polarity in order, the equivalent emf is given as

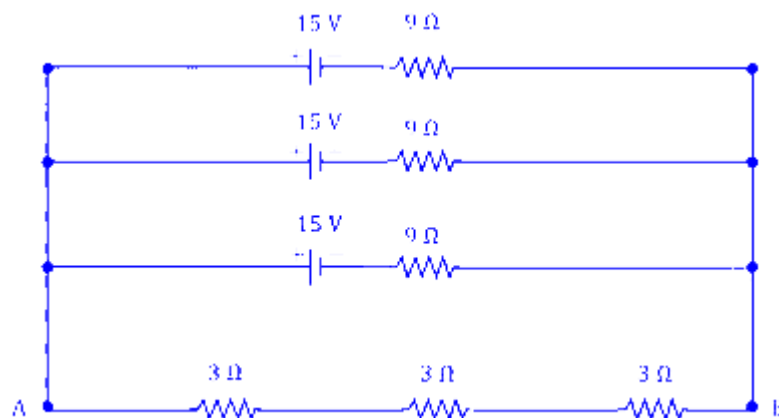
$$E_{\text{eq(p)}} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2} + \dots + \frac{E_n}{r_n}}{\frac{1}{r_1} + \frac{1}{r_2} + \dots + \frac{1}{r_n}}$$

The equivalent internal resistance is given as $\frac{1}{r_{\text{eq(p)}}} = \frac{1}{r_1} + \frac{1}{r_2} + \dots + \frac{1}{r_n}$.

For each branch, total emf is, $E_{\text{eq}} = (5 + 5 + 5)\text{V} = 15 \text{ V}$

Total internal resistance is, $r_{\text{eq}} = (3 + 3 + 3)\Omega = 9\Omega$

So, the circuit simplifies to;



Now all three branches containing cell are in parallel.

Using the parallel combination of cells;

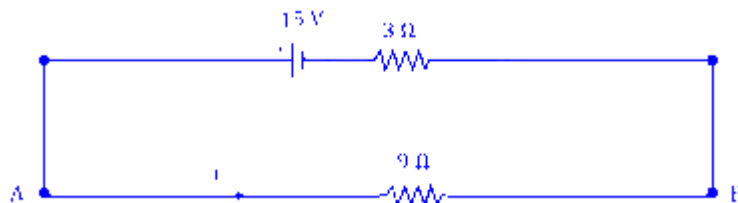
$$E_{\text{total}} = \frac{\frac{15}{9} + \frac{15}{9} + \frac{15}{9}}{\frac{1}{9} + \frac{1}{9} + \frac{1}{9}} = \frac{45}{3} = 15 \text{ V}$$

The total internal resistance is,

$$\frac{1}{r_{\text{total}}} = \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{1}{3} \Rightarrow r_{\text{total}} = 3\Omega$$

And the total external resistance between A and B is $R_{\text{total}} = (3 + 3 + 3)\Omega = 9\Omega$

So, the circuit simplifies to,



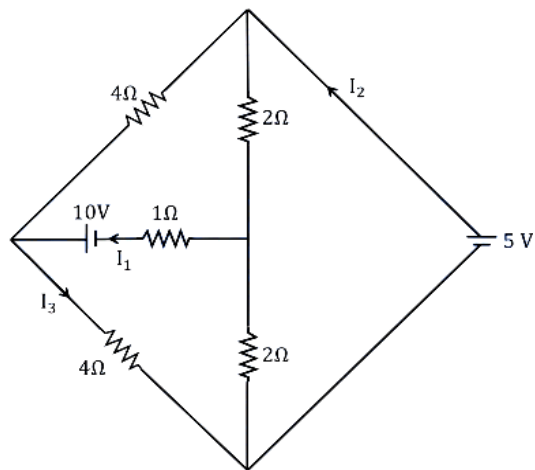
Using Ohm's Law ($I = \frac{V}{R}$) for the entire circuit, the current through A and B will be

$$i = \frac{E_{\text{total}}}{R_{\text{net}}} = \frac{15}{(3+9)} = \frac{15}{12} \text{ A} = 1.25 \text{ A}$$

Thus, the correct option is (B).

Question21

Refer to the figure given below. The values of I_1 , I_2 and I_3 are _____



JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

$$I_1 = 2.5 \text{ A}, I_2 = 1.875 \text{ A}, I_3 = 1.875 \text{ A}$$

B. $I_1 = 1.875 \text{ A}, I_2 = 2.5 \text{ A}, I_3 = 1.875 \text{ A}$

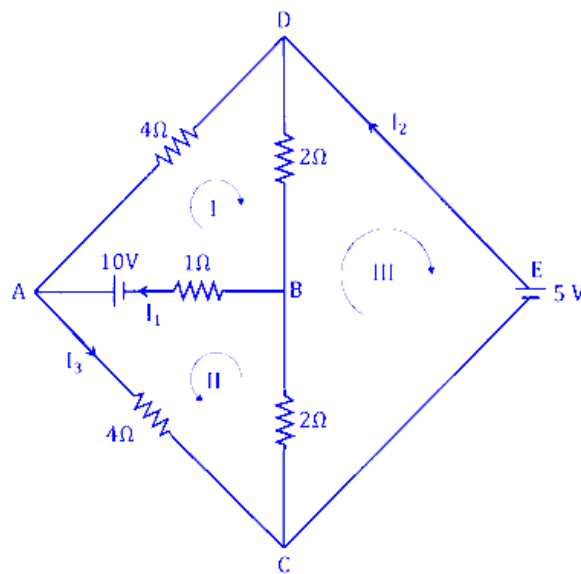
C. $I_1 = 1.875 \text{ A}, I_2 = 1.875 \text{ A}, I_3 = 2.5 \text{ A}$

D.

$I_1 = 2.5 \text{ A}, I_2 = 2.5 \text{ A}, I_3 = 1.875 \text{ A}$

Answer: A

Solution:



Using Kirchhoff's current law, the net accumulation of current at a junction is zero.

So, at junction A, $I_1 - I_3 - I_{AD} = 0 \Rightarrow I_{AD} = I_1 - I_3$

And at junction D, $I_{AD} + I_2 - I_{DB} = 0 \Rightarrow I_1 - I_3 + I_2 - I_{DB} = 0 \Rightarrow I_{DB} = I_1 - I_3 + I_2$

Applying Kirchhoff's voltage law in closed loop I,

$$-4(I_1 - I_3) - 2(I_1 + I_2 - I_3) - I_1 + 10 = 0$$

$$-7I_1 - 2I_2 + 6I_3 = -10 \dots (i)$$

In loop (II)

$$I_1 + 10 - 4I_3 - 2(I_3 - I_2) = 0$$

$$2I_2 - 6I_3 - I_1 = -10 \dots (ii)$$

In loop III,

$$-5 - 2(I_3 - I_2) + 2(I_1 + I_2 - I_3) = 0$$

$$2I_1 + 4I_2 - 4I_3 = 5 \dots (iii)$$

From equation (iii),

$$I_1 = \frac{5-4I_2+4I_3}{2} \dots (iv)$$

On substituting equation (iv) in equation (ii);

$$2I_2 - 6I_3 - \frac{5-4I_2+4I_3}{2} = -10$$

$$\Rightarrow 4I_2 - 12I_3 - 5 + 4I_2 - 4I_3 = -20$$

$$\Rightarrow 8I_2 - 16I_3 = -15$$

$$\Rightarrow I_2 - 2I_3 = -\frac{15}{8}$$

$$\Rightarrow I_2 = 2I_3 - \frac{15}{8} \dots (v)$$

On substituting equation (v) in equation (iv),

$$I_1 = \frac{5-4(2I_3-\frac{15}{8})}{2} = \frac{5-8I_3+\frac{15}{2}+4I_3}{2} = \frac{\frac{25}{2}-4I_3}{2} = \frac{25}{4} - 2I_3$$

$$\Rightarrow I_1 = \frac{25}{4} - 2I_3 \dots (vi)$$

Substituting equation (v) and (vi) in equation (i);

$$-7\left(\frac{25}{4} - 2I_3\right) - 2\left(2I_3 - \frac{15}{8}\right) + 6I_3 = -10$$

$$-\frac{175}{4} + 14I_3 - 4I_3 + \frac{15}{4} + 6I_3 = -10$$

$$\frac{-175 + 15}{4} + 16I_3 = -10$$

$$\frac{-160}{4} + 16I_3 = -10$$

$$-40 + 16I_3 = -10$$

$$16I_3 = 30 \Rightarrow I_3 = \frac{30}{16} \text{ A}$$

$$\Rightarrow I_3 = 1.875 \text{ A}$$

From equation (v)

$$I_2 = 2\left(\frac{15}{8}\right) - \frac{15}{8} = \frac{15}{8} = 1.875 \text{ A}$$

From equation (vi),

$$I_1 = \frac{25}{4} - 2\left(\frac{15}{8}\right) = \frac{25}{4} - \frac{15}{4} = \frac{10}{4} \text{ A} = 2.5 \text{ A}$$

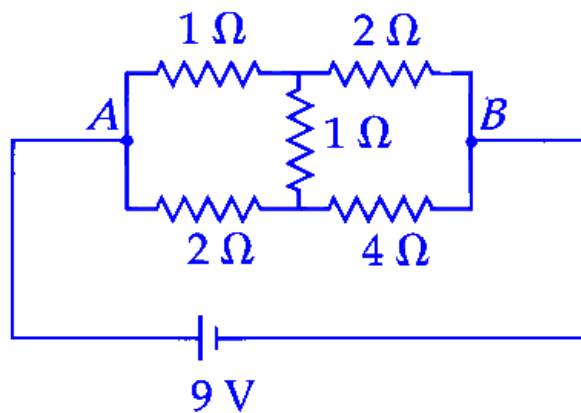
Therefore, the values of currents in the circuit are; $I_1 = 2.5 \text{ A}$, $I_2 = 1.875 \text{ A}$, $I_3 = 1.875 \text{ A}$.

Thus, the correct option is (A).

Question22

The heat generated in 1 minute between points A and B in the given circuit, when a battery of 9 V with internal resistance of 1Ω is

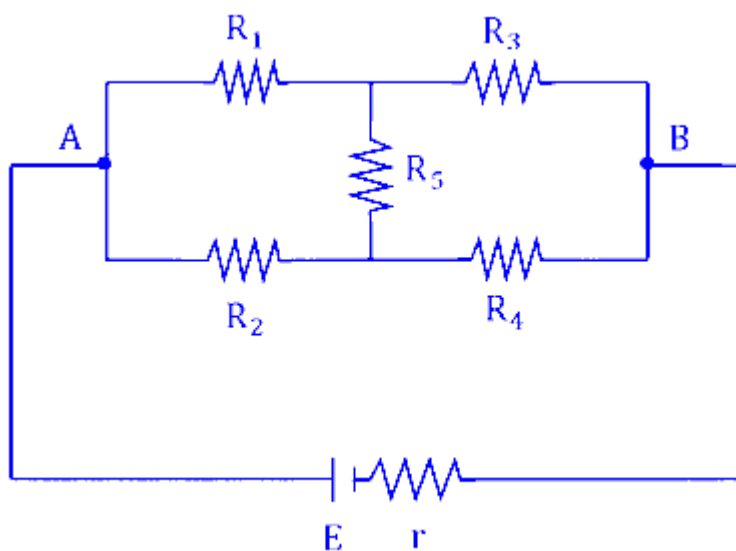
connected across these points is ____ J.



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Answer: 1080

Solution:



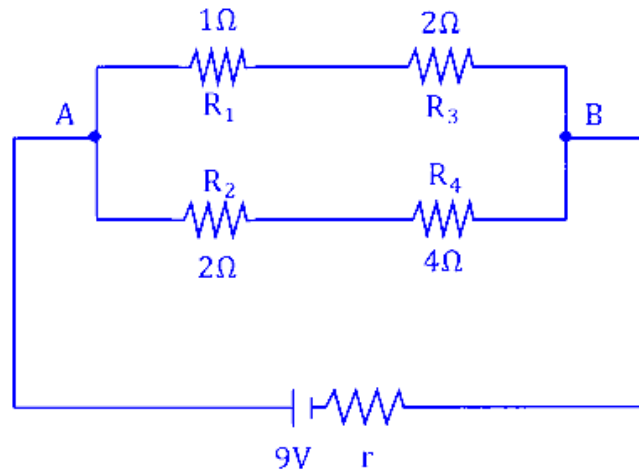
A bridge is balanced if the ratio of the resistances in the two arms is equal :

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

So, from the given circuit, $\frac{R_1}{R_2} = \frac{1}{2}$ and $\frac{R_3}{R_4} = \frac{2}{4} = \frac{1}{2} \Rightarrow \frac{R_1}{R_2} = \frac{R_3}{R_4}$

Since the ratio is equal $\left(\frac{1}{2} = \frac{1}{2}\right)$, the bridge is balanced.

Because the bridge is balanced, no current flows through the central 1Ω resistor. The circuit now simplifies to two parallel branches :



Here, R_1 and R_3 are in series, $R_{13} = 1 + 2 = 3\Omega$

Also, R_2 and R_4 are in series, $R_{24} = 2 + 4 = 6\Omega$

Now, R_{13} and R_{24} are in parallel.

So, the equivalent resistance between A and B (R_{AB}) is :

$$\frac{1}{R_{AB}} = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{1}{2}$$

$$\Rightarrow R_{AB} = 2\Omega$$

The battery has an emf of 9 V and an internal resistance (r) of 1Ω . The total resistance (R_{total}) is:

$$R_{total} = R_{AB} + r = 2 + 1 = 3\Omega$$

Using Ohm's Law for the whole circuit:

$$I = \frac{V}{R_{total}} = \frac{9}{3} = 3 \text{ Amperes}$$

So, the power dissipated between points A and B is,

$$P_{AB} = I^2 \cdot R_{AB}$$

$$\Rightarrow P_{AB} = (3)^2 \times 2 = 9 \times 2 = 18 \text{ Watts}$$

So, the total heat (H) generated in $t = 1 \text{ minute} = 60 \text{ seconds}$ is :

$$H = P_{AB} \times t$$

$$\Rightarrow H = 18 \times 60 = 1080 \text{ J}$$

Therefore, the heat generated in one minute is 1080 J between points A and B .

Hence, the correct answer is 1080.

Question23

A cylindrical conductor of length 2 m and area of cross-section 0.2 mm^2 carries an electric current of 1.6 A when its ends are connected to a 2 V battery. Mobility of electrons in the conductor is $\alpha \times 10^{-3} \text{ m}^2/\text{V.s}$. The value of α is :

(electron concentration $= 5 \times 10^{28}/\text{m}^3$ and electron charge $= 1.6 \times 10^{-19} \text{ C}$)

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Answer: 1

Solution:

The mobility (μ) of free electrons in a conductor is defined as the magnitude of their drift velocity (v_d) per unit electric field (E).

$$\mu = \frac{v_d}{E}$$

The electric field (E) established across a conductor of length L connected to a potential difference V is given by:

$$E = \frac{V}{L}$$

The electric current (I) flowing through a conductor with cross-sectional area A and free electron number density n is:

$$I = neAv_d$$

$$\Rightarrow v_d = \frac{1}{neA}$$

Using mobility formula,

$$\mu = \frac{\left(\frac{1}{neA}\right)}{\left(\frac{V}{L}\right)}$$

$$\Rightarrow \mu = \frac{IL}{neAV}$$

We have given,

- Length, $L = 2 \text{ m}$
- Current, $I = 1.6 \text{ A}$

- Voltage, $V = 2 \text{ V}$
- Electron concentration, $n = 5 \times 10^{28} \text{ m}^{-3}$
- Electron charge, $e = 1.6 \times 10^{-19} \text{ C}$
- Area of cross-section, $A = 0.2 \text{ mm}^2 = 0.2 \times 10^{-6} \text{ m}^2 = 2 \times 10^{-7} \text{ m}^2$

Substituting the values,

$$\mu = \frac{1.6 \times 2}{(5 \times 10^{28}) \times (1.6 \times 10^{-19}) \times (2 \times 10^{-7}) \times 2}$$

$$\Rightarrow \mu = \frac{3.2}{32 \times 10^2}$$

$$\Rightarrow \mu = 0.001 \text{ m}^2/\text{V} \cdot \text{s}$$

$$\Rightarrow \mu = 1 \times 10^{-3} \text{ m}^2/\text{V} \cdot \text{s}$$

$$\Rightarrow 1 \times 10^{-3} = \alpha \times 10^{-3} \Rightarrow \alpha = 1$$

Question24

In a meter bridge experiment to determine the value of unknown resistance, first the resistances 2Ω and 3Ω are connected in the left and right gaps of the bridge and the null point is obtained at a distance l cm from the left. Now when an unknown resistance $x\Omega$ is connected in parallel to 3Ω resistance, the null point is shifted by 10 cm to the right of wire. The value of unknown resistance x is

_____ Ω .

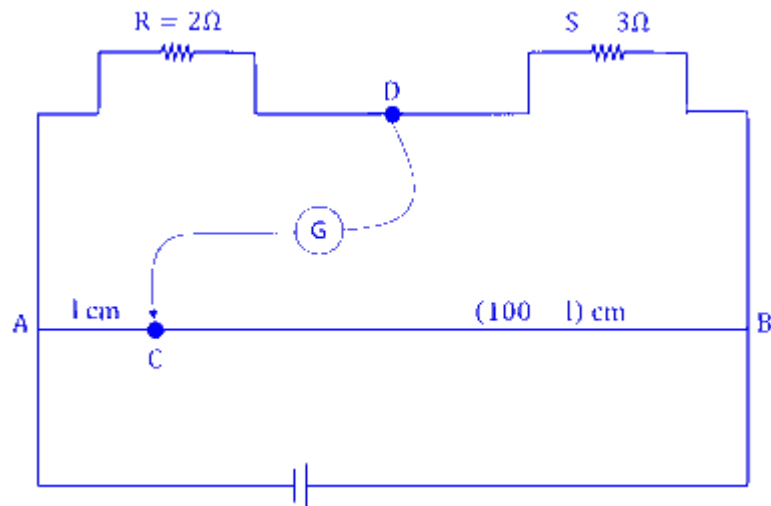
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Answer: 6

Solution:

A meter bridge works on the principle of a balanced Wheatstone bridge. The ratio of the resistances in the gaps is equal to the ratio of the lengths of the wire segments.

Case 1 :



Null point from left is l cm . So, the remaining length from right $(100 - l)$ cm

At balanced condition

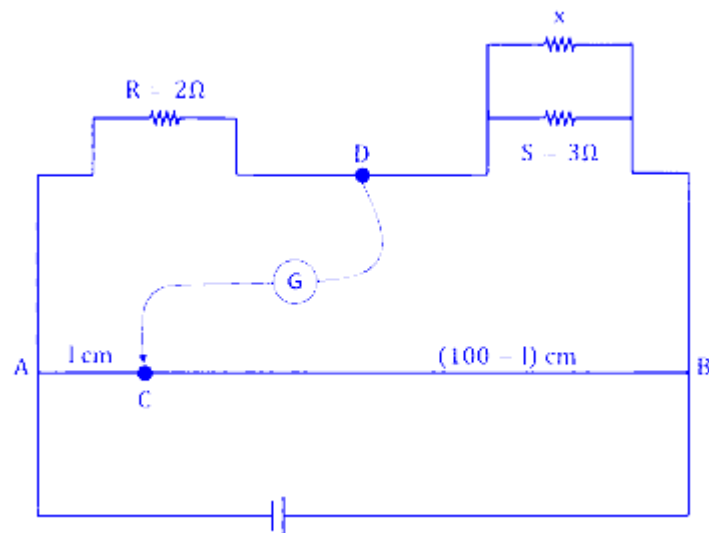
$$\frac{2}{3} = \frac{l}{100-l} \Rightarrow 2(100 - l) = 3l$$

$$\Rightarrow 200 - 2l = 3l$$

$$\Rightarrow 5l = 200 \Rightarrow l = 40 \text{ cm}$$

Case 2 :

When an unknown resistance x is connected in parallel with the 3Ω resistor in the right gap. This changes the equivalent resistance of the right gap.



$$\text{New right gap resistance } S' = \frac{3x}{3+x}$$

$$\text{New null point } l' = l + 10 = 40 + 10 = 50 \text{ cm}$$

$$\text{So, the remaining length from right} = 100 - 50 = 50 \text{ cm}$$

At balanced condition for the second case:

$$\frac{R}{S'} = \frac{l'}{100-l'}$$

$$\Rightarrow \frac{2}{\left(\frac{3x}{3+x}\right)} = \frac{50}{50} \Rightarrow \frac{6+2x}{3x} = 1$$

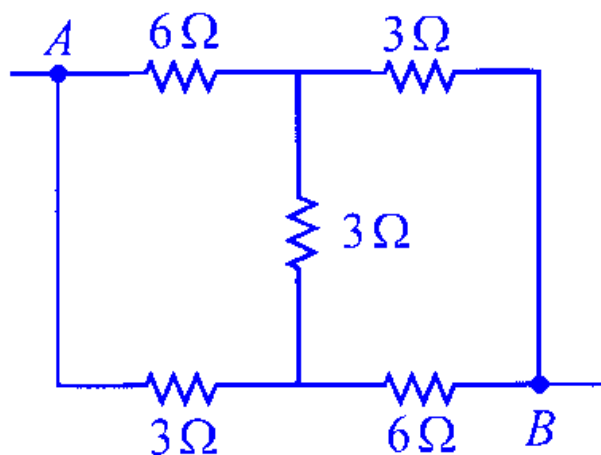
$$\Rightarrow 3x = 6 + 2x \Rightarrow x = 6\Omega$$

The value of unknown resistance x is 6Ω .

Hence, the correct answer is 6.

Question25

The equivalent resistance between the points A and B in the following circuit is $\frac{x}{5}\Omega$. The value of x is ____ .



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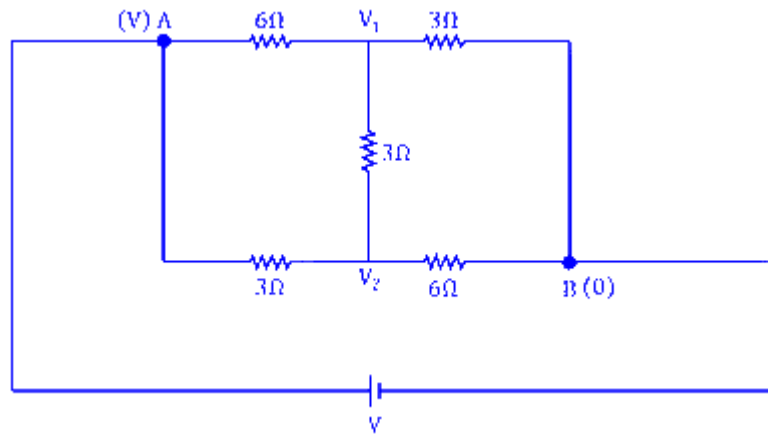
Answer: 21

Solution:

The given circuit is a symmetric unbalanced bridge.

To find the equivalent resistance (R_{AB}), we apply a potential difference V across terminals A and B and determine the total current I entering the circuit.

Let the potential at node A be V (input terminal) and the potential at node B be 0.



Using Kirchhoff's Current Law (KCL) the sum of currents leaving each junction is zero.

At node V_1 :

$$\frac{(V_1 - V)}{6} + \frac{(V_1 - 0)}{3} + \frac{(V_1 - V_2)}{3} = 0$$

$$\Rightarrow (V_1 - V) + 2V_1 + 2(V_1 - V_2) = 0$$

$$\Rightarrow 5V_1 - 2V_2 = V \dots (i)$$

At node V_2 :

$$\frac{V_2 - V}{3} + \frac{V_2 - 0}{6} + \frac{V_2 - V_1}{3} = 0$$

$$\Rightarrow 2(V_2 - V) + V_2 + 2(V_2 - V_1) = 0$$

$$\Rightarrow -2V_1 + 5V_2 = 2V \dots (ii)$$

Solving both the equations, (i) $\times 5$ + (ii) $\times 2$

$$(5V_1 - 2V_2) \times 5 + (-2V_1 + 5V_2) \times 2 = V \times 5 + 2V \times 2$$

$$\Rightarrow (25V_1 - 10V_2) + (-4V_1 + 10V_2) = 5V + 4V$$

$$\Rightarrow 21V_1 = 9V \Rightarrow V_1 = \frac{9}{21}V = \frac{3}{7}V$$

Substituting V_1 into Equation (i) to find V_2 :

$$5 \left(\frac{3}{7} V \right) - 2V_2 = V$$

$$2V_2 = \frac{15}{7} V - V = \frac{8}{7} V \Rightarrow V_2 = \frac{4}{7} V$$

The total current I entering from terminal A is the sum of currents in the two left branches :

$$I = \frac{V - V_1}{6} + \frac{V - V_2}{3}$$

$$\Rightarrow I = \frac{V - \frac{3}{7}V}{6} + \frac{V - \frac{4}{7}V}{3}$$

$$\Rightarrow I = \frac{\frac{4V}{7}}{6} + \frac{\frac{3V}{7}}{3} = \frac{2V}{21} + \frac{V}{7}$$

$$I = \frac{2V + 3V}{21} = \frac{5V}{21}$$

The equivalent resistance R_{AB} is :

$$R_{AB} = \frac{V}{I} = \frac{V}{5V/21} = \frac{21}{5} \Omega$$

Given that $R_{AB} = \frac{x}{5} \Omega$:

$$\frac{x}{5} = \frac{21}{5} \Rightarrow x = 21$$

Therefore, the value of x is 21.

Question26

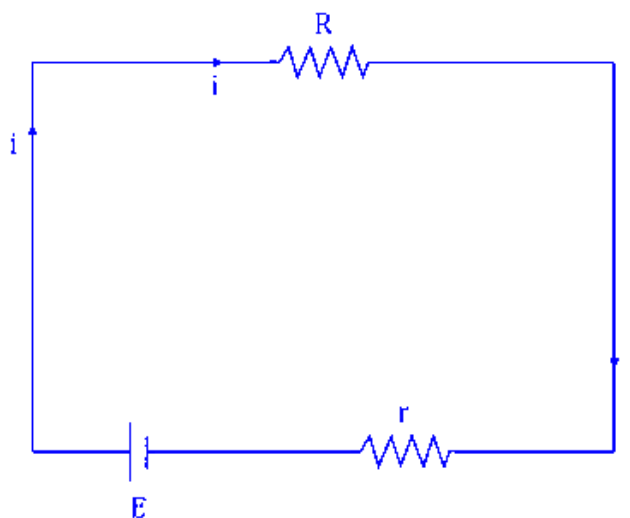
When an external resistance of 5Ω is connected across terminals of a cell, a current of 0.25 A flows through it. When the 5Ω resistor is replaced by a 2Ω resistor, a current of 0.5 A flows through it. The internal resistance of the cell is _____ Ω .

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Answer: 1

Solution:

For a cell with EMF E and internal resistance r connected to an external resistance R .



The total resistance in the circuit is

$$R_{\text{total}} = R + r$$

According to Ohm's law, the current I is :

$$I = \frac{E}{R_{\text{total}}} = \frac{E}{R+r}$$

$$\Rightarrow E = I(R + r)$$

Case 1 when the external resistance connected is $R_1 = 5\Omega$ and the current through it is $I_1 = 0.25 \text{ A}$.

$$E = 0.25(5 + r) \dots (i)$$

Case 2 when the external resistance connected is $R_2 = 20\Omega$ and the current through it is $I_2 = 0.1 \text{ A}$.

$$E = 0.5(2 + r) \dots (ii)$$

From both the equations,

$$0.25(5 + r) = 0.5(2 + r)$$

$$\Rightarrow \frac{1}{4}(5 + r) = \frac{1}{2}(2 + r)$$

$$\Rightarrow 5 + r = 2(2 + r)$$

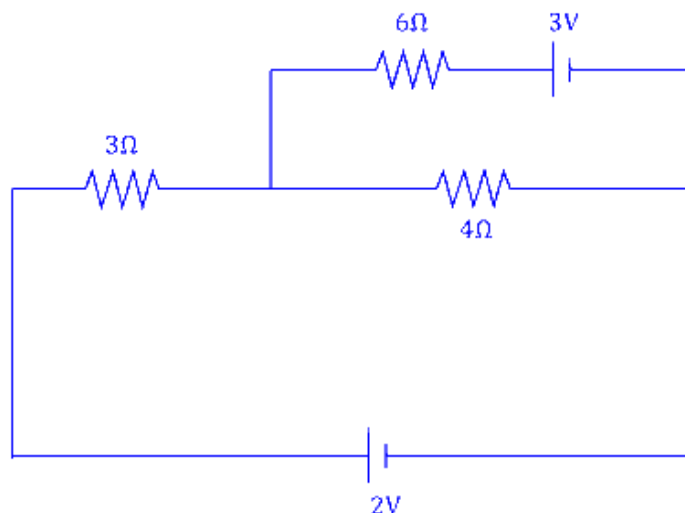
$$\Rightarrow 5 + r = 4 + 2r$$

$$\Rightarrow r = 1\Omega$$

Therefore, the internal resistance of the cell is 1Ω .

Question27

Refer to the circuit diagram given below. The heat generated across the 6Ω resistance in 100 second is $\frac{\alpha}{100} \text{ J}$. The value of α is ____ .
(Nearest integer)



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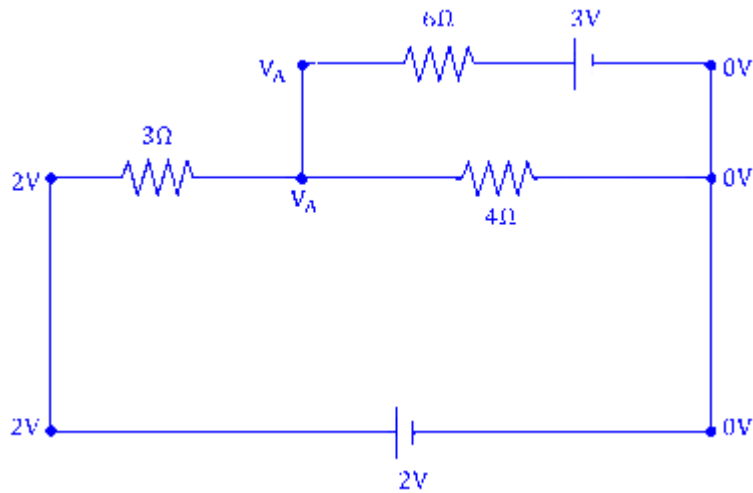
Answer: 3477

Solution:

The heat (H) generated in a resistor (R) due to a current (i) flowing for a time (t) is :

$$H = i^2 R t$$

Let V_A be the potential at the junction A . Using Kirchhoff's current law, the sum of currents = 0 :



$$\frac{V_A - 3}{6} + \frac{V_A - 0}{4} + \frac{V_A - 2}{3} = 0$$

$$\Rightarrow 2(V_A - 3) + 3V_A + 4(V_A - 2) = 0$$

$$\Rightarrow 2V_A - 6 + 3V_A + 4V_A - 8 = 0$$

$$\Rightarrow 9V_A = 14 \Rightarrow V_A = \frac{14}{9} \text{ V}$$

So, the current through the 6Ω resistor is,

$$i = \frac{3 - V_A}{6} = \frac{3 - 14/9}{6} = \frac{27 - 14}{9 \times 6} = \frac{13}{54} \text{ A}$$

The time is $t = 100 \text{ s}$.

Hence, total heat generated in 6Ω is;

$$H = \left(\frac{13}{54}\right)^2 \times 6 \times 100 = \frac{169}{2916} \times 600 \approx 34.77 \text{ J}$$

It is given that $H = \frac{\alpha}{100}$, then $\alpha = H \times 100$:

$$\Rightarrow \alpha \approx 3477$$

Therefore, the correct answer is 3477.

Question28

Two cells of emfs 1 V and 2 V and internal resistance 2Ω and 1Ω , respectively connected in parallel, gave a current of 1 A through an external resistance. If the polarity of one cell is reversed, then value of current through the external resistance will be $\frac{\alpha}{5}$ A. The value of α is _____.

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Answer: 3

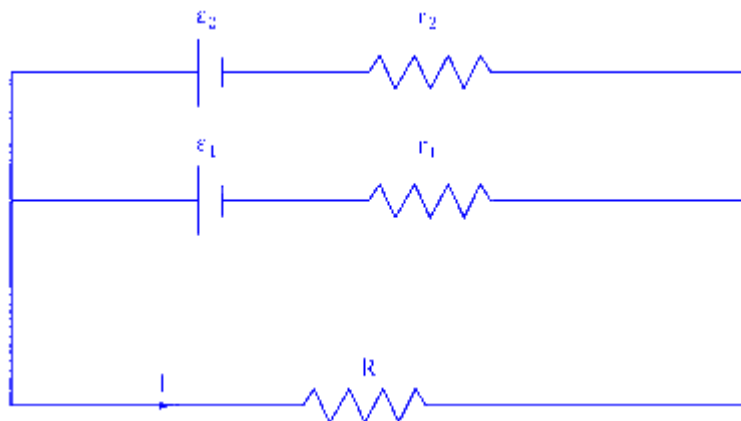
Solution:

When two cells with emfs $\varepsilon_1, \varepsilon_2$ and internal resistances r_1, r_2 are connected in parallel, the equivalent emf (ε_{eq}) and equivalent internal resistance (r_{eq}) are :

$$\varepsilon_{eq} = \frac{\frac{\varepsilon_1 + \varepsilon_2}{\frac{1}{r_1} + \frac{1}{r_2}}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

Case 1 : Cells are connected with same polarity



The voltages and internal resistances of two cells are $\varepsilon_1 = 1$ V, $r_1 = 2\Omega$ and $\varepsilon_2 = 2$ V, $r_2 = 1\Omega$.

$$\varepsilon_{eq} = \frac{(1 \times 1) + (2 \times 2)}{2 + 1} = \frac{1 + 4}{3} = \frac{5}{3} \text{ V}$$

The equivalent internal resistance is,

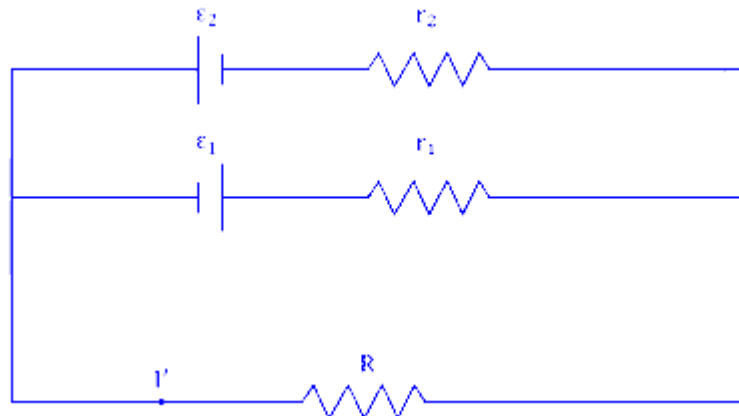
$$r_{eq} = \frac{2 \times 1}{2 + 1} = \frac{2}{3} \Omega$$

The current I through external resistance R is :

$$I = \frac{\varepsilon_{eq}}{R + r_{eq}}$$

$$\Rightarrow 1 = \frac{5/3}{R + \frac{2}{3}} \Rightarrow R + \frac{2}{3} = \frac{5}{3} \Rightarrow R = 1\Omega$$

Case 2 : Polarity of One Cell Reversed



If we reverse the 1 V cell, its emf becomes -1 V .

The equivalent emf of combination is,

$$\varepsilon'_{eq} = \frac{(-1 \times 1) + (2 \times 2)}{2 + 1} = \frac{-1 + 4}{3} = \frac{3}{3} = 1 \text{ V}$$

The equivalent internal resistance r_{eq} remains $\frac{2}{3}\Omega$.

So, the new current I' across the external resistance is:

$$I' = \frac{\varepsilon'_{eq}}{R + r_{eq}} = \frac{1}{1 + \frac{2}{3}} = \frac{1}{5/3} = \frac{3}{5} \text{ A}$$

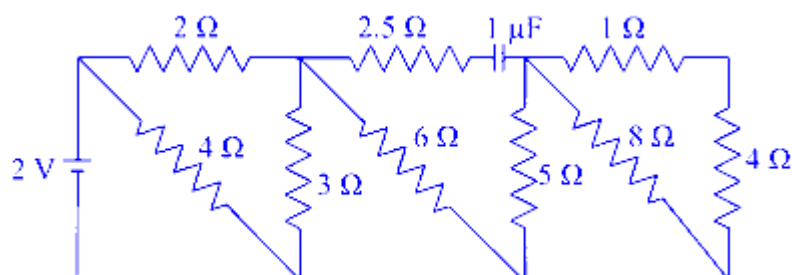
It is given that $I' = \frac{\alpha}{5}$:

$$\frac{3}{5} = \frac{\alpha}{5} \Rightarrow \alpha = 3$$

Therefore, the value of α is 3 .

Question29

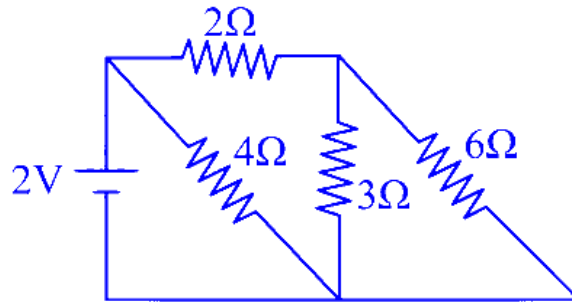
The net current flowing in the given circuit is _____ A.



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Answer: 1

Solution:



$$R_{\text{eq}} = 2\Omega$$

$$I = \frac{2}{2} = 1 \text{ A}$$

Question30

A wire of resistance 9Ω is bent to form an equilateral triangle. Then the equivalent resistance across any two vertices will be _____ ohm.

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Answer: 2

Solution:

Each side of the triangle has a resistance of $R_{\text{side}} = \frac{9\Omega}{3} = 3\Omega$.

When measuring the equivalent resistance between any two vertices, there are two paths:

A direct path along one side with resistance:

$$R_1 = 3 \Omega.$$

An indirect path passing through the other two sides in series:

$$R_2 = 3 \Omega + 3 \Omega = 6 \Omega.$$

These two paths are in parallel, so the equivalent resistance is calculated by:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{3 \Omega} + \frac{1}{6 \Omega} = \frac{2}{6 \Omega} + \frac{1}{6 \Omega} = \frac{3}{6 \Omega}.$$

Thus,

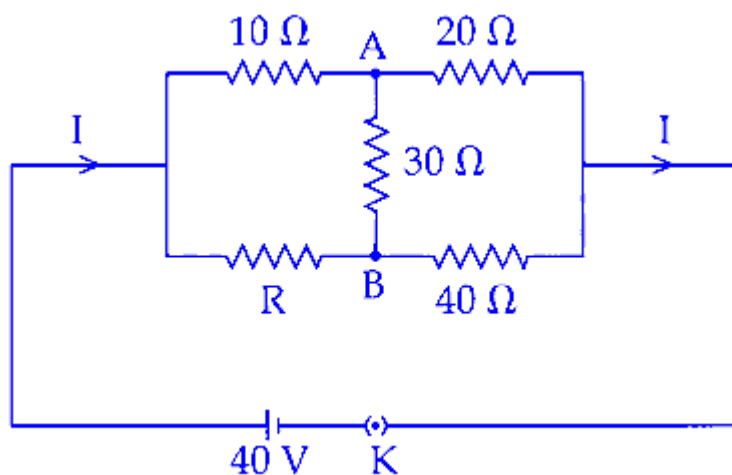
$$R_{eq} = \frac{6 \Omega}{3} = 2 \Omega.$$

The equivalent resistance across any two vertices of the equilateral triangle is therefore:

$$2 \Omega.$$

Question31

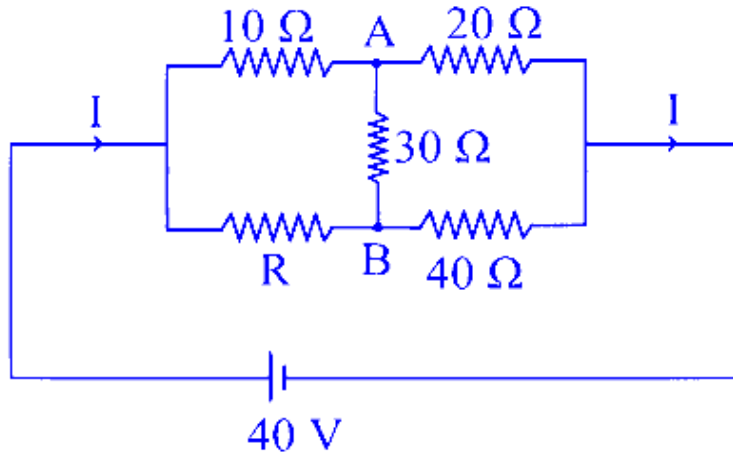
The value of current I in the electrical circuit as given below, when potential at A is equal to the potential at B, will be _____ A.



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Answer: 2

Solution:



$V_A = V_B \Rightarrow$ the bridge is balanced

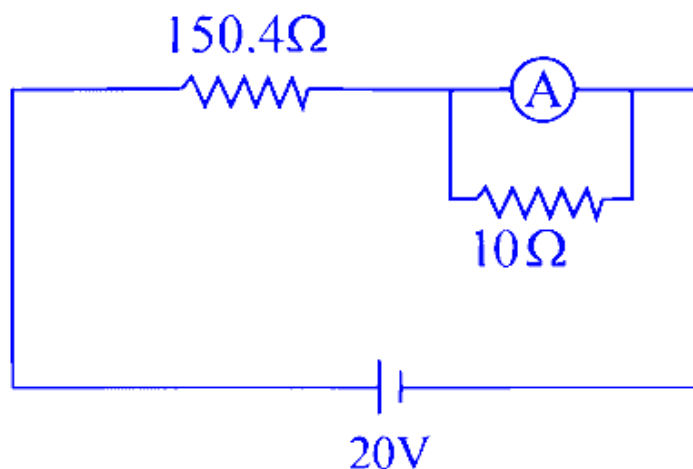
$$\Rightarrow \frac{10}{R} = \frac{20}{40}$$

$$R = 20\Omega$$

$$I = \frac{40}{20} = 2 \text{ A}$$

Question32

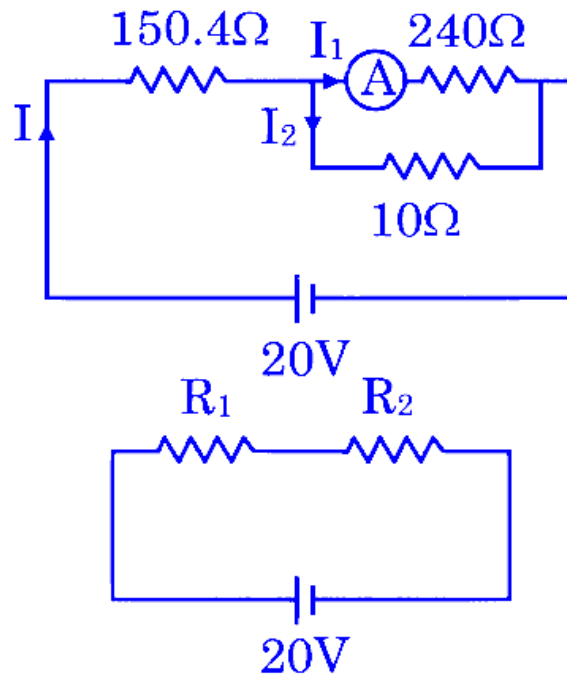
In the figure shown below, a resistance of 150.4Ω is connected in series to an ammeter A of resistance 240Ω . A shunt resistance of 10Ω is connected in parallel with the ammeter. The reading of the ammeter is _____ mA .



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Answer: 5

Solution:



$$R_{eq} = R_1 + R_2$$

$$R_{eq} = 150.4 + \frac{240 \times 10}{250}$$
$$= 150.4 + 9.6 = 160\Omega$$

$$I_1 = \frac{IR_2}{240}$$

$$I_1 = \frac{I \times 9.6}{240}$$

$$= \frac{20}{160} \times \frac{9.6}{2400} = \frac{1}{200} = 5 \times 10^{-3} \text{ A} = 5 \text{ mA}$$

Question33

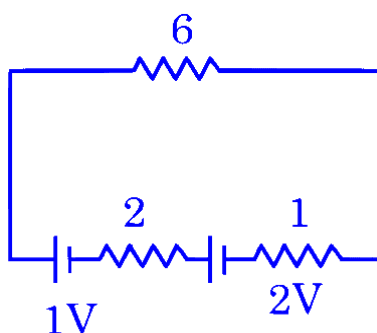
Two cells of emfs 1 V and 2 V and internal resistances 2Ω and 1Ω , respectively, are connected in series with an external resistance of 6Ω . The total current in the circuit is I_1 . Now the same two cells in parallel configuration are connected to same external resistance. In

this case, the total current drawn is I_2 . The value of $\left(\frac{I_1}{I_2}\right)$ is $\frac{x}{3}$. The value of x is _____.

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Answer: 4

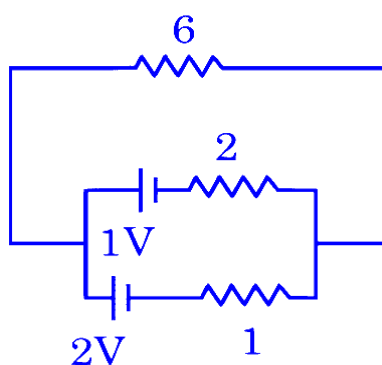
Solution:



$$\varepsilon_{\text{cq}} = 3$$

$$R_{\text{cq}} = 9$$

$$i_1 = \frac{3}{9} = \frac{1}{3}$$



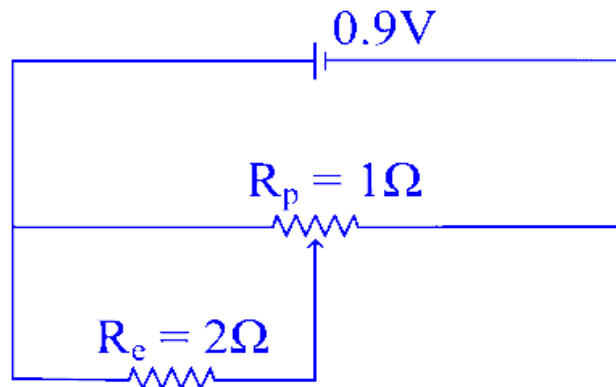
$$\varepsilon_{\text{eq}} = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}$$

$$\varepsilon_{\text{eq}} = \frac{\frac{1}{2} + \frac{2}{1}}{\frac{1}{2} + \frac{1}{1}} = \frac{5}{3}$$

$$r_{\text{equ}} = \frac{2 \times 1}{3} + 6 = \frac{20}{3}$$

$$i_2 = \frac{1}{4} \Rightarrow \frac{i_1}{i_2} = \frac{4}{3}$$

Question34



Sliding contact of a potentiometer is in the middle of the potentiometer wire having resistance $R_p = 1\Omega$ as shown in the figure. An external resistance of $R_e = 2\Omega$ is connected via the sliding contact. The electric current in the circuit is :

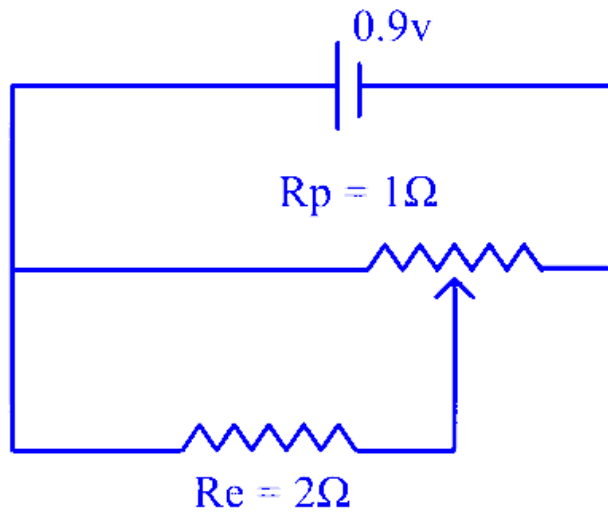
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Options:

- A. 1.35 A
- B. 0.9 A
- C. 1.0 A
- D. 0.3 A

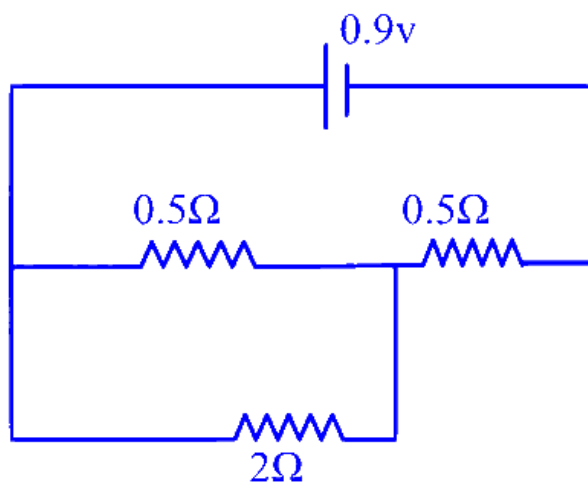
Answer: C

Solution:



We know, $R = \rho \frac{l}{A} \Rightarrow R \propto l$

As sliding contact of a potentiometer is in the middle of the potentiometer wire.



$$\therefore R_{eq} = 0.5 + \frac{0.5 \times 2}{0.5 + 2} = 0.5 + \frac{1}{2.5}$$

$$= \frac{1}{2} + \frac{2}{5} = \frac{5+4}{10} = \frac{9}{10} = 0.9\Omega$$

$$\text{Hence, } I = \frac{v}{R_{eq}} = \frac{0.9v}{0.9\Omega} \Rightarrow I = 1A$$

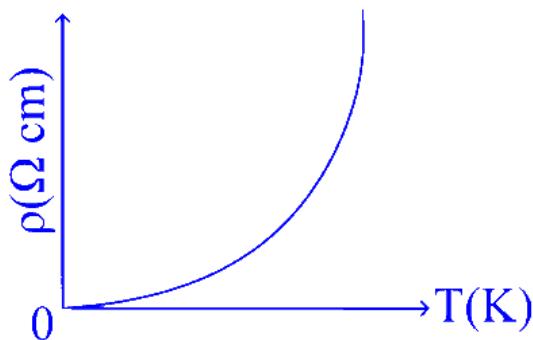
Question35

Which of the following resistivity (ρ) v/s temperature (T) curves is most suitable to be used in wire bound standard resistors?

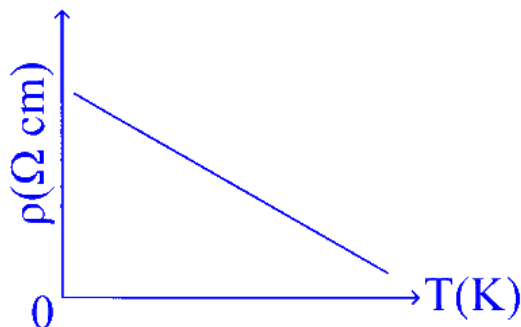
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Options:

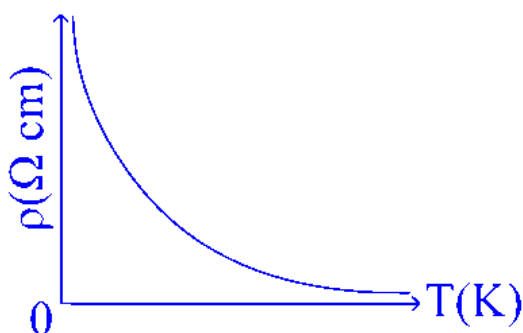
A.



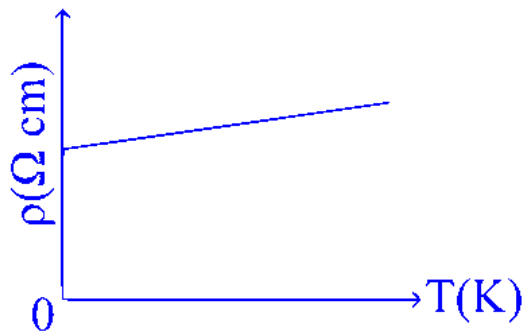
B.



C.



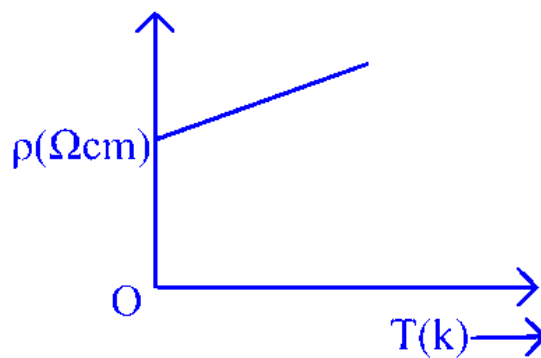
D.



Answer: D

Solution:

We use Nichrome, manganin and constantan to make standard (reference) wire bound resistors their resistivity is less dependent on temperature. So ρ vs T curve will not show any sharp change.



Hence, option 4 is correct.

Question36

Given below are two statements :

Statement-I : The equivalent emf of two nonideal batteries connected in parallel is smaller than either of the two emfs.

Statement-II : The equivalent internal resistance of two nonideal batteries connected in parallel is smaller than the internal resistance

of either of the two batteries.

In the light of the above statements, choose the correct answer from the options given below.

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

- A. Statement-I is false but Statement-II is true
- B. Statement-I is true but Statement-II is false
- C. Both Statement-I and Statement-II are false
- D. Both Statement-I and Statement-II are true

Answer: A

Solution:

When two non-ideal batteries are connected in parallel, their equivalent emf is

$$E_{eq} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} \Rightarrow E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

If emf of both cells is same say E,

$$E_{eq} = \frac{E r_2 + E r_1}{r_1 + r_2} = \frac{E(r_1 + r_2)}{(r_1 + r_2)} = E$$

Hence, equivalent emf is equal to emf of the single cell.

Also, when two non-ideal batteries are connected in parallel, their equivalent resistance is $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$

Hence, the equivalent internal resistance is smaller than either of the two internal resistances. Therefore 1 is correct.

Question37

Consider a moving coil galvanometer (MCG):

A. The torsional constant in moving coil galvanometer has dimensions $[ML^2 T^{-2}]$

B. Increasing the current sensitivity may not necessarily increase the voltage sensitivity.

C. If we increase number of turns (N) to its double (2 N), then the voltage sensitivity doubles.

D. MCG can be converted into an ammeter by introducing a shunt resistance of large value in parallel with galvanometer.

E. Current sensitivity of MCG depends inversely on number of turns of coil.

Choose the correct answer from the options given below:

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Options:

A. A, D Only

B. A, B Only

C. A, B, E Only

D. B, D, E Only

Answer: B

Solution:

Let's analyze each statement step by step.

Statement A: "The torsional constant in moving coil galvanometer has dimensions $[ML^2T^{-2}]$."

The torsional constant (often denoted by k) relates the restoring torque to the angular displacement via $\tau = k\theta$. Since torque has dimensions of force $\times$ distance with dimensions $[MLT^{-2} \times L = ML^2T^{-2}]$ and the angular displacement (in radians) is dimensionless, the torsional constant indeed has the dimensions $[ML^2T^{-2}]$.

Thus, Statement A is correct.

Statement B: “Increasing the current sensitivity may not necessarily increase the voltage sensitivity.”

The current sensitivity of a galvanometer is defined as its deflection per unit current (i.e. θ/I).

The voltage sensitivity, on the other hand, is the deflection per unit applied voltage when the galvanometer is used as a voltmeter. This sensitivity is proportional to the current sensitivity divided by the galvanometer’s internal resistance.

Often, if you try to increase current sensitivity (for instance, by increasing the number of turns or enhancing the magnetic field), the internal resistance may also change (typically increase when number of turns is increased). Therefore, the improvement in current sensitivity does not automatically translate to a proportional increase in voltage sensitivity.

Hence, Statement B is true.

Statement C: “If we increase number of turns (N) to its double ($2N$), then the voltage sensitivity doubles.”

The deflection (current sensitivity) is proportional to N since $\theta \propto NI$.

However, when you double N , the length of the wire increases, and so does the internal resistance (approximately doubling it, assuming the wire diameter and material remain the same).

Since voltage sensitivity is roughly given by $\frac{\text{current sensitivity}}{\text{internal resistance}} \propto \frac{N}{R_g}$, if both N and R_g double, the ratio remains essentially unchanged.

Therefore, doubling N does not double the voltage sensitivity, so Statement C is incorrect.

Statement D: “MCG can be converted into an ammeter by introducing a shunt resistance of large value in parallel with galvanometer.”

To convert a galvanometer into an ammeter, a shunt resistor is used to bypass most of the current so that only a small fraction (the full-scale deflection current) passes through the galvanometer.

For this to work, the shunt resistance should be very low compared to the galvanometer’s resistance, not of “large value.”

Therefore, Statement D is false.

Statement E: “Current sensitivity of MCG depends inversely on number of turns of coil.”

As mentioned before, the electromagnetic torque in the coil is given by $\tau_e \propto NI$, so the deflection per unit current is $\theta/I \propto N$.

This shows that the current sensitivity increases with N (i.e. it is directly proportional to N), not inversely.

Hence, Statement E is false.

Only Statements A and B are correct.

Looking at the given options, the correct choice is:

Option B: A, B Only.

Question38

A galvanometer having a coil of resistance 30Ω need 20 mA of current for full-scale deflection. If a maximum current of 3 A is to be measured using this galvanometer, the resistance of the shunt to be added to the galvanometer should be $\frac{30}{X}\Omega$, where X is

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Options:

A. 447

B. 298

C. 149

D. 596

Answer: C

Solution:

Given:

Galvanometer resistance:

$$R_g = 30\Omega$$

Full-scale deflection current:

$$I_g = 20\text{ mA} = 0.02\text{ A}$$

Maximum current to be measured:

$$I = 3\text{ A}$$

When using the galvanometer to measure 3 A , the extra current passing through the shunt resistor (R_s) is:

$$I_s = I - I_g = 3 - 0.02 = 2.98\text{ A}.$$

Since the galvanometer and the shunt resistor are connected in parallel, their voltage drops must be equal. Therefore:

$$I_g R_g = I_s R_s.$$

Substitute the given values:

$$0.02 \times 30 = 2.98 \times R_s \quad \Rightarrow \quad R_s = \frac{0.6}{2.98} \approx 0.2013\Omega.$$

The shunt resistance is represented as:

$$R_s = \frac{30}{X} \Omega.$$

Equate the two expressions:

$$\frac{30}{X} = 0.2013.$$

Solving for X :

$$X = \frac{30}{0.2013} \approx 149.$$

Thus, the value of X is approximately 149.

Question39

A wire of resistance R is bent into an equilateral triangle and an identical wire is bent into a square. The ratio of resistance between the two end points of an edge of the triangle to that of the square is

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Options:

A. $8/9$

B. $9/8$

C. $32/27$

D. $27/32$

Answer: C

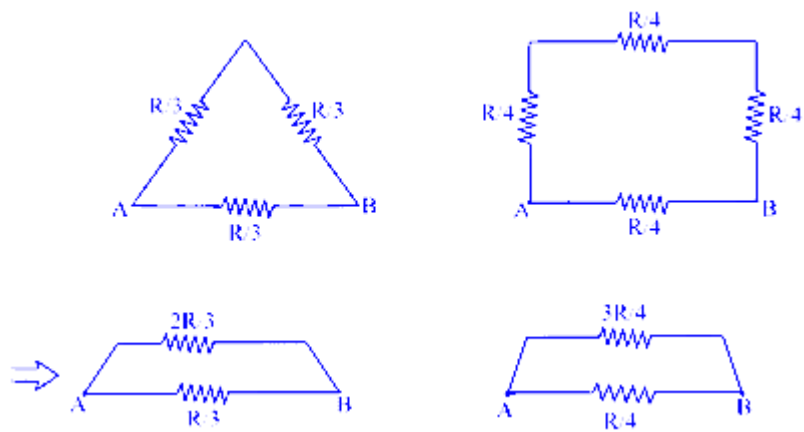
Solution:

We know, $R = \rho \frac{l}{A} \Rightarrow R \propto l$

Side length of triangle = $\frac{l}{3}$

Side length of square = $\frac{l}{4}$

So,



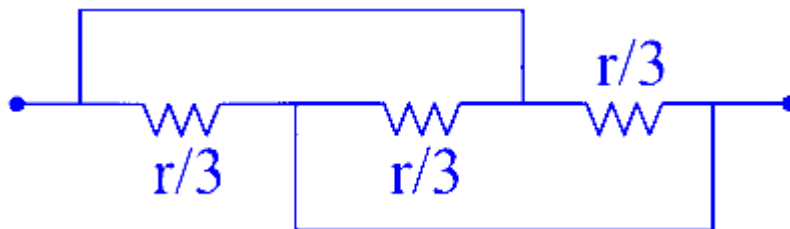
$$\Rightarrow (R_{eq})_1 = \frac{\frac{2R}{3} \times \frac{R}{3}}{\frac{2R}{3} + \frac{R}{3}} \text{ and } (R_{eq})_2 = \frac{\frac{3R}{4} \times \frac{R}{4}}{\frac{3R}{4} + \frac{R}{4}}$$

$$= \frac{2R^2}{9} \times \frac{3}{3R} = \frac{2R}{9} \text{ and } (R_{eq})_2 = \frac{3R^2}{16} - R = \frac{3R}{16}$$

$$\text{Hence, } \frac{(R_{eq})_1}{(R_{eq})_2} = \frac{\frac{2R}{9}}{\frac{3R}{16}} = \frac{32}{27}$$

Question40

Find the equivalent resistance between two ends of the following circuit



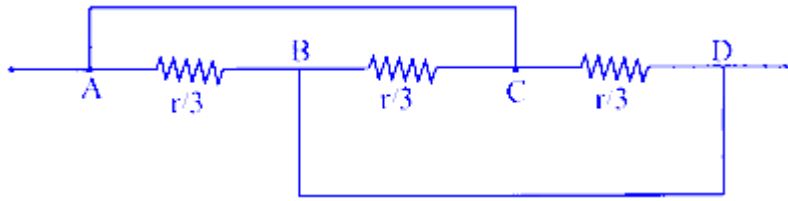
JEE Main 2025 (Online) 28th January Morning Shift

Options:

- A. r
- B. $\frac{r}{6}$
- C. $\frac{r}{3}$
- D. $\frac{r}{9}$

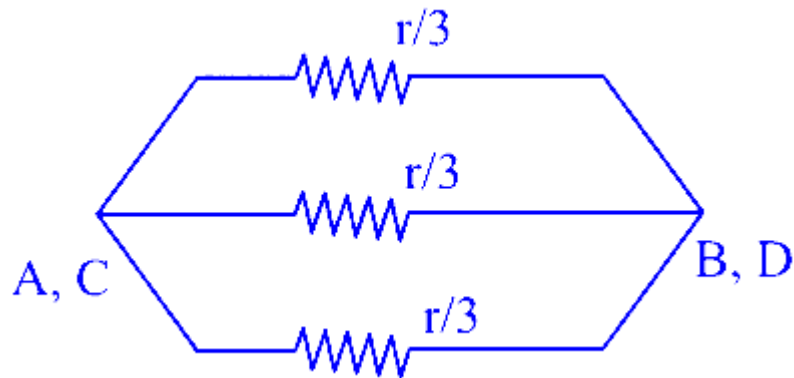
Answer: D

Solution:



Here, $V_A = V_C$ & $V_B = V_D$

So,



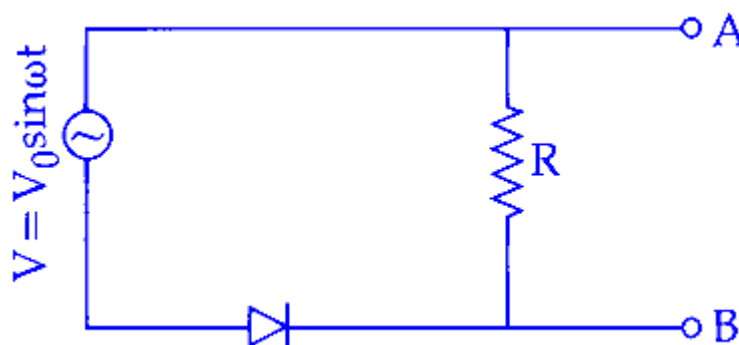
So resistances are in parallel combination.

We know, $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$\Rightarrow \frac{1}{R_p} = \frac{3}{r} + \frac{3}{r} + \frac{3}{r} = \frac{9}{r}$$

$$\Rightarrow R_p = \frac{r}{9}$$

Question41

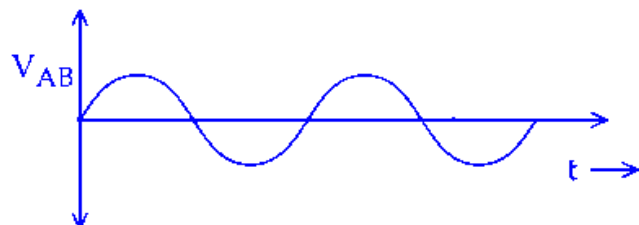


In the circuit shown here, assuming threshold voltage of diode is negligibly small, then voltage V_{AB} is correctly represented by:

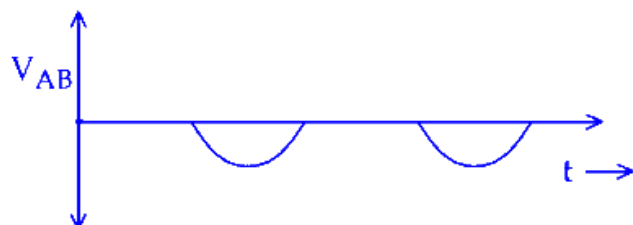
JEE Main 2025 (Online) 28th January Evening Shift

Options:

A.



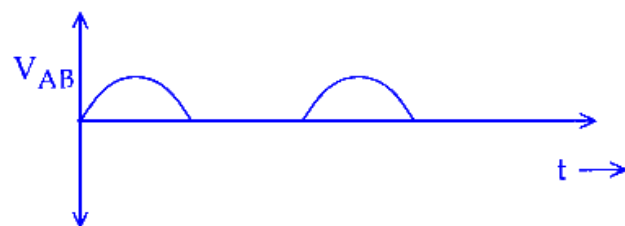
B.



C.

V_{AB} would be zero at all times.

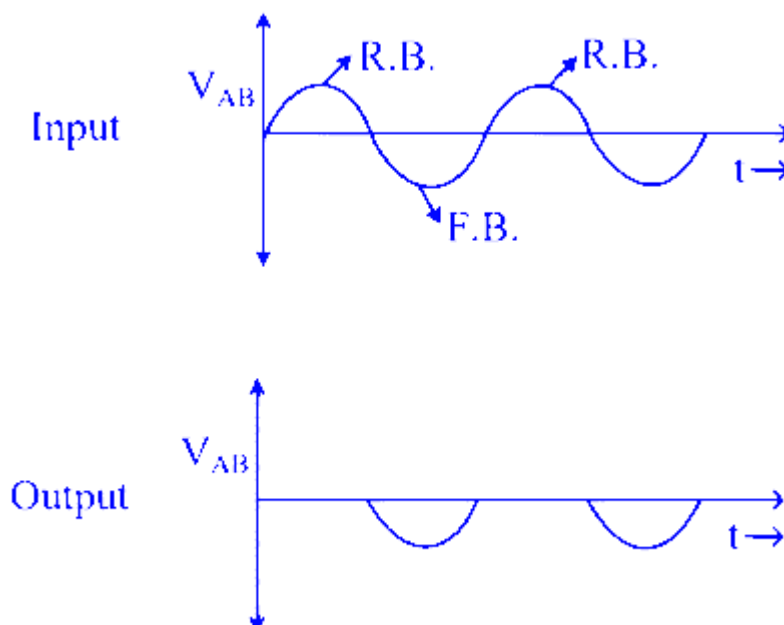
D.



Answer: B

Solution:

$$V = V_0 \sin \omega t$$



Question42

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Choke coil is simply a coil having a large inductance but a small resistance. Choke coils are used with fluorescent mercury-tube fittings. If household electric power is directly connected to a mercury tube, the tube will be damaged.

Reason (R): By using the choke coil, the voltage across the tube is reduced by a factor $\left(R/\sqrt{R^2 + \omega^2 L^2} \right)$, where ω is frequency of the supply across resistor R and inductor L . If the choke coil were not used, the voltage across the resistor would be the same as the applied voltage.

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

A.

(A) is true but (R) is false

B.

Both (A) and (R) are true and (R) is the correct explanation of (A) 3.

C.

Both (A) and (R) are true but (R) is not the correct explanation of (A)

D.

(A) is false but (R) is true

Answer: B

Solution:

Assertion (A):

A choke coil is simply a coil having large inductance but small resistance.

Choke coils are used with fluorescent mercury-tube fittings.

If household electric power is directly connected to a mercury tube, the tube will be damaged.

A choke (or ballast) **does** have high inductance and low resistance.

In a fluorescent lamp (or mercury-vapor lamp) circuit, the choke limits the **current** through the tube.

If we connect the tube directly to the full line voltage **without** any current-limiting device, excessive current will flow and damage the tube.

Therefore, (A) **is true**.

Reason (R):

By using the choke coil, the voltage across the tube is reduced by a factor

$$\frac{R}{\sqrt{R^2 + (\omega L)^2}},$$

where ω is the supply frequency, and R and L are the resistance and inductance in series.

If the choke coil were not used, the voltage across the resistor (tube) would be the same as the applied voltage.

In a simple **series R–L** circuit (where R represents the lamp's effective resistance once conducting, and L is the choke's inductance), the total impedance is

$$Z = \sqrt{R^2 + (\omega L)^2}.$$

The current is the same through R and L . The fraction of the **total supply voltage** appearing across R alone is

$$\frac{V_R}{V_{\text{total}}} = \frac{IR}{IZ} = \frac{R}{\sqrt{R^2 + (\omega L)^2}}.$$

This shows that using a large inductance L (the choke) can substantially reduce (or limit) the voltage—and hence the current—through the lamp.

Without the choke coil, the tube (treated here as a resistor once it starts conducting) would indeed see nearly the **full line voltage**, causing a large current that could destroy it.

Therefore, **(R) is also true**.

Does (R) correctly explain (A)?

(A) states *why* a choke coil is needed (to prevent the lamp from being damaged by excessive current).

(R) shows *how* the choke coil limits the voltage/current through the lamp—by giving the fraction of the supply voltage that appears across the lamp in a series R–L circuit.

Thus, (R) **does** provide the correct (physics-based) explanation: the choke's inductive reactance drops a significant portion of the supply voltage, thereby protecting the lamp.

Hence, the correct option is:

Option B: Both (A) and (R) are true and (R) is the correct explanation of (A).

Question43

The battery of a mobile phone is rated as 4.2 V, 5800 mAh. How much energy is stored in it when fully charged?

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. 87.7 kJ

B. 43.8 kJ

C. 48.7 kJ

D. 24.4 kJ

Answer: A

Solution:

To find the energy stored in the battery, follow these steps:

Convert mAh to Ah:

The battery capacity is given as 5800 mAh. Since

$$1 \text{ Ah} = 1000 \text{ mAh},$$

we have

$$5800 \text{ mAh} = 5.8 \text{ Ah}.$$

Calculate energy in watt-hours (Wh):

The energy in Wh is given by the product of the voltage and the capacity in Ah. Thus,

$$\text{Energy (Wh)} = 4.2 \text{ V} \times 5.8 \text{ Ah} = 24.36 \text{ Wh}.$$

Convert watt-hours to joules (J):

Since

$$1 \text{ Wh} = 3600 \text{ J},$$

the total energy in joules is:

$$24.36 \text{ Wh} \times 3600 \text{ J/Wh} = 87,696 \text{ J} \approx 87.7 \text{ kJ}.$$

Thus, the energy stored in the battery when fully charged is about 87.7 kJ.

The correct answer is Option A: 87.7 kJ.

Question44

A wire of length 25 m and cross-sectional area 5 mm^2 having resistivity of $2 \times 10^{-6} \Omega \text{ m}$ is bent into a complete circle. The resistance between diametrically opposite points will be

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. 100Ω

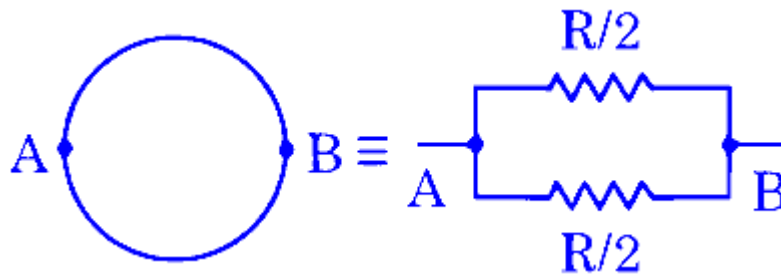
B. 2.5Ω

C. 12.5Ω

D. 50Ω

Answer: B

Solution:



$$L = 25 \text{ m}, A = 5 \text{ mm}^2 = 5 \times 10^{-6} \text{ m}^2$$

$$\rho = 2 \times 10^{-6} \Omega \text{ m}$$

$$R_{\text{wire}} = \frac{\rho L}{A} = \frac{2 \times 10^{-6} \times 25}{5 \times 10^{-6}} = 10$$

$$R_{\text{eq}} = \frac{R}{4} = \frac{10}{4} = 2.5 \Omega$$

Answer does not match with NTA option.

Question45

A motor operating on 100 V draws a current of 1 A . If the efficiency of the motor is 91.6%, then the loss of power in units of cal/s is

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. 6.2

B. 2

C. 8.4

D. 4

Answer: B

Solution:

The motor operates at 100 V and draws 1 A, so the total electrical power input is:

$$P_{\text{in}} = IV = 100 \text{ V} \times 1 \text{ A} = 100 \text{ W}.$$

With an efficiency of 91.6%, only 91.6% of the input power is used for the motor's work, while the rest is lost as heat. The output power is:

$$P_{\text{out}} = 0.916 \times 100 \text{ W} = 91.6 \text{ W}.$$

The power lost is:

$$P_{\text{loss}} = P_{\text{in}} - P_{\text{out}} = 100 \text{ W} - 91.6 \text{ W} = 8.4 \text{ W}.$$

Since $1 \text{ W} = 1 \text{ J/s}$ and $1 \text{ cal} \approx 4.2 \text{ J}$, we convert the lost power into calories per second:

$$P_{\text{loss}} \text{ (in cal/s)} = \frac{8.4 \text{ J/s}}{4.2 \text{ J/cal}} = 2 \text{ cal/s}.$$

Thus, the correct answer is:

Option B: 2

Question46

Current passing through a wire as function of time is given as $I(t) = 0.02t + 0.01 \text{ A}$. The charge that will flow through the wire from $t = 1 \text{ s}$ to $t = 2 \text{ s}$ is

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. 0.02 C

B. 0.07 C

C. 0.06 C

D. 0.04 C

Answer: D

Solution:

To find the total charge q that flows through the wire from $t = 1 \text{ s}$ to $t = 2 \text{ s}$, we can integrate the current function $I(t) = 0.02t + 0.01 \text{ A}$ over this interval.

The formula for the charge q flowing through the wire is given by the integral of the current with respect to time:

$$q = \int I(t) dt$$

We integrate the function from $t = 1$ to $t = 2$:

$$q = \int_1^2 (0.02t + 0.01) dt$$

To solve this, first find the antiderivative:

$$q = \left[0.02 \frac{t^2}{2} + 0.01t \right]_1^2$$

Calculate the expression at the upper and lower bounds:

$$= \left[0.01t^2 + 0.01t \right]_1^2$$

Plug in the values:

For $t = 2$:

$$= 0.01(2)^2 + 0.01(2) = 0.04 + 0.02 = 0.06$$

For $t = 1$:

$$= 0.01(1)^2 + 0.01(1) = 0.01 + 0.01 = 0.02$$

Subtract the result at $t = 1$ from the result at $t = 2$:

$$q = 0.06 - 0.02 = 0.04 \text{ C}$$

Therefore, the charge that flows through the wire from $t = 1$ s to $t = 2$ s is 0.04 C.

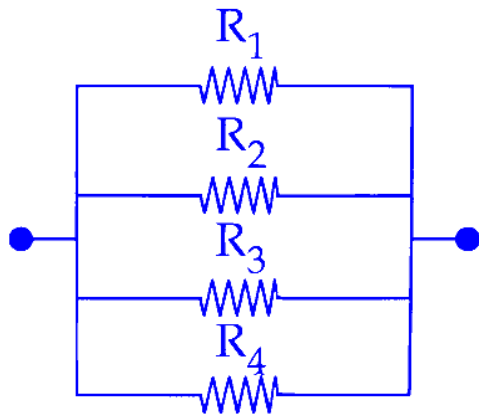
Question47

From the combination of resistors with resistances values $R_1 = R_2 = R_3 = 5\Omega$ and $R_4 = 10\Omega$, which of the following combination is the best circuit to get an equivalent resistance of 6Ω ?

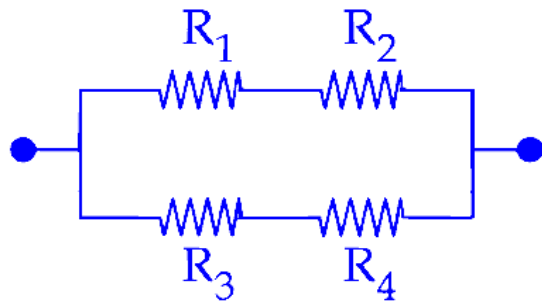
JEE Main 2025 (Online) 4th April Evening Shift

Options:

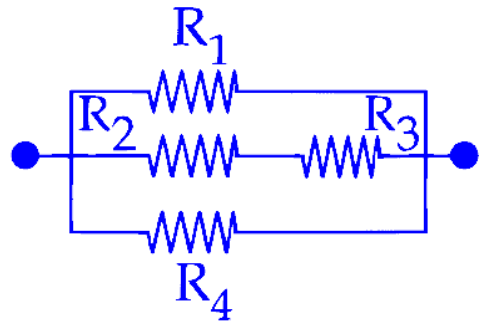
A.



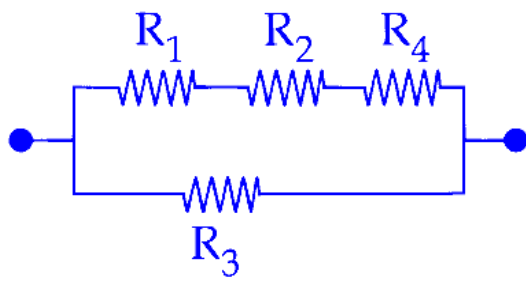
B.



C.



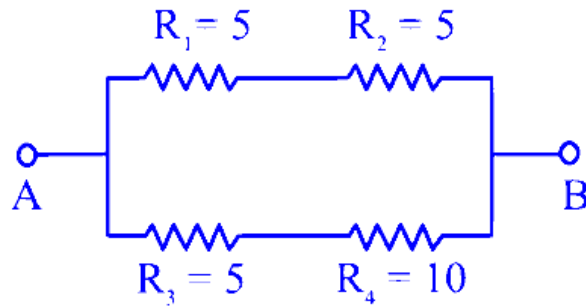
D.



Answer: B

Solution:

$$\frac{1}{R_P} = \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{1}{6}$$



Question48

There are ' n ' number of identical electric bulbs, each is designed to draw a power p independently from the mains supply. They are now joined in series across the mains supply. The total power drawn by the combination is :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. np

B. p

C. $\frac{p}{n^2}$

D. $\frac{p}{n}$

Answer: D

Solution:

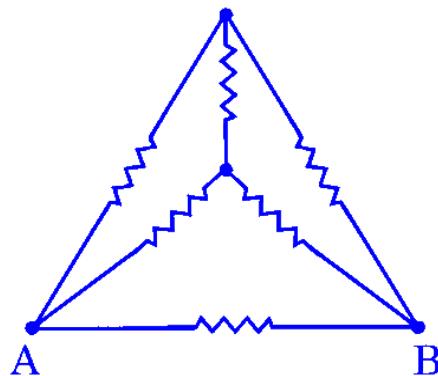
$$R_s = R_1 + R_2 + R_3 + \dots + R_n$$

$$\frac{V^2}{P_s} = \frac{V^2}{P} + \frac{V^2}{P} + \dots + \frac{V^2}{P_n}$$

$$P_s = \frac{P}{n}$$

Question49

A wire of resistance R is bent into a triangular pyramid as shown in figure with each segment having same length. The resistance between points A and B is R/n . The value of n is :



JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. 16

B. 10

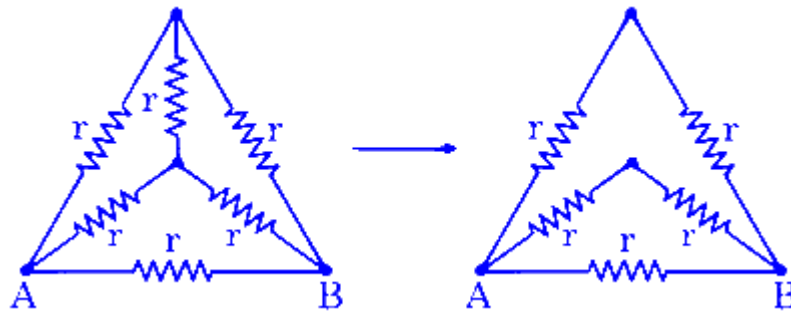
C. 12

D. 14

Answer: C

Solution:

$$\text{As } r = \frac{R}{6}$$



(As balanced wheat stone bridge is formed) Now, Equivalent resistance between A and B can be written as

$$\frac{1}{R_{AB}} = \frac{1}{2r} + \frac{1}{2r} + \frac{1}{r} = \frac{2}{r}$$
$$R_{AB} = \frac{R}{12}$$

Question50

A wire of length 10cm and radius $\sqrt{7} \times 10^{-4}\text{m}$ connected across the right gap of a meter bridge. When a resistance of 4.5Ω is connected on the left gap by using a resistance box, the balance length is found to be at 60cm from the left end. If the resistivity of the wire is $R \times 10^{-7}\Omega\text{m}$, then value of R is :

[27-Jan-2024 Shift 1]

Options:

A.

63

B.

70

C.

66

D.

35

Answer: C

Solution:

For null point,

$$\frac{4.5}{60} = \frac{R}{40}$$

$$\text{Also, } R = \frac{\rho \ell}{A} = \frac{\rho \ell}{\pi r^2}$$

$$4.5 \times 40 = \rho \times \frac{0.1}{\pi \times 7 \times 10^{-8}} \times 60$$

$$\rho = 66 \times 10^{-7} \Omega \times \text{m}$$

Question51

A wire of resistance R and length L is cut into 5 equal parts. If these parts are joined parallelly, then resultant resistance will be :

[27-Jan-2024 Shift 1]

Options:

A.

$$\frac{1}{25}R$$

B.

$$\frac{1}{5}R$$

C.

$$25R$$

D.

$$5R$$

Answer: A

Solution:

$$\text{Resistance of each part} = \frac{R}{5}$$

$$\text{Total resistance} = \frac{1}{5} \times \frac{R}{5} = \frac{R}{25}$$

Question52

A current of $200\mu\text{A}$ deflects the coil of a moving coil galvanometer through 60° . The current to cause deflection through $\pi/10$ radian is :

[27-Jan-2024 Shift 2]

Options:

A.

$$30\mu\text{A}$$

B.

$$120\mu\text{A}$$

C.

$60\mu\text{A}$

D.

$180\mu\text{A}$

Answer: D

Solution:

$i \propto \theta$ (angle of deflection)

$$\therefore \frac{i_2}{i_1} = \frac{\theta_2}{\theta_1} \Rightarrow \frac{i_2}{200\mu\text{A}} = \frac{\pi/10}{\pi/3} = \frac{3}{10}$$

$$\Rightarrow i_2 = 60\mu\text{A}$$

Question53

Wheatstone bridge principle is used to measure the specific resistance (S_1) of given wire, having length L , radius r . If X is

the resistance of wire, then specific resistance is : $S_1 = X \left(\frac{\pi r^2}{L} \right)$. If the length of the wire gets doubled then the value of specific resistance will be :

[27-Jan-2024 Shift 2]

Options:

A.

$S_1/4$

B.

$2S_1$

C.

$$S_1/2$$

D.

$$S_1$$

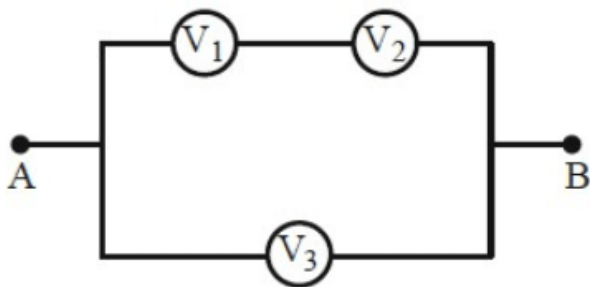
Answer: D

Solution:

As specific resistance does not depend on dimension of wire so, it will not change.

Question 54

Three voltmeters, all having different internal resistances are joined as shown in figure. When some potential difference is applied across A and B, their readings are V_1, V_2 and V_3 . Choose the correct option.



[27-Jan-2024 Shift 2]

Options:

A.

$$V_1 = V_2$$

B.

$$V_1 \neq V_3 - V_2$$

C.

$$V_1 + V_2 > V_3$$

D.

$$V_1 + V_2 = V_3$$

Answer: D

Solution:

From KVL,

$$V_1 + V_2 - V_3 = 0 \Rightarrow V_1 + V_2 = V_3$$

Question 55

The electric current through a wire varies with time as $I = I_0 + \beta t$, where $I_0 = 20\text{A}$ and $\beta = 3\text{A/s}$. The amount of electric charge crossed through a section of the wire in 20 s is :

[29-Jan-2024 Shift 1]

Options:

A.

80C

B.

1000C

C.

800C

D.

1600C

Answer: B

Solution:

Given that

$$\text{Current } I = I_0 + \beta t$$

$$I_0 = 20\text{A}$$

$$\beta = 3\text{A/s}$$

$$I = 20 + 3t$$

$$\frac{dq}{dt} = 20 + 3t$$

$$\int_0^q dq = \int_0^{20} (20 + 3t) dt$$

$$q = \int_0^{20} 20 dt + \int_0^{20} 3t dt$$

$$q = \left[20t + \frac{3t^2}{2} \right]_0^{20} = 1000\text{C}$$

Question56

A galvanometer having coil resistance 10Ω shows a full scale deflection for a current of 3mA . For it to measure a current of 8A , the value of the shunt should be:

[29-Jan-2024 Shift 1]

Options:

A.

$$3 \times 10^{-3}\Omega$$

B.

$$4.85 \times 10^{-3}\Omega$$

C.

$$3.75 \times 10^{-3}\Omega$$

D.

$$2.75 \times 10^{-3}\Omega$$

Answer: C

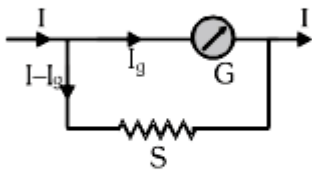
Solution:

Given $G = 10\Omega$

$$I_g = 3 \text{ mA}$$

$$I = 8 \text{ A}$$

In case of conversion of galvanometer into ammeter.



We have $I_g G = (I - I_g)S$

$$S = \frac{I_g G}{I - I_g}$$

$$S = \frac{(3 \times 10^{-3})10}{8 - 0.003} = 3.75 \times 10^{-3} \Omega$$

Question57

The deflection in moving coil galvanometer falls from 25 divisions to 5 division when a shunt of 24Ω is applied. The resistance of galvanometer coil will be :

[29-Jan-2024 Shift 1]

Options:

A.

$$12\Omega$$

B.

$$96\Omega$$

C.

$$48\Omega$$

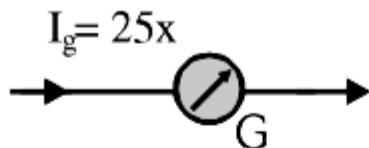
D.

100Ω

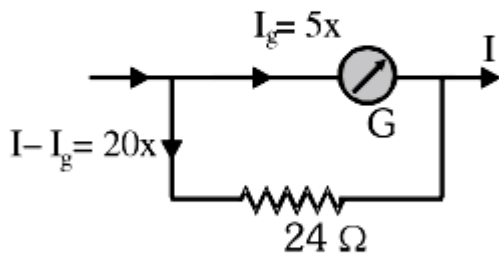
Answer: B

Solution:

Let x = current/division



After applying shunt



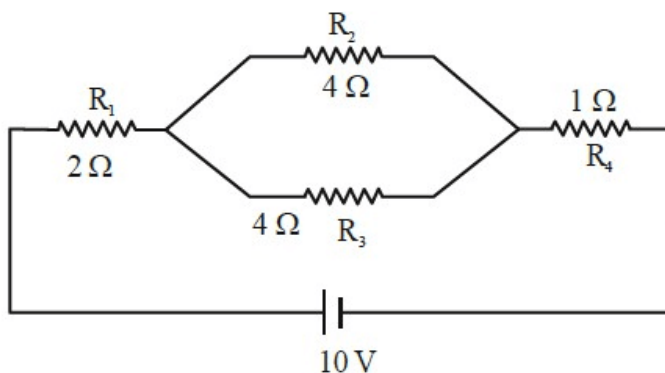
$$\text{Now } 5x \times G = 20x \times 24$$

$$G = 4 \times 24$$

$$G = 96\Omega$$

Question58

In the given circuit, the current in resistance R_3 is :



[29-Jan-2024 Shift 2]

Options:

A.

1A

B.

1.5A

C.

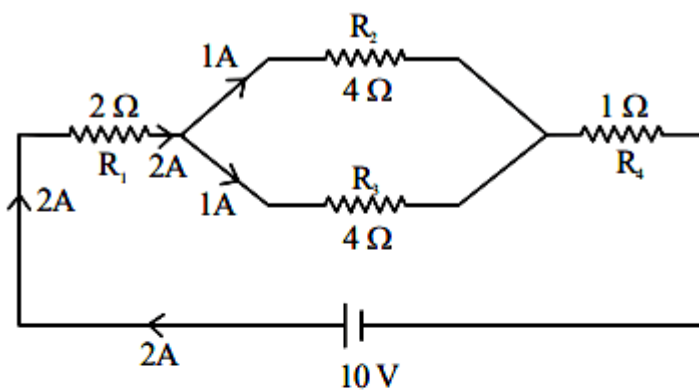
2A

D.

2.5A

Answer: A

Solution:



$$R_{eq} = 2\Omega + 2\Omega + 1\Omega = 5\Omega$$

$$i = \frac{V}{R_{eq}} = \frac{10}{5} = 2A$$

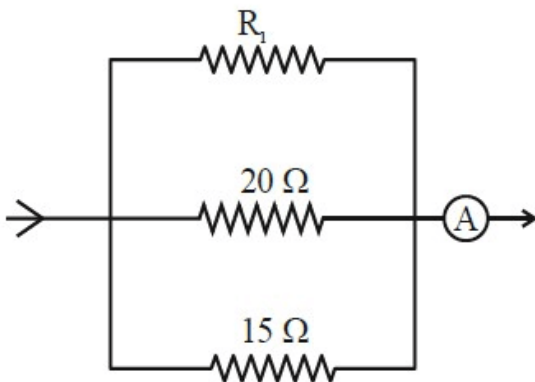
$$\text{Current in resistance } R_3 = 2 \times \left(\frac{4}{4+4} \right)$$

$$= 2 \times \frac{4}{8}$$

$$= 1A$$

Question59

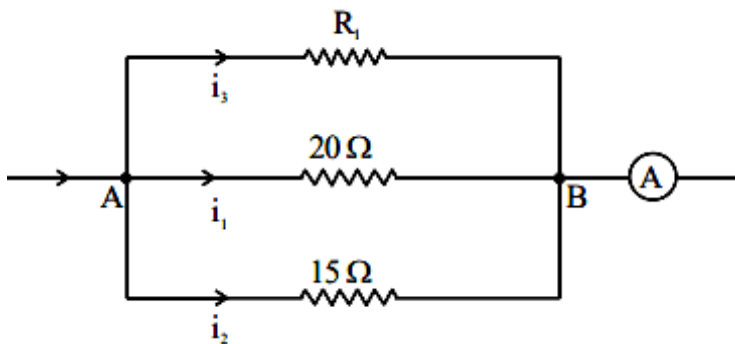
In the given circuit, the current flowing through the resistance 20Ω is 0.3A , while the ammeter reads 0.9A . The value of R_1 is _____ Ω .



[29-Jan-2024 Shift 2]

Answer: 30

Solution:



Given, $i_1 = 0.3\text{A}$, $i_1 + i_2 + i_3 = 0.9\text{A}$

$$\text{So, } V_{AB} = i_1 \times 20\Omega = 20 \times 0.3\text{V} = 6\text{V}$$

$$i_2 = \frac{6\text{V}}{15\Omega} = \frac{2}{5}\text{A}$$

$$i_1 + i_2 + i_3 = \frac{9}{10}\text{A}$$

$$\frac{3}{10} + \frac{2}{5} + i_3 = \frac{9}{10}$$

$$\frac{7}{10} + i_3 = \frac{9}{10}$$

$$i_3 = 0.2\text{A}$$

$$\text{So, } i_3 \times R_1 = 6\text{V}$$

$$(0.2)R_1 = 6$$

$$R_1 = \frac{6}{0.2} = 30\Omega$$

Question60

An electric toaster has resistance of 60Ω at room temperature (27°C). The toaster is connected to a 220V supply. If the current flowing through it reaches 2.75A , the temperature attained by toaster is around : (if $\alpha = 2 \times 10^{-4}/^\circ\text{C}$)

[30-Jan-2024 Shift 1]

Options:

A.

694°C

B.

1235°C

C.

1694°C

D.

1667°C

Answer: C

Solution:

$$R_{T=27} = 60\Omega, R_T = \frac{220}{2.75} = 80\Omega$$

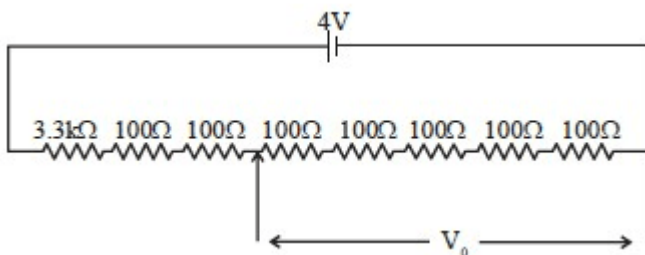
$$R = R_0(1 + \alpha \Delta T)$$

$$80 = 60[1 + 2 \times 10^{-4}(T - 27)]$$

$$T \approx 1694^\circ\text{C}$$

Question61

A potential divider circuit is shown in figure. The output voltage V_0 is



[30-Jan-2024 Shift 1]

Options:

A.

4V

B.

2mV

C.

0.5V

D.

12mV

Answer: C

Solution:

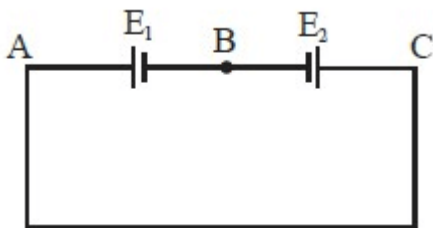
$$R_{eq} = 4000\Omega$$

$$i = \frac{4}{4000} = \frac{1}{1000}\text{A}$$

$$V_0 = i \cdot R = \frac{1}{1000} \times 500 = 0.5\text{V}$$

Question62

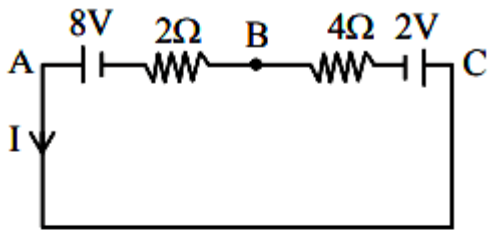
Two cells are connected in opposition as shown. Cell E_1 is of 8V emf and 2Ω internal resistance; the cell E_2 is of 2V emf and 4Ω internal resistance. The terminal potential difference of cell E_2 is:



[30-Jan-2024 Shift 1]

Answer: 6

Solution:



$$I = \frac{8-2}{2+4} = \frac{6}{6} = 1\text{A}$$

Applying Kirchhoff from C to B

$$V_C - 2 - 4 \times 1 = V_B$$

$$V_C - V_B = 6\text{V}$$

$$= 6\text{V}$$

Question63

When a potential difference V is applied across a wire of resistance R , it dissipates energy at a rate W . If the wire is cut into two halves and these halves are connected mutually parallel across the same supply, the energy dissipation rate will become:

[30-Jan-2024 Shift 2]

Options:

A.

$$\frac{1}{4}W$$

B.

$$\frac{1}{2}W$$

C.

$$2W$$

D.

4W

Answer: D

Solution:

$$\frac{v^2}{R} = W \dots\dots(i)$$

$$\frac{\frac{v^2}{\frac{1}{2}\left(\frac{R}{2}\right)}}{R} = W' \dots\dots(ii)$$

From (i) & (ii), we get

$$W' = 4W$$

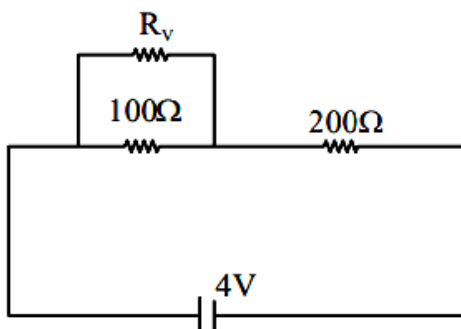
Question64

Two resistance of 100Ω and 200Ω are connected in series with a battery of 4V and negligible internal resistance. A voltmeter is used to measure voltage across 100Ω resistance, which gives reading as 1V. The resistance of voltmeter must be _____ Ω

[30-Jan-2024 Shift 2]

Answer: 200

Solution:



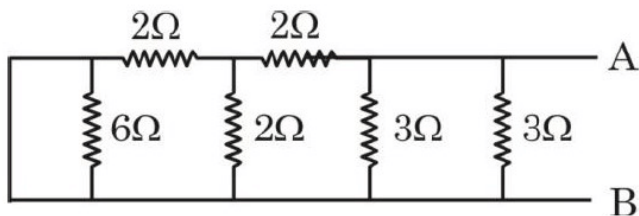
$$\frac{R_v 100}{R_v + 100} = \frac{200}{3}$$

$$3R_v = 2R_v + 200$$

$$R_v = 200$$

Question 65

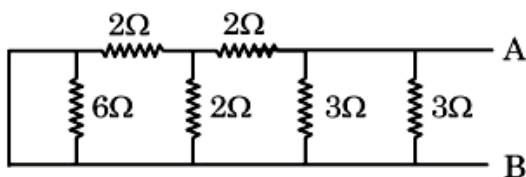
Equivalent resistance of the following network is Ω .



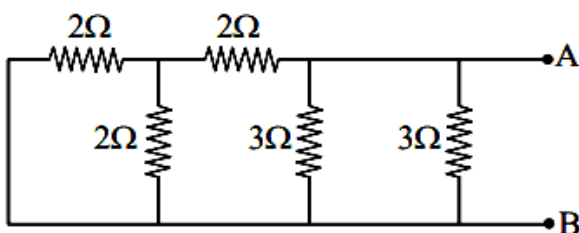
[31-Jan-2024 Shift 1]

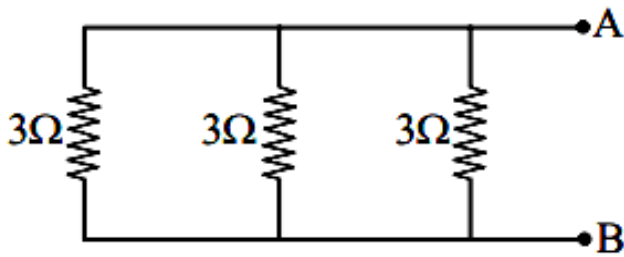
Answer: 1

Solution:



6Ω is short circuit





$$R_{eq} = 3 \times \frac{1}{3} = 1\Omega$$

Question66

The resistance per centimeter of a meter bridge wire is r , with $X\Omega$ resistance in left gap. Balancing length from left end is at 40cm with 25Ω resistance in right gap. Now the wire is replaced by another wire of $2r$ resistance per centimeter. The new balancing length for same settings will be at

[31-Jan-2024 Shift 2]

Options:

A.

20 cm

B.

10 cm

C.

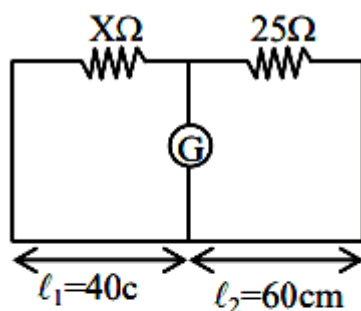
80 cm

D.

40 cm

Answer: D

Solution:



$$\frac{25}{r\ell_1} = \frac{X}{r\ell_2} \dots\dots\dots(i)$$

$$\frac{25}{2r\ell_1'} = \frac{X}{2r\ell_2'} \dots\dots\dots(ii)$$

From (i) and (ii)

$$\ell_2' = \ell_2 = 40 \text{ cm}$$

Question67

By what percentage will the illumination of the lamp decrease if the current drops by 20% ?

[31-Jan-2024 Shift 2]

Options:

A.

46%

B.

26%

C.

36%

D.

56%

Answer: C

Solution:

$$P = i^2 R$$

$$P_{\text{int}} = I_{\text{int}}^2 R$$

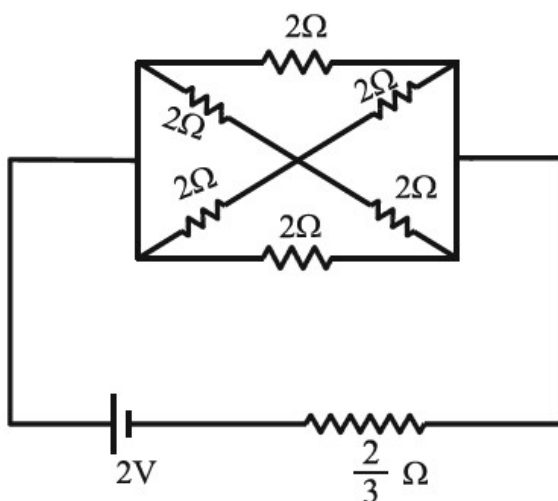
$$P_{\text{final}} = (0.8 I_{\text{int}})^2 R$$

% change in power =

$$\frac{P_{\text{final}} - P_{\text{int}}}{P_{\text{int}}} \times 100 = (0.64 - 1) \times 100 = -36\%$$

Question68

In the following circuit, the battery has an emf of 2 V and an internal resistance of $\frac{2}{3}\Omega$. The power consumption in the entire circuit is _____ W.



[31-Jan-2024 Shift 2]

Answer: 3

Solution:

$$R_{eq} = \frac{4}{3}\Omega$$

$$\therefore P = \frac{V^2}{R_{eq}} = \frac{4}{4/3} = 3W$$

Question69

A galvanometer has a resistance of 50Ω and it allows maximum current of $5mA$. It can be converted into voltmeter to measure upto $100V$ by connecting in series a resistor of resistance

[1-Feb-2024 Shift 1]

Options:

A.

5975Ω

B.

20050Ω

C.

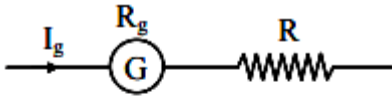
19950Ω

D.

19500Ω

Answer: C

Solution:



$$R = \frac{V}{I_g} - R_g = \frac{100}{5 \times 10^{-3}} - 50$$

$$= 20000 - 50$$

$$= 19950 \Omega$$

Question70

The current in a conductor is expressed as $I = 3t^2 + 4t^3$, where I is in Ampere and t is in second. The amount of electric charge that flows through a section of the conductor during $t = 1$ s to $t = 2$ s is _____ C.

[1-Feb-2024 Shift 1]

Answer: 22

Solution:

$$q = \int_1^2 i \, dt = \int_1^2 (3t^2 + 4t^3) \, dt$$

$$q = (t^3 + t^4)_1^2$$

$$q = 22C$$

Question71

In an ammeter, 5% of the main current passes through the galvanometer. If resistance of the galvanometer is G , the resistance of ammeter will be :

[1-Feb-2024 Shift 2]

Options:

A.

$G/20$

B.

$G/199$

C.

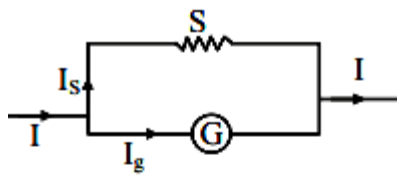
$199G$

D.

$200G$

Answer: A

Solution:



$$I_S S = I_G G$$

$$\frac{95}{100} I S = \frac{5I}{100} G$$

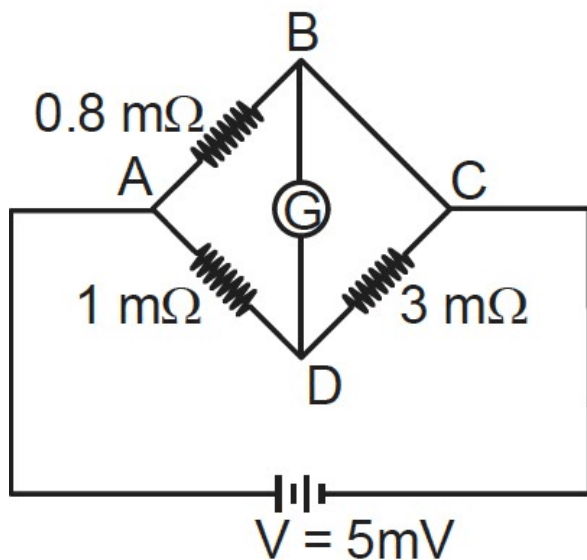
$$S = \frac{G}{19}$$

$$R_A = \frac{SG}{S+G} = \frac{\frac{G^2}{19}}{\frac{20G}{19}}$$

$$R_A = \frac{G}{20}$$

Question 72

To measure the temperature coefficient of resistivity α of a semiconductor, an electrical arrangement shown in the figure is prepared. The arm BC is made up of the semiconductor. The experiment is being conducted at 25°C and resistance of the semiconductor arm is $3\text{m}\Omega$. Arm BC is cooled at a constant rate of $2^\circ\text{C}/\text{s}$. If the galvanometer G shows no deflection after 10 s, then α is :



[1-Feb-2024 Shift 2]

Options:

A.

$$-2 \times 10^{-2}^\circ\text{C}^{-1}$$

B.

$$-1.5 \times 10^{-2}^\circ\text{C}^{-1}$$

C.

$$-1 \times 10^{-2}^\circ\text{C}^{-1}$$

D.

$$-2.5 \times 10^{-2}^\circ\text{C}^{-1}$$

Answer: C

Solution:

For no deflection $\frac{0.8}{1} = \frac{R}{3}$

$$\Rightarrow R = 2.4 \text{ m}\Omega$$

Temperature fall in 10 s = 20°C

$$\Delta R = R \alpha \Delta t$$

$$\alpha = \frac{\Delta R}{R \Delta t} = \frac{-0.6}{3 \times 20}$$

$$= -10^{-2} \text{C}^{-1}$$

Question 73

In a metre-bridge when a resistance in the left gap is 2Ω and unknown resistance in the right gap, the balance length is found to be 40cm. On shunting the unknown resistance with 2Ω , the balance length changes by :

[1-Feb-2024 Shift 2]

Options:

A.

22.5 cm

B.

20 cm

C.

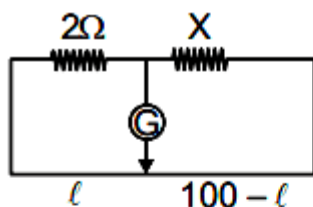
62.5 cm

D.

65 cm

Answer: A

Solution:



First case $\frac{2}{40} = \frac{X}{60} \Rightarrow X = 3\Omega$

In second case $X' = \frac{2 \times 3}{2 + 3} = 1.2\Omega$

$$\frac{2}{\ell} = \frac{1.2}{100 - \ell}$$

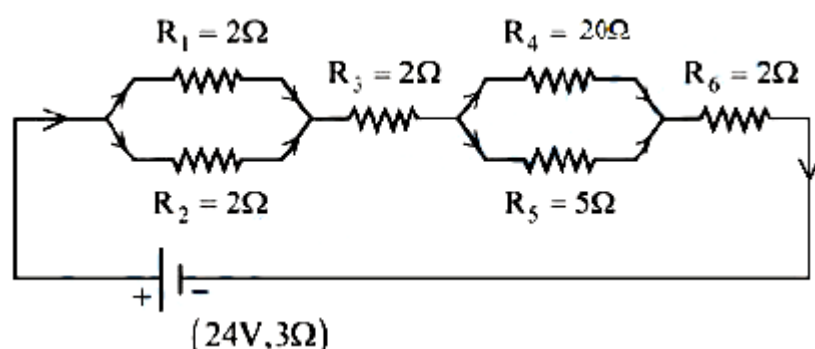
$$200 - 2\ell = 1.2\ell$$

$$\ell = \frac{200}{3.2} = 62.5 \text{ cm}$$

Balance length changes by 22.5 cm

Question 74

As shown in the figure, a network of resistors is connected to a battery of 24V with an internal resistance of 3Ω . The currents through the resistors R_4 and R_5 are I_4 and I_5 respectively. The values of I_4 and I_5 are :



(24V, 3Ω)

[24-Jan-2023 Shift 1]

Options:

A. $I_4 = \frac{8}{5}\text{A}$ and $I_5 = \frac{2}{5}\text{A}$

B. $I_4 = \frac{24}{5}\text{A}$ and $I_5 = \frac{6}{5}\text{A}$

C. $I_4 = \frac{6}{5}A$ and $I_5 = \frac{24}{5}A$

D. $I_4 = \frac{2}{5}A$ and $I_5 = \frac{8}{5}A$

Answer: D

Solution:

Equivalent resistance of circuit

$$R_{eq} = 3 + 1 + 2 + 4 + 2 \\ = 12\Omega$$

Current through battery $i = \frac{24}{12} = 2A$

$$I_4 = \frac{R_5}{R_4 + R_5} \times 2 = \frac{5}{20 + 5} \times 2 = \frac{2}{5}A$$

$$I_5 = 2 - \frac{2}{5} = \frac{8}{5}A$$

Question 75

A hollow cylindrical conductor has length of 3.14 m, while its inner and outer diameters are 4 mm and 8 mm respectively. The resistance of the conductor is $n \times 10^{-3}\Omega$.

If the resistivity of the material is $2.4 \times 10^{-8}\Omega m$. The value of n is _____

[24-Jan-2023 Shift 1]

Answer: 2

Solution:

$R = \rho \frac{\ell}{A}$, the cross-sectional area is $\pi(b^2 - a^2)$

$$R = \rho \frac{\ell}{\pi(b^2 - a^2)} = \frac{2.4 \times 10^{-8} \times 3.14}{3.14 \times (4^2 - 2^2) \times 10^{-6}}$$

$$= 2 \times 10^{-3} \Omega$$

$$\rightarrow n = 2$$

Question 76

A cell of emf 90V is connected across series combination of two resistors each of 100Ω resistance. A voltmeter of resistance 400Ω is used to measure the potential difference across each resistor. The reading of the voltmeter will be :
[24-Jan-2023 Shift 2]

Options:

A. 40V

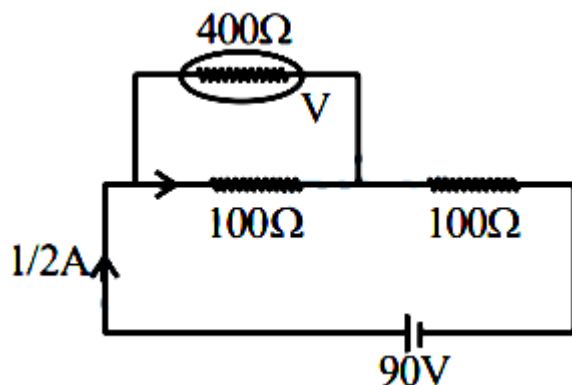
B. 45V

C. 80V

D. 90V

Answer: A

Solution:



$$R_{eq} = \frac{400 \times 100}{500} + 100$$

$$= 180\Omega$$

$$i = \frac{90}{180} = \frac{1}{2}A$$

$$\text{Reading} = \frac{1}{2} \times \frac{400}{500} \times 100$$

$$= 40 \text{ volt}$$

Question77

If a copper wire is stretched to increase its length by 20%. The percentage increase in resistance of the wire is ____ %.

[24-Jan-2023 Shift 2]

Answer: 44

Solution:

As volume is constant,

So resistance $\propto (\text{length})^2$

$$\Rightarrow \% \text{ change in resistance} = 20 + 20 + \frac{400}{100} = 44\%$$

Question78

A uniform metallic wire carries a current 2A. when 3.4V battery is connected across it. The mass of uniform metallic wire is 8.92×10^{-3} kg. density is $8.92 \times 10^3 \text{ kg/m}^3$ and resistivity is $1.7 \times 10^{-8} \Omega - \text{m}$. The length of wire is :

[25-Jan-2023 Shift 1]

Options:

A. $l = 6.8\text{m}$

B. $l = 10\text{m}$

C. $l = 5\text{m}$

D. $l = 100\text{m}$

Answer: B

Solution:

$$I = 2\text{A}$$

$$\Delta V = 3.4\text{V}$$

Using Ohm's Law

$$R = \frac{3.4}{2} = 1.7\Omega$$

$$1.7 = \frac{\rho L}{A}$$

$$L = \frac{1.7(A)}{\rho}$$

$$M = (\text{density volume})$$

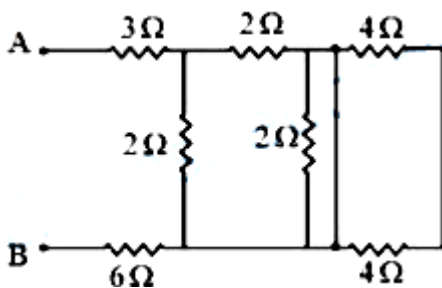
$$\text{Volume} = \frac{8.92 \times 10^{-3}}{8.92 \times 10^3} = 10^{-6}$$

$$L^2 = \frac{1.7}{\rho}(10^{-6}) = \frac{1.7}{1.7} \times 10^2$$

$$L = 10\text{m}$$

Question 79

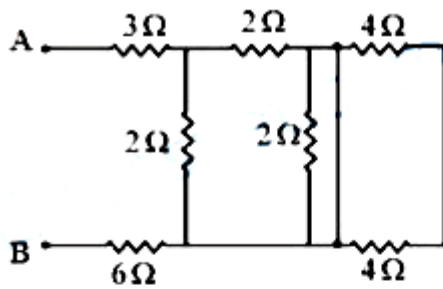
In the given circuit, the equivalent resistance between the terminal A and B is _____ Ω .



[25-Jan-2023 Shift 1]

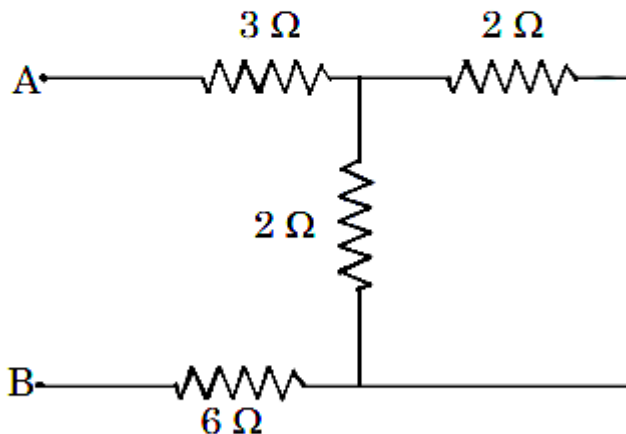
Answer: 10

Solution:



Both 4Ω resistance gets short.

Remove the resistors that have no current.



$$R_{eq} = 3 - (2 \parallel 2) + 6$$

$$R_{eq} = 3 - 1 + 6$$

$$R_{eq} = 10\Omega$$

Question80

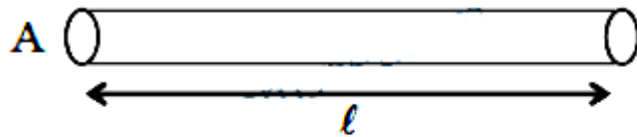
The resistance of a wire is 5Ω . It's new resistance in ohm if stretched to 5 times of it's original length will be :
[25-Jan-2023 Shift 2]

Options:

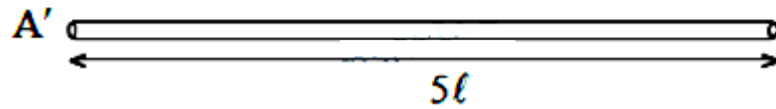
- A. 625
- B. 5
- C. 125
- D. 25

Answer: C

Solution:



$$R_{\text{initial}} = \frac{\rho \ell}{A} = 5\Omega$$



$\therefore$ Volume of wire is constant in stretching

$$V_i = V_f$$

$$A_i \ell_i = A_f \ell_f$$

$$A \ell = A' (5\ell)$$

$$A' = \frac{A}{5}$$

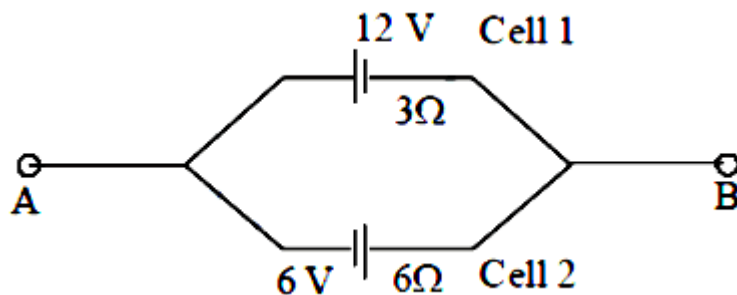
$$R_f = \frac{\rho \ell_f}{A_f} = \frac{\rho(5\ell)}{\left(\frac{A}{5}\right)}$$

$$= 25 \left(\frac{\rho \ell}{A} \right)$$

$$= 25 \times 5 = 125\Omega$$

Question81

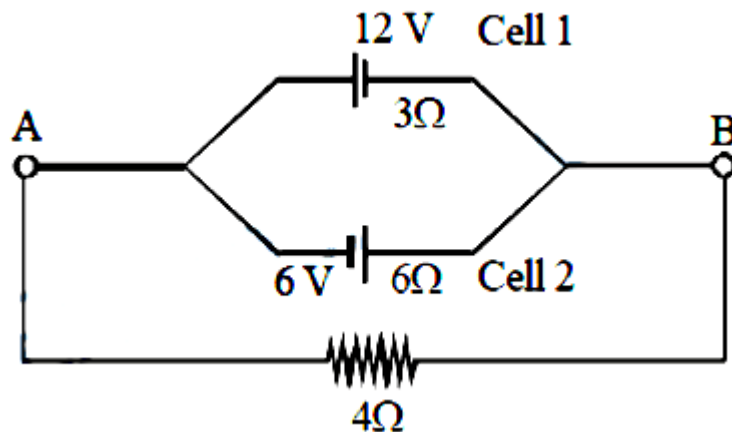
Two cells are connected between points A and B as shown. Cell 1 has emf of 12V and internal resistance of 3Ω . Cell 2 has emf of 6V and internal resistance of 6Ω . An external resistor R of 4Ω is connected across A and B. The current flowing through R will be _____ A.



[25-Jan-2023 Shift 2]

Answer: 1

Solution:

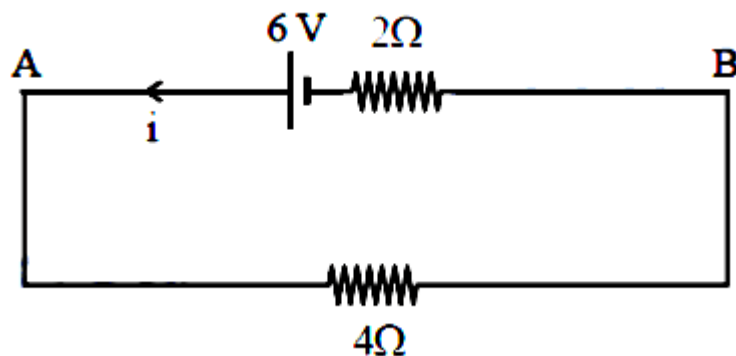


$$E_{eq} = \frac{\frac{12}{3} - \frac{6}{6}}{\frac{1}{3} + \frac{1}{6}}$$

$$E_{eq} = 6V$$

$$r_{eq} = 2\Omega$$

$$R = 4\Omega$$



$$\text{So, } i = \frac{6}{2+4} = 1A$$

Question82

With the help of potentiometer, we can determine the value of emf of a given cell. The sensitivity of the potentiometer is
(A) directly proportional to the length of the potentiometer wire
(B) directly proportional to the potential gradient of the wire
(C) inversely proportional to the potential gradient of the wire
(D) inversely proportional to the length of the potentiometer wire

Choose the correct option for the above statements:
[29-Jan-2023 Shift 2]

Options:

- A. B and D only
- B. A and C only
- C. A only
- D. C only

Answer: B

Solution:

Sensitivity of potentiometer wire is inversely proportional to potential gradient.

Question83

When two resistance R_1 and R_2 connected in series and introduced into the left gap of a meter bridge and a resistance of 10Ω is introduced into the right gap, a null point is found at 60 cm from left side. When R_1 and R_2 are connected in parallel and introduced into the left gap, a resistance of 3Ω is introduced into the right-gap to get null point at 40 cm from left end. The product of $R_1 R_2$ is _____ Ω^2

[29-Jan-2023 Shift 2]

Answer: 30

Solution:

$$\frac{R_1 + R_2}{10} = \frac{60}{40} = \frac{3}{2} \Rightarrow R_1 + R_2 = 15$$

$$\text{Now } \frac{R_1 R_2}{(R_1 + R_2) \times 3} = \frac{40}{60} = \frac{2}{3} \Rightarrow R_1 R_2 = 30$$

Question84

The charge flowing in a conductor changes with time as $Q(t) = \alpha t - \beta t^2 + \gamma t^3$. Where α , β and γ are constants. Minimum value of current is :

[30-Jan-2023 Shift 1]

Options:

A. $\alpha - \frac{3\beta^2}{\gamma}$

B. $\alpha - \frac{\gamma^2}{3\beta}$

C. $\beta - \frac{\alpha^2}{3\gamma}$

D. $\alpha - \frac{\beta^2}{3\gamma}$

Answer: D

Solution:

$$Q = (\alpha t - \beta t^2 + \gamma t^3)$$

$$i = \frac{dQ}{dt} = (\alpha - 2\beta t + 3\gamma t^2)$$

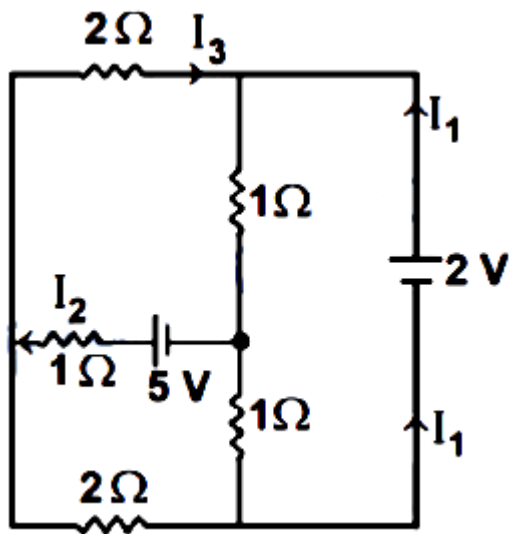
$$\frac{di}{dt} = (3\gamma t - 2\beta) = 0$$

$$\Rightarrow t = \frac{\beta}{3\gamma}$$

$$i = (\alpha - 2\beta t + 3\gamma t^2) = \left(\alpha - \frac{\beta^2}{3\gamma} \right)$$

Question85

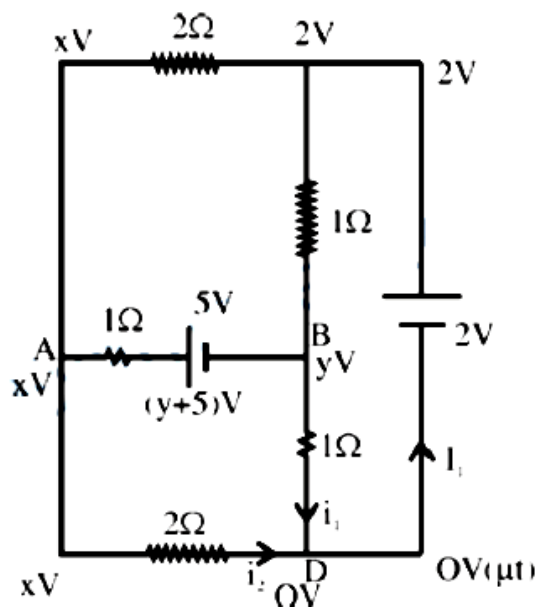
In the following circuit, the magnitude of current I_1 , is _____ A.



[30-Jan-2023 Shift 1]

Answer: 2

Solution:



Junction law at A,

$$\frac{x - (y + 5)}{1} + \frac{x - 2}{2} + \frac{x - 0}{2} = 0 \dots (1)$$

Junction law at B,

$$\frac{y + 5 - x}{1} + \frac{y - 0}{1} + \frac{y - 2}{1} = 0 \dots (2)$$

On solving equation (1) and Equation (2)

$$x = 3$$

$$\& y = 0$$

At D junction

$$I_1 = i_1 + i_2$$

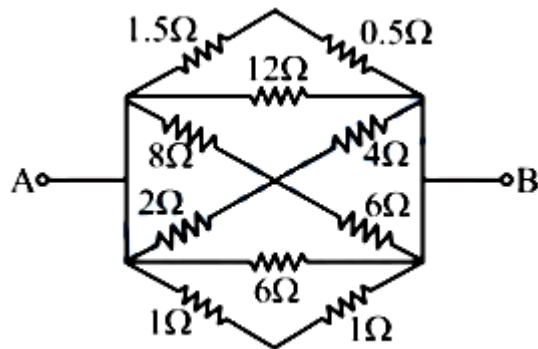
$$I_1 = \frac{y-0}{1} + \frac{x-0}{2}$$

$$= \frac{0-0}{1} + \frac{3-0}{2}$$

$$I_1 = 1.5\text{A}$$

Question86

The equivalent resistance between A and B is



[30-Jan-2023 Shift 2]

Options:

A. $\frac{2}{3}\Omega$

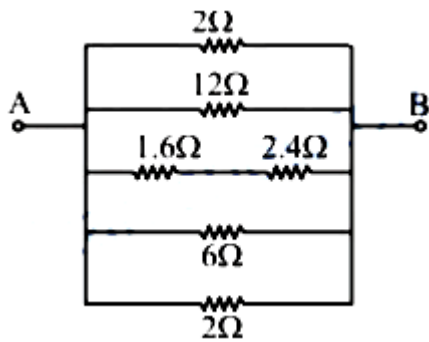
B. $\frac{1}{2}\Omega$

C. $\frac{3}{2}\Omega$

D. $\frac{1}{3}\Omega$

Answer: A

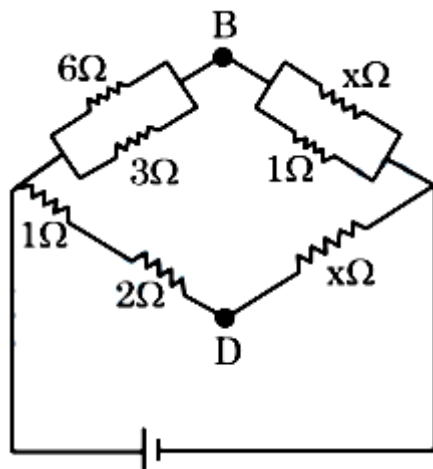
Solution:



$$\begin{aligned}\frac{1}{R_{eq}} &= \frac{1}{2} + \frac{1}{12} + \frac{1}{4} + \frac{1}{6} + \frac{1}{2} \\ &= \frac{6+1+3+2+6}{12} = \frac{18}{12} = \frac{3}{2} \\ \Rightarrow R_{eq} &= \frac{2}{3} \Omega\end{aligned}$$

Question87

If the potential difference between B and D is zero, the value of x is $\frac{1}{n}\Omega$. The value of n is _____



[30-Jan-2023 Shift 2]

Answer: 2

Solution:

$$\begin{aligned}\frac{2}{3} &= \frac{\frac{x}{x+1}}{x} \\ \Rightarrow \frac{2}{3} &= \frac{1}{x+1}\end{aligned}$$

$$\Rightarrow x = 0.5 = \frac{1}{2}$$

$$n = 2$$

Question88

The drift velocity of electrons for a conductor connected in an electrical circuit is V_d . The conductor is now replaced by another conductor with same material and same length but double the area of cross section. The applied voltage remains same. The new drift velocity of electrons will be
[31-Jan-2023 Shift 1]

Options:

A. V_d

B. $\frac{V_d}{2}$

C. $\frac{V_d}{4}$

D. $2V_d$

Answer: A

Solution:

$$V_d = \frac{eE}{m}\tau \text{ that is independent of area}$$

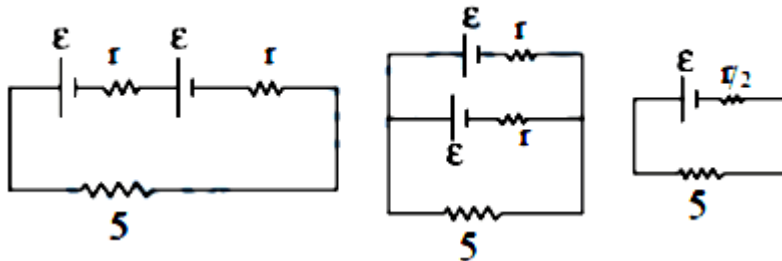
Question89

Two identical cells, when connected either in parallel or in series gives same current in an external resistance 5Ω . The internal resistance of each cell will be _____ Ω .
[31-Jan-2023 Shift 1]

Answer: 5

Solution:

Parallel



$$i = \frac{2\epsilon}{5+2r} \quad (1) \quad i = \frac{\epsilon}{\frac{r}{2}+5} \quad (2)$$

Equating (1) and (2)

$$\frac{2\epsilon}{5+2r} = \frac{\epsilon}{\frac{r}{2}+5} \Rightarrow r+10 = 5+2r$$

$$r = 5 \text{ Ans. } 5$$

Question90

The H amount of thermal energy is developed by a resistor in 10 s when a current of 4A is passed through it. If the current is increased to 16A, the thermal energy developed by the resistor in 10 s will be:

[31-Jan-2023 Shift 2]

Options:

A. H

B. 16H

C. $\frac{H}{4}$

D. 4H

Answer: B

Solution:

$H \propto i^2$ for $t = \text{constant}$

$$\Rightarrow \frac{H}{H'} = \left(\frac{4}{16} \right)^2$$

$$\Rightarrow H' = 16H$$

Question91

The number of turns of the coil of a moving coil galvanometer is increased in order to increase current sensitivity by 50%. The percentage change in voltage sensitivity of the galvanometer will be:

[31-Jan-2023 Shift 2]

Options:

- A. 100%
- B. 50%
- C. 75%
- D. 0%

Answer: D

Solution:

Current sensitivity = Voltage sensitivity $\times R$

Current sensitivity is made 1.5 times.

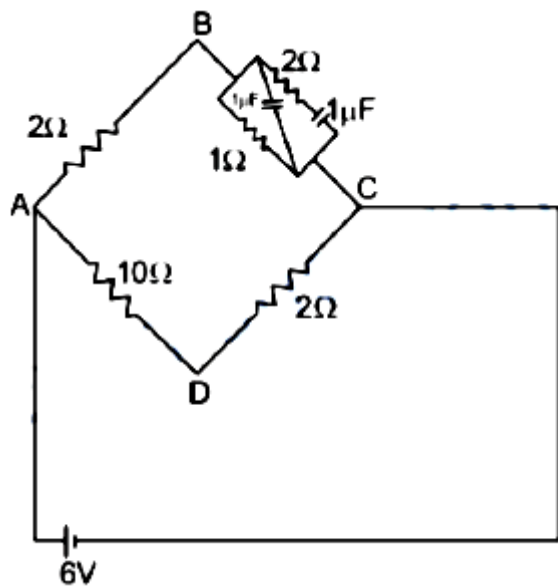
R also increase 1.5 times.

$$\text{Hence voltage sensitivity} = \frac{1.5 \times \text{current sensitivity}}{1.5 \times R}$$

= no change

Question92

For the given circuit, in the steady state,
 $|V_B - V_D| =$

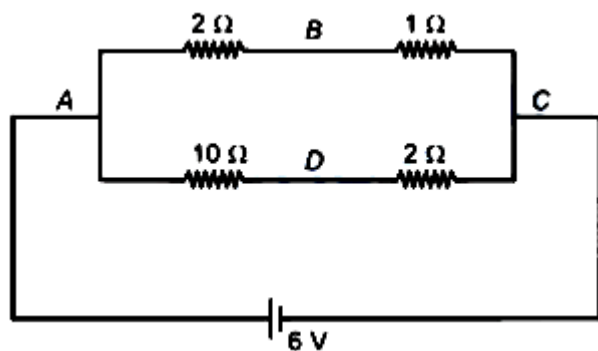


[31-Jan-2023 Shift 2]

Answer: 1

Solution:

In steady state, capacitor behaves as an open circuit. Circuit is :



$$\Rightarrow i_{AB} = \frac{6}{3} = 2A$$

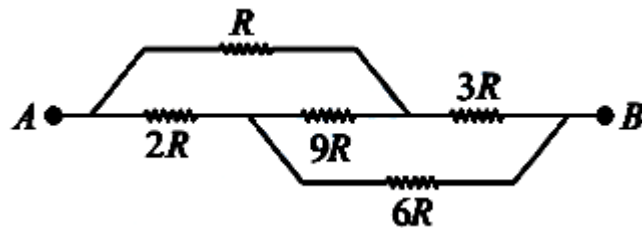
$$i_{AD} = \frac{6}{12} = 0.5A$$

$$\Rightarrow V_B + 2 \times 2 - 10 \times 0.5 = V_D$$

$$\Rightarrow V_B - V_D = 1 \text{ volt}$$

Question93

The equivalent resistance between A and B of the network shown in figure:



[1-Feb-2023 Shift 1]

Options:

A. $11 \frac{2R}{3}$

B. $14R$

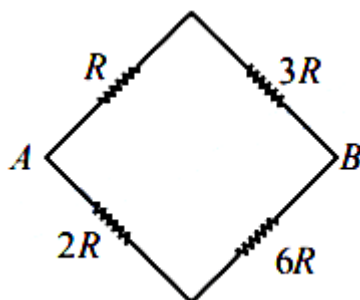
C. $21R$

D. $\frac{8}{3}R$

Answer: D

Solution:

Wheat stone bridge is in balanced condition.



$$\frac{1}{R_{eq}} = \frac{1}{4R} + \frac{1}{8R}$$

$$R_{eq} = \frac{8R}{3}$$

Question94

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : For measuring the potential difference across a

resistance of 600Ω , the voltmeter with resistance 1000Ω will be preferred over voltmeter with resistance 4000Ω .

Reason R : Voltmeter with higher resistance will draw smaller current than voltmeter with lower resistance.

In the light of the above statements, choose the most appropriate answer from the options given below.

[1-Feb-2023 Shift 2]

Options:

- A. A is not correct but R is correct
- B. Both A and R are correct and R is the correct explanation of A
- C. Both A and R are correct but R is not the correct explanation of A
- D. A is correct but R is not correct

Answer: A

Solution:

Error of voltmeter decreases with increase in its resistance.

Question95

Equivalent resistance between the adjacent corners of a regular n-sided polygon of uniform wire of resistance R would be:

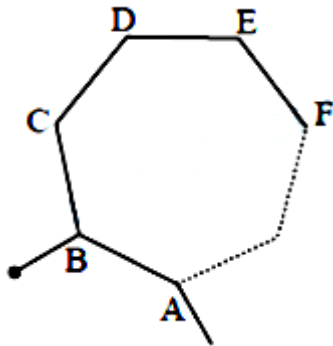
[1-Feb-2023 Shift 2]

Options:

- A. $\frac{(n-1)R}{n^2}$
- B. $\frac{(n-1)R}{(2n-1)}$
- C. $\frac{n^2R}{n-1}$
- D. $\frac{(n-1)R}{n}$

Answer: A

Solution:



Suppose resistance of each arm is r , then $r = R/n$

$$R_{eq(AB)} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\frac{r(n-1)r}{r + (n-1)r}$$

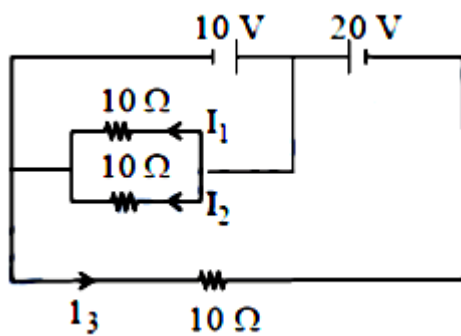
$$= \frac{r(n-1)r}{nr}$$

$$= \frac{n-1}{n}r$$

$$= \frac{(n-1)R}{n^2}$$

Question96

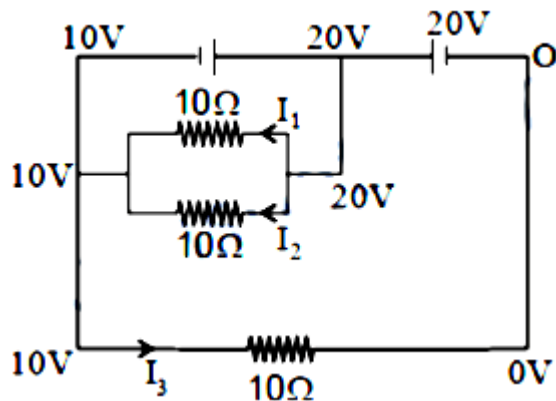
In the given circuit the value of $\left| \frac{I_1 + I_3}{I_2} \right|$ is:



[1-Feb-2023 Shift 2]

Answer: 2

Solution:



$$I_1 = I_2 = \frac{20 - 10}{10} = 1\text{A}$$

$$I_3 = 1\text{A}$$

$$\left| \frac{I_1 + I_3}{I_2} \right| = 2$$

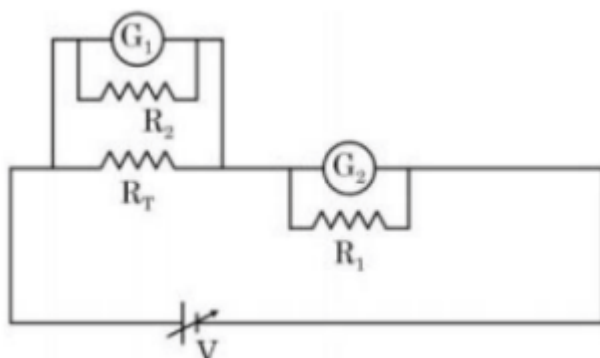
Question97

A student is provided with a variable voltage source V , a test resistor $R_T = 10\Omega$, two identical galvanometers G_1 and G_2 and two additional resistors, $R_1 = 10\text{M}\Omega$ and $R_2 = 0.001\Omega$. For conducting an experiment to verify ohm's law, the most suitable circuit is :

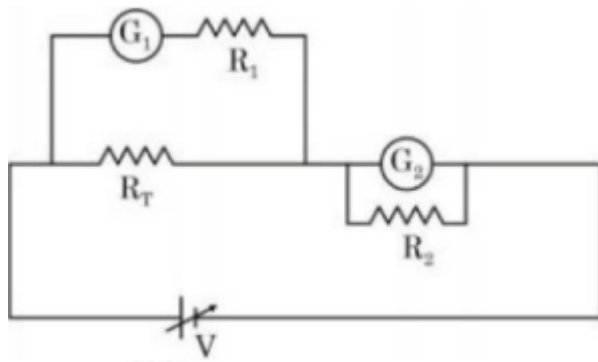
[6-Apr-2023 shift 2]

Options:

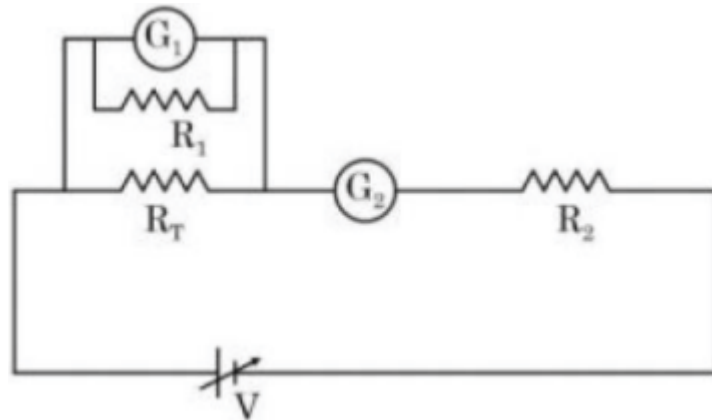
A.



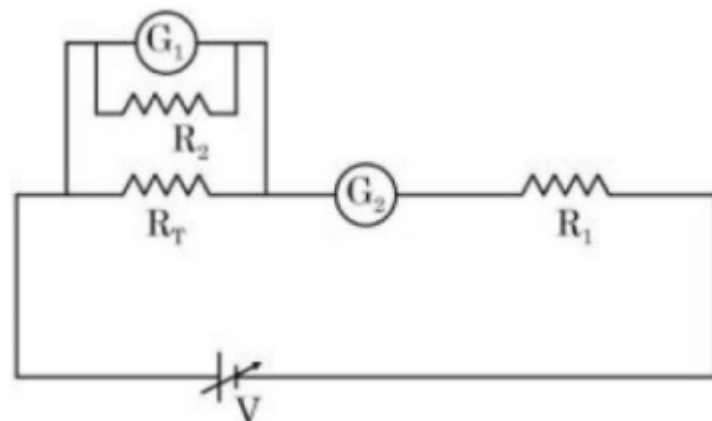
B.



C.



D.



Answer: B

Solution:

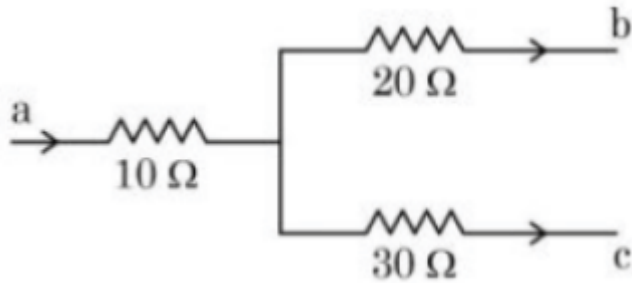
This question is based on the conceptual clarity that we should connect ammeter in series and voltmeter in parallel to measure current and potential difference, respectively. Also, when we use a galvanometer to create an ammeter, shunt resistance should be very small and should be in parallel.

When we create a voltmeter shunt should be large and in series with galvanometer.

All these criteria are satisfied in option (2)

Question98

Figure shows a part of an electric circuit. The potentials at points a, b and c are 30V, 12V and 2V respectively. The current through the 20Ω resistor will be



[6-Apr-2023 shift 2]

Options:

- A. 1.0A
- B. 0.2A
- C. 0.4A
- D. 0.6A

Answer: C

Solution:

Let potential of the junction be x volts using junction law $i_1 + i_2 + i_3 = 0$ or

$$\frac{x-30}{10} + \frac{x-12}{20} + \frac{x-2}{30} = 0$$

$$\text{or } \frac{1}{60}[6x - 180 + 3x - 36 + 2x - 4] = 0$$

$$\text{or } \frac{1}{60}[11x - 220] = 0$$

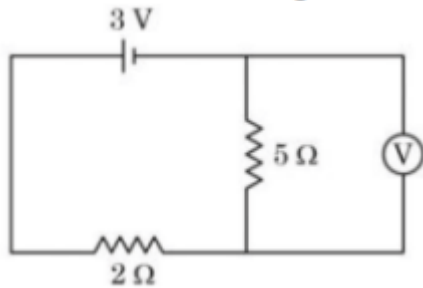
$$\text{or } x = \frac{220}{11} = 20\text{V}$$

$$\text{current through } 20\Omega \text{ is } = \frac{x-12}{20}$$

$$i_2 = \frac{20-12}{20} = 0.4\text{A}$$

Question99

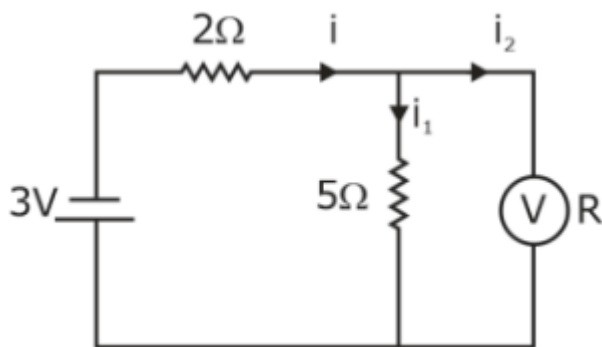
As shown in the figure, the voltmeter reads 2V across 5Ω resistor. The resistance of the voltmeter is _____ Ω .



[6-Apr-2023 shift 2]

Answer: 20

Solution:



Method-I:

$$R_{eq} = 2 + \frac{5R}{5+R} = \frac{10+7R}{5+R}$$

$$i = \frac{3}{R_{eq}} = \frac{3(5+R)}{10+7R}$$

$$i_1 = \frac{2}{5}, i_2 = \frac{2}{R}$$

$$i = i_1 + i_2$$

$$\frac{3(5+R)}{10+7R} = \frac{2}{5} + \frac{2}{R} = \frac{2(5+R)}{5R}$$

$$15R(5+R) = 2(5+R)(10+7R)$$

$$75R + 15R^2 = 2(50 + 35R + 10R + 7R^2)$$

$$15R^2 + 75R = 14R^2 + 90R + 100$$

$$R^2 - 15R - 100 = 0$$

$$R = \frac{15 \pm \sqrt{225 + 400}}{2}$$

$$= \frac{15 \pm \sqrt{625}}{2} = \frac{15 \pm 25}{2}$$

$$R = 20\Omega$$

Method-II:

Given potential across 5Ω and voltmeter is $2V$. To find resistance R of voltmeter.

Let current in 5Ω be i_1 , and in R be i_2 .

$$i_1 = \frac{2}{5} \text{ and } i_2 = \frac{2}{R}$$

V across 2Ω will be 1 volt and $i = \frac{1}{2}A$.

Using junction law: $i = i_1 + i_2$

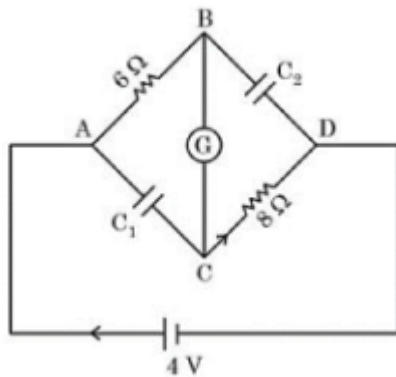
$$\frac{1}{2} = \frac{2}{5} + \frac{2}{R}$$

$$\frac{2}{R} = \frac{1}{2} - \frac{2}{5} = \frac{1}{10}$$

$$R = 20\Omega$$

Question100

In this figure the resistance of the coil of galvanometer G is 2Ω . The emf of the cell is $4V$. The ratio of potential difference across C_1 and C_2 is:



[8-Apr-2023 shift 1]

Options:

A. $\frac{5}{4}$

B. 1

C. $\frac{4}{5}$

D. $\frac{3}{4}$

Answer: C

Solution:

At steady state current will not be in the capacitor branch.

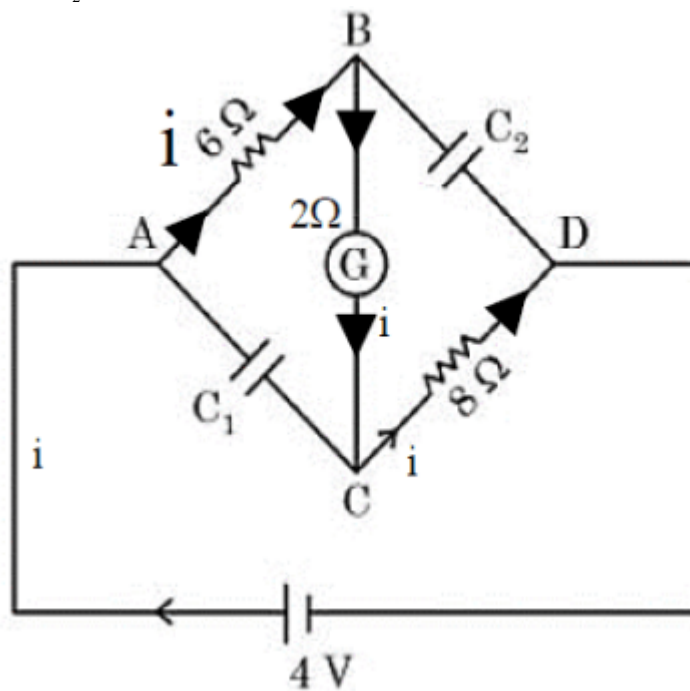
$$i = \frac{4}{6+2+8}$$

$$i = \frac{1}{4} \text{ A}$$

$$\Delta V_{C_1} = i(6+2)$$

$$\Delta V_{C_2} = i(2+8)$$

$$\frac{\Delta V_{C_1}}{\Delta V_{C_2}} = \frac{4}{5}$$



Question101

A current of 2A through a wire of cross-sectional area 25.0mm^2 . The number of free electrons in a cubic meter are 2.0×10^{28} . The drift velocity of the electrons is _____ $\times 10^{-6}\text{ms}^{-1}$ (given, charge on electron = $1.6 \times 10^{-19}\text{C}$).
[8-Apr-2023 shift 1]

Answer: 25

Solution:

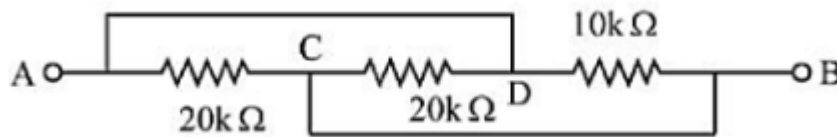
$$I = neAV_d$$

$$V_d = \frac{I}{neA} \Rightarrow V_d = \frac{2}{2 \times 10^{28} \times 1.6 \times 10^{-19} \times 25 \times 10^{-6}}$$

$$V_d = 25 \text{ m/s}$$

Question 102

The equivalent resistance between A and B as shown in figure is:



[8-Apr-2023 shift 2]

Options:

A. $20\text{k}\Omega$

B. $30\text{k}\Omega$

C. $5\text{k}\Omega$

D. $10\text{k}\Omega$

Answer: C

Solution:

Potential difference across all resistor is same

So they are in parallel

$$\frac{1}{R} = \frac{1}{20} + \frac{1}{20} + \frac{1}{10}$$

$$R_{eq} = 5\text{k}\Omega$$

Question 103

The number density of free electrons in copper is nearly

$8 \times 10^{28} \text{ m}^{-3}$. A copper wire has its area of cross section

$= 2 \times 10^{-6} \text{ m}^2$ and is carrying a current of 3.2A. The drift speed

of the electrons is _____ $\times 10^{-6} \text{ms}^{-1}$
[8-Apr-2023 shift 2]

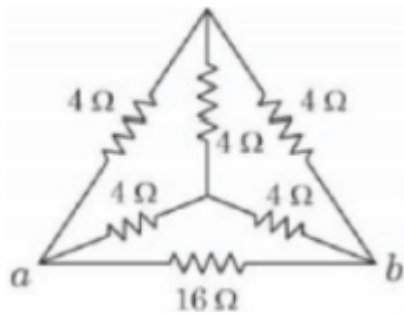
Answer: 125

Solution:

$$\begin{aligned} I &= neAv_d \\ \Rightarrow 3.2 &= 8 \times 10^{28} \times 1.6 \times 10^{-19} \times 2 \times 10^{-6} (v_d) \\ \Rightarrow v_d &= \frac{1}{8 \times 10^{-6} \times 10^9} \\ \Rightarrow v_d &= 125 \times 10^{-6} \text{m/s} \end{aligned}$$

Question104

The equivalent resistance of the circuit shown below between points a and b is :



[10-Apr-2023 shift 1]

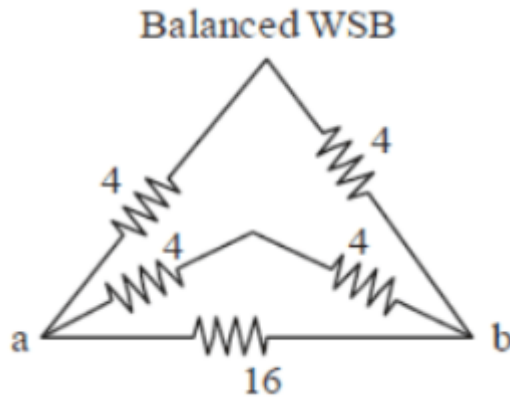
Options:

- A. 20Ω
- B. 16Ω
- C. 24Ω
- D. 3.2Ω

Answer: D

Solution:

$$\begin{aligned}\frac{1}{R_{ab}} &= \frac{1}{16} + \frac{1}{8} + \frac{1}{8} \\ &= \frac{1}{R_{ab}} = \frac{1+2+2}{16} = \frac{5}{16} \\ R_{ab} &= \frac{16}{5} = 3.2\end{aligned}$$



Question105

10 resistors each of resistance 10Ω can be connected in such as to get maximum and minimum equivalent resistance. The ratio of maximum and minimum equivalent resistance will be _____. [10-Apr-2023 shift 1]

Answer: 100

Solution:

$$R_{\max} \Rightarrow \text{in series} \Rightarrow 10R = 10 \times 10 = 100\Omega$$

$$R_{\min} \Rightarrow \text{in parallel} = \frac{R}{10} = \frac{10}{10} = 1\Omega$$

$$\frac{R_{\max}}{R_{\min}} = \frac{100}{1} = 100 \text{ Ans.}$$

$$R_{\min} \Rightarrow \frac{100}{1} = 100 \text{ Ans.}$$

Question106

In a metallic conductor, under the effect of applied electric field, the free electrons of the conductor

[10-Apr-2023 shift 2]

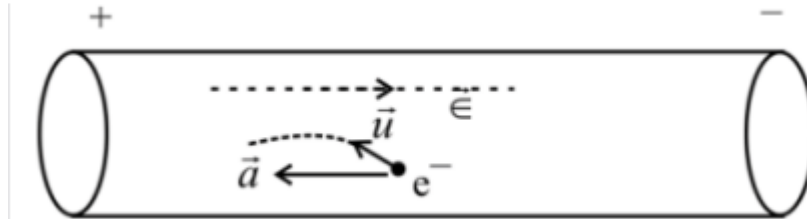
Options:

- A. Move with the uniform velocity throughout from lower potential to higher potential
- B. Move in the curved paths from lower potential to higher potential
- C. Move in the straight line paths in the same direction
- D. Drift from higher potential to lower potential.

Answer: B

Solution:

Electrons moves in curved path because there velocity $\vec{u}$ may make any angle θ with acceleration $\vec{a}$ between time interval of two successive collisions.



Also electron moves from lower potential to higher potential.

Question107

A rectangular parallelepiped is measured as

$1\text{ cm} \times 1\text{ cm} \times 100\text{ cm}$. If its specific resistance is $3 \times 10^{-7}\Omega\text{m}$, then the resistance between its tow opposite rectangular faces will be

_____ $\times 10^{-7}\Omega$

[10-Apr-2023 shift 2]

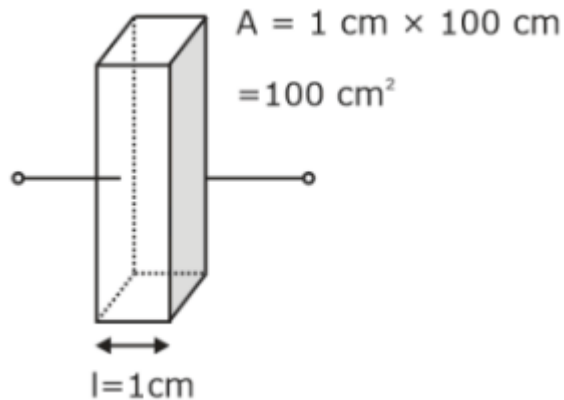
Answer: 3

Solution:

$$\rho = 3 \times 10^{-7} \Omega - \text{cm}$$

$$R = \rho \cdot \frac{1}{A}$$

$$= \frac{3 \times 10^{-7} \times (10^{-2} \text{m})}{(100 \times 10^{-4} \text{m}^2)} = 3 \times 10^{-7}$$



Question108

Two identical heater filaments are connected first in parallel and then in series. At the same applied voltage, the ratio of heat produced in same time for parallel to series will be :

[11-Apr-2023 shift 1]

Options:

A. 1 : 2

B. 4 : 1

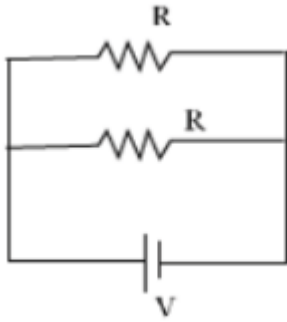
C. 1 : 4

D. 2 : 1

Answer: B

Solution:

$$H_1 = \frac{V^2}{(R/2)}t = \frac{2V^2}{R} \dots (1)$$



$$H_2 = \frac{V^2}{2R}t$$

$$\frac{H_1}{H_2} = \left(\frac{2V^2t}{R} \right) \times \frac{2R}{V^2t} = \frac{4}{1}$$

Question109

**The current sensitivity of moving coil galvanometer is increased by 25%. This increase is achieved only by changing in the number of turns of coils and area of cross section of the wire while keeping the resistance of galvanometer coil constant. The percentage change in the voltage sensitivity will be :
[11-Apr-2023 shift 1]**

Options:

- A. +25%
- B. -25%
- C. -50%
- D. Zero

Answer: A

Solution:

$\tau = mB$ $A =$ area of coil

$K\theta = IANB$ $B =$ magnetic field

$\frac{\theta}{I} = \frac{ANB}{K}$ Current sensitivity

$$1.25 \left(\frac{\theta}{I} \right)_2 = \left(\frac{\theta}{I} \right)_1 \dots \dots \dots (1)$$

$$1.25 \left[\frac{AN_2B}{K} \right] = \left[\frac{AN_1B}{K} \right]$$

$$1.25 = \frac{N_1}{N_2} = \frac{5}{4} \dots\dots (2)$$

$$\Rightarrow R = \frac{\delta \ell}{a} = \text{const.}$$

$$\Rightarrow \ell = a$$

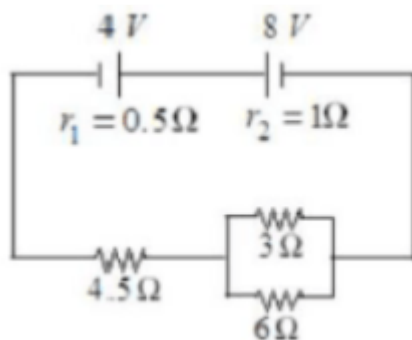
$$\text{Voltage sensitivity} = \frac{\theta}{V} = \frac{\theta}{R} = \frac{\text{Current sensitivity}}{R}$$

$R = \text{constant}$

Voltage sensitivity $\propto$ current sensitivity

Question110

In the circuit diagram shown in figure given below, the current flowing through resistance 3Ω is $\frac{x}{3}\text{A}$. The value of x is _____



[11-Apr-2023 shift 1]

Answer: 1

Solution:

$$R_{eq.} = 0.5 + 1 + 4.5 + \left(\frac{3 \cdot 6}{9} \right)$$

$$R_{eq.} = 6 + 2 = 8\Omega$$

$$I = \frac{8-4}{8} = \frac{1}{2}\text{A} = 0.5\text{A}$$

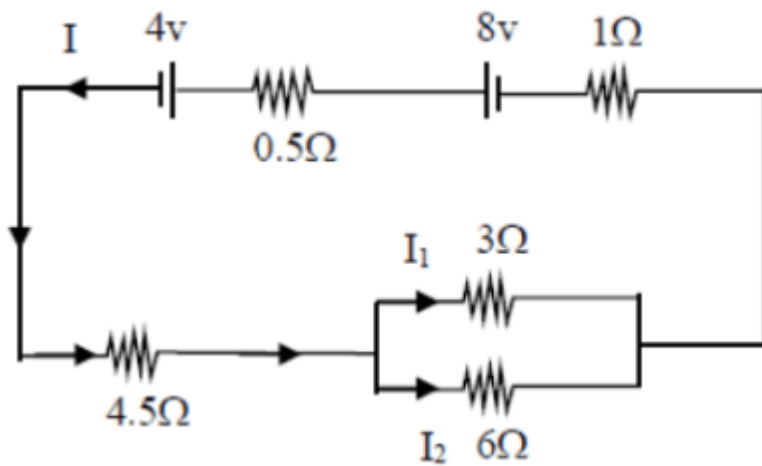
$$I_1 : I_2 = \frac{1}{3} : \frac{1}{6}$$

$$I_1 : I_2 = 2 : 1$$

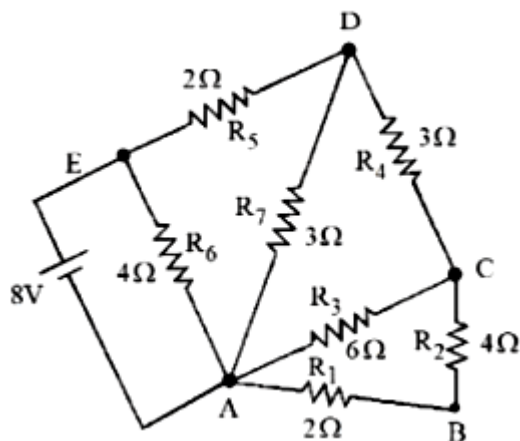
$$\text{and } I_1 + I_2 = 0.5\text{A}$$

$$I_1 = \frac{2}{3} \times 0.5 = \frac{1}{3} \text{A}$$

$$\text{So } \frac{1}{3} = \frac{x}{3} \Rightarrow x = 1$$



Question111



The current flowing through R_2 is:
[11-Apr-2023 shift 2]

Options:

A. $\frac{1}{3} \text{A}$

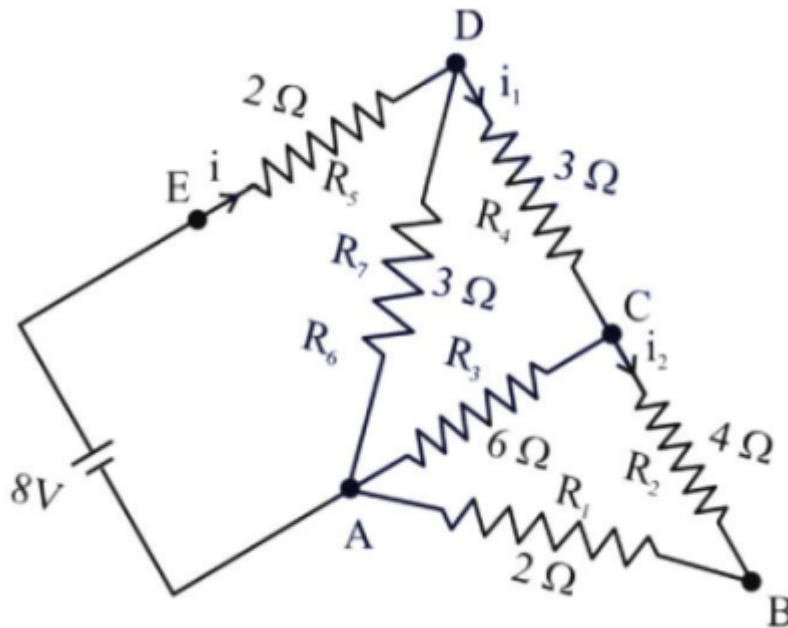
B. $\frac{1}{4} \text{A}$

C. $\frac{2}{3} \text{A}$

D. $\frac{1}{2} \text{A}$

Answer: A

Solution:



$$R_{eq} = 4\Omega$$

$$i = \frac{8}{4} = 2A$$

$$i_1 = \frac{2 \times 3}{3+6} = \frac{2}{3}A$$

$$i_2 = \frac{2/3}{2} = \frac{1}{3}A$$

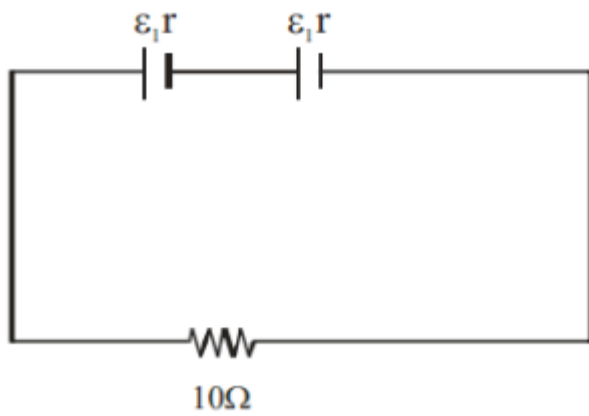
Question112

Two identical cells each of emf 1.5V are connected in series across a 10Ω resistance. An ideal voltmeter connected across 10Ω resistance reads 1.5V. The internal resistance of each cell is _____ Ω .

[11-Apr-2023 shift 2]

Answer: 5

Solution:



$$I = \frac{2\mathcal{E}}{10 + 2r} = \frac{1.5}{10}$$

$$20\mathcal{E} = 15 + 3r$$

$$\Rightarrow 20 \times 1.5 = 15 + 3r$$

$$\Rightarrow 30 = 15 + 3r$$

$$r = 5\Omega$$

Question113

A wire of resistance 160Ω is melted and drawn in a wire of one-fourth of its length. The new resistance of the wire will be [12-Apr-2023 shift 1]

Options:

A. 640Ω

B. 40Ω

C. 10Ω

D. 16Ω

Answer: C

Solution:

Volume remain same

$$A\ell = A'\ell'$$

$$A' = \frac{A\ell}{\ell'} = \frac{A\ell}{\ell/4}$$

$$A' = 4A$$

$$L' = \frac{L}{4}$$

$$\frac{R'}{R} = \frac{\rho \frac{L'}{A}}{\frac{L}{A}}$$

$$R' = R \left[\frac{L'}{L} \times \frac{A}{A} \right]$$

$$R' = 160 \left[\frac{L}{4L} \times \frac{A}{4A} \right]$$

$$R' = 10\Omega$$

Question114

The current flowing through a conductor connected across a source is 2A and 1.2A at 0°C and 100°C respectively. The current flowing through the conductor at 50°C will be _____ $\times 10^2$ mA

[12-Apr-2023 shift 1]

Answer: 15

Solution:

$$i_0 R_0 = i_{100} R_{100} \quad (\text{For same source})$$

$$\Rightarrow 2R_0 = 1.2R(1 + 100\alpha)$$

$$\Rightarrow 1 + 100\alpha = \frac{5}{3} \Rightarrow 100\alpha = \frac{2}{3}$$

$$\Rightarrow 50\alpha = \frac{1}{3}$$

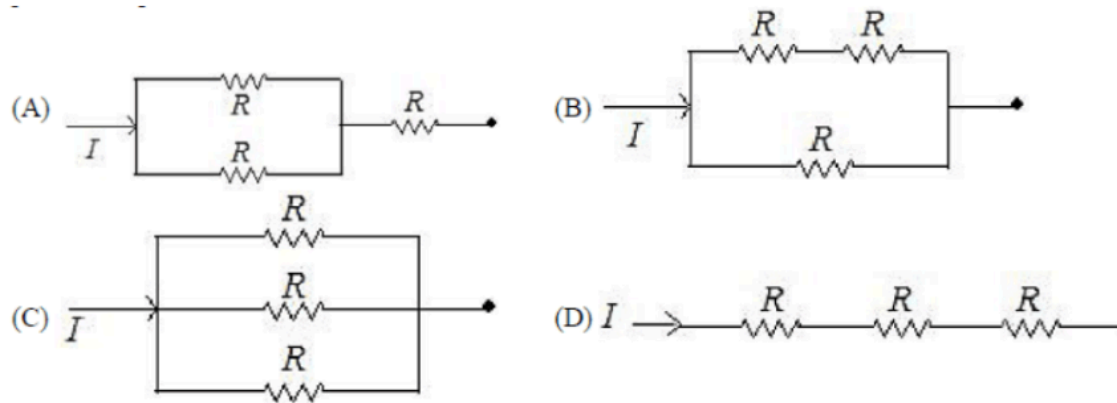
$$\therefore i_{50} R_{50} = i_0 R_0$$

$$\Rightarrow i_{50} = \frac{i_0 R_0}{R_{50}} = \frac{2 \times R_0}{R_0(1 + 50\alpha)} = \frac{2}{1 + \frac{1}{3}} = 1.5A$$

$$= 15 \times 10^2 \text{ mA}$$

Question115

Different combination of 3 resistors of equal resistance R are shown in the figures. The increasing order for power dissipation is:



[13-Apr-2023 shift 1]

Options:

- A. $P_C < P_B < P_A < P_D$
- B. $P_C < P_D < P_A < P_B$
- C. $P_B < P_C < P_D < P_A$
- D. $P_A < P_B < P_C < P_D$

Answer: A

Solution:

Power dissipation, $P = I^2 R$

$$(A) R_{eq} = \frac{R}{2} + R = \frac{3R}{2}$$

$$(B) R_{eq} = \frac{(2R)(R)}{2R + R} = \frac{2R}{3}$$

$$(C) R_{eq} = \frac{R}{3}$$

$$(D) R_{eq} = 3R$$

$$R_D > R_A > R_B > R_C$$

Since, $P \propto R_{eq}$

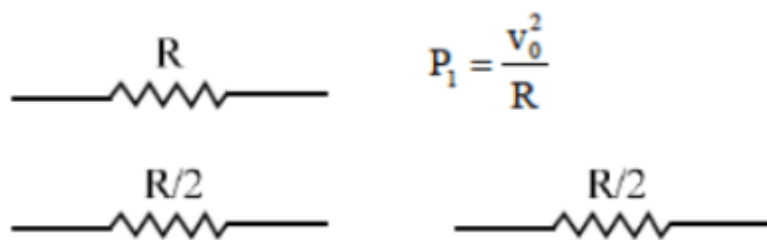
$$P_D > P_A > P_B > P_C$$

Question116

A potential V_0 is applied across a uniform wire of resistance R . The power dissipation is P_1 . The wire is then cut into two equal halves and a potential of V_0 is applied across the length of each half. The total power dissipation across two wires is P_2 . The ratio $P_2 : P_1$ is $\sqrt{x} : 1$. The value of x is _____
[13-Apr-2023 shift 1]

Answer: 16

Solution:



$$P_2 = \frac{V_0^2}{\left(\frac{R}{2}\right)} + \frac{V_0^2}{\left(\frac{R}{2}\right)}$$

$$P_2 = 4P_1$$

$$\frac{P_2}{P_1} = \frac{4}{1} = \frac{\sqrt{x}}{1}$$

$$x = 16$$

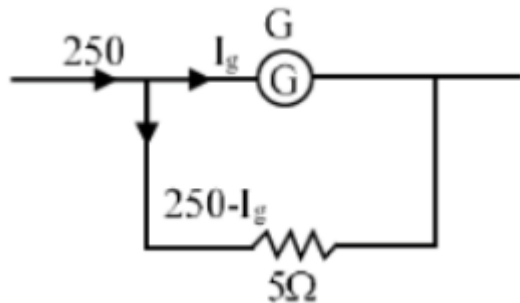
Question117

When a resistance of 5Ω is shunted with a moving coil galvanometer, it shows a full scale deflection for a current of 250 mA , however when 1050Ω resistance is connected with it in series, it gives full scale deflection for 25 volt . The resistance of

galvanometer is _____ Ω .
[13-Apr-2023 shift 1]

Answer: 50

Solution:



$$I_g(G) = (250 - I_g)5$$

$$I_g = \frac{1250}{5 + G} \text{ mA}$$

For voltmeter,

$$V = I_g R$$

$$25 = I_g(G + 1050)$$

From equation (1),

$$25 = \frac{1250 \times 10^{-3}}{G + 5}(G + 1050)$$

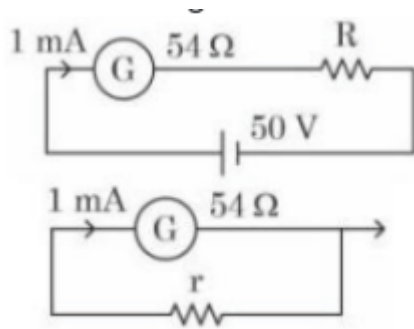
$$20(G + 5) = G + 1050$$

$$19G = 1050 - 100$$

$$G = \frac{950}{19} = 50 \Omega$$

Question118

For designing a voltmeter of range 50V and an ammeter of range 10 mA using a galvanometer which has a coil of resistance 54 Ω showing a full scale deflection for 1 mA as in figure.



(A) for voltmeter $R \approx 50\text{k}\Omega$

(B) for ammeter $r \approx 0.2\Omega$

(C) for ammeter $r \approx 6\Omega$

(D) for voltmeter $R \approx 5\text{k}\Omega$

(E) for voltmeter $R \approx 500\Omega$

Choose the correct answer from the options given below :
[15-Apr-2023 shift 1]

Options:

A. (C) and (D)

B. (A) and (B)

C. (C) and (E)

D. (A) and (C)

Answer: D

Solution:

For voltmeter,

$$I = \frac{50}{R + 54} = 0.001\text{A}$$

$$R = 50\text{k}\Omega$$

For Ammeter,

$$I_r = 10 - 1 = 9\text{ mA}$$

$$V_G = V_r$$

$$1\text{ mA} \times 54 = 9\text{ mA} \times r$$

$$r = 6\Omega$$

Question119

Given below are two statements :

Statement I: The equivalent resistance of resistors in a series combination is smaller than least resistance used in the combination.

Statement II: The resistivity of the material is independent of temperature.

In the light of the above statements, choose the correct answer from the options given below :

[15-Apr-2023 shift 1]

Options:

- A. Both Statement I and Statement II are true
- B. Both Statement I and Statement II are false
- C. Statement I is false but Statement II is true
- D. Statement I is true but Statement II is false

Answer: B

Solution:

In series,

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

$$R_{eq} > R_{\text{Greatest}}$$

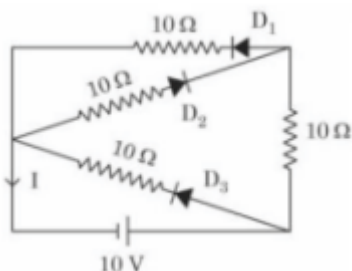
Hence, statement-I is false.

Resistivity of conductor increases with temperature.

Hence, statement-II is also false

Question120

In the given circuit, the current (I) through the battery will be



[15-Apr-2023 shift 1]

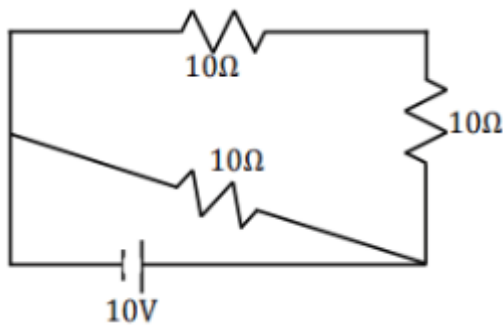
Options:

- A. 1A
- B. 1.5A
- C. 2A
- D. 2.5A

Answer: B

Solution:

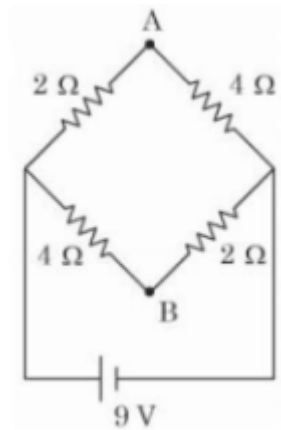
In given figure,
 $D_1, D_3 \rightarrow$ Forward Biased
 $D_2 \rightarrow$ Reversed Biased



$$R_{eq} = \frac{(20)(10)}{30} = \frac{20}{3} \Omega$$
$$I = \frac{V}{R} = \frac{10}{\left(\frac{20}{3}\right)} = 1.5A$$

Question121

A network of four resistances is connected to 9V battery, as shown in figure. The magnitude of voltage difference between the points A and B is _____ V.



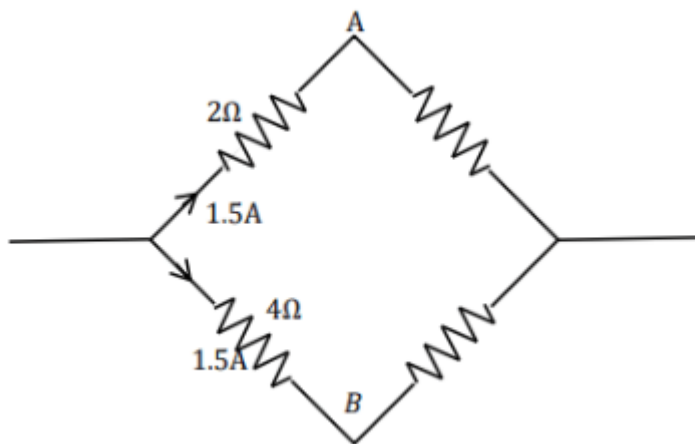
[15-Apr-2023 shift 1]

Answer: 3

Solution:

$$R_{eq} = \frac{6}{2} = 3\Omega$$

$$I = \frac{V}{R} = \frac{9}{3} = 3A$$



$$V_A + (1.5)(2) - (1.5)4 = V_B$$

$$V_A - V_B = 3V$$

Question122

The current density in a cylindrical wire of radius 4 mm is $4 \times 10^6 \text{ Am}^{-2}$. The current through the outer portion of the wire

between radial distances $\frac{R}{2}$ and R is _____ πA
[27-Jun-2022-Shift-1]

Answer: 48

Solution:

$$\begin{aligned} i &= A \times j \\ &= \pi \left(R^2 - \frac{R^2}{4} \right) j \\ &= \frac{3\pi R^2}{4} \times j \\ &= \frac{3\pi \times (4 \times 10^{-3})^2}{4} \times 4 \times 10^6 \\ &= 48\pi \end{aligned}$$

Question123

The current density in a cylindrical wire of radius $r = 4.0$ mm is $1.0 \times 10^6 \text{ A/m}^2$. The current through the outer portion of the wire between radial distances $\frac{r}{2}$ and r is $x\pi A$; where x is _____
[27-Jun-2022-Shift-2]

Answer: 12

Solution:

$$\begin{aligned} i &= A \times j \\ &= \pi \left(R^2 - \frac{R^2}{4} \right) j \\ &= \frac{3\pi R^2}{4} \times j \end{aligned}$$

$$= \frac{3\pi \times (4 \times 10^{-3})^2}{4} \times 1.0 \times 10^6$$

$$= 12\pi$$

Question124

In the given circuit 'a' is an arbitrary constant. The value of m for which the equivalent circuit resistance is minimum, will be

$\sqrt{\frac{x}{2}}$. The value of x is _____

[27-Jun-2022-Shift-2]

Answer: 3

Solution:

$$R_{\text{net}} = \frac{ma}{3} + \frac{a}{2m}$$

$$= a \left[\frac{m}{3} + \frac{1}{2m} - \frac{2}{\sqrt{6}} + \frac{2}{\sqrt{6}} \right]$$

$$= a \left[\left(\sqrt{\frac{m}{3}} - \frac{1}{\sqrt{2m}} \right)^2 + \sqrt{\frac{2}{3}} \right]$$

This will be minimum when

$$\sqrt{\frac{m}{3}} = \frac{1}{\sqrt{2m}}$$

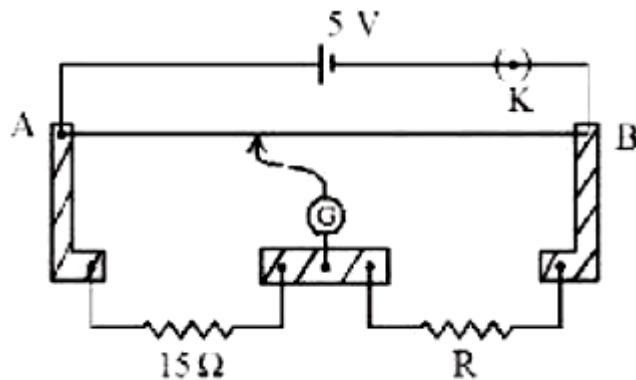
$$\text{or } m = \sqrt{\frac{3}{2}}$$

$$\text{so } x = 3$$

Question125

A meter bridge setup is shown in the figure. It is used to determine an unknown resistance R using a given resistor of 15Ω . The galvanometer (G) shows null deflection when tapping key is at 43 cm mark from end A. If the end correction for end A

is 2 cm, then the determined value of R will be _____Ω.



[28-Jun-2022-Shift-1]

Answer: 19

Solution:

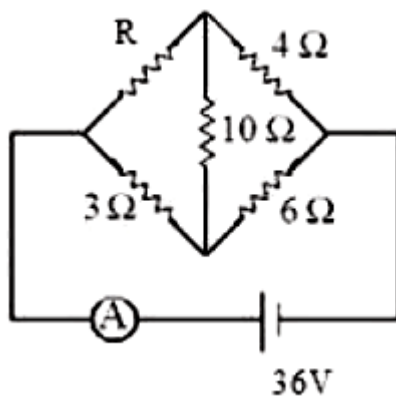
Using the conditions of a balanced wheat stone bridge and adding the end correction.

$$\frac{15}{(43+2)} = \frac{R}{(102-45)} \Rightarrow R = \frac{57}{45} \times 15$$

$$R = 19\Omega$$

Question126

Current measured by the ammeter (A) in the reported circuit when no current flows through 10Ω resistance, will be _____A.



[28-Jun-2022-Shift-1]

Answer: 10

Solution:

Using the condition of a balanced wheat stone bridge,

$$\Rightarrow \frac{R}{3} = \frac{4}{6} \Rightarrow R = 2\Omega$$

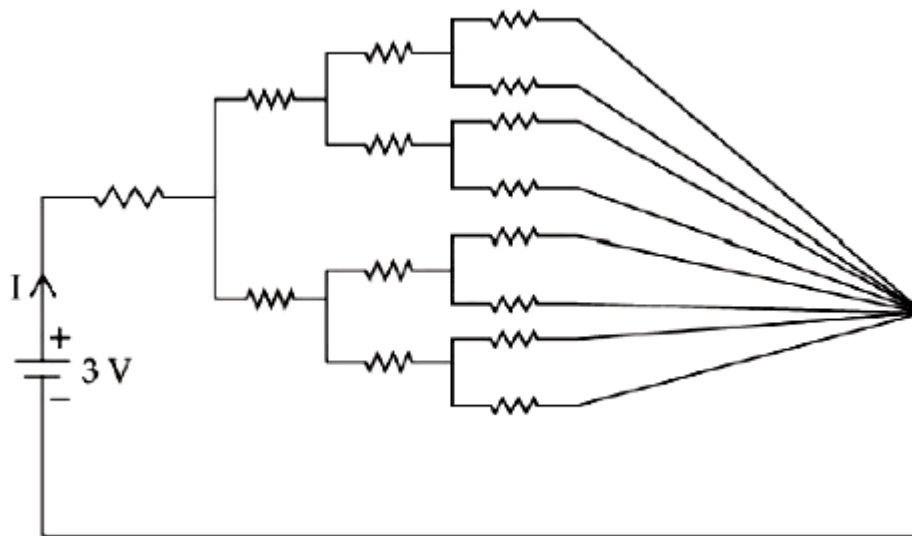
So the effective resistance of the circuit is

$$R_{eq} = \frac{6 \times 9}{6+9} = \frac{18}{5}\Omega$$

$$i = \frac{36}{R_{eq}} = 10A$$

Question127

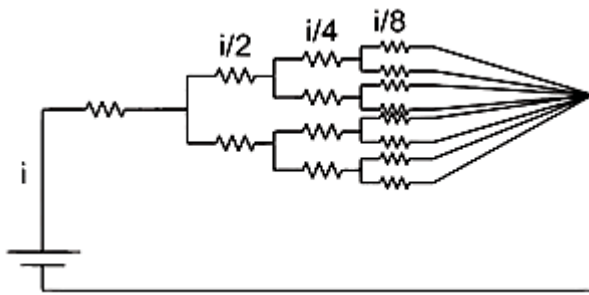
All resistances in figure are 1Ω each. The value of current 'I' is $\frac{a}{5}A$. The value of a is _____



[28-Jun-2022-Shift-2]

Answer: 8

Solution:



Let the current is i

Using Kirchhoff's law

$$iR + \frac{i}{2}R + \frac{i}{4}R + \frac{i}{8}R = 3$$

$$i = \frac{3 \times 8}{15} = \frac{8}{5} \text{ A}$$

So $a = 8$

Question128

Two coils require 20 minutes and 60 minutes respectively to produce same amount of heat energy when connected separately to the same source. If they are connected in parallel arrangement to the same source; the time required to produce same amount of heat by the combination of coils, will be _____ min.

[29-Jun-2022-Shift-1]

Answer: 15

Solution:

$$H = \frac{V^2}{R} \cdot \Delta t$$

$$\Rightarrow H = \frac{V^2}{R_1} \cdot 20 = \frac{V^2}{R_2} \cdot 60 \dots (i)$$

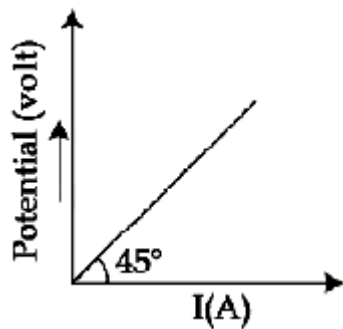
$$\text{Also, } H = \frac{V^2}{\left[\frac{R_1 R_2}{R_1 + R_2} \right]} \cdot \Delta t$$

$$= \frac{4}{3} \cdot \frac{V^2}{R_1} \cdot \Delta t [\because R_2 = 3R_1]$$

$$\Rightarrow \Delta t = 15$$

Question129

The variation of applied potential and current flowing through a given wire is shown in figure. The length of wire is 31.4 cm. The diameter of wire is measured as 2.4 cm. The resistivity of the given wire is measured as $x \times 10^{-3} \Omega \text{ cm}$. The value of x is _____
[Take $\pi = 3.14$]



[29-Jun-2022-Shift-1]

Answer: 144

Solution:

$$\text{Resistance} = \tan 45^\circ = 1 \Omega$$

$$\Rightarrow 1 = \frac{\rho l}{A}$$

$$\Rightarrow \rho = \frac{\pi (1.2 \text{ cm})^2}{31.4 \text{ cm}} = 1.44 \times 10^{-1} \Omega \text{ cm}$$

$$\Rightarrow x = 144$$

Question130

For the network shown below, the value of $V_B - V_A$ is _____ V.
[29-Jun-2022-Shift-1]

Answer: 10

Solution:

$$\begin{aligned} V_B - V_A &= i \times 2 \\ &= \frac{15}{1+2} \times 2 \\ \Rightarrow V_B - V_A &= 10 \text{ volts} \end{aligned}$$

Question131

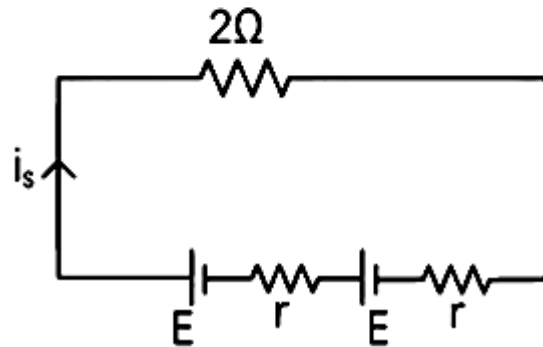
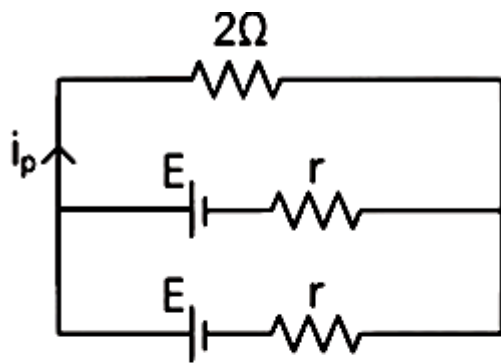
The combination of two identical cells, whether connected in series or parallel combination provides the same current through an external resistance of 2Ω . The value of internal resistance of each cell is
[29-Jun-2022-Shift-2]

Options:

- A. 2Ω
- B. 4Ω
- C. 6Ω
- D. 8Ω

Answer: A

Solution:



From diagram

$$i_p = \frac{E}{2 + \frac{r}{2}} \text{ and } i_s = \frac{2E}{2 + 2r}$$

given $i_p = i_s$

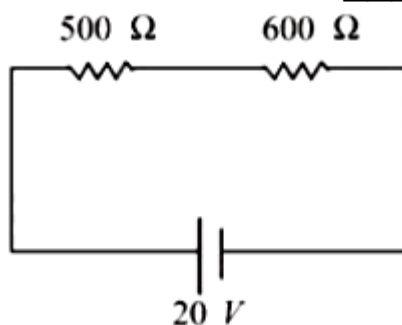
$$\frac{1}{2 + \frac{r}{2}} = \frac{1}{1 + r}$$

$$1 + r = 2 + \frac{r}{2}$$

$$r = 2\Omega$$

Question132

Two resistors are connected in series across a battery as shown in figure. If a voltmeter of resistance 2000Ω is used to measure the potential difference across 500Ω resistor, the reading of the voltmeter will be _____ V.



[29-Jun-2022-Shift-2]

Answer: 8

Solution:

$$\text{New } R_{\text{eff}} = \frac{2000 \times 500}{2500} + 600\Omega = 1000\Omega$$

$$\Rightarrow \text{Reading of voltmeter} = \frac{400}{1000} \times 20 = 8 \text{ volts}$$

Question133

Two identical cells each of emf 1.5V are connected in parallel across a parallel combination of two resistors each of resistance 20Ω. A voltmeter connected in the circuit measures 1.2V. The internal resistance of each cell is :

[24-Jun-2022-Shift-1]

Options:

A. 2.5Ω

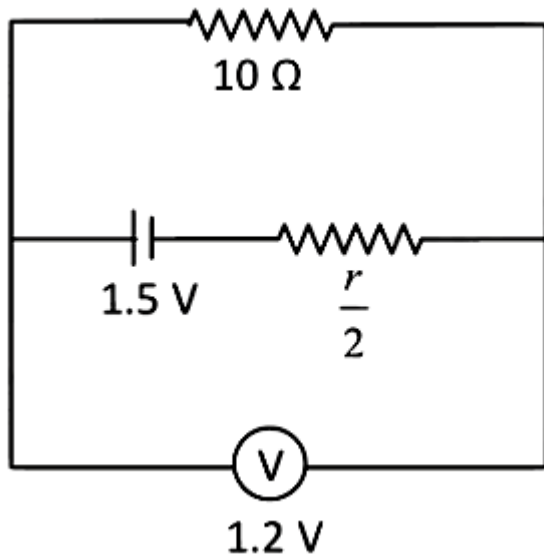
B. 4Ω

C. 5Ω

D. 10Ω

Answer: C

Solution:



$$\frac{1.5 \times 10}{10 + \frac{r}{2}} = 1.2$$

$$\Rightarrow r = 5\Omega$$

Question134

In a potentiometer arrangement, a cell gives a balancing point at 75 cm length of wire. This cell is now replaced by another cell of unknown emf. If the ratio of the emf's of two cells respectively is 3 : 2, the difference in the balancing length of the potentiometer wire in above two cases will be ____ cm.

[24-Jun-2022-Shift-1]

Answer: 25

Solution:

At balancing point, we know that emf is proportional to the balancing length. i.e.,
 $\text{emf} \propto \text{balancing length}$

Now, let the emf's be 3ε and 2ε .

$$\Rightarrow 3\varepsilon = k(75) \dots (1)$$

$$\text{and } 2\varepsilon = k(I) \dots (2)$$

$$\Rightarrow I = 50 \text{ cm}$$

$$\Rightarrow \text{Difference is } (75 - 50) \text{ cm} = 25 \text{ cm.}$$

Question135

What will be the most suitable combination of three resistors $A = 2\Omega$, $B = 4\Omega$, $C = 6\Omega$ so that $\left(\frac{22}{3}\right)\Omega$ is equivalent resistance of combination?

[24-Jun-2022-Shift-2]

Options:

- A. Parallel combination of A and C connected in series with B.
- B. Parallel combination of A and B connected in series with C.
- C. Series combination of A and C connected in parallel with B.
- D. Series combination of B and C connected in parallel with A.

Answer: B

Solution:

$$R_{eq} = \frac{2 \times 4}{2 + 4} + 6 = \frac{22}{3}$$

$\Rightarrow$ A and B are in parallel and C is in series.

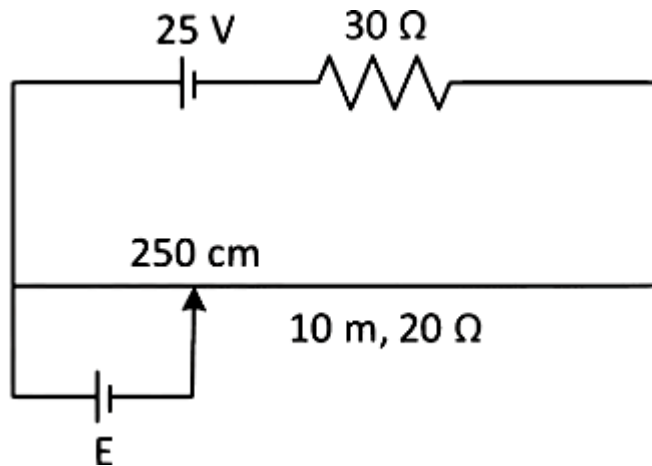
Question136

A potentiometer wire of length 10m and resistance 20Ω is connected in series with a 25V battery and an external resistance 30Ω . A cell of emf E in secondary circuit is balanced by 250 cm long potentiometer wire. The value of E (in volt) is $\frac{x}{10}$. The value of x is _____

[24-Jun-2022-Shift-2]

Answer: 25

Solution:



$$\begin{aligned}\therefore E &= I \times \left(\frac{20}{4} \right) = \frac{25}{(30+20)} \times \left(\frac{20}{4} \right) \\ &= \frac{1}{2} \times 5 = 2.5 \text{ volts} = \frac{25}{10} \text{ volts}\end{aligned}$$

Question137

A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer. Which of the below is true for measuring value of G ?

[25-Jun-2022-Shift-1]

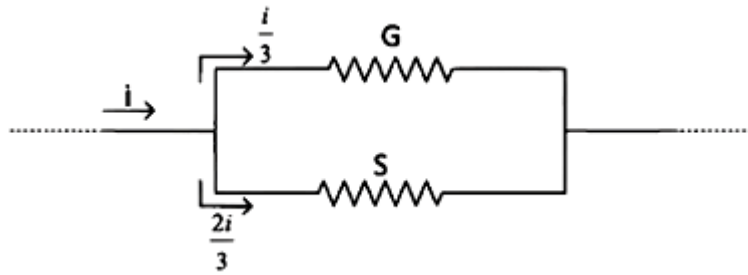
Options:

- A. $\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.
- B. $\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistances.
- C. $\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistances.
- D. $\frac{1}{3}$ deflection method can be used and in this case the G value equals to the shunt resistances.

Answer: B

Solution:

The circuit for the given situation is:



Since G and S are in parallel,

$$\Rightarrow \frac{i}{3} \times G = \frac{2i}{3} \times S$$

$$\Rightarrow G = 2S$$

$\Rightarrow G$ equals twice the value of shunt resistance.

Question138

A resistor develops 300J of thermal energy in 15s, when a current of 2A is passed through it. If the current increases to 3A, the energy developed in 10s is _____ J.

[25-Jun-2022-Shift-1]

Answer: 450

Solution:

$$300 = I^2 R \times 15$$

$$\Rightarrow R = 5\Omega$$

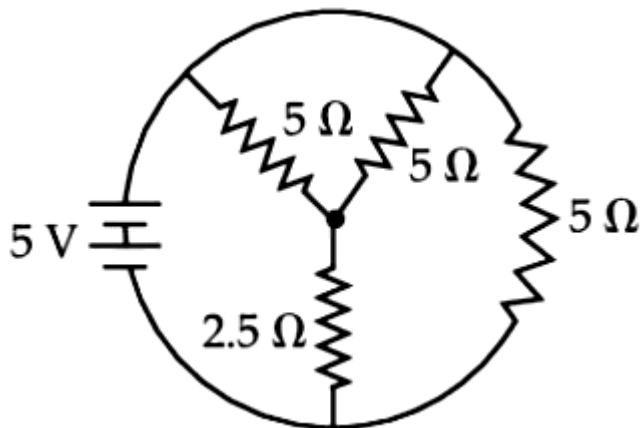
$$\text{Now } I_2^2 R t_2$$

$$= 9 \times 5 \times 10$$

$$= 450\text{J}$$

Question139

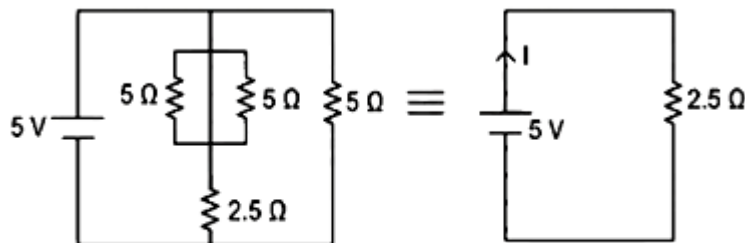
The total current supplied to the circuit as shown in figure by the 5V battery is _____A.



[25-Jun-2022-Shift-1]

Answer: 2

Solution:



$$\Rightarrow I = \frac{5}{2.5} = 2A$$

Question140

Two cells of same emf but different internal resistances r_1 and r_2 are connected in series with a resistance R . The value of resistance R , for which the potential difference across second cell is zero, is :

[25-Jun-2022-Shift-2]

Options:

A. $r_2 - r_1$

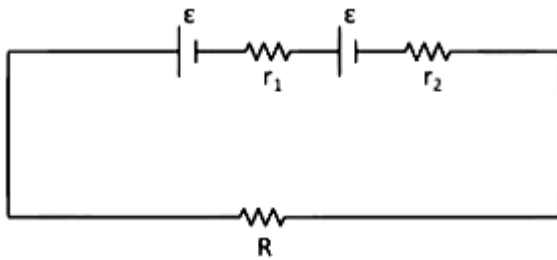
B. $r_1 - r_2$

C. r_1

D. r_2

Answer: A

Solution:



$$I = \frac{2\epsilon}{R + r_1 + r_2}$$

As per the question,

$$\frac{2\epsilon}{R + r_1 + r_2} \times r_2 - \epsilon = 0$$

$$\Rightarrow R = r_2 - r_1$$

Question141

If n represents the actual number of deflections in a converted galvanometer of resistance G and shunt resistance S . Then the total current I when its figure of merit is K will be:

[25-Jun-2022-Shift-2]

Options:

A. $\frac{KS}{(S+G)}$

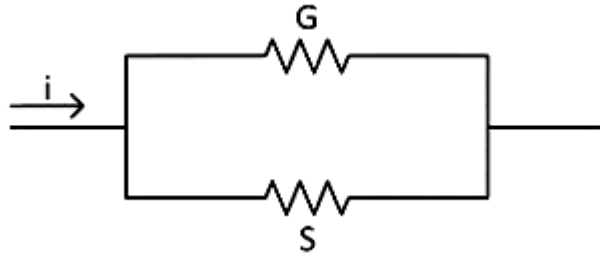
B. $\frac{(G+S)}{nKS}$

C. $\frac{nKS}{(G+S)}$

D. $\frac{nK(G+S)}{S}$

Answer: D

Solution:



$$\Rightarrow \frac{S}{S+G}i = nK$$
$$\Rightarrow i = \frac{nK(S+G)}{S}$$

Question142

The length of a given cylindrical wire is increased to double of its original length. The percentage increase in the resistance of the wire will be _____%.

[25-Jun-2022-Shift-2]

Answer: 300

Solution:

Volume is constant so on length doubled

Area is halved so $R = \rho \frac{l}{A}$ and $R' = \rho \frac{2l}{\frac{A}{2}} = 4\rho \frac{l}{A} = 4R$

So percentage increase will be

$$R\% = \frac{4R - R}{R} \times 100 = 300\%$$

Question143

**An aluminium wire is stretched to make its length, 0.4% larger.
The percentage change in resistance is :
[26-Jun-2022-Shift-1]**

Options:

A. 0.4%

B. 0.2%

C. 0.8%

D. 0.6%

Answer: C

Solution:

$$R = \frac{\rho l}{A}$$

Also volume will remain constant

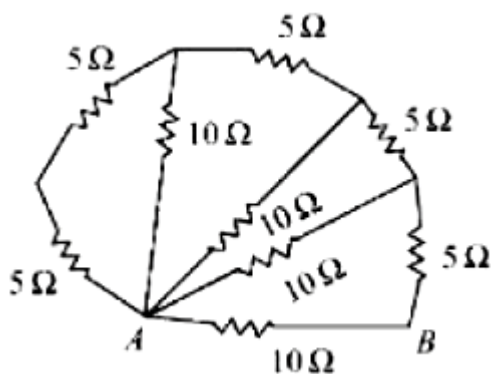
$$\text{i.e., } Al = \text{constant} \Rightarrow A \propto \frac{1}{l}$$

$$\therefore R \propto l^2$$

$$\frac{\Delta R}{R} = 2 \frac{\Delta l}{l} = 0.8$$

Question144

The equivalent resistance between points A and B in the given network is :



[26-Jun-2022-Shift-2]

Options:

A. 65Ω

B. 20Ω

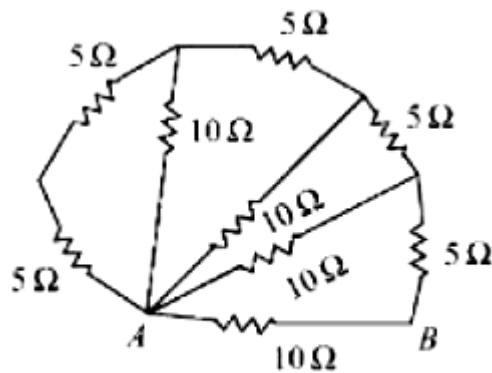
C. 5Ω

D. 2Ω

Answer: C

Solution:

Initially 5Ω and 5Ω are in series and then in parallel with 10Ω this pattern continues thus



$$R_{\text{net}} = 5\Omega$$

Question145

A 72Ω galvanometer is shunted by a resistance of 8Ω . The percentage of the total current which passes through the galvanometer is :

[27-Jun-2022-Shift-1]

Options:

A. 0.1%

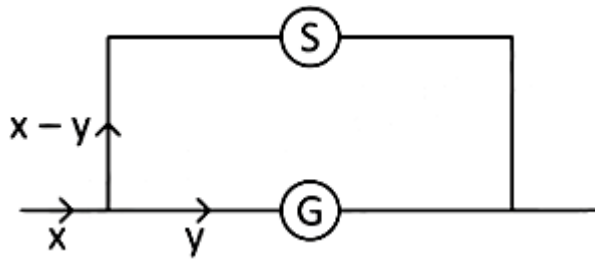
B. 10%

C. 25%

D. 0.25%

Answer: B

Solution:



From the given setup

$$y \times R_G = (x - y)(R_S)$$

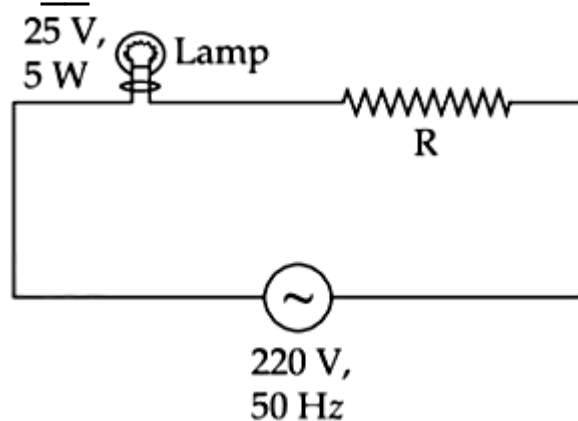
$$\Rightarrow y \times 72 = (x - y) \times 8$$

$$\Rightarrow 9y = x - y$$

$$\Rightarrow y = \frac{x}{10} \text{ or } 10\% \text{ of } x$$

Question146

A 220V, 50 Hz AC source is connected to a 25V, 5W lamp and an additional resistance R in series (as shown in figure) to run the lamp at its peak brightness, then the value of R (in ohm) will be _____



[27-Jun-2022-Shift-1]

Answer: 975

Solution:

$$R_b = \frac{(25)^2}{5} = 125\Omega$$

$$I_{\text{rms}} = \sqrt{\frac{5}{125}} = \frac{1}{5}\text{A}$$

$$\Rightarrow \frac{220}{R + 125} = \frac{1}{5}$$

$$\Rightarrow R = 1100 - 125 \\ = 975\Omega$$

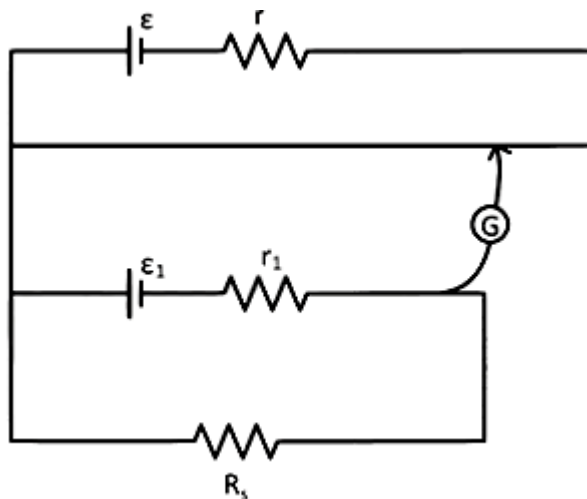
Question147

A cell, shunted by a 8Ω resistance, is balanced across a potentiometer wire of length 3m. The balancing length is 2m when the cell is shunted by 4Ω resistance. The value of internal resistance of the cell will be ____ Ω .

[27-Jun-2022-Shift-1]

Answer: 8

Solution:



$$\frac{\varepsilon_1 8}{r_1 + 8} = 3c$$

$$\frac{\varepsilon_1 4}{r_1 + 4} = 2c$$

$$\Rightarrow \frac{2(r_1 + 4)}{r_1 + 8} = \frac{3}{2}$$

$$\Rightarrow r_1 = 8\Omega$$

Question148

Which of the following physical quantities have the same dimensions?

[25-Jul-2022-Shift-1]

Options:

- A. Electric displacement ($\vec{D}$) and surface charge density
- B. Displacement current and electric field
- C. Current density and surface charge density
- D. Electric potential and energy

Answer: A

Solution:

Electric displacement ($\vec{D}$) = $\epsilon_0 \vec{E}$

$$\Rightarrow [\vec{D}] = [\epsilon_0][\vec{E}]$$

$$= M^{-1}L^{-3}T^4A^2][M^1L^1A^{-1}T^{-3}]$$

$$[\vec{D}] = [L^{-2}T^1A^1]$$

$$[\text{Surface charge density}] = \frac{[Q]}{[A]}$$

$$[\sigma] = [AT L^{-2}]$$

$\Rightarrow \vec{D}$ and $[\sigma]$ have same dimensions

Question149

In the given figure, the value of V_0 will be V.

[25-Jul-2022-Shift-1]

Answer: 4

Solution:

Using Kirchhoff's junction rule.

$$\frac{2 - V_0}{1} + \frac{4 - V_0}{1} + \frac{6 - V_0}{1} = 0$$

$$12 - 3V_0 = 0$$

$$V_0 = 4V$$

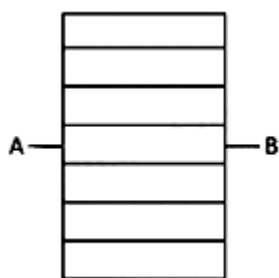
Question 150

Eight copper wire of length l and diameter d are joined in parallel to form a single composite conductor of resistance R . If a single copper wire of length $2l$ have the same resistance (R) then its diameter will be d .

[25-Jul-2022-Shift-1]

Answer: 4

Solution:



$$R_{AB} = R$$

$$R = \frac{1}{8} (\text{Resistance of one wire}) = \frac{1}{8} \rho \frac{l}{\pi \frac{d^2}{4}} = \frac{\rho l}{2\pi d^2}$$

Resistance of copper wire of length $2l$ and diameter $x = R$

$$\rho \frac{2l}{\pi \frac{x^2}{4}} = R$$

$$\frac{8\rho l}{\pi x^2} = \frac{\rho l}{2\pi d^2}$$

$$16d^2 = x^2$$

$$x = 4d$$

Question151

In AM modulation, a signal is modulated on a carrier wave such that maximum and minimum amplitudes are found to be 6V and 2V respectively. The modulation index is:
[25-Jul-2022-Shift-2]

Options:

- A. 100%
- B. 80%
- C. 60%
- D. 50%

Answer: D

Solution:

$$A_{\max} = 6V$$

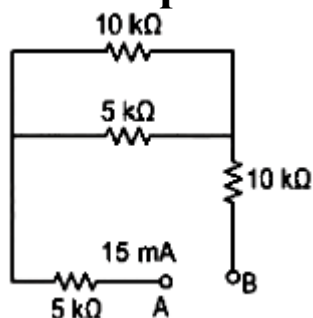
$$A_{\min} = 2V$$

$$\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} = \frac{6 - 2}{6 + 2} = 0.5$$

$$\mu = 50\%$$

Question152

A current of 15mA flows in the circuit as shown in figure. The value of potential difference between the points A and B will be:



[25-Jul-2022-Shift-2]

Options:

A. 50V

B. 75V

C. 150V

D. 275V

Answer: D

Solution:

$$\begin{aligned}\text{Effective } R &= \left[5 + \frac{5 \times 10}{5 + 10} + 10 \right] \text{k}\Omega \\ &= \frac{275}{15} \text{k}\Omega \\ \Rightarrow \Delta V_{AB} &= 15 \text{mA} \times \frac{275}{15} \text{k}\Omega \\ &= 275 \text{V}\end{aligned}$$

Question153

**In a potentiometer arrangement, a cell of emf 1.20V gives a balance point at 36cm length of wire. This cell is now replaced by another cell of emf 1.80V . The difference in balancing length of potentiometer wire in above conditions will be ____ cm.
[25-Jul-2022-Shift-2]**

Answer: 18

Solution:

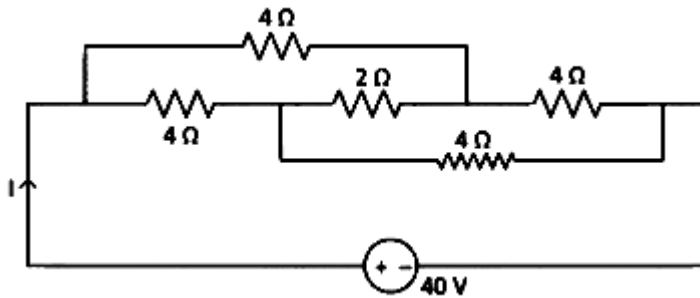
$$\begin{aligned}E &\propto l \\ \frac{1.2}{1.8} &= \frac{36}{l'}\end{aligned}$$

$$I' = \frac{3}{2} \times 36 = 54 \text{ cm}$$

$$\Delta I = I' - I = 54 - 36 = 18 \text{ cm}$$

Question 154

The current I in the given circuit will be :



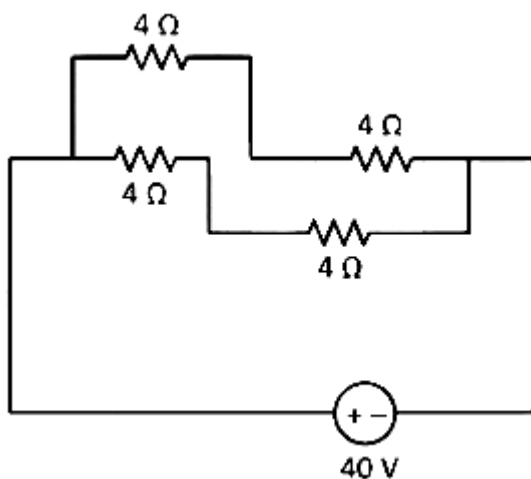
[26-Jul-2022-Shift-1]

Options:

- A. 10A
- B. 20A
- C. 4A
- D. 40A

Answer: A

Solution:

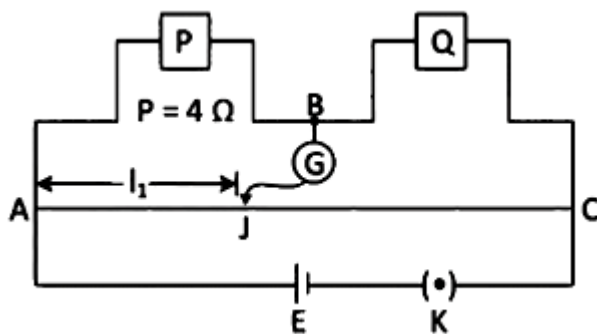


The grouping of resistance is a wheatstone bridge
 So, $R_{\text{net}} = 4\Omega$

$$\text{So, } i = \frac{V}{R_{\text{net}}} = 10\text{A}$$

Question155

Resistances are connected in a meter bridge circuit as shown in the figure. The balancing length l_1 is 40 cm. Now an unknown resistance x is connected in series with P and new balancing length is found to be 80 cm measured from the same end. Then the value of x will be _____ Ω .



[26-Jul-2022-Shift-1]

Answer: 20

Solution:

$$\frac{P}{40} = \frac{Q}{60} \dots\dots (1)$$

$$\frac{P+x}{80} = \frac{Q}{20} \dots\dots (2)$$

$$\frac{P}{P+x} \times \frac{80}{40} = \frac{20}{60}$$

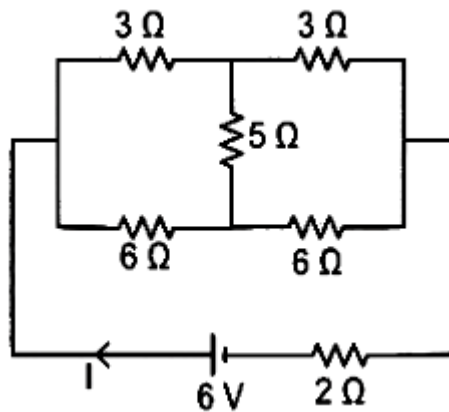
$$\frac{4}{4+x} \times 2 = \frac{1}{3}$$

$$24 = 4 + x$$

$$x = 20$$

Question156

A battery of 6V is connected to the circuit as shown below. The current I drawn from the battery is:



[26-Jul-2022-Shift-2]

Options:

- A. 1A
- B. 2A
- C. $\frac{6}{11}A$
- D. $\frac{4}{3}A$

Answer: A

Solution:

Balanced wheat stone bridge in circuit so there is no current in 5Ω resistor so it can be removed from the circuit.

$$R_{eq} = \frac{6 \times 12}{6 + 12} + 2$$

$$= \frac{6 \times 12}{18} + 2$$

$$R_{eq} = 6\Omega$$

$$I = \frac{V}{R_{eq}} = \frac{6}{6} = 1 \text{ Amp.}$$

Question157

A potentiometer wire of length 300 cm is connected in series with a resistance 780Ω and a standard cell of emf 4V. A constant

current flows through potentiometer wire. The length of the null point for cell of emf 20 mV is found to be 60 cm. The resistance of the potentiometer wire is _____ Ω
[26-Jul-2022-Shift-2]

Answer: 20

Solution:

Let resistance of potentiometers wire is R i = $\frac{4}{R + 780}$

Potential difference across AB

$$= \frac{4R}{R + 780}$$

Potential difference across AC

$$= \frac{4R \times 60}{(R + 780) \times 300} = \frac{4R}{5(R + 780)}$$

This should be equal to 20 mV

$$\frac{4R}{5(R + 780)} = 20 \times 10^{-3} = 2 \times 10^{-2}$$

$$4R = 10^{-1}(R + 780)$$

$$4R = \frac{R}{10} + 78$$

$$4R - \frac{R}{10} = 78$$

$$\frac{39R}{10} = 78$$

$$R = 20\Omega$$

Question158

Two sources of equal emfs are connected in series. This combination is connected to an external resistance R . The internal resistances of the two sources are r_1 and r_2 ($r_1 > r_2$). If the potential difference across the source of internal resistance r_1 is zero, then the value of R will be :
[27-Jul-2022-Shift-1]

Options:

A. $r_1 - r_2$

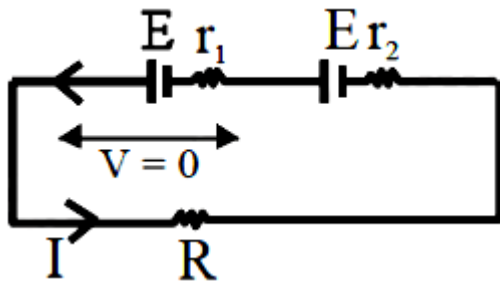
B. $\frac{r_1 r_2}{r_1 + r_2}$

C. $\frac{r_1 + r_2}{2}$

D. $r_2 - r_1$

Answer: A

Solution:



$$I = \frac{2E}{r_1 + r_2 + R}$$

$$IR = E - Ir_2$$

$$I(R + r_2) = E$$

$$I = \frac{E}{R + r_2}$$

$$\frac{2E}{r_1 + r_2 + R} = \frac{E}{R + r_2}$$

$$2R + 2r_2 = r_1 + r_2 + R$$

$$R = r_1 - r_2$$

Question159

A direct current of 4A and an alternating current of peak value 4A flow through resistance of 3Ω and 2Ω respectively. The ratio of heat produced in the two resistances in same interval of time will be :

[27-Jul-2022-Shift-1]

Options:

A. 3 : 2

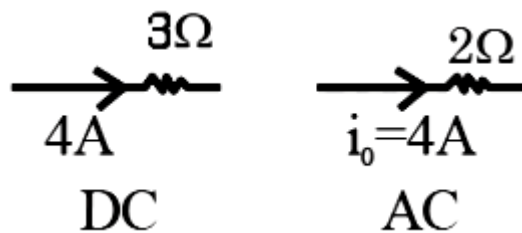
B. 3 : 1

C. 3 : 4

D. 4 : 3

Answer: B

Solution:



$$H_1 = i^2 R_1 t \quad H_2 = i_{\text{rms}}^2 R_2 t \quad \left\{ i_{\text{rms}} = \frac{i_0}{\sqrt{2}} \right\}$$

$$H_1 = 16(3)t \quad H_2 = \frac{i_0^2}{2} R_2 t$$

$$H_2 = 16t$$

$$H_1 : H_2 = 3 : 1$$

Question160

A 1m long copper wire carries a current of 1A. If the cross section of the wire is 2.0mm^2 and the resistivity of copper is $1.7 \times 10^{-8}\Omega\text{m}$, the force experienced by moving electron in the wire is _____ $\times 10^{-23}\text{N}$.

(charge on electron = $1.6 \times 10^{-19}\text{C}$)

[27-Jul-2022-Shift-1]

Answer: 136

Solution:

$$l = 1\text{m}$$

$$i = 1\text{A}$$

$$\text{Area} = 2 \times 10^{-6}$$

$$\rho = 1.7 \times 10^{-8}$$

$$R = \frac{\rho l}{A} = \frac{1.7 \times 10^{-8} \times 1}{2 \times 10^{-5}} = \frac{1.7}{2} \times 10^{-2}$$

$$v = \frac{1.7}{2} \times 10^{-2}$$

$$F = 1.6 \times 10^{-19} \times \frac{1.7}{2} \times 10^{-2}$$

$$= 1.36 \times 10^{-21}$$

$$= 136 \times 10^{-23}$$

Question161

(A) The drift velocity of electrons decreases with the increase in the temperature of conductor.

(B) The drift velocity is inversely proportional to the area of cross-section of given conductor.

(C) The drift velocity does not depend on the applied potential difference to the conductor.

(D) The drift velocity of electron is inversely proportional to the length of the conductor.

(E) The drift velocity increases with the increase in the temperature of conductor.

Choose the correct answer from the options given below :

[27-Jul-2022-Shift-2]

Options:

A. (A) and (B) only

B. (A) and (D) only

C. (B) and (E) only

D. (B) and (C) only

Answer: B

Solution:

$$\text{Drift velocity} = \left(\frac{e\tau}{m} \right) E$$

$$v_d = \left(\frac{e\tau}{m} \right) \left(\frac{\Delta V}{\ell} \right)$$

ΔV = Potential difference applied across the wire

As temperature increases, relaxation time decreases, hence V_d decreases.

As per formula, $V_d \propto \frac{1}{\ell}$

$v_d = \frac{I}{neA}$, as it is not mentioned that current is at steady state neither it is mentioned that n is constant for given conductor. So it can't be said that v_d is inversely proportional to A .

$$I = neAv_d = \frac{V}{R} = \frac{V}{\rho \ell} A$$

$$v_d = \frac{V}{\rho \ell ne} \left(E = \frac{V}{\ell} \right)$$

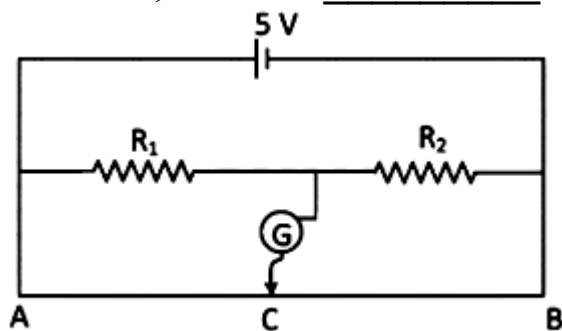
$$v_d = \frac{eE\tau}{m}$$

τ decrease with temperature increase.

First and fourth statements are correct.

Question162

In the given figure of meter bridge experiment, the balancing length AC corresponding to null deflection of the galvanometer is 40 cm. The balancing length, if the radius of the wire AB is doubled, will be _____ cm.



[27-Jul-2022-Shift-2]

Answer: 40

Solution:

Even if the radius of wire is doubled, the balancing point would not change as $\frac{x}{1-x} = \frac{R_1}{R_2}$, which is not including a term of area.

Question163

A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is:

[28-Jul-2022-Shift-1]

Options:

A. 9 : 1

B. 1 : 9

C. 4 : 1

D. 3 : 1

Answer: A

Solution:

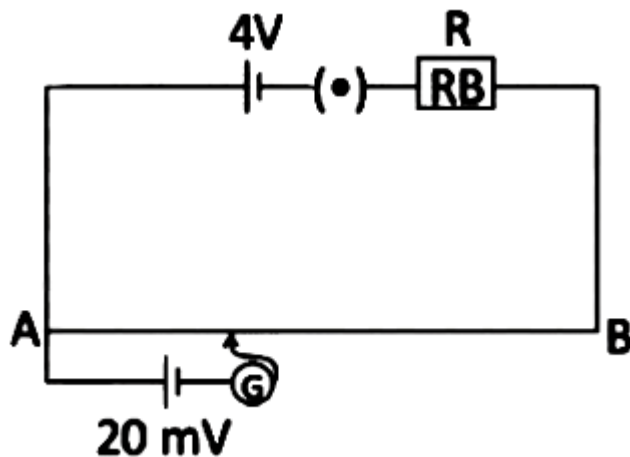
$$R_1 = \rho \frac{L_1}{A_1}$$

$$R_2 = \rho \left(\frac{3L_1}{A_1/3} \right) = 9\rho \frac{L_1}{A_1}$$

$$\therefore \frac{R_2}{R_1} = 9$$

Question164

As shown in the figure, a potentiometer wire of resistance 20Ω and length 300 cm is connected with resistance box (R.B.) and a standard cell of emf 4 V. For a resistance ' R ' of resistance box introduced into the circuit, the null point for a cell of 20 mV is found to be 60 cm. The value of ' R ' _____ is Ω .



[28-Jul-2022-Shift-1]

Answer: 780

Solution:

$$E = \frac{AC}{AB}(V_A - V_B)$$

$$\therefore 20 \times 10^{-3} = \frac{60}{300} \times \frac{4 \times 20}{R + 20}$$

$$\therefore R = 780$$

Question165

Given below are two statements :

Statement I : A uniform wire of resistance 80Ω is cut into four equal parts. These parts are now connected in parallel. The equivalent resistance of the combination will be 5Ω .

Statement II: Two resistances $2R$ and $3R$ are connected in parallel in a electric circuit. The value of thermal energy developed in $3R$ and $2R$ will be in the ratio $3 : 2$.

In the light of the above statements, choose the most appropriate answer from the option given below

[28-Jul-2022-Shift-2]

Options:

- A. Both statement I and statement II are correct
- B. Both statement I and statement II are incorrect
- C. Statement I is correct but statement II is incorrect
- D. Statement I is incorrect but statement II is correct

Answer: C

Solution:

$$\text{Statement I: } R_{1 \text{ part}} = \frac{80}{4} = 20\Omega$$

$$\Rightarrow R_{\text{eff}} = \frac{20}{4} = 5\Omega$$

$$\begin{aligned} \text{Statement II : Ratio} &= \frac{\frac{(\Delta V)^2}{3R}}{\frac{(\Delta V)^2}{2R}} \\ &= \frac{2}{3} \end{aligned}$$

Question166

An electrical bulb rated 220V, 100W, is connected in series with another bulb rated 220V, 60W. If the voltage across combination is 220V, the power consumed by the 100W bulb will be about _____ W.

[28-Jul-2022-Shift-2]

Answer: 14

Solution:

$$R_1 = \frac{V^2}{P} = \frac{220^2}{100} = 484$$

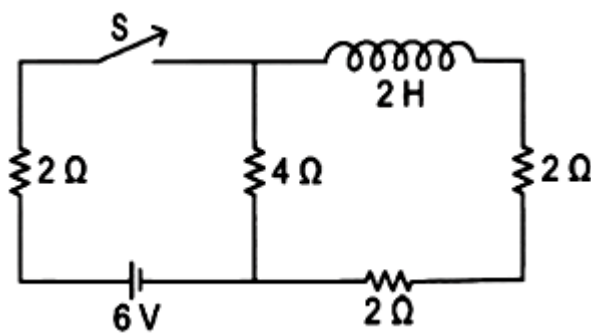
$$R_2 = \frac{V^2}{P} = \frac{220^2}{60} = 484 \left(\frac{10}{6} \right)$$

$$I = \frac{220}{484 + 484 \times \frac{10}{6}}$$

$$P_1 = I^2 R_1 = 14.06 \text{ W}$$

Question 167

For the given circuit the current through battery of 6V just after closing the switch 'S' will be _____ A.



[28-Jul-2022-Shift-2]

Answer: 1

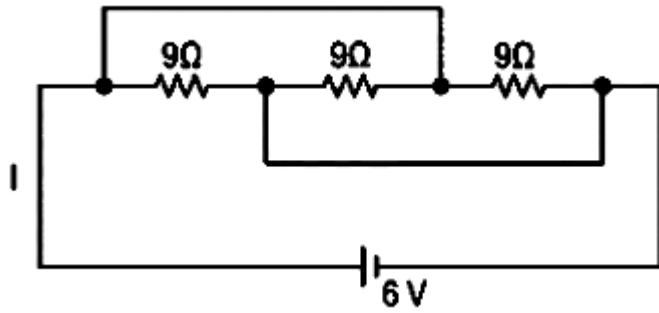
Solution:

Just after closing the switch S, inductor behaves like an open circuit.

$$I = \frac{6}{2+4} = 1 \text{ A}$$

Question 168

The current I flowing through the given circuit will be _____ A.



[29-Jul-2022-Shift-1]

Answer: 2

Solution:

All 9Ω resistances are in parallel

$$R_{eq} = 3\Omega$$

$$I = \frac{6}{3}A = 2A$$

Question169

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Alloys such as constantan and manganin are used in making standard resistance coils.

Reason R: Constantan and manganin have very small value of temperature coefficient of resistance.

In the light of the above statements, choose the correct answer from the options given below.

[29-Jul-2022-Shift-2]

Options:

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Answer: A

Solution:

Since they have low temperature coefficient of resistance, their resistance remains almost constant.

Question170

A 1m long wire is broken into two unequal parts X and Y. The X part of the wire is stretched into another wire W. Length of W is twice the length of X and the resistance of W is twice that of Y. Find the ratio of length of X and Y.
[29-Jul-2022-Shift-2]

Options:

A. 1 : 4

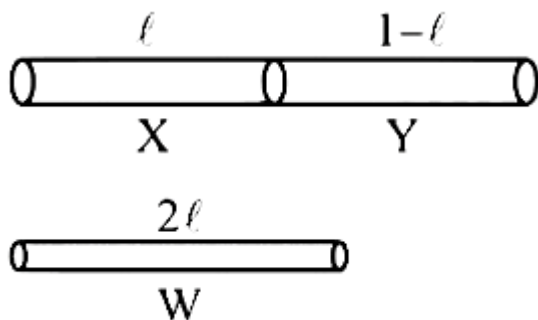
B. 1 : 2

C. 4 : 1

D. 2 : 1

Answer: B

Solution:



$$\frac{R_X}{R_Y} = \frac{\ell_X}{\ell_Y}$$

When wire is stretched to double of its length, then resistance becomes 4 times

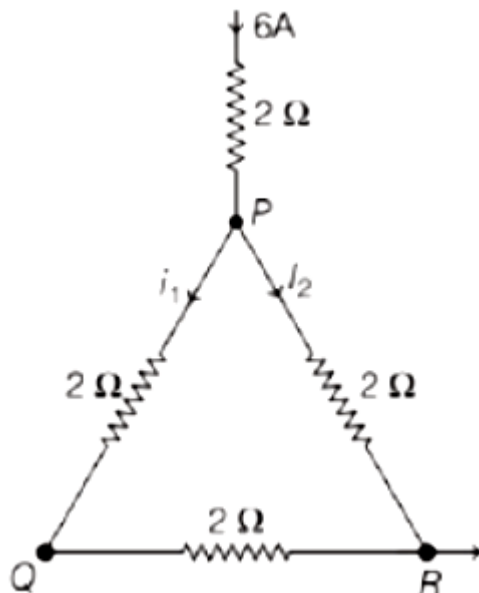
$$R_W = 4R_X = 2R_Y$$

$$\frac{R_X}{R_Y} = \frac{1}{2}$$

$$\text{So, } \frac{\ell_x}{\ell_y} = \frac{1}{2}$$

Question171

A current of 6A enters one corner P of an equilateral triangle PQR having three wires of resistance 2Ω each and leaves by the corner R. The currents i , in ampere is



[25 Feb 2021 Shift 2]

Answer: 2

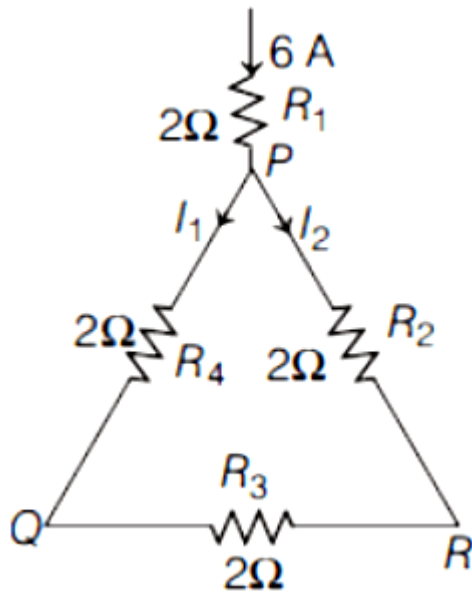
Solution:

Let resistances be R_1, R_2, R_3 and R_4 and I_1 current is passing through R_4 as shown in figure

$\therefore I_2 = (6 - I_1)$ is passing through R_2

As, same current is flowing through R_4 and R_3 .

$\therefore R_4$ and R_3 are in series.



and series equivalent resistance,

$$R_{eq} = R_4 + R_3$$

$$\therefore R_{eq} = 2 + 2 = 4\Omega$$

Voltage through R_{eq} and R_2 will be same.

$$\Rightarrow I_1 R_{eq} = I_2 R_2 \Rightarrow I_1 4 = (6 - I_1) 2$$

$$\Rightarrow 2I_1 = 6 - I_1 \Rightarrow I_1 = 2A$$

Question172

A cylindrical wire of radius 0.5mm and conductivity $5 \times 10^7 \text{ S/m}$ is subjected to an electric field of 10mV /m. The expected value of current in the wire will be $x^3 \pi \text{ mA}$. The value of x is
[24 Feb 2021 Shift 2]

Answer: 5

Solution:

Given, radius of cylindrical wire, $r = 0.5\text{mm} = 0.5 \times 10^{-3}\text{m}$

Conductivity, $\sigma = 5 \times 10^7 \text{ S/m}$

Electric field, $E = 10\text{mV /m} = 10 \times 10^{-3} \text{ V /m}$

We know that current density,

$$\therefore J = \sigma E$$

$$= 5 \times 10^7 \times 10 \times 10^{-3} = 5 \times 10^5 \text{ A/m}^2$$

Also, $J = I / A \Rightarrow I = J A$

$$\Rightarrow I = 5 \times 10^5 \times \pi \times (0.5 \times 10^{-3})^2$$

$$= 5 \times 10^5 \times \pi \times 25 \times 10^{-8} = 125\pi \times 10^{-3}$$

$$\Rightarrow x^3 \pi \text{mA} = 125\pi \text{mA} \Rightarrow x^3 = 5^3$$

$$\Rightarrow x = 5$$

Question173

In an electrical circuit, a battery is connected to pass 20C of charge through it in a certain given time. The potential difference between two plates of the battery is maintained at 15V . The work done by the battery is J.

[26 Feb 2021 Shift 1]

Answer: 300

Solution:

Given, charge passing through circuit, $q = 20\text{C}$

Potential difference between two plates,

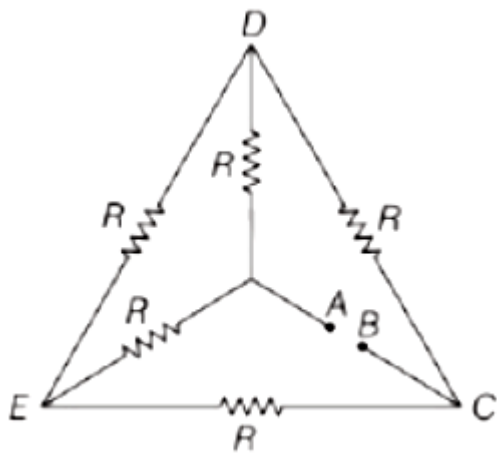
$$V = 15\text{V}$$

Let W be the amount of work done by battery.

$$\therefore W = qV = 20 \times 15 = 300\text{J}$$

Question174

Five equal resistances are connected in a network as shown in figure. The net resistance between the points A and B is



[26 Feb 2021 Shift 1]

Options:

A. $2R$

B. $R/2$

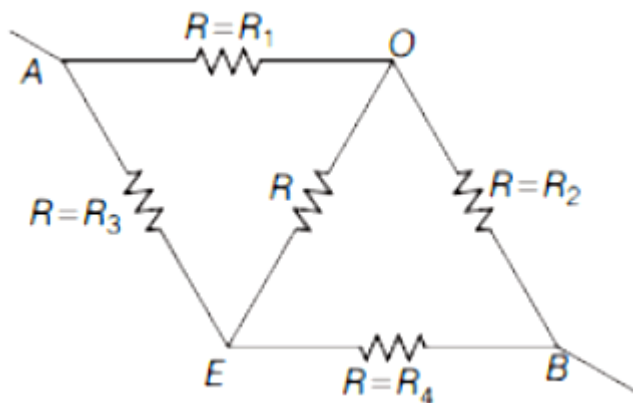
C. $\frac{3R}{2}$

D. R

Answer: D

Solution:

Given all resistances have same resistance R .
Now, we can redraw the circuit as below



Let resistances be R_1 , R_2 , R_3 and R_4 .

$$\therefore \frac{R_1}{R_3} = \frac{R_2}{R_4}$$

So, circuit will behave as a Wheatstone bridge and no current will flow through middle resistor.

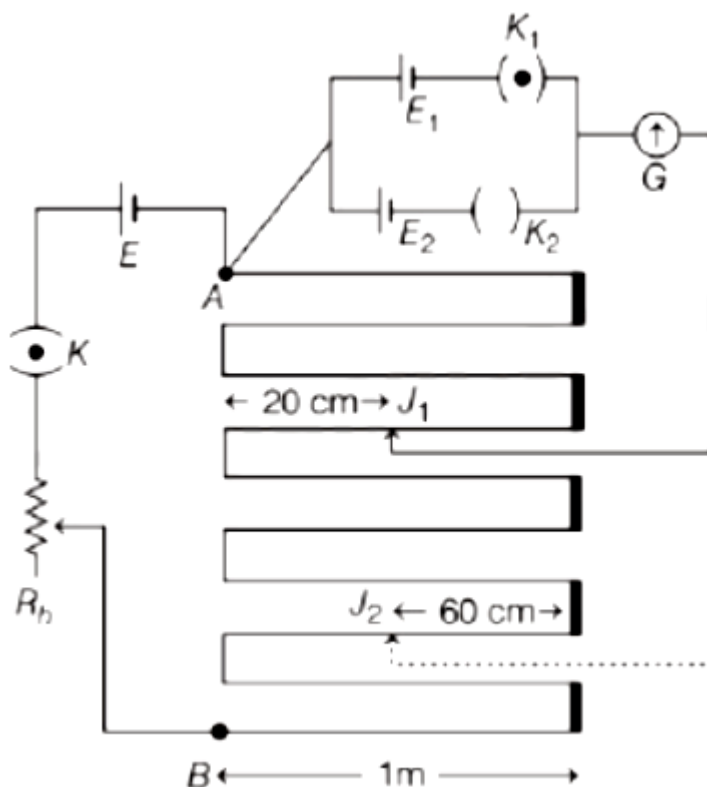
$$\therefore R_{eq} = \frac{(R_1 + R_2)(R_3 + R_4)}{(R_1 + R_2) + (R_3 + R_4)}$$

$$= \frac{(R+R)(R+R)}{(R+R) + (R+R)}$$

$$= R$$

Question 175

In the given circuit of potentiometer, the potential difference E across AB (10 m length) is larger than E_1 and E_2 as well. For key K_1 (closed), the jockey is adjusted to touch the wire at point J_1 , so that there is no deflection in the galvanometer. Now, the first battery (E_1) is replaced by second battery (E_2) for working by making K_1 open and K_2 closed. The galvanometer gives then null deflection at J_2 . The value of $\frac{E_1}{E_2}$ is $\frac{a}{b}$, where $a = \dots\dots\dots$.



[25 Feb 2021 Shift 1]

Answer: 1

Solution:

Given, length of AB = 10m = 1000cm

Length of one arm = $\frac{1000}{10} = 100\text{cm}$

For no deflection,

In first case, $l_1 = 3 \times 100 + 80 = 380\text{cm}$

In 2 nd case, $l_2 = 7 \times 100 + 60 = 760\text{cm}$

As we know that in balanced potentiometer,

$$\Rightarrow \frac{a}{b} = \frac{380}{760} = \frac{1}{2}$$

$$\therefore a = 1$$

Question176

A current through a wire depends on time as $i = \alpha_0 t + \beta t^2$ where $\alpha_0 = 20\text{A/s}$ and $\beta = 8\text{As}^{-2}$. Find the charge crossed through a section of the wire in 15s.
[24feb2021shift1]

Options:

A. 2250C

B. 11250C

C. 2100C

D. 260C

Answer: B

Solution:

Given, $i = \alpha_0 t + \beta t^2$

Put $\alpha_0 = 20$ and $\beta = 8$

We get $i = 20t + 8t^2$

Current, $i = \frac{dq}{dt}$

$$\Rightarrow \int dq = \int i dt$$

$$\Rightarrow q = \int_0^{15} (20t + 8t^2) dt$$

$$\Rightarrow q = \left(\frac{20t^2}{2} + \frac{8t^3}{3} \right)_0^{15}$$

$$\Rightarrow q = 20 \times \left(\frac{15^2 - 0^2}{2} \right) + \frac{8}{3}(15^3 - 0^3)$$

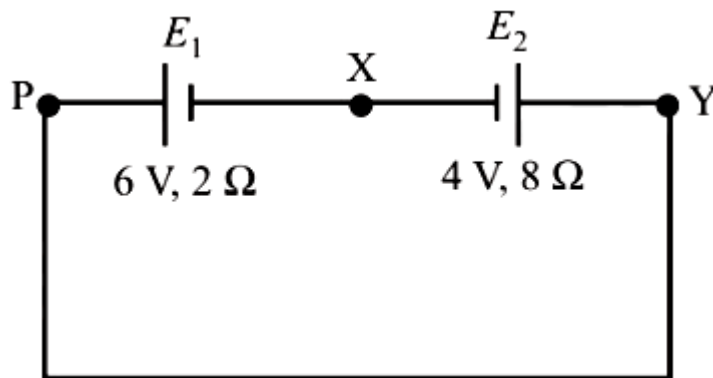
$$\Rightarrow q = 10 \times (15)^2 + \frac{8(15)^3}{3}$$

$$\Rightarrow q = 2250 + 9000$$

$$\Rightarrow q = 11250 \text{C}$$

Question 177

A cell E_1 of emf 6V and internal resistance 2Ω is connected with another cell E_2 of emf 4V and internal resistance 8Ω (as shown in the figure). The potential difference across points X and Y is :



[24feb2021shift1]

Options:

A. 10.0V

B. 3.6V

C. 5.6V

D. 2.0V

Answer: C

Solution:

$$I = \frac{6 - 4}{10} = \frac{1}{5} \text{A}$$

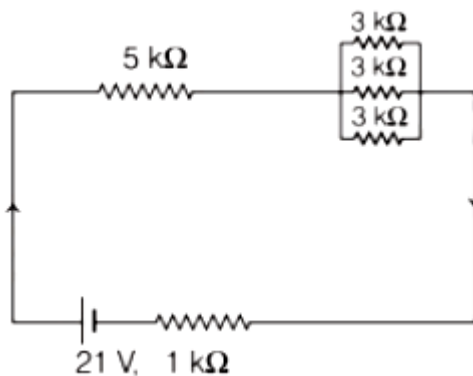
$$V_x + 4 + 8 \times \frac{1}{5} - V_y = 0$$

$$V_x - V_y = -5.6$$

$$\Rightarrow |V_x - V_y| = 5.6V$$

Question 178

In the figure given, the electric current flowing through the $5k\Omega$ resistor is xmA .



The value of x to the nearest Integer is ...3...
[16 Mar 2021 Shift 1]

Answer: 3

Solution:

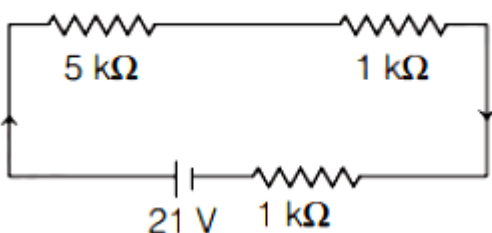
According to the figure given in question, all 3Ω resistances are in parallel combination. So, their equivalent resistance is

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3}$$

$$\frac{1}{R_{\text{parallel}}} = \frac{3}{3}$$

$$\Rightarrow R_{\text{parallel}} = 1k\Omega$$

Equivalent circuit



$5\text{ k}\Omega$ and $1\text{ k}\Omega$ resistance are in series to the equivalent of all $3\text{ }\Omega$ resistances.

$$\therefore R_{\text{net}} = 5 + 1 + R_{\text{parallel}} \dots(i)$$

$$\Rightarrow R_{\text{net}} = (5 + 1 + 1)\text{k}\Omega = 7\text{k}\Omega = 7 \times 10^3 \Omega$$

$\therefore$ The value of electric current flowing through $5\text{k}\Omega$ resistor will be

$$I = \frac{V}{R_{\text{net}}} = \frac{21}{7 \times 10^3} = 3 \times 10^{-3} \text{A}$$

$$= 3\text{mA}$$

Comparing with the given value in the question i.e., $x\text{mA}$, the value of $x = 3$.

Question 179

Two wires of same length and thickness having specific resistances $6\Omega - \text{cm}$ and $3\Omega - \text{cm}$ respectively are connected in parallel. The effective resistivity is $\rho\Omega - \text{cm}$. The value of ρ to the nearest integer, is
[18 Mar 2021 Shift 2]

Answer: 4

Solution:

Given, specific resistance for wire 1 ,

$$\rho_1 = 6\Omega - \text{cm}$$

Specific resistance for wire 2,

$$\rho_2 = 3\Omega - \text{cm}$$

Resistance,

$$R = \frac{\rho l}{A}$$

For parallel connections,

$$R_{\text{net}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\Rightarrow \frac{\rho l}{2A} = \frac{\frac{\rho_1 l}{A} \times \frac{\rho_2 l}{A}}{\frac{\rho_1 l}{A} + \frac{\rho_2 l}{A}}$$

$$\Rightarrow \frac{\rho}{2} = \frac{\rho_1 \rho_2}{\rho_1 + \rho_2}$$

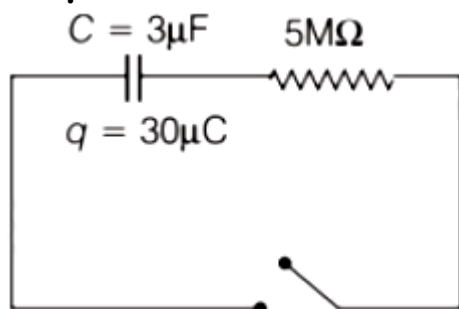
$$\Rightarrow \frac{\rho}{2} = \frac{6 \times 3}{6 + 3}$$

$$\rho = 4\Omega - \text{cm}$$

Hence, the value of ρ to the nearest integer is 4 .

Question180

The circuit shown in the figure consists of a charged capacitor of capacity $3\mu\text{F}$ and a charge of $30\mu\text{C}$. At time $t = 0$, when the key is closed, the value of current flowing through the $5\text{M}\Omega$ resistor is $x\mu\text{A}$. The value of x to the nearest integer is



[18 Mar 2021 Shift 1]

Answer: 2

Solution:

According to given circuit diagram,
At $t = 0$, the key is in closed position.
Current through the resistor will be maximum.
Using Ohm's law,

$$I_{\max} = \frac{V}{R} \Rightarrow I_{\max} = \left(\frac{Q}{C} \right) \times \frac{1}{R}$$

$$\Rightarrow I_{\max} = \left(\frac{30 \times 10^{-6}}{3 \times 10^{-6}} \right) \times \frac{1}{5 \times 10^6}$$

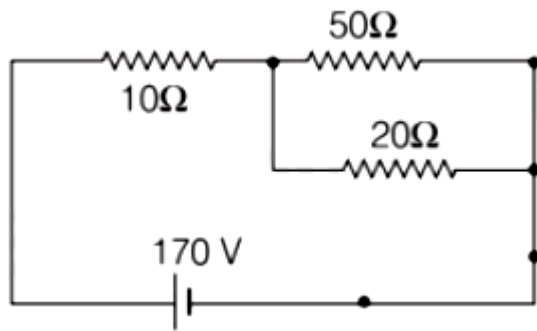
$$I_{\max} = 2 \times 10^{-6} \text{A}$$

$$I_{\max} = 2\mu\text{A}$$

The value of the current flowing through the 5Ω resistor is $2\mu\text{A}$.
Hence, the value of the x to the nearest integer is 2 .

Question181

The voltage across the 10Ω resistor in the given circuit is x volt.

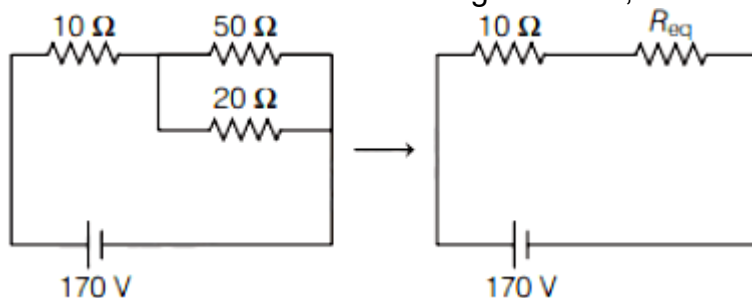


The value of x to the nearest integer is
[18 Mar 2021 Shift 1]

Answer: 70

Solution:

Electrical circuit is shown in the diagram. Now, let's draw the equivalent circuit.



Equivalent resistance of the circuit,

$$R_{eq} = \frac{50 \times 20}{50 + 20}$$

$$R_{eq} = \frac{100}{7} \Omega$$

According to voltage division rule,
voltage across the 10Ω resistance of the circuit,

$$V_{10\Omega} = 170 \times \left(\frac{10}{10 + \frac{100}{7}} \right)$$

$$V_{10\Omega} = 70V$$

The value of the x to the nearest integer is 70 .

Question182

A current of 10A exists in a wire of cross sectional area of 5mm^2 with a drift velocity of $2 \times 10^{-3}\text{ms}^{-1}$. The number of free electrons in each cubic metre of the wire is
[17 Mar 2021 Shift 1]

Options:

A. 2×10^6

B. 625×10^{25}

C. 2×10^{25}

D. 1×10^{23}

Answer: B

Solution:

Given, current, $I = 10\text{A}$

Cross-sectional area, $A = 5\text{mm}^2 = 5 \times 10^{-6}\text{m}^2$

Drift velocity, $v_d = 2 \times 10^{-3}\text{ms}^{-1}$

The value of current flowing through a conductor can be given by

$$I = neAv_d \dots (i)$$

where, n = number of free electrons

and e = charge on an electron

Putting all the given values in Eq. (i) we get

$$10 = n \times 1.6 \times 10^{-19} \times 5 \times 10^{-6} \times 2 \times 10^{-3}$$

$$\Rightarrow n = \frac{10}{1.6 \times 10^{-19} \times 5 \times 10^{-6} \times 2 \times 10^{-3}}$$

$$= 0.625 \times 10^{28} = 625 \times 10^{25}$$

Question183

The equivalent resistance of series combination of two resistors is s. When they are connected in parallel, the equivalent resistance is p. If $s = np$, then the minimum value for n is (Round off to the nearest integer)

[17 Mar 2021 Shift 1]

Answer: 4

Solution:

Let two resistors have resistances R_1 and R_2 , respectively.

As per question, equivalent resistance of series combination is s

$$\Rightarrow s = R_1 + R_2 \dots (i)$$

and equivalent resistance of parallel combination is p

$$\Rightarrow p = \frac{R_1 R_2}{R_1 + R_2} \dots (ii)$$

According to the question, $s = np \dots (iii)$

From Eqs. (i), (ii) and (iii), we get

$$R_1 + R_2 = n \frac{R_1 R_2}{(R_1 + R_2)} \Rightarrow n(R_1 R_2) = (R_1 + R_2)^2$$

$$\Rightarrow n = \frac{(R_1 + R_2)^2}{R_1 R_2}$$

For n to be minimum, $R_1 = R_2 = R$

$$\Rightarrow n = \frac{(R + R)^2}{R \cdot R} = \frac{(2R)^2}{R^2} = \frac{4R^2}{R^2} \Rightarrow n = 4$$

Question184

A conducting wire of length l , area of cross-section A and electric resistivity ρ is connected between the terminals of a battery. A potential difference V is developed between its ends, causing an electric current. If the length of the wire of the same material is doubled and the area of cross-section is halved, the resultant current would be

[16 Mar 2021 Shift 1]

Options:

A. $\frac{1}{4} \frac{VA}{\rho l}$

B. $\frac{3}{4} \frac{VA}{\rho l}$

C. $\frac{1}{4} \frac{\rho l}{VA}$

D. $4 \frac{V A}{\rho l}$

Answer: A

Solution:

Initially, the resistance of wire is $R_1 = \rho L/A$

In second case,

Length, $l' = 2l$

Area, $A' = \frac{A}{2}$

$$\therefore R_2 = \frac{\rho l'}{A'} = \frac{\rho(2l)}{(A/2)} = \frac{4\rho l}{A}$$

According to Ohm's law, $I = \frac{V}{R_2}$

$$\Rightarrow I = \frac{V}{4\rho l/A} = \frac{1}{4} \frac{V A}{\rho l}$$

This is the required value of resultant current.

Question185

The energy dissipated by a resistor is 10mJ in 1s when an electric current of 2mA flows through it. The resistance is Ω . (Round off to the nearest integer)
[16 Mar 2021 Shift 2]

Answer: 2500

Solution:

Since, $\lambda v = c = \text{constant}$

where, $\lambda = \text{wavelength of light}$

and $v = \text{frequency of light}$.

Red light and blue light have different wavelengths and different frequencies but same speed.

$$\Rightarrow R = \frac{H}{I^2 T} \dots\dots(i)$$

Substituting the given values in Eq. (i), we get

$$R = \frac{10 \times 10^{-3}}{(2 \times 10^{-3})^2 \times 1}$$

$$\Rightarrow R = \frac{10^{-2}}{4 \times 10^{-6}} \Rightarrow R = 0.25 \times 10^4$$

$$\Rightarrow R = 2500\Omega$$

Question 186

A resistor develops 500J of thermal energy in 20s, when a current of 1.5A is passed through it. If the current is increased from 1.5A to 3A, what will be the energy developed in 20 s?
[16 Mar 2021 Shift 2]

Options:

- A. 1500J
- B. 1000J
- C. 500J
- D. 2000J

Answer: D

Solution:

Given,

Heat energy, $H_1 = 500\text{J}$

Initial current, $I_1 = 1.5\text{A}$, final current, $I_2 = 3\text{A}$

and time, $t = 20\text{s}$

According to Joule's law of heating,

$$H = I^2 R t$$

$$\Rightarrow H_1 = I_1^2 R t \dots (i)$$

$$\text{and } H_2 = I_2^2 R t \dots (ii)$$

[Resistance and time being the same in both cases]

Dividing Eq. (i) by Eq. (ii), we get

$$\Rightarrow \frac{H_1}{H_2} = \frac{I_1^2 R t}{I_2^2 R t} \Rightarrow \frac{H_1}{H_2} = \left(\frac{I_1}{I_2} \right)^2$$

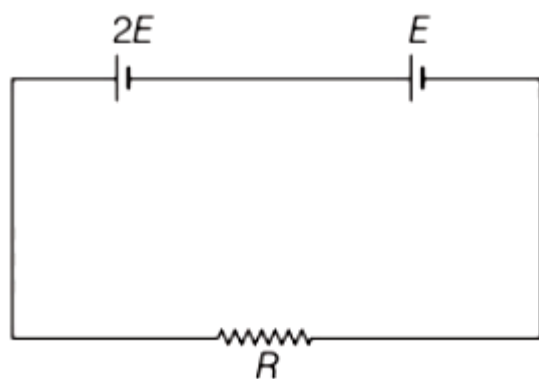
$$\Rightarrow \frac{H_1}{H_2} = \left(\frac{1.5}{3} \right)^2 = \left(\frac{15}{30} \right)^2 \Rightarrow \frac{H_1}{H_2} = \left(\frac{1}{2} \right)^2$$

$$\Rightarrow \frac{H_1}{H_2} = \frac{1}{4} \Rightarrow \frac{500}{H_2} = \frac{1}{4} \Rightarrow H_2 = 500 \times 4$$

$$\Rightarrow H_2 = 2000\text{J}$$

Question 187

Two cells of emf $2E$ and E with internal resistance r_1 and r_2 respectively are connected in series to an external resistor R (see figure). The value of R , at which the potential difference across the terminals of the first cell becomes zero is



[17 Mar 2021 Shift 2]

Options:

A. $r_1 + r_2$

B. $\frac{r_1}{2} - r_2$

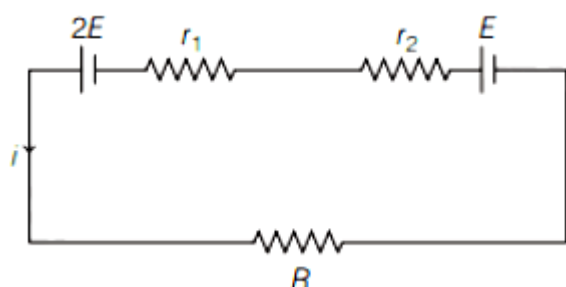
C. $\frac{r_1}{2} + r_2$

D. $r_1 - r_2$

Answer: B

Solution:

The given circuit can be drawn as



Since in series combination, the current through each resistance remains same. So, equivalent resistance of the circuit is given as

$$R_{\text{equivalent}} = R + r_1 + r_2$$

$$\text{and equivalent emf, } E_{\text{equivalent}} = 2E + E = 3E$$

$$\text{From Ohm's law, } I = \frac{E_{\text{equivalent}}}{R_{\text{equivalent}}} \Rightarrow I = \frac{3E}{R + r_1 + r_2}$$

When potential difference is zero across the first cell, then potential positive terminal is equal to the potential at negative terminal.

$$V_P = V_N$$

$$2E = I r_1$$

Substituting the values in the above equation, we get

$$2E = \frac{3E}{R + r_1 + r_2} r_1$$

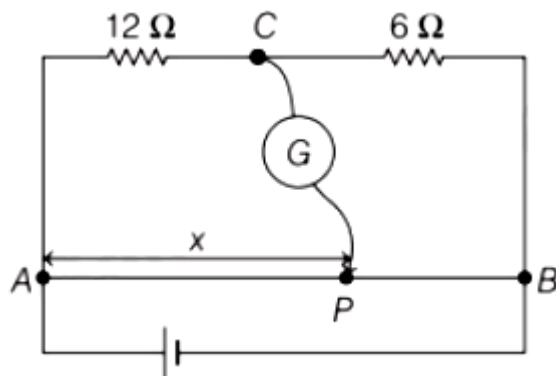
$$2R + 2r_1 + 2r_2 = 3r_1 \Rightarrow R = \frac{r_1 - 2r_2}{2}$$

$$\Rightarrow R = \frac{r_1}{2} - r_2$$

Question188

Consider a 72cm long wire AB as shown in the figure. The galvanometer jockey is placed at P on AB at a distance xcm from A. The galvanometer shows zero deflection.

The value of x, to the nearest integer, is _____.



[18 Mar 2021 Shift 2]

Answer: 48

Solution:

At the balanced condition of the

Wheatstone bridge, $\frac{R}{S} = \frac{L_1}{L - L_1}$

$$\Rightarrow \frac{12}{6} = \frac{x}{72 - x}$$

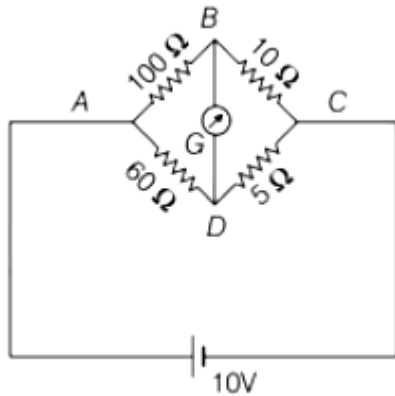
$$\Rightarrow x = 48\text{cm}$$

$\therefore$ The galvanometer jockey is placed at P on AB at a distance of 48 cm from the A.

So, the value of the x to the nearest integer is 48 .

Question189

The four arms of a Wheatstone bridge have resistances as shown in the figure. A galvanometer of 15Ω resistance is connected across BD. Calculate the current through the galvanometer when a potential difference of 10V is maintained across AC.



[17 Mar 2021 Shift 2]

Options:

A. $2.44\mu\text{A}$

B. 2.44mA

C. 4.87mA

D. $4.87\mu\text{A}$

Answer: C

Solution:

As, A is directly connected to the positive terminal of the battery, $V_A = 10\text{V}$ and $V_C = 0$

By nodal analysis at B,

$$\frac{V_B - 10}{100} + \frac{V_B - V_D}{15} + \frac{V_B - 0}{10} = 0$$

$$53V_B - 20V_D = 30 \dots (i)$$

By nodal analysis at D,

$$\frac{V_D - 10}{60} + \frac{V_D - V_B}{15} + \frac{V_D - 0}{5} = 0$$

$$-4V_B + 17V_D = 10 \dots (ii)$$

Solving Eqs. (i) and (ii) by substitution method, we get

$$V_D = 0.792V \Rightarrow V_B = 0.865V$$

The current through the galvanometer,

$$I = \frac{V_B - V_D}{R}$$

Substituting the values in the above equation, we get

$$I = \frac{0.865 - 0.792}{15} \Rightarrow I = 4.87\text{mA}$$

Question190

In the experiment of Ohm's law, a potential difference of 5.0V is applied across the end of a conductor of length 10.0cm and diameter of 5.00mm. The measured current in the conductor is 2.00A. The maximum permissible percentage error in the resistivity of the conductor is
[18 Mar 2021 Shift 1]

Options:

A. 3.9

B. 8.4

C. 7.5

D. 3.0

Answer: A

Solution:

Given, the potential difference applied across the ends of the conductor, $V = 5V$

The length of the conductor, $L = 10\text{cm}$

The measured value of the current in the conductor, $I = 2A$

The diameter of the conductor, $d = 5\text{mm}$

As we know that, $R = \frac{\rho l}{A}$

Using Ohm's law,

$$V = IR \Rightarrow R = \frac{V}{I}$$

$$\frac{\rho l}{A} = \frac{V}{I} \Rightarrow \rho = \frac{V}{Il} \left(\frac{\pi d^2}{4} \right)$$

In error form,

$$\frac{\Delta \rho}{\rho} = \frac{\Delta V}{V} + \frac{\Delta l}{l} + \frac{\Delta l}{l} + 2 \frac{\Delta d}{d}$$

$$\Rightarrow \frac{\Delta \rho}{\rho} = \frac{0.1}{5} + \frac{0.01}{2} + \frac{0.1}{10} + 2 \frac{(0.01)}{(5)} \Rightarrow \frac{\Delta \rho}{\rho} = 0.039$$

$$\frac{\Delta \rho}{\rho} \times 100 = 3.9\%$$

The maximum permissible percentage error in the resistivity of the conductor is 3.9%.

Question191

The resistance of a conductor at 15°C is 16Ω and at 100°C is 20Ω. What will be the temperature coefficient of resistance of the conductor?

[27 Jul 2021 Shift 2]

Options:

A. 0.010°C^{-1}

B. 0.033°C^{-1}

C. 0.003°C^{-1}

D. 0.042°C^{-1}

Answer: C

Solution:

$$16 = R_0[1 + \alpha(15 - T_0)]$$

$$20 = R_0[1 + \alpha(100 - T_0)]$$

Assuming $T_0 = 0^\circ\text{C}$, as a general convention.

$$\Rightarrow \frac{16}{20} = \frac{1 + \alpha \times 15}{1 + \alpha \times 100}$$

$$\Rightarrow \alpha = 0.003^\circ\text{C}^{-1}$$

Question192

In Bohr's atomic model, the electron is assumed to revolve in a circular orbit of radius $0.5\text{\AA}$. If the speed of electron is $2.2 \times 10^6 \text{ m/s}$, then the current associated with the electron will be _____ $\times 10^{-2} \text{ mA}$. [Take π as $\frac{22}{7}$]
[27 Jul 2021 Shift 1]

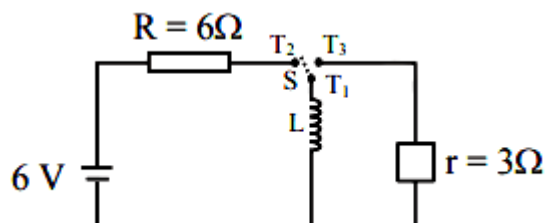
Answer: 112

Solution:

$$I = \frac{e}{T} = \frac{e\omega}{2\pi} = \frac{eV}{2\pi r}$$
$$I = \frac{1.6 \times 10^{-19} \times 2.2 \times 10^6 \times 7}{2 \times 22 \times 0.5 \times 10^{-10}}$$
$$= 1.12 \text{ mA}$$
$$112 \times 10^{-2} \text{ mA}$$

Question193

Consider an electrical circuit containing a two way switch 'S'. Initially S is open and then T_1 is connected to T_2 . As the current in $R = 6\Omega$ attains a maximum value of steady state level, T_1 is disconnected from T_2 and immediately connected to T_3 . Potential drop across $r = 3\Omega$ resistor immediately after T_1 is connected to T_3 is _____ V.
(Round off to the Nearest Integer)



[27 Jul 2021 Shift 1]

Answer: 3

Solution:

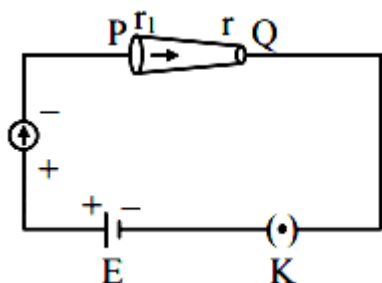
When T_1 and T_2 are connected, then the steady state current in the inductor $I = \frac{6}{6} = 1A$

When T_1 and T_3 are connected then current through inductor remains same. So potential difference across 3Ω

$$V = Ir = 1 \times 3 = 3\text{volt}$$

Question194

In the given figure, a battery of emf E is connected across a conductor PQ of length ' l ' and different area of cross-sections having radii r_1 and r_2 ($r_2 < r_1$). Choose the correct option as one moves from P to Q :



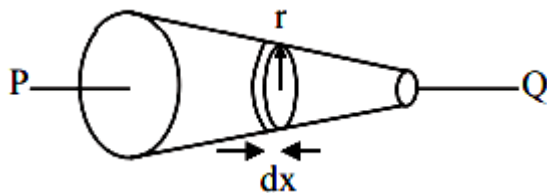
[27 Jul 2021 Shift 1]

Options:

- A. Drift velocity of electron increases.
- B. Electric field decreases.
- C. Electron current decreases.
- D. All of these

Answer: A

Solution:



Current is constant in conductor $i = \text{constant}$

$$\text{Resistance of element } dR = \frac{\rho dx}{\pi r^2}$$

$$dV = i dR = \frac{i \rho dx}{\pi r^2}$$

$$E = \frac{dV}{dx} = \frac{i \rho}{\pi r^2}$$

$$\& V_d = \frac{eE\tau}{m}$$

$$\therefore V_d \propto E$$

$$\rightarrow E \propto \frac{1}{r^2}$$

if r decreases, E will increase $\therefore V_d$ will increase

Question195

For the circuit shown, the value of current at time $t = 3.2$ s will be _____ A.

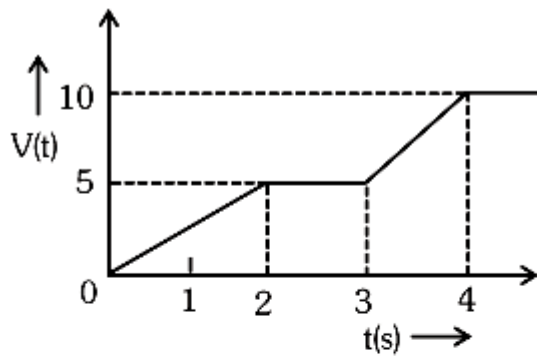


Figure 1

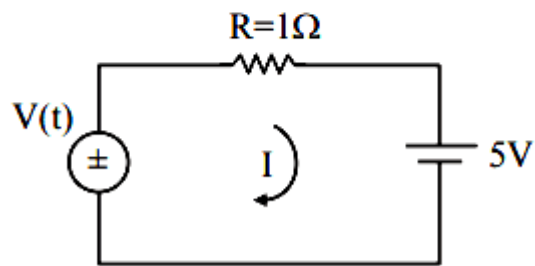


Figure-2

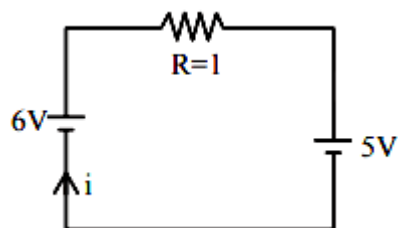
[Voltage distribution $V(t)$ is shown by Fig. (1) and the circuit is shown in Fig. (2)]

[27 Jul 2021 Shift 2]

Answer: 1

Solution:

From graph voltage at $t = 3.2$ sec is 6 volt.



$$i = \frac{6-5}{1}$$

$$i = 1\text{A}$$

Question196

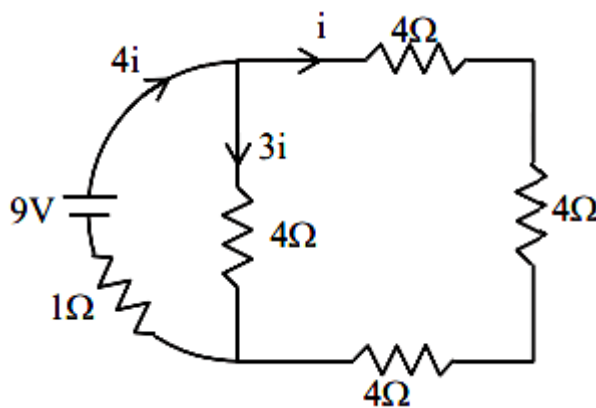
A 16Ω wire is bent to form a square loop. A $9V$ supply having internal resistance of 1Ω is connected across one of its sides. The potential drop across the diagonals of the square loop is

 $\times 10^{-1}V$
 [25 Jul 2021 Shift 2]

Answer: 45

Solution:

here assume current as



By KVL in outer loop

$$9 - 12i - 4i = 0$$

$$16i = 9$$

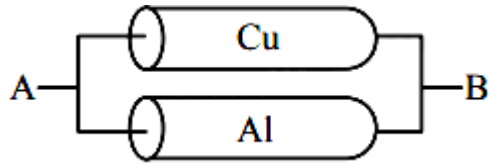
$$8i = \frac{9}{2} = 4.5$$

$$= 45 \times 10^{-1}$$

Question197

A Copper (Cu) rod of length $25cm$ and cross sectional area $3mm^2$ is joined with a similar Aluminum (Al) rod as shown in figure. Find the resistance of the combination between the ends A and B.

(Take Resistivity of Copper $= 1.7 \times 10^{-8}\Omega m$ Resistivity of Aluminium $= 2.6 \times 10^{-8}\Omega m$)



[22 Jul 2021 Shift 2]

Options:

- A. 2.170 mΩ
- B. 1.420 mΩ
- C. 0.0858 mΩ
- D. 0.858 mΩ

Answer: D

Solution:

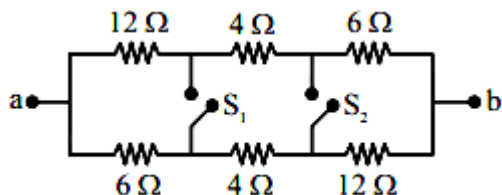
$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{1}{A} \cdot \frac{\rho_1 \rho_2}{\rho_1 + \rho_2}$$

$$R = \frac{25 \times 10^{-2}}{3 \times 10^{-6}} \times \frac{1.7 \times 2.6 \times 10^{-16}}{4.3 \times 10^{-8}}$$

$$R = 0.858 \text{ m}\Omega$$

Question198

In the given figure switches S_1 and S_2 are in open condition. The resistance across ab when the switches S_1 and S_2 are closed is _____ Ω .

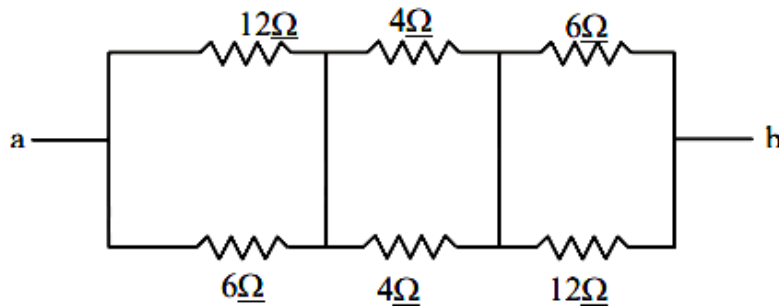


[20 Jul 2021 Shift 2]

Answer: 10

Solution:

when switch S_1 and S_2 are closed



$$\frac{12 \times 6}{12 + 6} + 2 + \frac{6 \times 12}{6 + 12}$$
$$\frac{72}{18} + 2 + \frac{72}{18} = 4 + 2 + 4 = 10\Omega$$

Question199

A current of 5A is passing through a non-linear magnesium wire of cross-section 0.04m^2 . At every point the direction of current density is at an angle of 60° with the unit vector of area of cross-section.

The magnitude of electric field at every point of the conductor is :

(Resistivity of magnesium $\rho = 44 \times 10^{-8}\Omega\text{m}$)

[20 Jul 2021 Shift 1]

Options:

A. $11 \times 10^{-2}\text{V} / \text{m}$

B. $11 \times 10^{-7}\text{V} / \text{m}$

C. $11 \times 10^{-5}\text{V} / \text{m}$

D. $11 \times 10^{-3}\text{V} / \text{m}$

Answer: C

Solution:

$$I = \vec{J} \cdot \vec{A} = J A \cos(\theta)$$

$$5 = J \left(\frac{4}{100} \right) \times \cos(60)$$

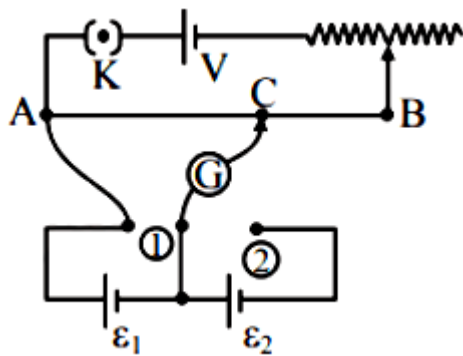
$$J = 5 \times 50 = 250 \text{ A/m}^2$$

$$\text{Now, } \vec{E} = \rho \cdot \vec{J}$$

$$= 44 \times 10^{-8} \times 250 = 11 \times 10^{-5} \text{ V/m}$$

Question 200

In the given potentiometer circuit arrangement, the balancing length AC is measured to be 250cm. When the galvanometer connection is shifted from point (1) to point (2) in the given diagram, the balancing length becomes 400cm. The ratio of the emf of two cells, $\frac{\epsilon_1}{\epsilon_2}$ is :



[25 Jul 2021 Shift 2]

Options:

A. $\frac{5}{3}$

B. $\frac{8}{5}$

C. $\frac{4}{3}$

D. $\frac{3}{2}$

Answer: A

Solution:

$$E_1 = k l_1 \dots\dots\dots(i)$$

$$E_1 + E_2 = k l_2 \dots\dots\dots(ii)$$

$$\frac{E_1}{E_1 + E_2} = \frac{l_1}{l_2} = \frac{250}{400} = \frac{5}{8}$$

$$8E_1 = 5E_1 + 5E_2$$

$$3E_1 = 5E_2$$

$$\frac{E_1}{E_2} = \frac{5}{3}$$

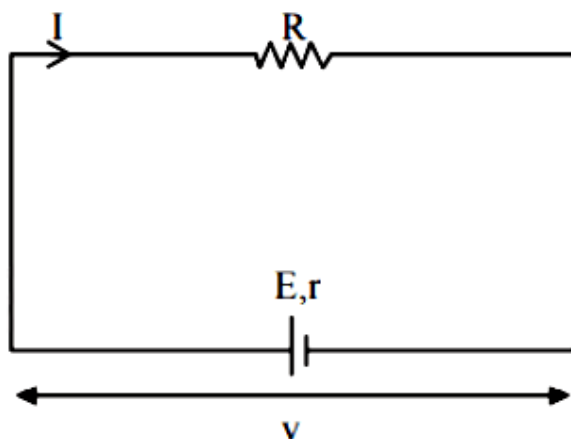
Question201

In an electric circuit, a cell of certain emf provides a potential difference of 1.25V across a load resistance of 5Ω. However, it provides a potential difference of 1V across a load resistance of 2Ω. The emf of the cell is given by $\frac{x}{10}$ V . Then the value of x is

_____.
[22 Jul 2021 Shift 2]

Answer: 15

Solution:



$$\text{Terminal voltage } V = iR = \frac{E R}{R + r}$$

$$1^{\text{st}} \rightarrow 1.25 = \frac{E (5)}{5 + r} \dots\dots\dots(i)$$

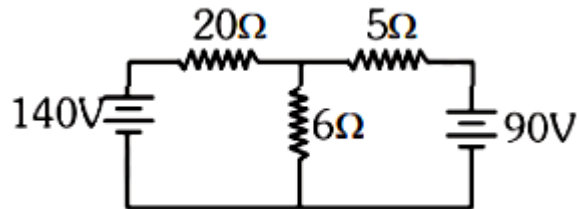
$$2^{\text{nd}} \rightarrow 1 = \frac{E (2)}{2 + r} \dots\dots\dots(ii)$$

By (i) and (ii)

$$r = 1\Omega, E = \frac{3}{2}V = \frac{15}{10}\text{ volt}$$

$$\Rightarrow x = 15$$

Question202



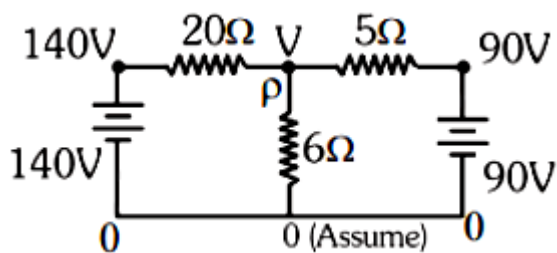
The value of current in the 6Ω resistance is :
[20 Jul 2021 Shift 1]

Options:

- A. 4A
- B. 8A
- C. 10A
- D. 6A

Answer: C

Solution:



Applying KCL at point P,

$$\frac{V - 0}{6} + \frac{V - 90}{5} + \frac{V - 140}{20} = 0$$

$$\Rightarrow 10V + 12V - 1080 + 3V - 420 = 0$$

$$\Rightarrow V = 60$$

$$\therefore \text{current in } 6\Omega = \frac{V - 0}{6} = 10\text{A}$$

Hence option 3.

Question203

A square shaped wire with resistance of each side 3Ω is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be
[31 Aug 2021 Shift 1]

Answer: 3

Solution:

Let the sides of square be a .

$\therefore$ Total length or perimeter of square $= 4a$

If the radius of shape of circle be r , then

$$\therefore 2\pi r = 4a$$

$$\Rightarrow r = \frac{4a}{2\pi}$$

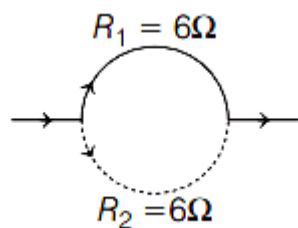
$$\Rightarrow r = \frac{2a}{\pi}$$

Since, resistance of each side of square $= 3\Omega$

$\therefore$ Total resistance of square $= 4 \times 3 = 12\Omega$

i.e. resistance of length $2\pi r = 12\Omega$

$\Rightarrow$ Resistance of $\pi r = 6\Omega$

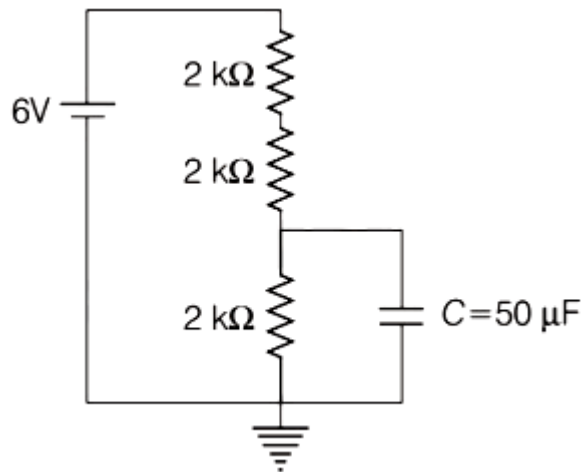


Now, equivalent resistance of circle diametrically opposite

$$(R_{eq}) = \frac{R_1 R_2}{R_1 + R_2} = \frac{6 \times 6}{6 + 6} = 3\Omega$$

Question204

A capacitor of $50\mu\text{F}$ is connected in a circuit as shown in figure.
The charge on the upper plate of the capacitor is..... μC .



[31 Aug 2021 Shift 1]

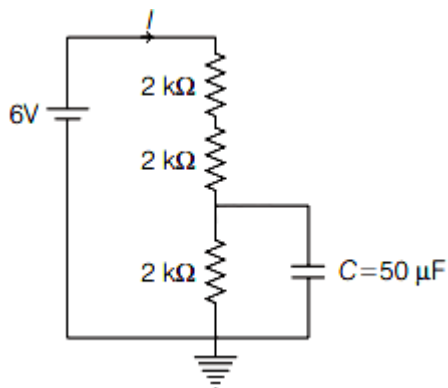
Answer: 100

Solution:

According to given circuit diagram,

Capacitance $= 50\text{ }\mu\text{F} = 50 \times 10^{-6}\text{ F}$

Supply voltage, $V = 6\text{ V}$



In steady state, capacitor will act as open circuit,

$\therefore$ Equivalent resistance $R_{\text{eq}} = (2 + 2 + 2)\text{ k}\Omega = 6\text{ k}\Omega$

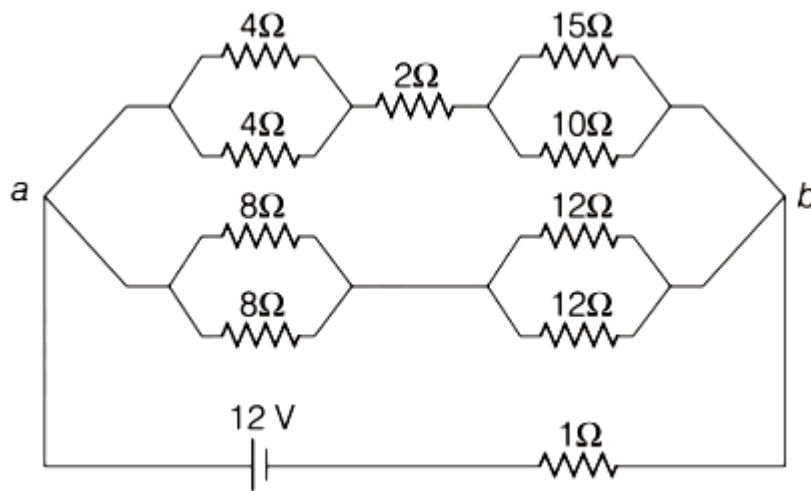
Circuit current, $I = \frac{V}{R_{\text{eq}}} = \frac{6}{6 \times 1000} = 10^{-3}\text{ A}$

$\therefore$ Voltage across $2\text{ k}\Omega = 1 \times 2 = 10^{-3} \times 2 \times 10^3 = 2\text{ V}$

Now, charge on capacitor, $q = CV = 50 \times 10^{-6} \times 2$
 $= 100 \times 10^{-6}\text{ C}$
 $= 100\text{ }\mu\text{C}$

Question205

The voltage drop across 15Ω resistance in the given figure will be V.

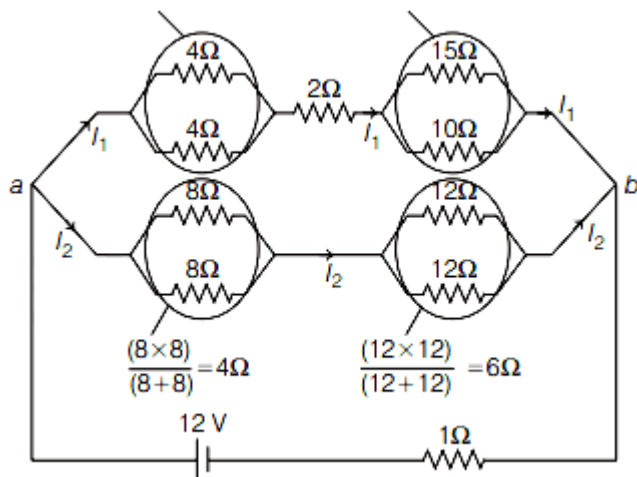


[31 Aug 2021 Shift 1]

Answer: 6

Solution:

According to given circuit diagram,



As we know that, parallel equivalent resistance,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

and series equivalent resistance,

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

Let the net resistance across a and b be R'

$$\therefore \frac{1}{R'} = \frac{1}{2+2+6} + \frac{1}{4+6}$$

$$a = \frac{1}{10} + \frac{1}{10} = \frac{2}{10}$$

$$R' = 5\Omega$$

Hence, total resistance, $R = R' + 1 = 5 + 1 = 6\Omega$

According to current division rule, current in upper branch,

$$I_1 = I \cdot \frac{4+6}{(2+2+6)+(4+6)}$$
$$= 1 \cdot \frac{10}{20} = \frac{1}{2} = \frac{1}{2} \cdot \frac{V}{R} = \frac{1}{2} \times \frac{12}{6} = 1\text{A}$$

Again, according to current division rule, current in 15Ω resistor

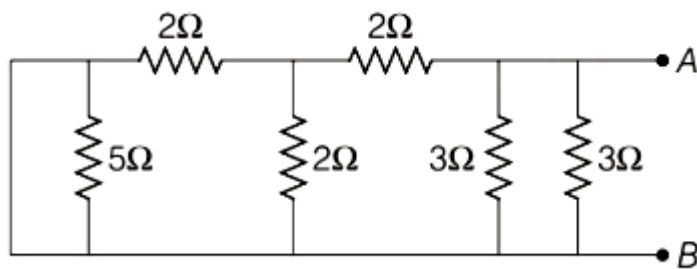
$$I_{15} = I_1 \cdot \frac{10}{10+15} = 1 \times \frac{2}{5} = 0.4\text{A}$$

$\therefore$ Voltage drop across 15Ω resistor,

$$V_{15} = I_{15} \times 15 = 0.4 \times 15 = 6\text{V}$$

Question 206

The equivalent resistance of the given circuit between the terminals A and B is



[31 Aug 2021 Shift 2]

Options:

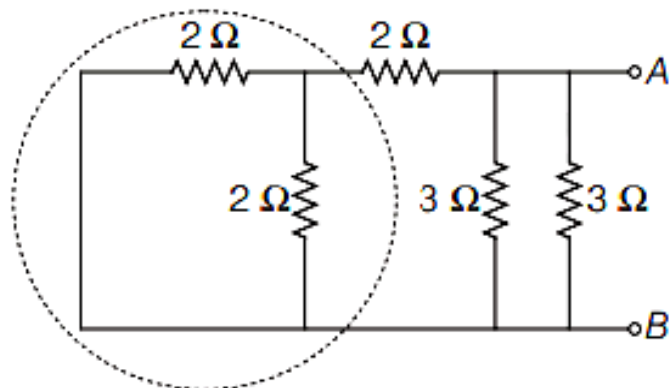
- A. 0
- B. 3Ω
- C. $\frac{9}{2}\Omega$
- D. 1Ω

Answer: D

Solution:

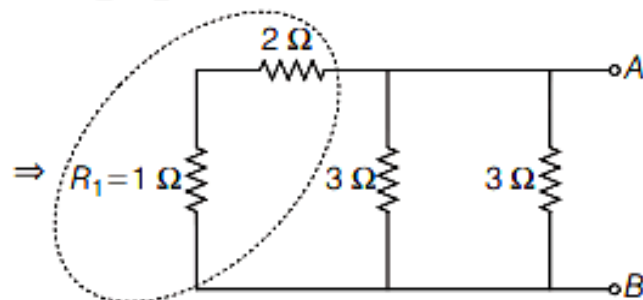
In the given circuit, $5\ \Omega$ resistance is shorted. So, it can be discarded.

Now, we get a resolved circuit as shown below



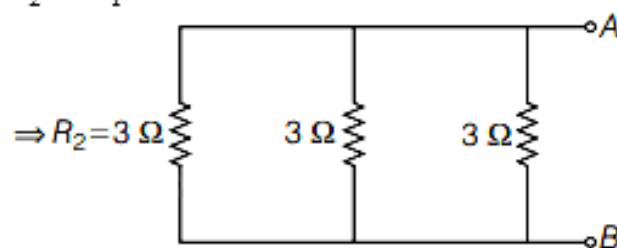
In parallel,

$$R_1 = \frac{2 \times 2}{2 + 2} = 1\ \Omega$$



In series,

$$R_2 = R_1 + 2 = 1 + 2 = 3\ \Omega$$



Here, all the three resistances($3\ \Omega$) are parallel.

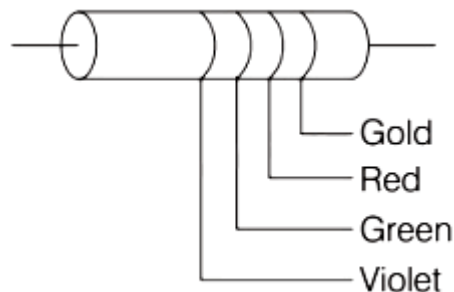
$\therefore$ Equivalent resistance across A and B,

$$\frac{1}{R_{eq}} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

$$\Rightarrow R_{eq} = 1\ \Omega$$

Question207

The colour coding on a carbon resistor is shown in the given figure. The resistance value of the given resistor is



[27 Aug 2021 Shift 2]

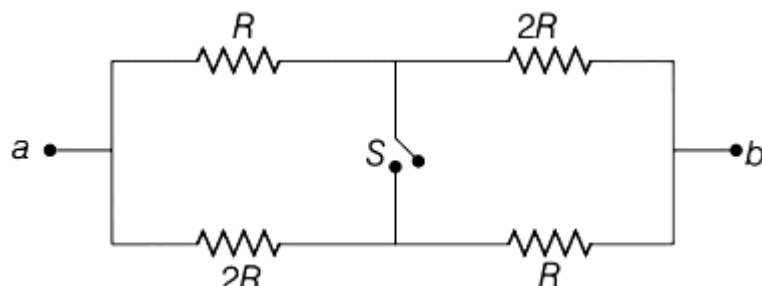
Options:

- A. $(5700 \pm 285)\Omega$
- B. $(7500 \pm 750)\Omega$
- C. $(5700 \pm 375)\Omega$
- D. $(7500 \pm 375)\Omega$

Answer: D

Question208

The ratio of the equivalent resistance of the network (shown in figure) between the points a and b when switch is open and switch is closed is $x : 8$. The value of x is



[27 Aug 2021 Shift 2]

Answer: 9

Solution:

According to given circuit diagram When switch is open, then combination R_1 and R_2 will be in series and also combination R_3 and R_4 will be in series and these branches will be in parallel.

∴ Equivalent resistance

$$(R_{eq}) = \frac{3R \cdot 3R}{3R + 3R} = \frac{9R^2}{6R} = \frac{3}{2}R$$

When switch is closed, then combination R_1 and R_3 will be in parallel and also, combination R_2 and R_4 will be in parallel. After that, both will be in series.

∴ Equivalent resistance,

$$\begin{aligned}(R'_{eq}) &= \frac{R \cdot 2R}{R + 2R} + \frac{R \cdot 2R}{R + 2R} = \frac{2R^2}{3R} + \frac{2R^2}{3R} \\ &= \frac{2R}{3} + \frac{2R}{3} = \frac{4R}{3}\end{aligned}$$

Now, dividing Eq. (i) by Eq. (ii), we get

$$c \frac{R_{eq}}{R'_{eq}} = \frac{3 \frac{R}{2}}{4 \frac{R}{3}} = \frac{9}{8}$$

$$x = 9$$

Question209

First, a set of n equal resistors of 10Ω each are connected in series to a battery of emf 20V and internal resistance 10Ω . A current / is observed to flow. Then, the n resistors are connected in parallel to the same battery. It is observed that the current is increased 20 times, then the value of n is

[27 Aug 2021 Shift 1]

Answer: 20

Solution:

Given, value of each resistance, $R = 10 \Omega$

Emf of battery, $e = 20V$

Internal resistance of battery, $r = 10 \Omega$

Current in parallel connection is 20 times current in series combination, $i_p = 20i_s$.

Net resistance in parallel combination will be given as

$$R_p = r + \left[\frac{1}{\frac{1}{R} + \frac{1}{R} + \frac{1}{R} \dots + n} \right]$$
$$= r + \frac{R}{n}$$

$$R_p = 10 + \frac{10}{n} (\because r = 10 \Omega \text{ and } R = 10 \Omega) \dots (i)$$

In series combination,

The net resistance of circuit will be equivalent to sum of all resistances as all are connected in series.

$$R_s = [R + R + \dots + n] + r = nR + r$$

$$R_s = 10n + 10 (\because r = 10 \Omega \text{ and } R = 10 \Omega) \dots (ii)$$

By Ohm's law, current flowing in circuit is given as

$$i = \frac{V}{R}$$

$$\text{As, } i_p = 20i_s$$

$$\frac{V_p}{R_p} = 20 \frac{V_s}{R_s}$$

$$\Rightarrow \frac{20}{10 + \frac{10}{n}} = \frac{20 \times 20}{10n + 10} [\because V_p = V_s = e = 20V]$$

$$\Rightarrow \frac{20n}{10n + 10} = \frac{400}{10n + 10}$$

$$\Rightarrow 20n = 400$$

$$n = 20$$

Thus, the value of number of resistances n is 20.

Question 210

If you are provided a set of resistances 2Ω , 4Ω , 6Ω and 8Ω . Connect these resistances, so as to obtain an equivalent resistance of $\frac{46}{3}\Omega$.

[26 Aug 2021 Shift 2]

Options:

A. 4Ω and 6Ω are in parallel with 2Ω and 8Ω in series.

B. 6Ω and 8Ω are in parallel with 2Ω and 4Ω in series.

C. 2Ω and 6Ω are in parallel with 4Ω and 8Ω in series.

D. 2Ω and 4Ω are in parallel with 6Ω and 8Ω in series.

Answer: D

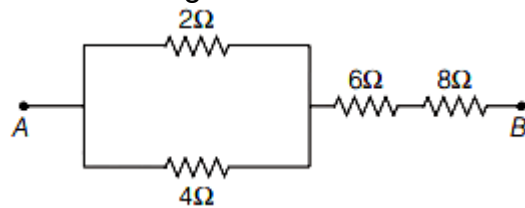
Solution:

The given value of resistances are 2Ω , 4Ω , 6Ω and 8Ω .

The required value of combination is $\frac{46}{3}\Omega$.

In order to achieve the above mentioned values of resistance from given resistances, we will connect 2Ω and 4Ω resistance in parallel, then join 6Ω and 8Ω resistance in series with the combination.

The circuit diagram for connection is shown below.



$$\therefore R_{eq} = (2 \parallel 4) + 6 + 8 = \frac{2 \times 4}{2 + 4} + 14 = \frac{46}{3}\Omega$$

Thus, resistance of 2Ω and 4Ω are in parallel with 6Ω and 8Ω in series combination.

Question211

What equal length of an iron wire and a copper-nickel alloy wire, each of 2 mm diameter connected parallel to give an equivalent resistance of 3Ω ?

(Given, resistivities of iron and copper-nickel alloy wire are $12\mu\Omega \text{ cm}$ and $51\mu\Omega \text{ cm}$ respectively)

[26 Aug 2021 Shift 1]

Options:

A. 82m

B. 97m

C. 110m

D. 90m

Answer: B

Solution:

Let the resistance of iron wire be R_1 and that of copper nickel alloy wire be R_2

$$r_1 = r_2 = 1 \text{ mm} = 10^{-3} \text{ m}$$

$$\rho_1 = 12 \mu\Omega \text{ cm}$$

$$= 12 \times 10^{-6} \Omega \text{ cm}$$

$$= 12 \times 10^{-8} \Omega \text{ m}$$

$$\rho_2 = 51 \mu\Omega \text{ cm}$$

$$= 51 \times 10^{-6} \Omega \text{ cm}$$

$$= 51 \times 10^{-8} \Omega \text{ m}$$

For parallel combination,

$$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$3 = \frac{\frac{\rho_1 l}{\pi r_1^2} \frac{\rho_2 l}{\pi r_2^2}}{\frac{\rho_1 l}{\pi r_1^2} + \frac{\rho_2 l}{\pi r_2^2}}$$

$$\Rightarrow 3 = \frac{(12 \times 10^{-8})l \times \frac{51 \times 10^{-8}l}{\pi \times 10^{-6}}}{\frac{12 \times 10^{-8}l}{\pi \times 10^{-6}} + \frac{51 \times 10^{-8}l}{\pi \times 10^{-6}}}$$

On solving, $l = 97 \text{ m}$

Question 212

A resistor dissipates 192 J of energy in 1 s when a current of 4A is passed through it. Now, when the current is doubled, the amount of thermal energy dissipated in 5 s is J.

[31 Aug 2021 Shift 2]

Answer: 3840

Solution:

Given that, initial current, $I_1 = 4\text{A}$

Final current, $I_2 = 2I_1 = 8\text{A}$

Initial heat dissipated, $H_1 = 192\text{J}$

Initial time, $t_1 = 1\text{s}$

Final time, $t_2 = 5\text{s}$

Let final heat dissipated = H_2

By Joule's law of heating,

$$H \propto I^2 RT$$

Since resistance remains same at initial and final condition,

$$\therefore \frac{H_2}{H_1} = \frac{I_2^2 R t_2}{I_1^2 R t_1} = \frac{I_2^2 t_2}{I_1^2 t_1}$$

Substituting the given values, we get

$$\frac{H_2}{192} = \left(\frac{8}{4}\right)^2 \times \frac{5}{1}$$

$$\Rightarrow H_2 = 3840 \text{ J}$$

Question 213

An electric bulb of 500W at 100V is used in a circuit having a 200V supply. Calculate the resistance R to be connected in series with the bulb, so that the power delivered by the bulb is 500W. [26 Aug 2021 Shift 2]

Options:

A. 20Ω

B. 30Ω

C. 5Ω

D. 10Ω

Answer: A

Solution:

Given, power rating of bulb, $P_B = 500\text{W}$

Voltage across bulb, $V_B = 100\text{V}$

Supply voltage, $V_S = 200\text{V}$

If a resistance R is attached in series with the bulb, then the voltage across resistance will be 100 V.

Now, current flowing in circuit when bulb delivers power of 500 W is given as

$$P_B = V_B I$$

$$\Rightarrow 500 = 100 \times I$$

$$\Rightarrow I = 5\text{A}$$

Same amount of current will flow from the resistance as it is connected in series.

Using Ohm's law,

$$V = IR$$

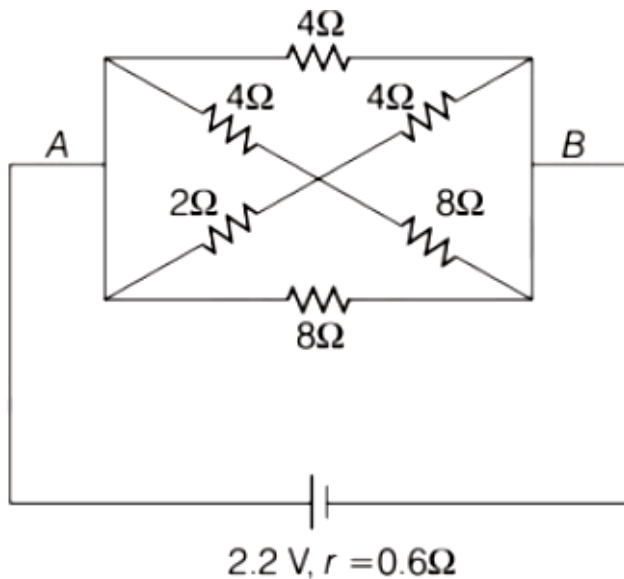
$$\Rightarrow 100 = 5 \times R$$

$$\Rightarrow R = 20\Omega$$

Thus, the resistance connected in series is 20Ω .

Question214

In the given figure, the emf of the cell is 2.2V and if internal resistance is 0.6Ω . Calculate the power dissipated in the whole circuit



[26 Aug 2021 Shift 1]

Options:

A. 1.32W

B. 0.65W

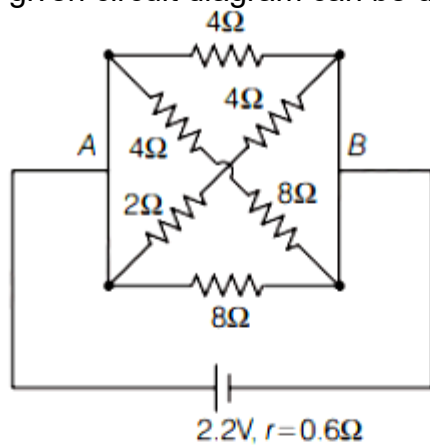
C. 2.2W

D. 4.4W

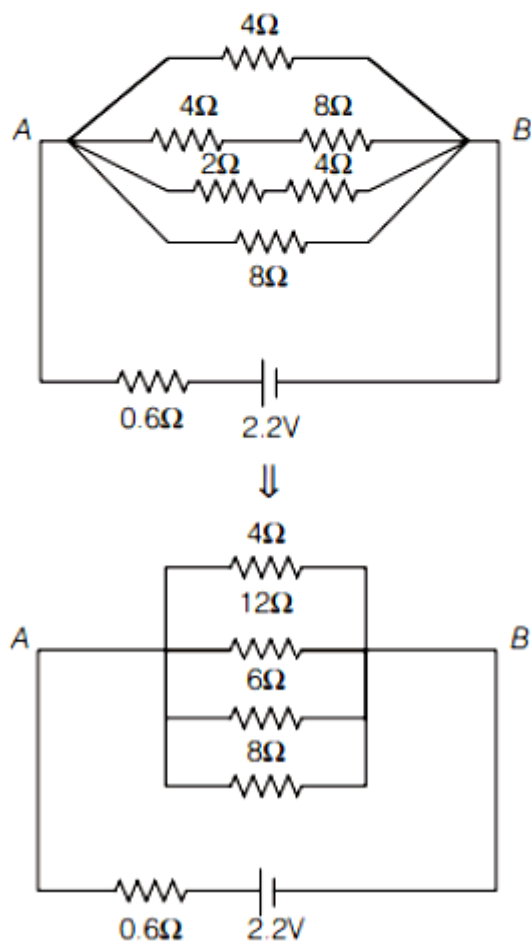
Answer: C

Solution:

The given circuit diagram can be drawn as

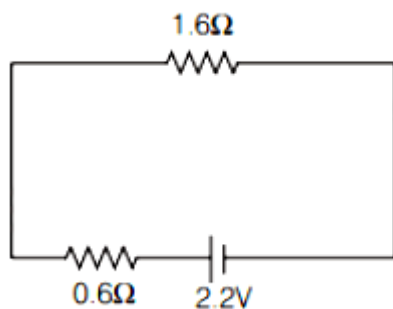


Redrawing the above circuit diagram as



Equivalent resistance of the circuit between point A and B is given as

$$\begin{aligned}\frac{1}{R_{eq}} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \\ &= \frac{1}{4} + \frac{1}{12} + \frac{1}{6} + \frac{1}{8} \\ &= \frac{6+2+4+3}{24} \\ &= \frac{15}{24} \\ &= \frac{24}{15} \Omega = 1.6 \Omega\end{aligned}$$



$$R_{\text{total}} = 1.6 + 0.6 = 2.2\Omega$$

$$P = \frac{V^2}{R_{\text{total}}} = \frac{2.2 \times 2.2}{2.2}$$

$$= 2.2 \text{ W}$$

Question215

Five identical cells each of internal resistance 1Ω and emf 5V are connected in series and in parallel with an external resistance R . For what value of R , current in series and parallel combination will remain the same ?
[27 Aug 2021 Shift 1]

Options:

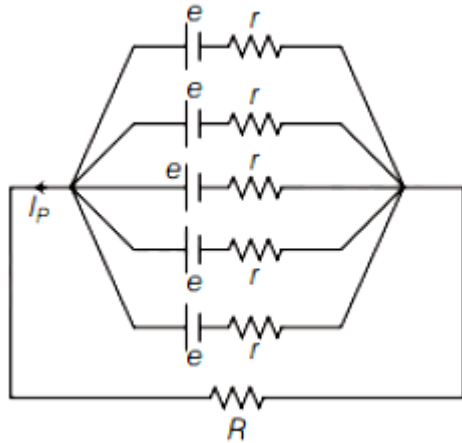
- A. 1Ω
- B. 25Ω
- C. 5Ω
- D. 10Ω

Answer: A

Solution:

Given, number of cells, $n = 5$
 Internal resistance of each cell, $r = 1\Omega$
 Emf of each cell, $e = 5 \text{ V}$

When all cells are connected in parallel as shown below.



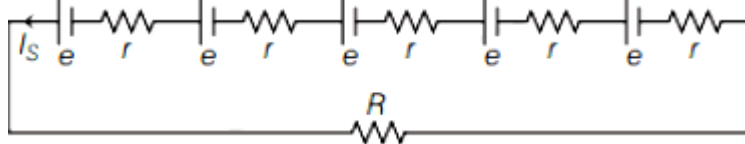
Potential will remain same as, $V_p = 5V$

Net resistance in parallel combination will be given as

$$R_p = R + \left[\frac{1}{\frac{1}{r} + \frac{1}{r} + \frac{1}{r} + \frac{1}{r} + \frac{1}{r}} \right] = R + \frac{r}{5}$$

$$R_p = R + \frac{1}{5} \quad (\because r = 1 \Omega) \dots (i)$$

When all cells are connected in series as shown below



The net potential will increase as cells are connected in series,

$$V_s = 5 + 5 + 5 + 5 + 5 = 25V$$

The net resistance of circuit will be equivalent of sum of all resistances as all are connected in series.

$$R_s = r + r + r + r + r + R = 5r + R$$

$$R_s = 5 + R \quad (\because r = 1\Omega) \dots (ii)$$

By Ohm's law, current flowing in circuit is given as

$$I = \frac{V}{R}$$

As current in both series and parallel combination is same,

$$\frac{V_p}{R_p} = \frac{V_s}{R_s}$$

$$\Rightarrow \frac{5}{R + \frac{1}{5}} = \frac{25}{5 + R} \quad [\text{From Eqs. (i) and (ii)}]$$

$$\Rightarrow 25 + 5R = 25R + 5$$

$$\Rightarrow R = 1 \Omega$$

Question216

A uniform heating wire of resistance 36Ω is connected across a potential difference of $240V$. The wire is then cut into half and potential difference of $240V$ is applied across each half

separately. The ratio of power dissipation in first case to the total power dissipation in the second case would be 1 : x , where x is

[1 Sep 2021 Shift 2]

Answer: 4

Solution:

For case I,

The potential difference of the uniform wire, $V = 240 \text{ V}$

The resistance of the uniform wire, $R_1 = 36 \Omega$

The power dissipation in the first case,

$$P_1 = \frac{V^2}{R_1} = \frac{(240)^2}{36}$$

For case II,

The resistance of each half, $R_2 = \frac{R_1}{2} = \frac{36}{2} = 18 \Omega$

$$P_2 = \frac{V^2}{R_2} + \frac{V^2}{R_2} = \frac{(240)^2}{18} + \frac{(240)^2}{18} = \frac{(240)^2}{9}$$

Thus, the ratio of the total power dissipation in the first case to the second case

$$\frac{P_1}{P_2} = \frac{(240)^2/36}{(240)^2/9}$$

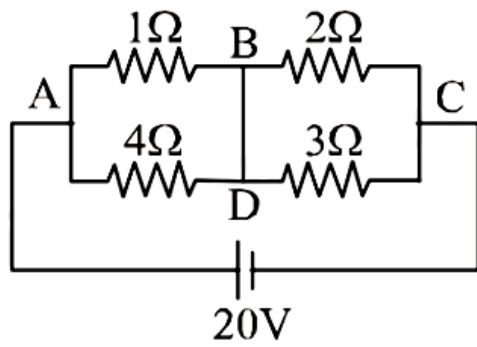
$$\Rightarrow \frac{P_1}{P_2} = \frac{1}{4}$$

Comparing with, $\frac{P_1}{P_2} = \frac{1}{x}$

The value of the $x = 4$.

Question 217

In the given circuit diagram, a wire is joining points B and D.
The current in this wire is:



[9 Jan. 2020 I]

Options:

A. 0.4A

B. 2A

C. 4A

D. zero

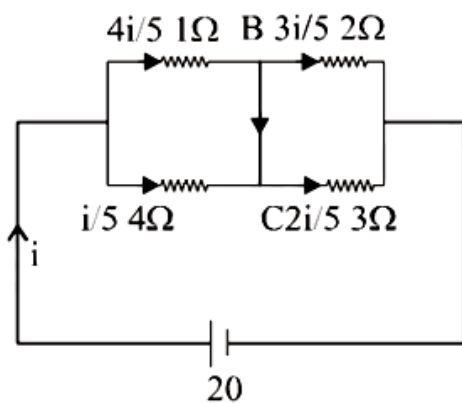
Answer: B

Solution:

From circuit diagram,

$$\frac{1}{R_1} = \frac{1}{1} + \frac{1}{4} \Rightarrow R_1 = \frac{4}{5}$$

$$\frac{1}{R_2} = \frac{1}{2} + \frac{1}{3} \Rightarrow R_2 = \frac{6}{5}$$



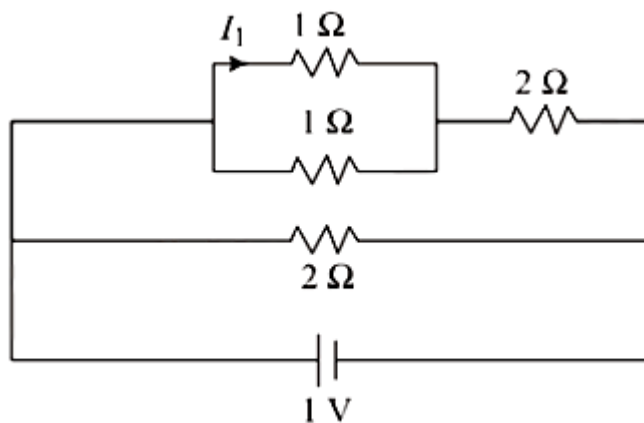
$$R_{\text{eff}} = R_1 + R_2 = \frac{4}{5} + \frac{6}{5} = 2\Omega$$

$$i = \frac{v}{R_{\text{eff}}} = \frac{20}{2} = 10\text{A}$$

$$\therefore I_{\text{BC}} = \frac{4i}{5} - \frac{3i}{5} = \frac{i}{5} = 2\text{A}$$

Question218

The current I_1 (in A) flowing through $1\ \Omega$ resistor in the following circuit is:



[7 Jan. 2020 I]

Options:

- A. 0.4
- B. 0.5
- C. 0.2
- D. 0.25

Answer: C

Solution:

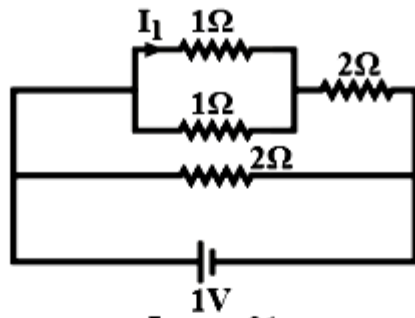


Image 01

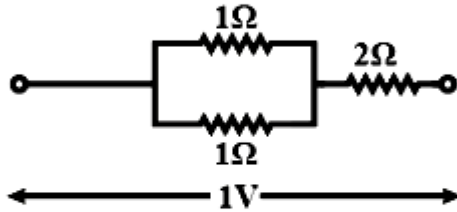


Image 02

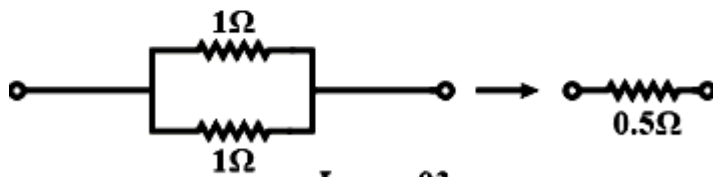


Image 03

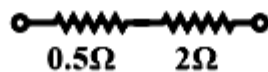


Image 04

Refer image .1

Potential difference across the top most branch is 1V

Refer image. 2

Equivalent resistance of the top most branch is given by the combinations of the 1Ω and the 2Ω resistor. The parallel combination of 1Ω resistor give 0.5Ω

Refer image. 3

∴ The equivalent circuit would be

Refer image. 4

$$R_{\text{total}} = 2 + 0.5$$

$$= 2.5\Omega$$

$$I_{\text{total}} = \frac{V}{R_{\text{total}}} = \frac{1}{2.5} = \frac{2}{5} = 0.4\text{A}$$

Now, using current division,

Refer image. 5

$$V = I_1 R_1 = I_2 R_2$$

$$T = I_1 + I_2$$

$$\therefore I_1 R_1 = (I - I_1) R_2$$

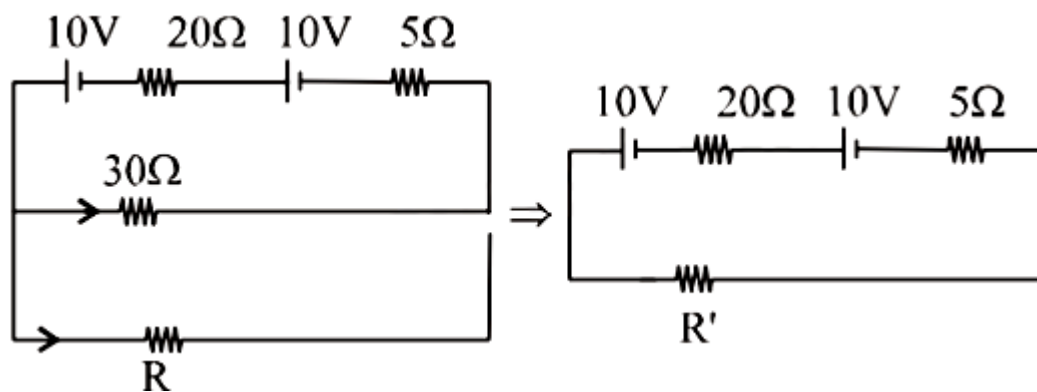
$$I_1 (R_1 + R_2) = I R_2$$

$$I_1 = \frac{I R_2}{R_1 + R_2} = \frac{0.4 \times 1}{1 + 1} = 0.2\text{A}$$

The series combination of two batteries, both of the same emf 10 V, but different internal resistance of $20\ \Omega$ and $5\ \Omega$, is connected to the parallel combination of two resistors $30\ \Omega$ and $R\ \Omega$. The voltage difference across the battery of internal resistance $20\ \Omega$ is zero, the value of R (in Ω) is _____.
[NA. 8 Jan. 2020 II]

Answer: 30

Solution:



The resistance of $30\ \Omega$ is in parallel with R . Their effective resistance $\frac{1}{R'} = \frac{1}{30} + \frac{1}{R}$

$$R' = \frac{30R}{30 + R} \dots\dots(i)$$

$$\text{Also, } V = IR \Rightarrow 10 = \frac{20 \times 20}{R' + 25}$$

$$\Rightarrow R' + 25 = 40 \Rightarrow R' = 15$$

$$R' = 15 = \frac{30R}{30 + R} \text{ Using (i)}$$

$$\Rightarrow 30 + R = 2R$$

$$\Rightarrow R = 30\ \Omega$$

Question220

In a building there are 15 bulbs of 45 W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of 1 kW. The voltage of electric main is 220 V. The minimum fuse capacity (rated value) of the building will be:
[7 Jan. 2020 II]

Options:

- A. 10 A
- B. 25 A
- C. 15 A
- D. 20 A

Answer: D

Solution:

Net Power, P

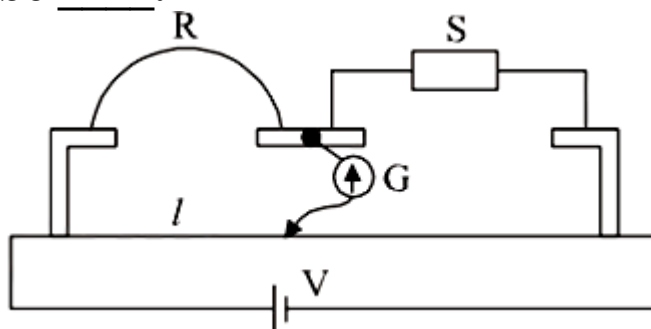
$$= 15 \times 45 + 15 \times 100 + 15 \times 10 + 2 \times 1000$$
$$= 15 \times 155 + 2000 \text{ W}$$

$$\text{Power, } P = V I \Rightarrow I = \frac{P}{V}$$

$$\therefore I_{\text{main}} = \frac{15 \times 155 + 2000}{220} = 19.66 \text{ A} \approx 20 \text{ A}$$

Question 221

In a meter bridge experiment S is a standard resistance. R is a resistance wire. It is found that balancing length is $l = 25 \text{ cm}$. If R is replaced by a wire of half length and half diameter that of R of same material, then the balancing distance l' (in cm) will now be _____.



[NA. 9 Jan. 2020 II]

Answer: 40

Solution:

For the given meter bridge

$$\frac{R}{S} = \frac{l_1}{100 - l_1} \text{ Where, } l_1 = \text{balancing length}$$

$$\Rightarrow \frac{R}{S} = \frac{25}{75} = \frac{1}{3} \dots\dots(i)$$

New resistance

$$R' = \frac{\rho \frac{l}{2}}{\frac{A}{4}} = \rho \frac{l \times 2}{A} \left(\because R = \rho \frac{l}{A} \right)$$

$$\Rightarrow R' = 2R$$

$$\frac{R'}{S} = \frac{l_2}{100 - l_2} \Rightarrow \frac{2R}{S} = \frac{l_2}{100 - l_2}$$

$$\Rightarrow 2 \times \frac{1}{3} = \frac{l_2}{100 - l_2} \text{ Using (i)}$$

$$\Rightarrow l_2 = 40\text{cm}$$

Question222

The length of a potentiometer wire is 1200 cm and it carries a current of 60 mA. For a cell of emf 5 V and internal resistance of 20Ω , the null point on it is found to be at 1000 cm. The resistance of whole wire is:

[8 Jan. 2020 I]

Options:

A. $80\ \Omega$

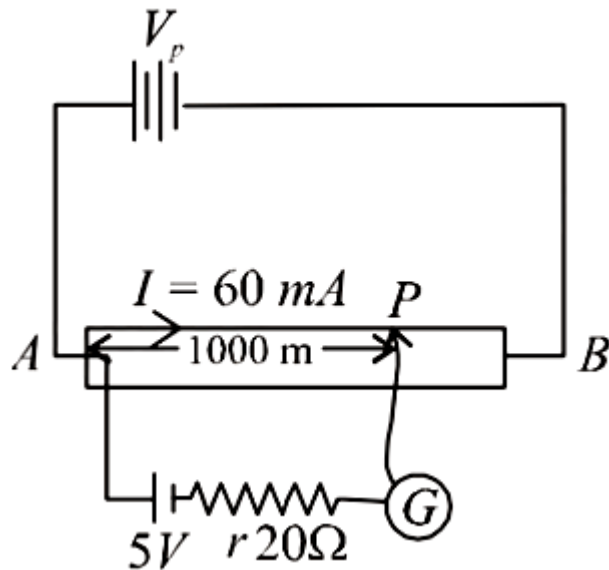
B. $120\ \Omega$

C. $60\ \Omega$

D. $100\ \Omega$

Answer: D

Solution:



Let R be the resistance of the whole wire
 Potential gradient for the potentiometer wire

$$AB' = -\frac{dV}{dl} = \frac{I \times R}{l} = \left[\frac{60 \times R}{1_{AB}} \right] \text{mv/m}$$

$$V_{AP} = \left(\frac{dV}{dl_{AB}} \right) l_{AP} = \frac{60 \times R}{1200} \times 1000 \text{mV}$$

$$\Rightarrow V_{AP} = 50R \text{mV}$$

Also, $V_{AP} = 5V$ (for balance point at P)

$$\therefore R = \frac{V_{AP}}{50 \times 10^{-3}} = \frac{5}{50 \times 10^{-3}} = 100\Omega$$

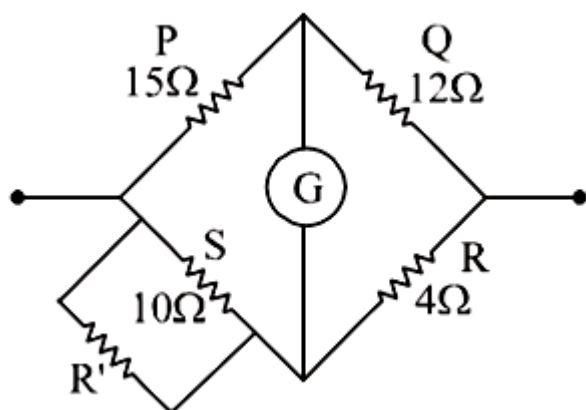
Question223

Four resistances of $15\ \Omega$, $12\ \Omega$, $4\ \Omega$ and $10\ \Omega$ respectively in cyclic order to form Wheatstone's network. The resistance that is to be connected in parallel with the resistance of $10\ \Omega$ to balance the network is _____ Ω .

[NA. 8 Jan. 2020 I]

Answer: 10

Solution:



As per Wheatstone bridge balance condition $\frac{P}{Q} = \frac{S}{R}$

Let resistance R' is connected in parallel with resistance S of 10Ω

$$\therefore \frac{15}{12} = \frac{10R'}{10 + R'} \Rightarrow 5 = \frac{10R'}{10 + R'}$$

$$\Rightarrow 50 + 5R' = 10R$$

$$\therefore R' = \frac{50}{5} = 10\Omega$$

Question224

The balancing length for a cell is 560 cm in a potentiometer experiment. When an external resistance of 10Ω is connected in parallel to the cell, the balancing length changes by 60 cm. If the internal resistance of the cell is $\frac{N}{10}\Omega$, where N is an integer then value of N is _____.

[NA. 7 Jan. 2020 II]

Answer: 12

Solution:

We know that

$E \propto l$ where l is the balancing length

$$\therefore E = k(560) \dots\dots(i)$$

When the balancing length changes by 60cm

$$\frac{E}{r + 10} 10 = k(500)$$

Dividing (i) by (ii) we get

$$\Rightarrow \frac{r+10}{10} = \frac{56}{50} \Rightarrow 50r + 500 = 560$$

$$\Rightarrow r = \frac{6}{5}\Omega = \frac{N}{10}\Omega \Rightarrow N = 12$$

Question225

A circuit to verify Ohm's law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit :

[Sep. 06, 2020 (II)]

Options:

- A. ammeter is always used in parallel and voltmeter is series
- B. Both ammeter and voltmeter must be connected in parallel
- C. ammeter is always connected in series and voltmeter in parallel
- D. Both, ammeter and voltmeter must be connected in series

Answer: C

Solution:

Ammeter : In series connection, the same current flows through all the components. It aims at measuring the current flowing through the circuit and hence, it is connected in series.

Voltmeter : A voltmeter measures voltage change between two points in a circuit. So we have to place the voltmeter in parallel with the circuit component.

Question226

Consider four conducting materials copper, tungsten, mercury and aluminium with resistivity ρ_C , ρ_T , ρ_M and ρ_A respectively.

Then :

[Sep. 02, 2020 (I)]

Options:

- A. $\rho_C > \rho_A > \rho_T$

B. $\rho_M > \rho_A > \rho_C$

C. $\rho_A > \rho_T > \rho_C$

D. $\rho_A > \rho_M > \rho_C$

Answer: B

Solution:

$$\rho_M = 98 \times 10^{-8}$$

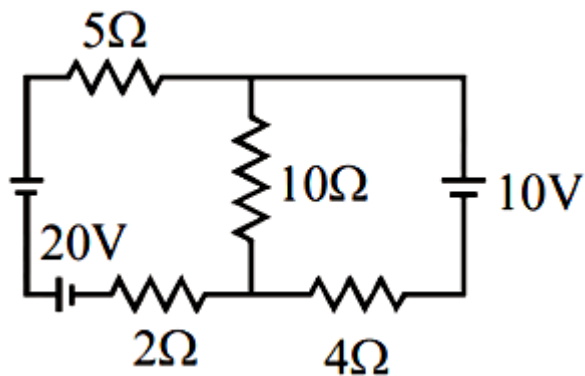
$$\rho_A = 2.65 \times 10^{-8}$$

$$\rho_C = 1.724 \times 10^{-8}$$

$$\rho_T = 5.65 \times 10^{-8}$$

$$\therefore \rho_M > \rho_T > \rho_A > \rho_C$$

Question227



**In the figure shown, the current in the 10 V battery is close to :
[Sep. 06, 2020 (II)]**

Options:

A. 0.71 A from positive to negative terminal

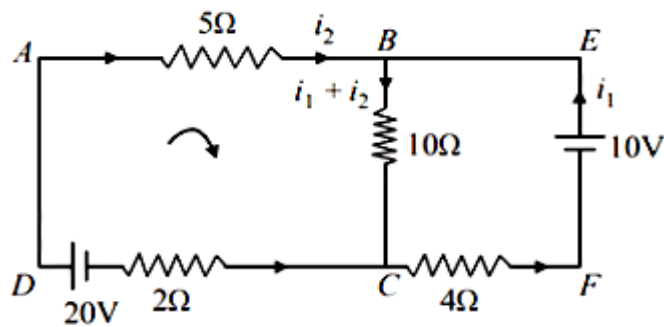
B. 0.42 A from positive to negative terminal

C. 0.21 A from positive to negative terminal

D. 0.36 A from negative to positive terminal

Answer: C

Solution:



Using Kirchhoff's loop law in loop ABCD

$$-5i_2 - 10(i_1 + i_2) - 2i_2 + 20 = 0$$

$$\Rightarrow -10i_1 - 17i_2 + 20 = 0 \dots\dots(i)$$

Using Kirchhoff's loop law in loop BEFC

$$\Rightarrow -10 + 4i_1 + 10(i_1 + i_2) = 0$$

$$\Rightarrow 14i_1 + 10i_2 + 10 = 0 \dots\dots(ii)$$

Multiplying equation (i) by 10, we have

$$(10i_1 + 17i_2 = 20) \times 10$$

$$\Rightarrow 100i_1 - 170i_2 = 200 \dots\dots(iii)$$

Multiplying equation (ii) by 17, we have

$$(14i_1 + 10i_2 = 10) \times 17$$

$$\Rightarrow 238i_1 - 170i_2 = 170 \dots\dots(iv)$$

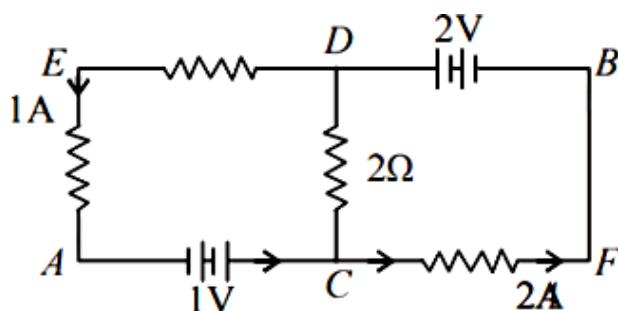
On solving equations (iii) and (iv), we get

$$-138i_1 = 30 \Rightarrow i_1 = -\frac{30}{138} = -0.217$$

i_1 is negative it means current flows from positive to negative terminal.

Question228

9. In the circuit, given in the figure currents in different branches and value of one resistor are shown. Then potential at point B with respect to the point A is :



[Sep. 05, 2020 (II)]

Options:

A. + 2 V

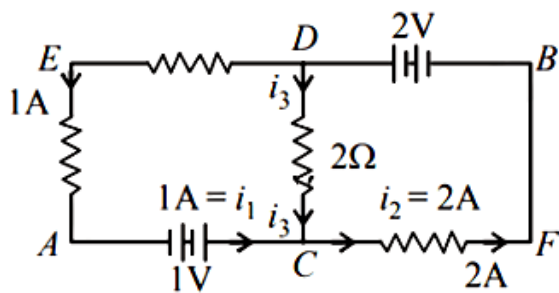
B. - 2 V

C. - 1 V

D. + 1 V

Answer: D

Solution:



Let us assume the potential at A = $V_A = 0$

Using Kirchhoff's junction rule at C, we get

$$i_1 + i_3 = i_2$$

$$1A + i_3 = 2A \Rightarrow i_3 = 2A$$

Now using Kirchhoff's loop law along ACDB

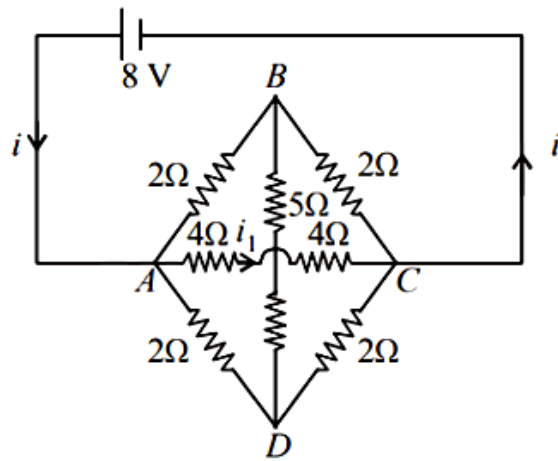
$$V_A + 1 + i_3(2) - 2 = V_B$$

$$\Rightarrow V_A + 1 + i_3(1) - 2 = V_B$$

$$\Rightarrow V_B - V_A = 3 - 2 = 1 \text{ volt}$$

Question229

The value of current i_1 flowing from A to C in the circuit diagram is



[Sep. 04, 2020 (II)]

Options:

A. 2 A

B. 4 A

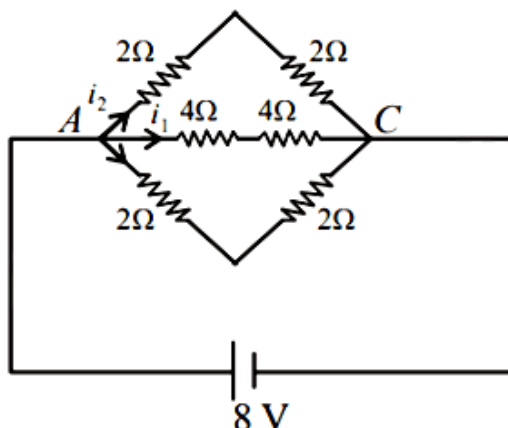
C. 1 A

D. 5 A

Answer: C

Solution:

The equivalent circuit can be drawn as

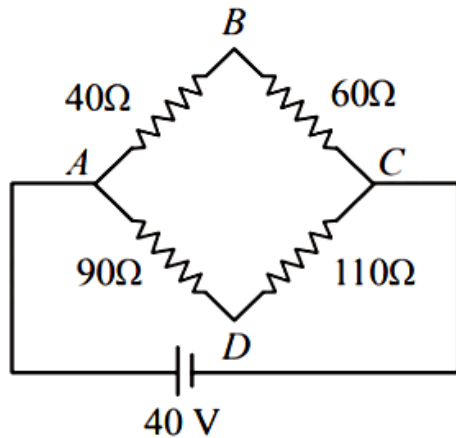


Voltage across AC = 8V

Resistance $R_{AC} = 4 + 4 = 8\Omega$

$$i_1 = \frac{V}{R_{AC}} = \frac{8}{4+4} = 1 \text{ Amp}$$

Question 230

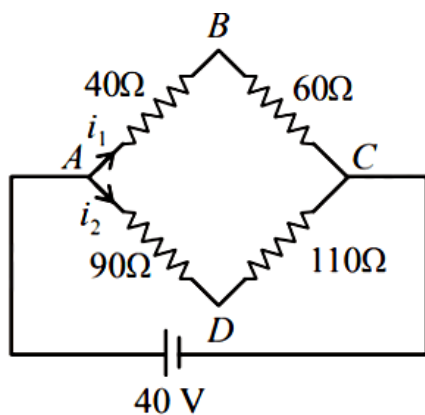


Four resistances $40\ \Omega$, $60\ \Omega$, $90\ \Omega$ and $110\ \Omega$ make the arms of a quadrilateral ABCD. Across AC is a battery of emf 40 V and internal resistance negligible. The potential difference across BD in V is _____.

[NA. Sep. 04, 2020 (II)]

Answer: 2

Solution:



Current through AB, $i_1 = \frac{40}{40 + 60} = 0.4$

Current through AD, $i_2 = \frac{40}{90 + 110} = \frac{1}{5}$

Using KVL in BAD loop

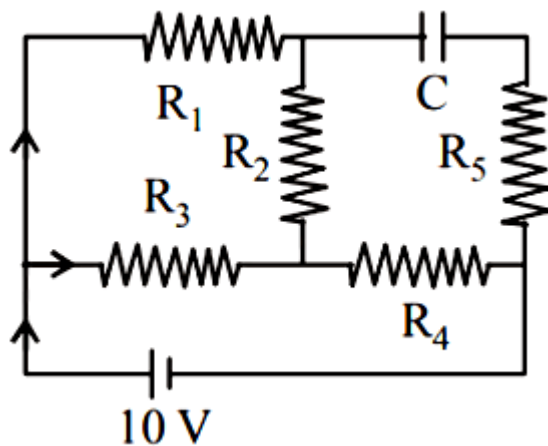
$$V_B + i_1(40) - i_2(90) = V_D$$

$$\Rightarrow V_B - V_D = \frac{1}{5}(90) - \frac{4}{10}(40)$$

$$\Rightarrow V_B - V_D = 18 - 16 = 2V$$

Question231

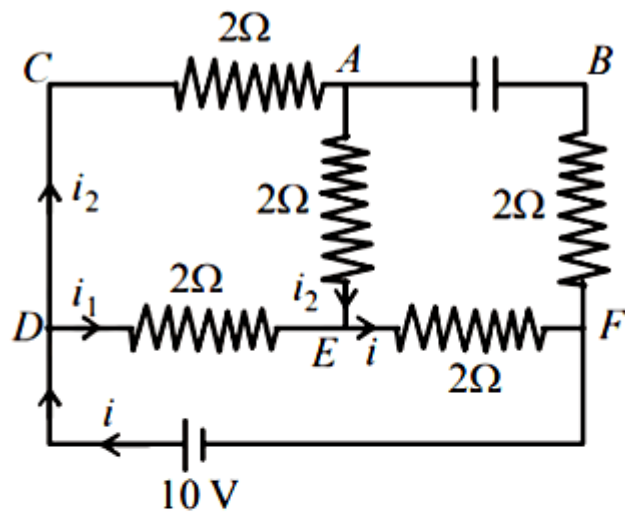
An ideal cell of emf 10 V is connected in circuit shown in figure. Each resistance is $2\ \Omega$. The potential difference (in V) across the capacitor when it is fully charged is _____.



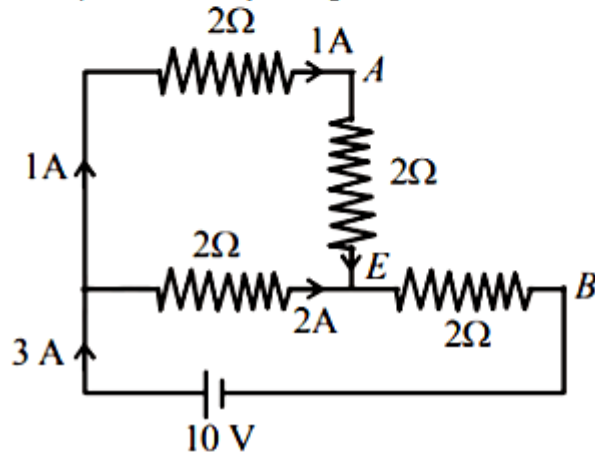
[Sep. 02, 2020 (II)]

Answer: 8

Solution:



As capacitor is fully charged no current will flow through it



We have the current distribution as shown in the figure.

Equivalent resistance, $R_{eq} = \left(\frac{4 \times 2}{4 + 2} \right) + 2$

Net current, $i = \frac{10}{\frac{4}{3} + 2} = \frac{10 \times 3}{10} = 3 \text{ Amp}$

$i_1 = 2\text{A}$ and $i_2 = 1\text{A}$

$V_{AEB} = 1 \times 2 + 3 \times 2 = 8\text{V}$

Question 232

An electrical power line, having a total resistance of 2Ω , delivers 1 kW at 220 V . The efficiency of the transmission line is approximately :
[Sep. 05, 2020 (I)]

Options:

A. 72%

B. 91%

C. 85%

D. 96%

Answer: D

Solution:

Given : Power, $P = 1\text{kW} = 1000\text{W}$ $R = 2\Omega$, $V = 220\text{V}$

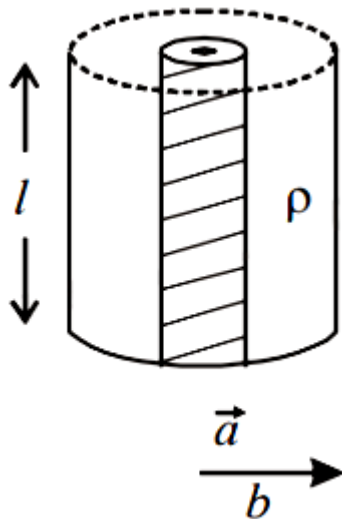
$$\text{Current, } I = \frac{P}{V} = \frac{1000}{220}$$

$$P_{\text{loss}} = I^2 R = \left(\frac{1000}{220} \right)^2 \times 2$$

$$\text{Efficiency} = \frac{1000}{1000 + P_{\text{loss}}} \times 100 = 96\%$$

Question233

Model a torch battery of length l to be made up of a thin cylindrical bar of radius ' a ' and a concentric thin cylindrical shell of radius ' b ' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R , the maximum Joule heating in R will take place for :



[Sep. 03, 2020 (I)]

Options:

A. $R = \frac{\rho}{2\pi l} \left(\frac{b}{a} \right)$

B. $R = \frac{\rho}{2\pi l} \ln\left(\frac{b}{a}\right)$

C. $R = \frac{\rho}{\pi l} \ln\left(\frac{b}{a}\right)$

D. $R = \frac{2\rho}{\pi l} \ln\left(\frac{b}{a}\right)$

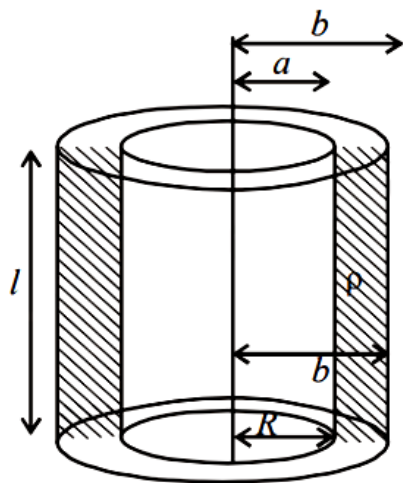
Answer: B

Solution:

Maximum power in external resistance is generated when it is equal to internal resistance of battery i.e., P_R maximum when $r = R$

The maximum Joule heating in R will take place for, the resistance of small element

$$\Delta R = \frac{\rho dr}{2\pi r l} \Rightarrow R = \frac{\rho}{2\pi l} \int_a^b \frac{dr}{r}$$



or, $R = \frac{\rho}{2\pi l} \ln \frac{b}{a}$

Question234

Two resistors 400Ω and 800Ω are connected in series across a 6 V battery. The potential difference measured by a voltmeter of $10\text{ k}\Omega$ across 400Ω resistor is close to:

[Sep. 03, 2020 (II)]

Options:

- A. 2 V
- B. 1.8 V
- C. 2.05 V
- D. 1.95 V

Answer: D

Solution:

The voltmeter of resistance $10\text{k}\Omega$ is parallel to the resistance of 400Ω . So, their equivalent resistance is

$$\begin{aligned}\frac{1}{R'} &= \frac{1}{10\text{k}\Omega} + \frac{1}{400\Omega} = \frac{1}{10000} + \frac{1}{400} \\ \Rightarrow \frac{1}{R'} &= \frac{1+25}{10000} = \frac{26}{10000} \\ \Rightarrow R' &= \frac{10000}{26}\Omega\end{aligned}$$

Using Ohm's law, current in the circuit

$$I = \frac{\text{Voltage}}{\text{Net Resistance}} = \frac{6}{\frac{10000}{26} + 800}$$

Potential difference measured by voltmeter

$$\begin{aligned}V = IR' &= \frac{6}{\frac{10000}{26} + 800} \times \frac{10000}{26} \\ \Rightarrow V &= \frac{150}{77} = 1.95 \text{ volt}\end{aligned}$$

Question235

**Which of the following will NOT be observed when a multimeter (operating in resistance measuring mode) probes connected across a component, are just reversed?
[Sep. 03, 2020 (II)]**

Options:

- A. Multimeter shows an equal deflection in both cases i.e. before and after reversing the probes if the chosen component is resistor.

B. Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is capacitor.

C. Multimeter shows a deflection, accompanied by a splash of light out of connected and NO deflection on reversing the probes if the chosen component is LED.

D. Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is metal wire.

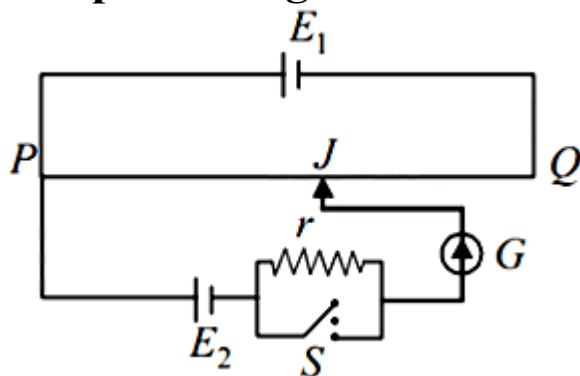
Answer: B

Solution:

Multimeter shows deflection in both cases i.e. before and after reversing the probes if the chosen component is capacitor.

Question236

A potentiometer wire PQ of 1m length is connected to a standard cell E_1 . Another cell E_2 of emf 1.02V is connected with a resistance 'r' and switch S (as shown in figure). With switch S open, the null position is obtained at a distance of 49cm from Q. The potential gradient in the potentiometer wire is :



[Sep. 02, 2020 (II)]

Options:

A. 0.02 V/cm

B. 0.01 V/cm

C. 0.03 V/cm

D. 0.04 V/cm

Answer: A

Solution:

Potential gradient, $x = \frac{\text{Potential drop}}{\text{length}}$

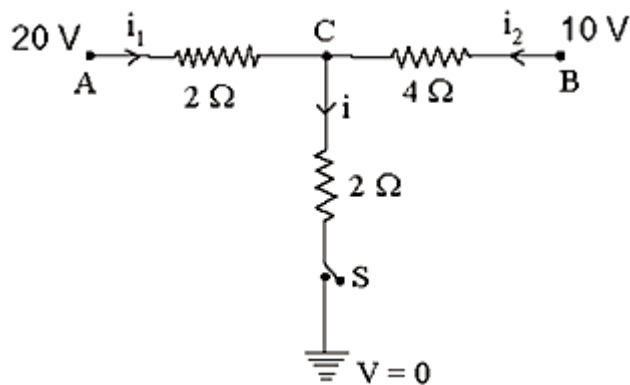
Here, Potential drop = 1.02

Balancing length from P = 100 – 49

$$\therefore x = \frac{1.02}{100 - 49} = 0.02 \text{ volt/cm}$$

Question237

When the switch S, in the circuit shown, is closed then the valued of current i will be:



[9 Jan. 2019 I]

Options:

A. 3A

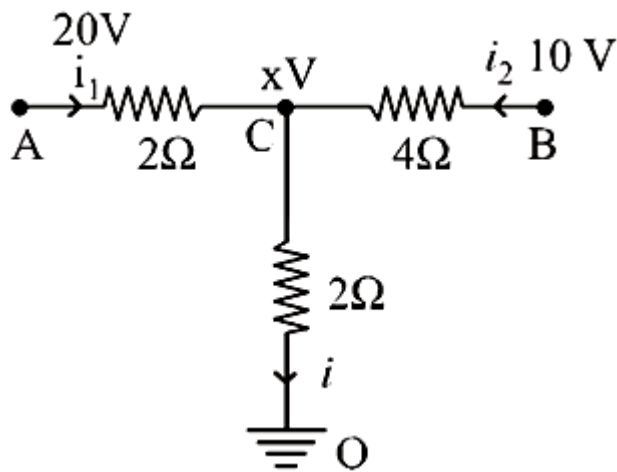
B. 5A

C. 4A

D. 2A

Answer: B

Solution:



Let voltage at C = xV
 From Kirchhoff's current law,
 KCL : $i_1 + i_2 = i$
 $\frac{20 - x}{2} + \frac{10 - x}{4} = \frac{x - 0}{2} \Rightarrow x = 10$
 $\therefore i = \frac{V}{R} = \frac{X}{R} = \frac{10}{2} = 5A$

Question 238

Two electric bulbs, rated at (25W , 220V) and (100W , 220V) are connected in series across a 220V voltage source. If the 25W and 100W bulbs draw powers P_1 and P_2 respectively, then:
 [12 Jan. 2019 I]

Options:

- A. $P_1 = 16W$, $P_2 = 4W$
- B. $P_1 = 16W$, $P_2 = 9W$
- C. $P_1 = 9W$, $P_2 = 16W$
- D. $P_1 = 4W$, $P_2 = 16W$

Answer: A

Solution:

$$\text{As } R = \frac{V^2}{P}, \text{ so } R_1 = \frac{220^2}{25} \text{ and } R_2 = \frac{220^2}{100}$$

$$\text{Current flow } i = \frac{220}{R_1 + R_2}$$

$$P_1 = i^2 R_1 = \frac{220^2}{\left(\frac{220^2}{25} + \frac{220^2}{100} \right)} \times \frac{220^2}{25} = 16W$$

$$\text{Similarly, } P_2 = i^2 R_2 = 4W$$

Question239

Two equal resistances when connected in series to a battery, consume electric power of 60 W. If these resistance are now connected in parallel combination to the same battery, the electric power consumed will be :

[11 Jan. 2019 I]

Options:

- A. 60 W
- B. 240 W
- C. 120 W
- D. 30 W

Answer: B

Solution:

When two resistances are connected in series,
 $R_{eq} = 2R$

$$\text{Power consumed, } P = \frac{\epsilon^2}{R_{eq}} = \frac{\epsilon^2}{2R}$$

In parallel condition, $R_{eq} = R/2$

$$\text{New power, } P' = \frac{\epsilon^2}{(R/2)}$$

$$\text{or } P' = 4P = 240W (\because P = 60W)$$

Question240

A 2 W carbon resistor is color coded with green, black, red and brown respectively. The maximum current which can be passed through this resistor is:

[10 Jan. 2019 I]

Options:

- A. 20 mA
- B. 100 mA
- C. 0.4 mA
- D. 63 mA

Answer: A

Solution:

Colour code for carbon resistor

Bl, Br, R, O, Y, G, Blue, V Gr, W

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Resistance, $R = AB \times C \pm D$

$\therefore$ Resistance, $R = 50 \times 10^2 \Omega$

Now using formula, Power, $P = i^2 R$

$$\therefore i = \sqrt{\frac{P}{R}} = \sqrt{\frac{2}{50 \times 10^2}} = 20 \text{ mA}$$

Question241

A current of 2 mA was passed through an unknown resistor which dissipated a power of 4.4 W. Dissipated power when an ideal power supply of 11 V is connected across it is:

[10 Jan. 2019 II]

Options:

- A. $11 \times 10^{-5} \text{ W}$
- B. $11 \times 10^{-3} \text{ W}$
- C. $11 \times 10^{-4} \text{ W}$

D. $11 \times 10^5 \text{W}$

Answer: A

Solution:

Power, $P = I^2 R$

$4.4 = 4 \times 10^{-6} \times R$

$\Rightarrow R = 1.1 \times 10^6 \Omega$

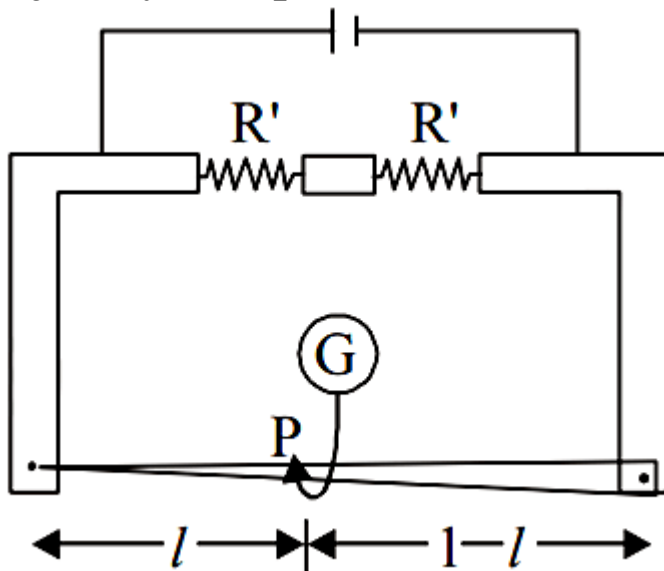
When supply of 11v is connected

Power, $P' = \frac{V^2}{R} = \frac{11^2}{1.1} \times \frac{11^2}{1.1} \times 10^{-6}$

$= 11 \times 10^{-5} \text{W}$

Question242

In a meter bridge, the wire of length 1m has a non-uniform cross-section such that, the variation $\frac{dR}{dl}$ of its resistance R with length l is $\frac{dR}{dl} \propto \frac{1}{\sqrt{l}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P. What is the length AP?



[12 Jan. 2019 I]

Options:

A. 0.2 m

B. 0.3 m

C. 0.25 m

D. 0.35 m

Answer: C

Solution:

We have given

$$\frac{dR}{dl} \propto \frac{1}{\sqrt{l}} \Rightarrow \frac{dR}{dl} = k \times \frac{1}{\sqrt{l}} \text{ (where k is constant)}$$

$$dR = k \frac{dl}{\sqrt{l}}$$

Let R_1 and R_2 be the resistance of AP and PB respectively. Using wheatstone bridge principle

$$\therefore \frac{R'}{R} = \frac{R_1}{R_2} \text{ or } R_1 = R_2$$

$$\text{Now, } \int dR = k \int \frac{dl}{\sqrt{l}}$$

$$\therefore R_1 = k \int_0^1 l^{-1/2} dl = k \cdot 2 \cdot \sqrt{l}$$

$$R_2 = k \int_1^2 l^{-1/2} dl = k \cdot (2 - 2\sqrt{l})$$

Putting $R_1 = R_2$

$$k2\sqrt{l} = k(2 - 2\sqrt{l})$$

$$\therefore 2\sqrt{l} = 1$$

$$\sqrt{l} = \frac{1}{2}$$

$$\text{i.e., } l = \frac{1}{4} \text{ m} \Rightarrow 0.25 \text{ m}$$

Question243

An ideal battery of 4 V and resistance R are connected in series in the primary circuit of a potentiometer of length 1 m and resistance 5Ω . The value of R, to give a potential difference of 5 mV across 10 cm of potentiometer wire is:

[12 Jan. 2019 I]

Options:

A. 490Ω

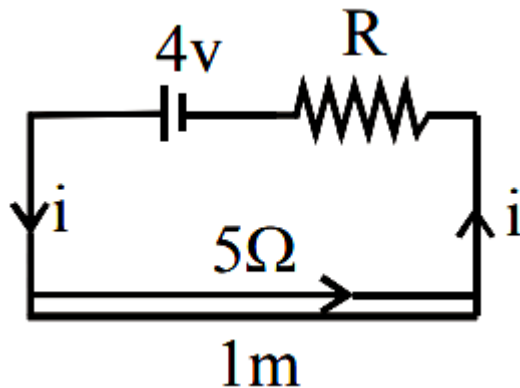
B. 480Ω

C. 395Ω

D. 495Ω

Answer: C

Solution:



Current flowing through the circuit (I) is given by

$$I = \left(\frac{4}{R+5} \right) \text{A}$$

Resistance of length 10cm of wire

$$= 5 \times \frac{10}{100} = 0.5\Omega$$

According to question,

$$5 \times 10^{-3} = \left(\frac{4}{R+5} \right) \cdot (0.5)$$

$$\therefore \frac{4}{R+5} = 10^{-2} \text{ or } R+5 = 400\Omega$$

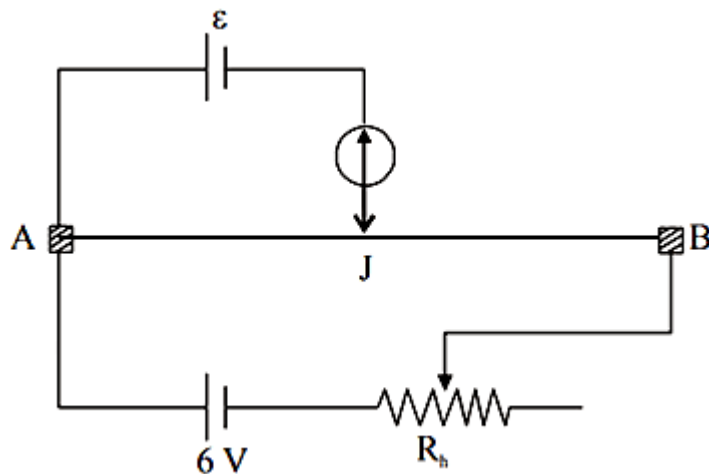
$$\therefore R = 395\Omega$$

Question244

The resistance of the meter bridge AB in given figure is 4Ω .

With a cell of emf $\varepsilon = 0.5\text{V}$ and rheostat resistance $R_h = 2\Omega$ the null point is obtained at some point J. When the cell is replaced by another one of emf $\varepsilon = \varepsilon_2$ the same null point J is found for

$R_h = 6\Omega$. The emf ε_2 is:



[11 Jan. 2019 I]

Options:

A. 0.4 V

B. 0.3 V

C. 0.6 V

D. 0.5 V

Answer: B

Solution:

Given, Emf of cell, $\varepsilon = 0.5\text{V}$

Rheostat resistance, $R_h = 2\Omega$

Potential gradient is

$$\frac{dV}{dL} = \left(\frac{6}{2+4} \right) \times \frac{4}{L}$$

Let null point be at 1cm when cell of emf $\varepsilon = 0.5\text{V}$ is used.

$$\text{thus } \varepsilon_1 = 0.5\text{V} = \left(\frac{6}{2+4} \right) \times \frac{4}{L} \times 1 \dots (i)$$

For resistance $R_h = 6\Omega$ new potential gradient is $\left(\frac{6}{4+6} \right) \times \frac{4}{L}$ and at null point

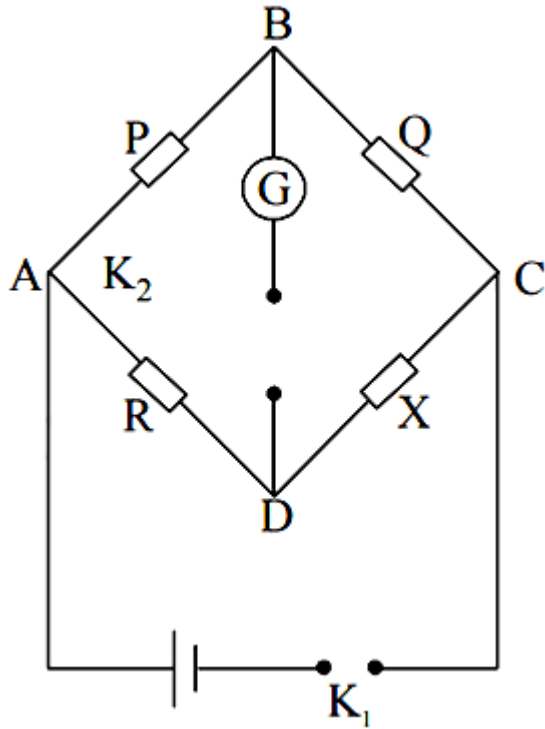
$$\left(\frac{6}{4+6} \right) \left(\frac{4}{L} \right) \times 1 = \varepsilon_2 \dots (ii)$$

Dividing equation (i) by (ii) we get

$$\frac{0.5}{\varepsilon_2} = \frac{10}{6} \text{ thus } \varepsilon_2 = 0.3\text{V}$$

Question245

In a Wheatstone bridge (see fig.), Resistances P and Q are approximately equal. When $R = 400\ \Omega$, the bridge is balanced. On interchanging P and Q, the value of R, for balance, is $405\ \Omega$. The value of X is close to :



[11 Jan. 2019 I]

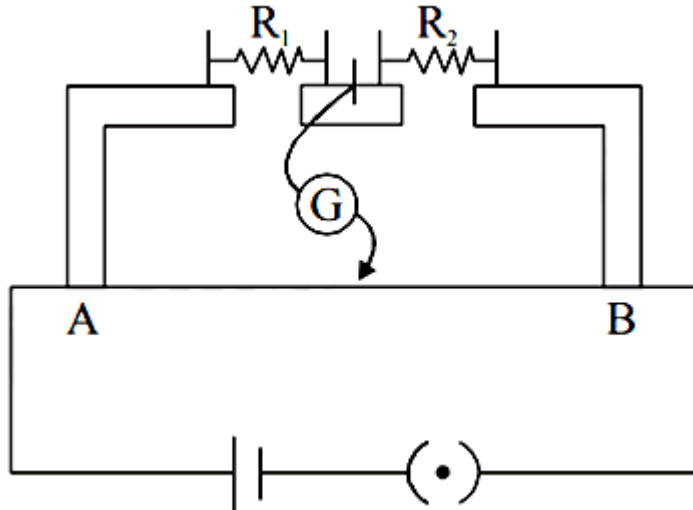
Options:

- A. 401.5 ohm
- B. 404.5 ohm
- C. 403.5 ohm
- D. 402.5 ohm

Answer: D

Question246

In the experimental set up of metre bridge shown in the figure, the null point is obtained at a distance of 40 cm from A. If a 10 Ω resistor is connected in series with R_1 , the null point shifts by 10 cm. The resistance that should be connected in parallel with $(R_1 + 10)\Omega$ such that the null point shifts back to its initial position is :



[11 Jan. 2019 II]

Options:

- A. 20 Ω
- B. 40 Ω
- C. 60 Ω
- D. 30 Ω

Answer: C

Solution:

Initially at null deflection $\frac{R_1}{R_2} = \frac{2}{3} \dots (i)$

Finally at null deflection, when null point is shifted

$$\frac{R_1 + 10}{R_2} = 1 \Rightarrow R_1 + 10 = R_2 \dots (ii)$$

Solving equations (i) and (ii) we get

$$\frac{2R_2}{3} + 10 = R_2$$

$$10 = \frac{R_2}{3} \Rightarrow R_2 = 30\Omega$$

$$\& R_1 = 20\Omega$$

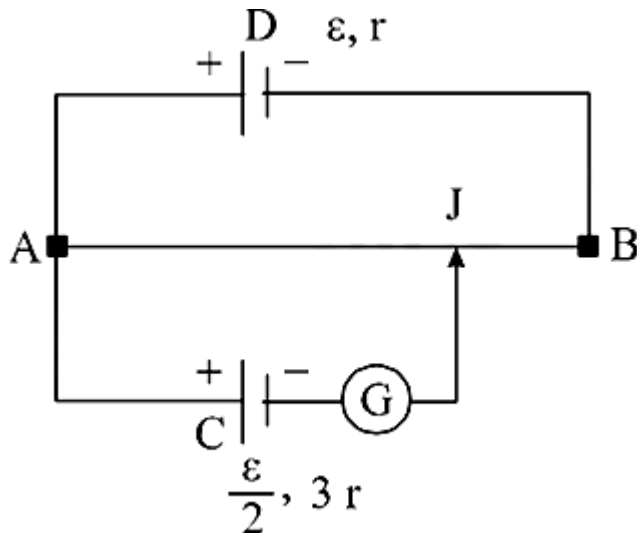
Now if required resistance is R then

$$\frac{30 \times R}{30 + R} = \frac{2}{3}$$

$$R = 60\Omega$$

Question247

A potentiometer wire AB having length L and resistance $12r$ is joined to a cell D of emf ε and internal resistance r . A cell C having emf $\varepsilon/2$ and internal resistance $3r$ is connected. The length AJ at which the galvanometer as shown in fig. shows no deflection is:



[10 Jan. 2019 I]

Options:

A. $\frac{11}{12}L$

B. $\frac{11}{24}L$

C. $\frac{13}{24}L$

D. $\frac{5}{12}L$

Answer: C

Solution:

Let x be the length AJ at which galvanometer shows null deflection current,

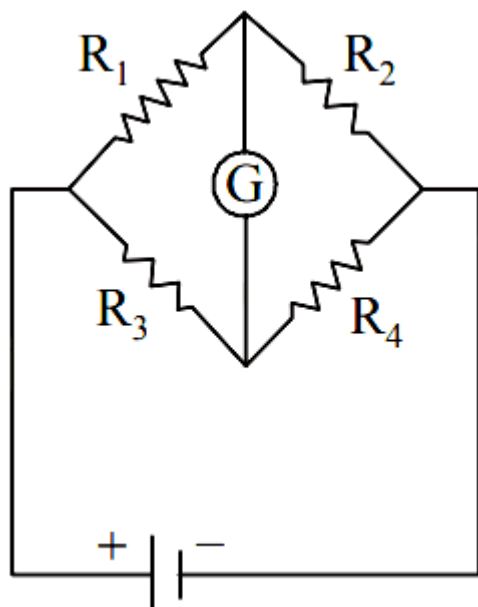
$$i = \frac{\varepsilon}{12r + r} = \frac{3}{13r} \text{ or, } i \left(\frac{x}{L} 12r \right) = \frac{\varepsilon}{2}$$

$$\Rightarrow \frac{\varepsilon}{13r} \left[\frac{x}{L} \cdot 12r \right] = \frac{\varepsilon}{2} \Rightarrow \frac{\varepsilon}{13r} \left[\frac{x}{L} \cdot 12r \right] = \frac{\varepsilon}{2}$$

$$\text{or, } x = \frac{13L}{24}$$

Question 248

The Wheatstone bridge shown in Fig. here, gets balanced when the carbon resistor used as R_1 has the colour code (Orange, Red, Brown). The resistors R_2 and R_4 are 80Ω and 40Ω , respectively. Assuming that the colour code for the carbon resistors gives their accurate values, the colour code for the carbon resistor, used as R_3 , would be:



[10 Jan. 2019 II]

Options:

- A. Brown, Blue, Brown
- B. Brown, Blue, Black
- C. Red, Green, Brown
- D. Grey, Black, Brown

Answer: A

Solution:

Given, colour code of resistance,

$R_1 = \text{Orange, Red and Brown}$

$$\therefore R_1 = 32 \times 10 = 320$$

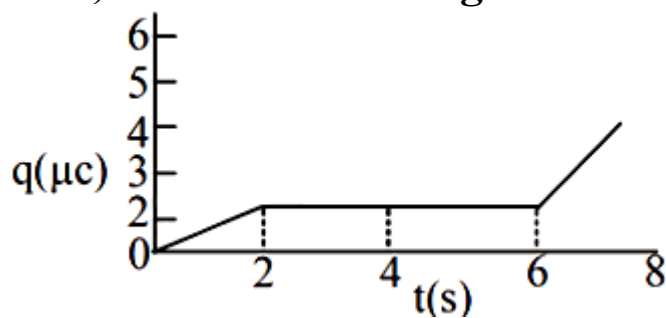
using balanced wheatstone bridge principle,

$$\frac{R_1}{R_3} = \frac{R_2}{R_4} \Rightarrow \frac{320}{R_3} = \frac{80}{40}$$

$\therefore R_3 = 160$ i.e. colour code for R_3 Brown, Blue and Brown

Question249

The charge on a capacitor plate in a circuit, as a function of time, is shown in the figure:



What is the value of current at $t = 4 \text{ s}$?
[12 Jan. 2019 II]

Options:

A. Zero

B. $3 \mu\text{A}$

C. $2 \mu\text{A}$

D. $1.5 \mu\text{A}$

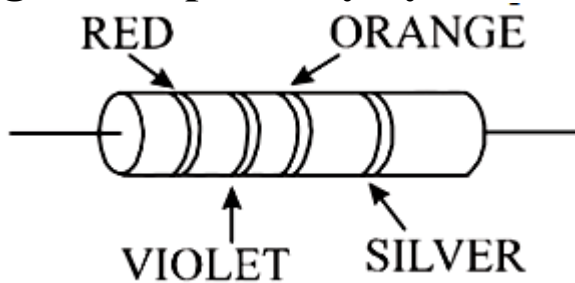
Answer: A

Solution:

Clearly, from graph Current, $I = \frac{dq}{dt} = 0$ at $t = 4\text{s}$ [Since q is constant]

Question250

A resistance is shown in the figure. Its value and tolerance are given respectively by:



[9 Jan. 2019 I]

Options:

- A. $270\ \Omega$, 10%
- B. $27\ \text{k}\Omega$, 10%
- C. $27\ \text{k}\Omega$, 20%
- D. $270\ \Omega$, 5%

Answer: B

Solution:

Color code :

Bl, Br, R, O, Y, G, B, V, Gr, W

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

$R = AB \times C \pm D\%$ where D = tolerance

$D_{\text{gold}} = \pm 5\%$, $D_{\text{silver}} = \pm 10\%$; $D_{\text{no colour}} = \pm 20\%$

Red violet orange silver

$R = 27 \times 10^3 \Omega \pm 10\% = 27\text{k}\Omega \pm 10\%$

Question251

Drift speed of electrons, when 1.5A of current flows in a copper wire of cross section 5mm^2 , is v. If the electron density in copper is $9 \times 10^{28}/\text{m}^3$ the value of v in mm/s close to (Take charge of

electron to be $= 1.6 \times 10^{-19} \text{C}$)

[9 Jan. 2019 I]

Options:

A. 0.02

B. 3

C. 2

D. 0.2

Answer: A

Solution:

Using, $I = neAv_d$

$\therefore$ Drift speed $v_d = \frac{1}{neA}$

$$\frac{1.5}{9 \times 10^{28} \times 1.6 \times 10^{-19} \times 5 \times 10^{-6}} = 0.02 \text{mms}^{-1}$$

Question252

A copper wire is stretched to make it 0.5% longer. The percentage change in its electrical resistance if its volume remains unchanged is:

[9 Jan. 2019 I]

Options:

A. 2.0%

B. 2.5%

C. 1.0%

D. 0.5%

Answer: C

Solution:

$$\text{Resistance, } R = \frac{\rho l}{A}$$

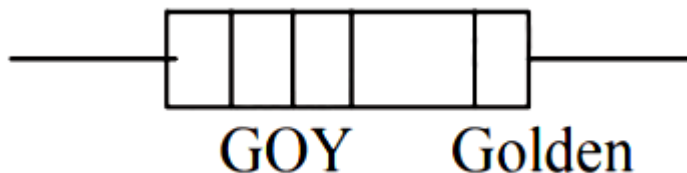
$$R = \rho \frac{1}{A} \times \frac{l}{1} = \frac{\rho l^2}{V} [\gg \text{Volume (V)} = A \times l]$$

Since resistivity and volume remains constant therefore % change in resistance

$$\frac{\Delta R}{R} = \frac{2\Delta l}{l} = 2 \times (0.5) = 1\%$$

Question 253

A carbon resistance has following colour code. What is the value of the resistance?



[9 Jan. 2019 II]

Options:

- A. $530\text{k}\Omega \pm 5\%$
- B. $5.3\text{k}\Omega \pm 5\%$
- C. $6.4\text{kM}\Omega \pm 5\%$
- D. $64\text{kM}\Omega \pm 10\%$

Answer: A

Solution:

Colour code for carbon resistor

Bl, Br, R, O, Y, G, Blue, V, Gr, W
0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Resistance, $R = AB \times C \pm D$

Bands A and B are the first two significant figures of resistance

B and C indicates the decimal multiplier or the number of zeros that follow A and B

B and D is tolerance: Gold = $\pm 5\%$

Silver = $\pm 10\%$ No colour = $\pm 20\%$

$$R = 53 \times 10^4 \pm 5\% = 530\text{k}\Omega \pm 5\%$$

Question254

A uniform metallic wire has a resistance of $18\ \Omega$ and is bent into an equilateral triangle. Then, the resistance between any two vertices of the triangle is:

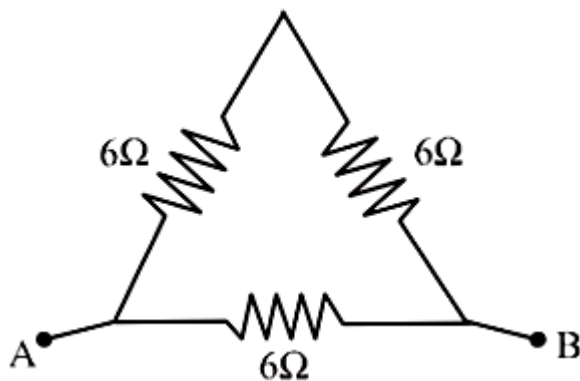
[10 Jan. 2019 I]

Options:

- A. $4\ \Omega$
- B. $8\ \Omega$
- C. $12\ \Omega$
- D. $2\ \Omega$

Answer: A

Solution:



Resistance, $R \propto l$ so resistance of each side of the equilateral triangle = $6\ \Omega$

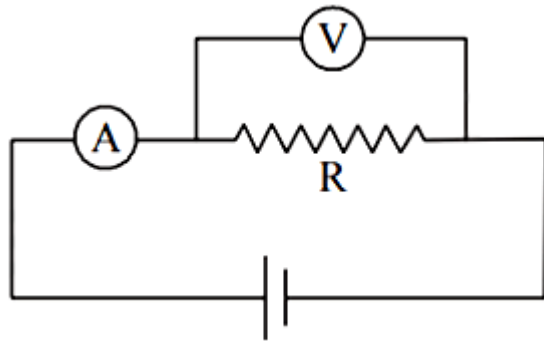
Resistance R_{eq} between any two vertices

$$\frac{1}{R_{eq}} = \frac{1}{12} + \frac{1}{6} \Rightarrow R_{eq} = 4\ \Omega$$

Question255

The actual value of resistance R , shown in the figure is $30\ \Omega$. This is measured in an experiment as shown using the standard formula $R = \frac{V}{I}$, where V and I are the reading of the voltmeter and ammeter, respectively. If the measured value of R is 5% less,

then the internal resistance of the voltmeter is:



[10 Jan. 2019 II]

Options:

A. $600\ \Omega$

B. $570\ \Omega$

C. $35\ \Omega$

D. $350\ \Omega$

Answer: B

Solution:

Using, $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$0.95R = \frac{RR_v}{R + R_v}$ (measured value 5% less than internal resistance of voltmeter)

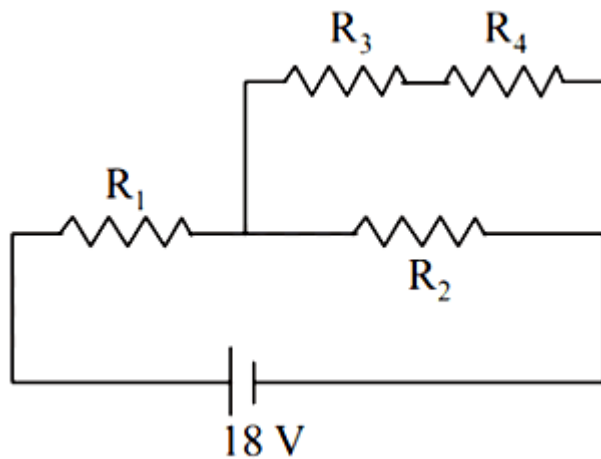
or, $0.95 \times 30 = 0.05R_v$

$\therefore R_v = 19 \times 30 = 570\ \Omega$

Question 256

In the given circuit the internal resistance of the 18V cell is negligible. If $R_1 = 400\ \Omega$, $R_3 = 100\ \Omega$ and $R_4 = 500\ \Omega$ and the reading of an ideal voltmeter across R_4 is 5V then the value of

R_2 will be:



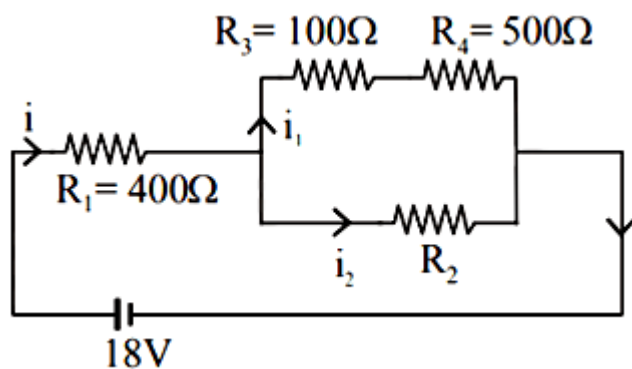
[9 Jan. 2019 II]

Options:

- A. $300\ \Omega$
- B. $450\ \Omega$
- C. $550\ \Omega$
- D. $230\ \Omega$

Answer: A

Solution:



Across R_4 reading of voltmeter, $V_4 = 5V$

$$\text{Current, } i_4 = \frac{V_4}{R_4} = 0.01A$$

$$V_3 = i_1 R_3 = 1V$$

$$V_3 + V_4 = 6V = V_2$$

$$V_1 + V_3 + V_4 = 18V$$

$$\Rightarrow V_1 = 12V$$

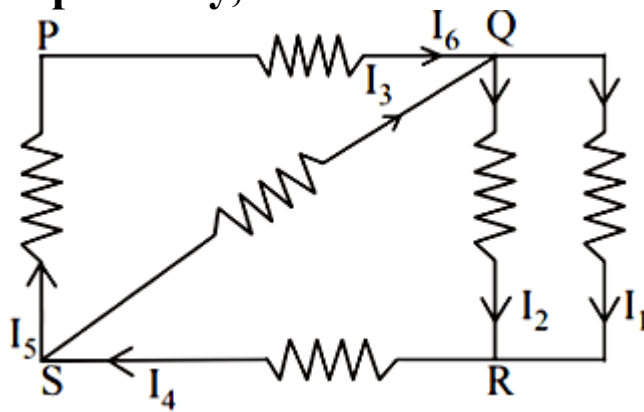
$$i = \frac{V_1}{R_1} = 0.03\text{A}$$

$$i = i_1 + i_2 \Rightarrow i_2 = i - i_1 = 0.03 - 0.01\text{A} = 0.02\text{A}$$

$$\therefore R_2 = \frac{V_2}{i_2} = \frac{6}{0.02} = 300\Omega$$

Question257

In the given circuit diagram, the currents, $I_1 = -0.3\text{A}$, $I_4 = 0.8\text{A}$ and $I_5 = 0.4\text{A}$, are flowing as shown. The currents I_2 , I_3 and I_6 , respectively, are



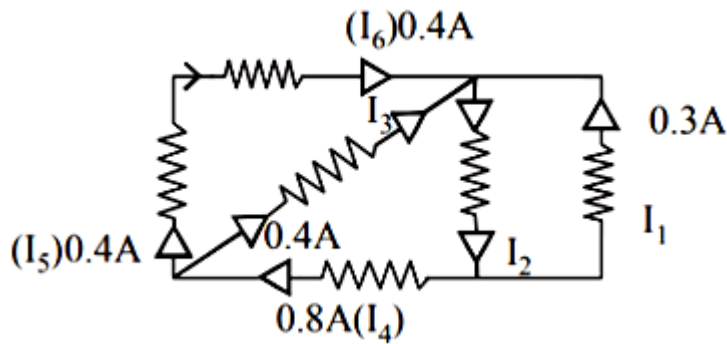
[12 Jan. 2019 II]

Options:

- A. 1.1 A, -0.4A , 0.4A
- B. 1.1 A, 0.4A , 0.4A
- C. 0.4A , 1.1 A, 0.4A
- D. -0.4A , 0.4A , 1.1 A

Answer: B

Solution:



From KCL, $I_3 = 0.8 - 0.4 = 0.4\text{A}$

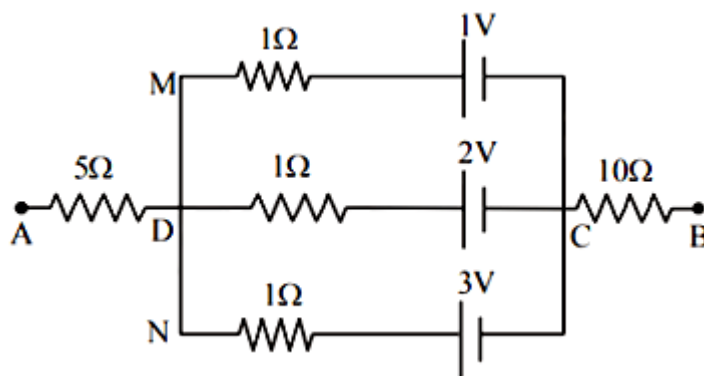
$$I_2 = 0.4 + 0.4 + 0.3$$

$$= 1.1\text{A}$$

$$\text{and } I_6 = 0.4\text{A}$$

Question 258

In the circuit shown, the potential difference between A and B is :



[11 Jan. 2019 II]

Options:

A. 1 V

B. 2 V

C. 3 V

D. 6 V

Answer: B

Solution:

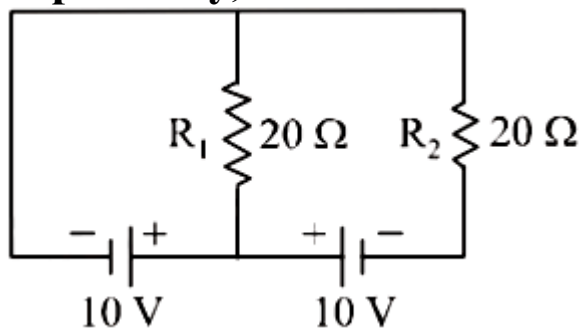
Given, $E_1 = 1\text{V}$, $E_2 = 2\text{V}$, $E_3 = 3\text{V}$, $r_1 = 1\Omega$

$r_2 = 1\Omega$ and $r_3 = 1\Omega$

$$V_{AB} = V_{CD} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2} + \frac{E_3}{r_3}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}} = \frac{\frac{1}{1} + \frac{2}{1} + \frac{3}{1}}{\frac{1}{1} + \frac{1}{1} + \frac{1}{1}} = \frac{6}{3} = 2\text{V}$$

Question 259

In the given circuit the cells have zero internal resistance. The currents (in Amperes) passing through resistance R_1 and R_2 respectively, are:



[10 Jan. 2019 I]

Options:

- A. 1, 2
- B. 2, 2
- C. 0.5, 0
- D. 0, 1

Answer: C

Solution:

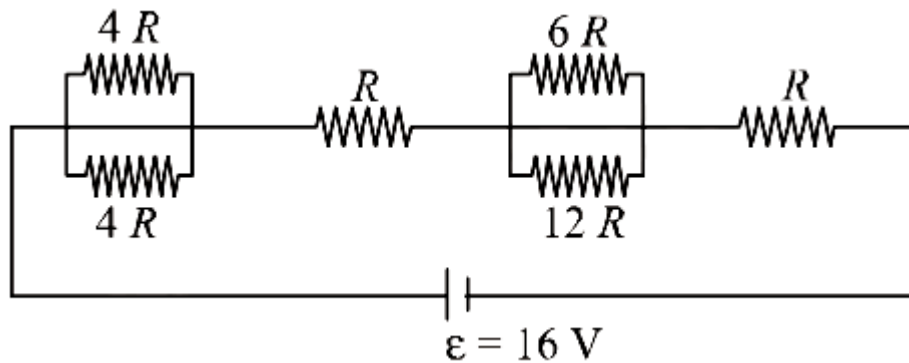
Current passing through resistance R_1 ,

$$i_1 = \frac{V}{R_1} = \frac{10}{20} = 0.5\text{A}$$

and, $i_2 = 0$

Question260

The resistive network shown below is connected to a D.C. source of 16 V. The power consumed by the network is 4 Watt. The value of R is :



[12 Apr. 2019 I]

Options:

- A. 6Ω
- B. 8Ω
- C. 1Ω
- D. 16Ω

Answer: B

Solution:

Equivalent resistance,

$$\begin{aligned} R_{\text{eq}} &= \frac{4R \times 4R}{4R + 4R} + R + \frac{6R \times 12R}{6R + 12R} + R \\ &= 2R + R + 4R + R \\ &= 8R \end{aligned}$$

$$\text{Using, } P = \frac{V^2}{R_{\text{eq}}} \Rightarrow 4 = \frac{16^2}{8R}$$

$$\therefore R = \frac{16^2}{4 \times 8} = 8\Omega$$

Question261

One kg of water, at 20°C, is heated in an electric kettle whose heating element has a mean (temperature averaged) resistance of 20 Ω. The rms voltage in the mains is 200 V. Ignoring heat loss from the kettle, time taken for water to evaporate fully, is close to :

[Specific heat of water = 4200 J/(kg°C), Latent heat of water = 2260 kJ/kg]

[12 Apr. 2019 II]

Options:

- A. 16 minutes
- B. 22 minutes
- C. 3 minutes
- D. 3 minutes

Answer: B

Solution:

$$Q = P \times t$$

$$Q = mc \Delta T + mL$$

$$P = \frac{V_{\text{rms}}^2}{R}$$

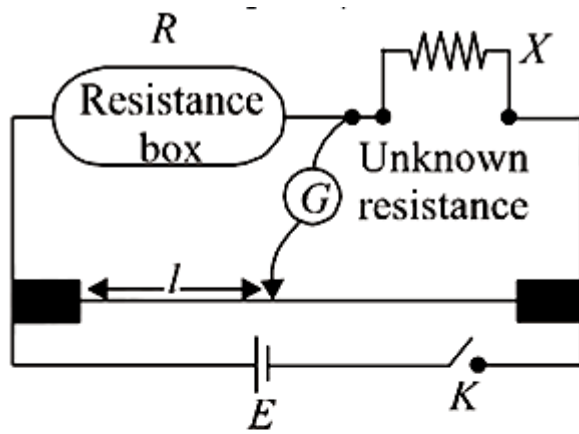
$$4200 \times 80 + 2260 \times 10^3 = \frac{(200)^2}{20} \times t$$

$$t = 1298 \text{ sec}$$

$$t \simeq 22 \text{ min}$$

Question 262

In a meter bridge experiment, the circuit diagram and the corresponding observation table are shown in figure.



Sl.No.	RΩ	l (cm)
1.	1000	60
2.	100	13
3.	10	1.5
4.	1	1.0

**Which of the reading is consistent ?
[10 Apr. 2019 I]**

Options:

- A. 3
- B. 2
- C. 4
- D. 1

Answer: C

Solution:

For a balanced bridge

$$\frac{R_1}{R_2} = \frac{l_2}{l_1}$$

$$\text{So } \frac{R}{X} = \frac{l}{100 - l}$$

Using the above expression

$$X = \frac{R(100 - l)}{l}$$

for observation (1) $X = \frac{100 \times 40}{60} = \frac{2000}{3} \Omega$

for observation(2) $X = \frac{100 \times 87}{13} = \frac{8700}{13} \Omega$

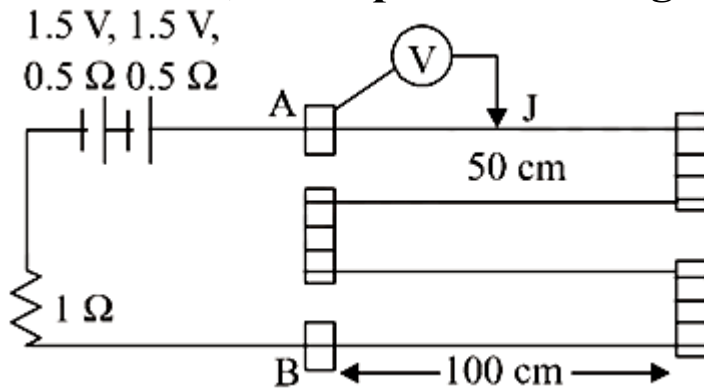
for observation(3) $X = 10 \times 98.51.5 = \frac{1970}{3} \Omega$

for observation(4) $X = 1 \times 991 = 99 \Omega$

Clearly we can see that the value of x calculated in observation (4) is inconsistent than other.

Question263

In the circuit shown, a four-wire potentiometer is made of a 400 cm long wire, which extends between A and B. The resistance per unit length of the potentiometer wire is $r = 0.01 \Omega/\text{cm}$. If an ideal voltmeter is connected as shown with jockey J at 50 cm from end A, the expected reading of the voltmeter will be :



[8 Apr. 2019 II]

Options:

A. 0.50 V

B. 0.75 V

C. 0.25 V

D. 0.20 V

Answer: C

Solution:

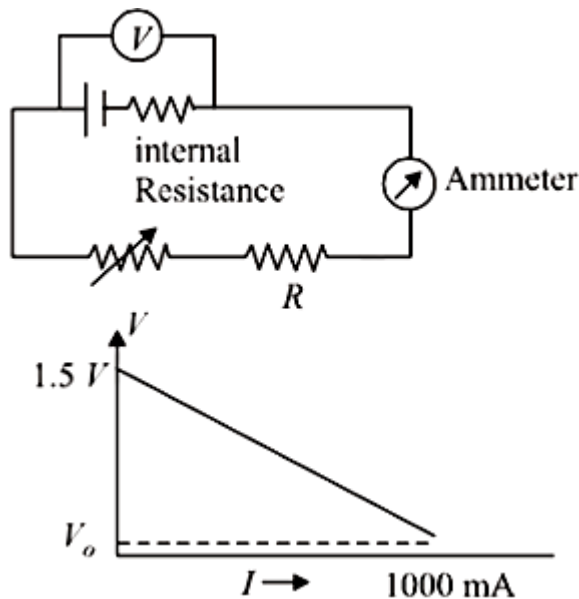
The resistance of potentiometer wire $R = 0.01 \times 400 = 4 \Omega$
 Current in the wire

$$i = \frac{V}{R_T} = \frac{3}{4 + 0.5 + 0.57 + 1} = \frac{1}{2} \text{ A}$$

$$\text{Now } V = iR_{AJ} = \frac{1}{2} \times (0.01 \times 50) = 0.25 \text{ V}$$

Question 264

To verify Ohm's law, a student connects the voltmeter across the battery as, shown in the figure. The measured voltage is plotted as a function of the current, and the following graph is obtained :



If V_0 is almost zero, identify the correct statement:
[12 Apr. 2019 I]

Options:

- A. The emf of the battery is 1.5 V and its internal resistance is 1.5Ω
- B. The value of the resistance R is 1.5Ω
- C. The potential difference across the battery is 1.5 V when it sends a current of 1000 mA
- D. The emf of the battery is 1.5 V and the value of R is 1.5Ω

Answer: A

Solution:

When $i = 0$, $V = \varepsilon = 1.5$ volt

Question265

A current of 5 A passes through a copper conductor (resistivity) $= 1.7 \times 10^{-8} \Omega\text{m}$) of radius of cross-section 5mm. Find the mobility of the charges if their drift velocity is $1.1 \times 10^{-3} \text{m/s}$. [10 Apr. 2019 I]

Options:

A. $1.8 \text{m}^2/\text{V s}$

B. $1.5 \text{m}^2/\text{V s}$

C. $1.3 \text{m}^2/\text{V s}$

D. $1.0 \text{m}^2/\text{V s}$

Answer: D

Solution:

Charge mobility

$$(\mu) = \frac{V_d}{E} \text{ [Where } V_d = \text{drift velocity]}$$

$$\text{and resistivity } (\rho) = \frac{E}{j} = \frac{E A}{I} \Rightarrow E = \frac{I(\rho)}{A}$$

$$\Rightarrow \mu = \frac{V_d}{E} = \frac{V_d A}{I \rho}$$

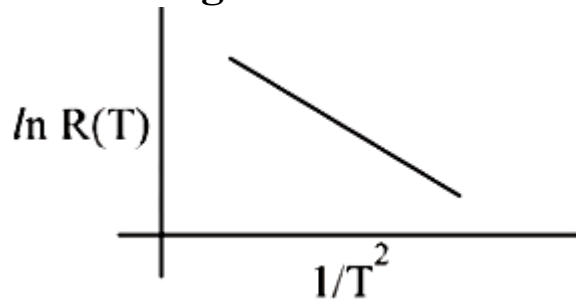
$$= \frac{1.1 \times 10^{-3} \times \pi \times (5 \times 10^{-3})^2}{5 \times 1.7 \times 10^{-8}}$$

$$\mu = 1.0 \frac{\text{m}^2}{\text{V s}}$$

Question266

In an experiment, the resistance of a material is plotted as a function of temperature (in some range). As shown in the figure,

it is a straight line.



[10 Apr. 2019 I]

Options:

A. $R(T) = \frac{R_0}{T^2}$

B. $R(T) = R_0 e^{-T_0^2/T^2}$

C. $R(T) = R_0 e^{-T^2/T_0^2}$

D. $R(T) = R_0 e^{T^2/T_0^2}$

Answer: B

Solution:

Equation of straight line from graph $y = -mx + c$

$$\Rightarrow \ln R = -m \left(\frac{1}{T^2} \right) + c$$

here, m & c are constants

$$R = e^{\left[-m \left(\frac{1}{T^2} \right) + c \right]} = e^{-m \left(\frac{1}{T^2} \right)} \times e^c$$

$$R(T) = R_0 e^{-\frac{T_0^2}{T^2}}$$

Question 267

Space between two concentric conducting spheres of radii a and b ($b > a$) is filled with a medium of resistivity ρ . The resistance between the two spheres will be :

[10 Apr. 2019 II]

Options:

A. $\frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$

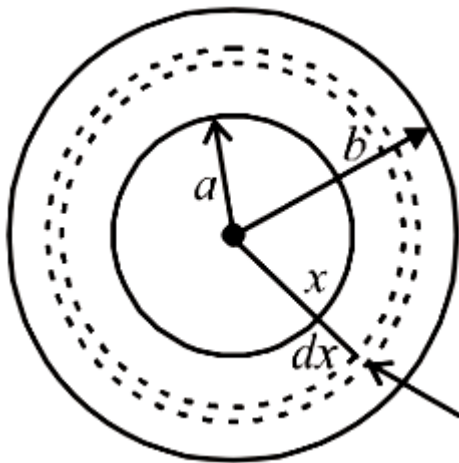
B. $\frac{\rho}{2\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$

C. $\frac{\rho}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$

D. $\frac{\rho}{4\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$

Answer: A

Solution:



$$dR = \frac{(\rho)(dx)}{4\pi x^2}$$

$$R = \int dR$$

$$\int dR = \rho \int_a^b \frac{dx}{4\pi x^2}$$

$$\Rightarrow R = \frac{\rho}{4\pi} \left[\frac{-1}{x} \right]_a^b$$

$$R = \left(\frac{\rho}{4\pi} \right) \cdot \left(\frac{1}{a} - \frac{1}{b} \right)$$

Question 268

In a conductor, if the number of conduction electrons per unit volume is $8.5 \times 10^{28} \text{ m}^{-3}$ and mean free time is 25 fs (femto

second), it's approximate resistivity is: ($m_e = 9.1 \times 10^{-31} \text{ kg}$)
[9 Apr. 2019 II]

Options:

A. $10^{-6} \Omega \text{ m}$

B. $10^{-7} \Omega \text{ m}$

C. $10^{-8} \Omega \text{ m}$

D. $10^{-5} \Omega \text{ m}$

Answer: C

Solution:

$$\begin{aligned}\rho &= \frac{m}{ne^2\tau} \\ &= \frac{9.1 \times 10^{-31}}{8.5 \times 10^{28} \times (1.6 \times 10^{-19})^2 \times 25 \times 10^{-15}} \\ &= 10^{-8} \Omega - \text{m}\end{aligned}$$

Question269

A 200Ω resistor has a certain color code. If one replaces the red color by green in the code, the new resistance will be :
[8 April 2019 I]

Options:

A. 100Ω

B. 400Ω

C. 300Ω

D. 500Ω

Answer: D

Solution:

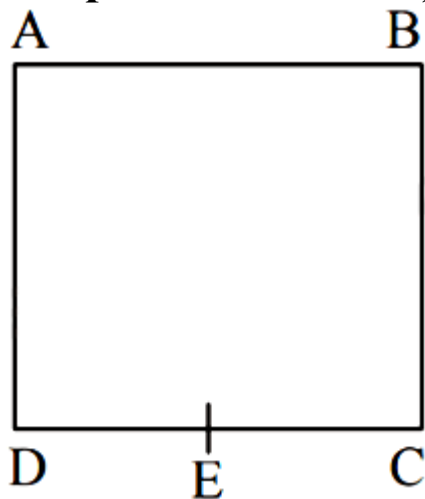
Number 2 is associated with the red colour. This colour is replaced by green.

∴ Colour code figure for green is 5

∴ New resistance = 500Ω

Question270

A wire of resistance R is bent to form a square ABCD as shown in the figure. The effective resistance between E and C is: (E is mid-point of arm CD)



[9 April 2019 I]

Options:

A. R

B. $\frac{7}{64}R$

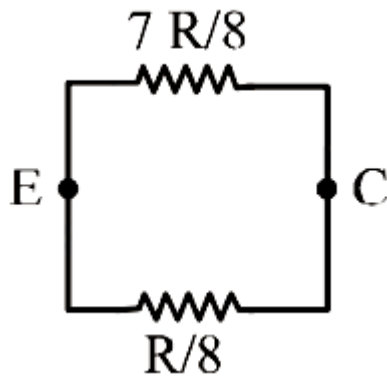
C. $\frac{3}{4}R$

D. $\frac{1}{16}R$

Answer: B

Solution:

$$R_{eq} = \frac{\left(\frac{7R}{8}\right)\left(\frac{R}{8}\right)}{R} = \frac{7R}{64}$$



Question271

A metal wire of resistance $3\ \Omega$ is elongated to make a uniform wire of double its previous length. This new wire is now bent and the ends joined to make a circle. If two points on the circle make an angle 60° at the centre, the equivalent resistance between these two points will be:

[9 April 2019 II]

Options:

A. $\frac{12}{5}\Omega$

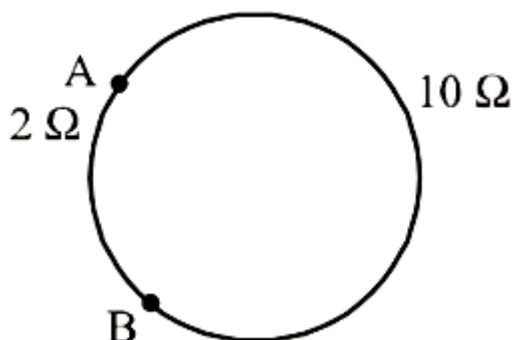
B. $\frac{5}{2}\Omega$

C. $\frac{5}{3}\Omega$

D. $\frac{7}{2}\Omega$

Answer: C

Solution:



When length becomes double its resistance becomes ($R \propto l^2$)

$$R = 4 \times 3 = 12\Omega$$

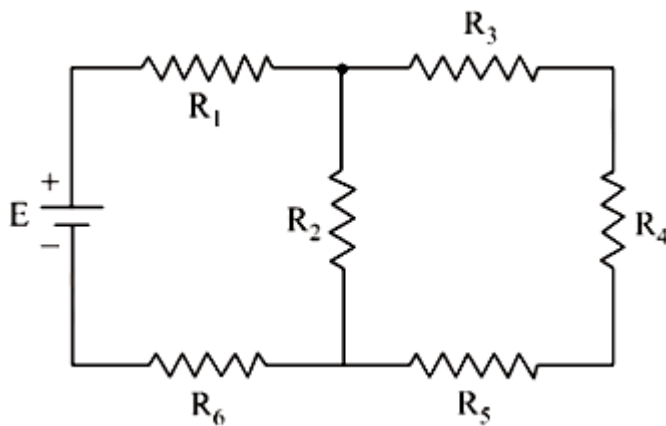
$$R_{eq} = \frac{2 \times 10}{12} = \frac{5}{3}\Omega$$

Question272

In the figure shown, what is the current (in Ampere) drawn from the battery? You are given :

$$R_1 = 15\Omega, R_2 = 10\Omega, R_3 = 20\Omega, R_4 = 5\Omega, R_5 = 25\Omega$$

$$R_6 = 30\Omega, E = 15V$$



[8 Apr. 2019 II]

Options:

A. $13/24$

B. $7/18$

C. $9/32$

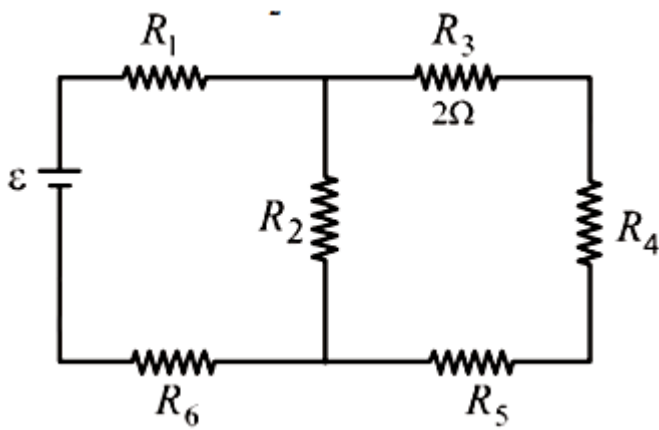
D. $20/3$

Answer: C

Solution:

R_3, R_4 and R_5 are in series so their equivalent $R = 20 + 5 + 25 = 50\Omega$

This is parallel with R_2 , and so net resistance of the circuit

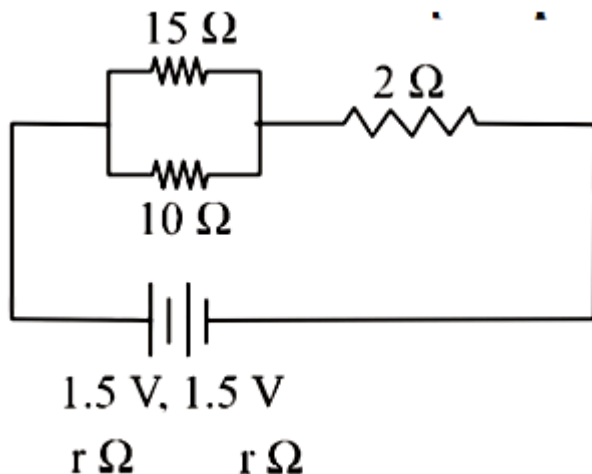


$$R_{eq} = \left(\frac{10 \times 50}{10 + 50} \right) + 15 + 30 = \frac{160}{3} \Omega$$

$$\text{So, } i = \frac{\varepsilon}{R_{eq}} = \frac{15}{(100/3)} = \frac{9}{32} \text{ A}$$

Question 273

In the given circuit, an ideal voltmeter connected across the 10Ω resistance reads 2 V . The internal resistance r , of each cell is :



[10 Apr. 2019 I]

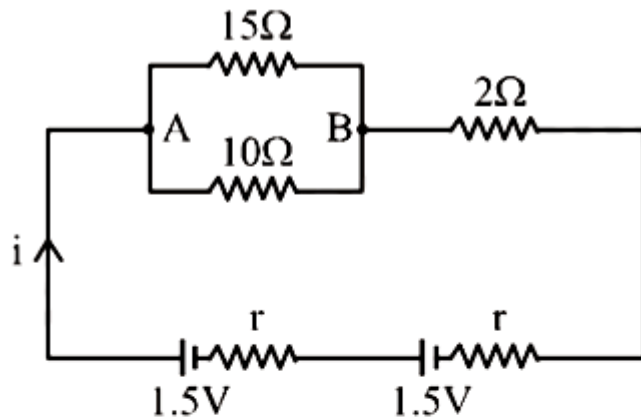
Options:

- A. 1Ω
- B. 0.5Ω
- C. 1.5Ω
- D. 0Ω

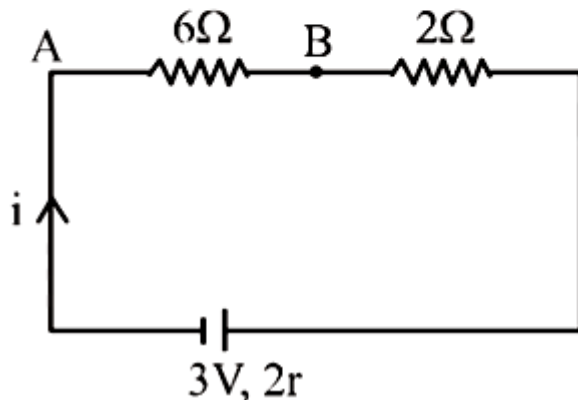
Answer: B

Solution:

For the given circuit



$$i = \frac{3}{8 + 2r}$$



Now voltage across AB

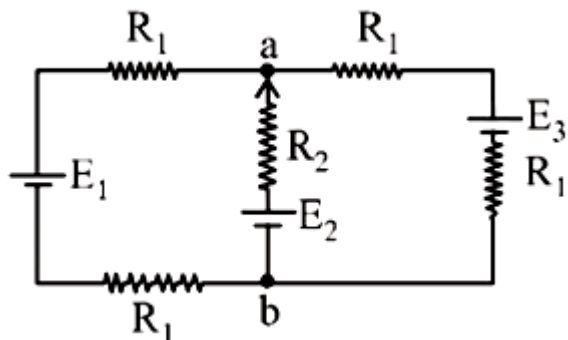
$$i \times 6 = \frac{3}{8 + 2r} \times 6 = 2$$

$$\Rightarrow 9 = 8 + 2r$$

$$\Rightarrow r = \frac{1}{2} \Omega$$

Question274

For the circuit shown, with $R_1 = 1.0\Omega$, $R_2 = 2.0 \Omega$, $E_1 = 2V$ and $E_2 = E_3 = 4V$, the potential difference between the points 'a' and 'b' is approximately (in V) :



[8 April 2019 I]

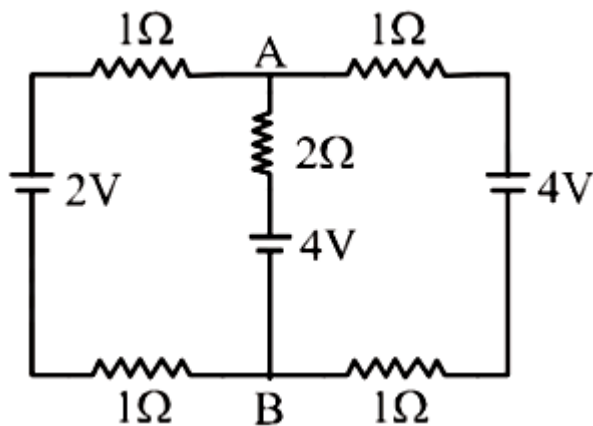
Options:

- A. 2.7
- B. 2.3
- C. 3.7
- D. 3.3

Answer: D

Solution:

Applying parallel combination of batteries $\frac{\frac{E_1}{1+1} + \frac{E_2}{2} + \frac{E_3}{1+1}}{\frac{1}{1+1} + \frac{1}{2} + \frac{1}{1+1}}$



$$\begin{aligned} \frac{\frac{2}{2} + \frac{4}{2} + \frac{4}{2}}{\frac{1}{2} + \frac{1}{2} + \frac{1}{2}} &= \frac{5 \times 2}{3} \\ &= \frac{10}{3} = 3.3 \text{ Volt} \end{aligned}$$

Question275

A cell of internal resistance r drives current through an external resistance R . The power delivered by the cell to the external resistance will be maximum when :

[8 Apr. 2019 II]

Options:

- A. $R = 0.001 r$

B. $R = 1000 r$

C. $R = 2r$

D. $R = r$

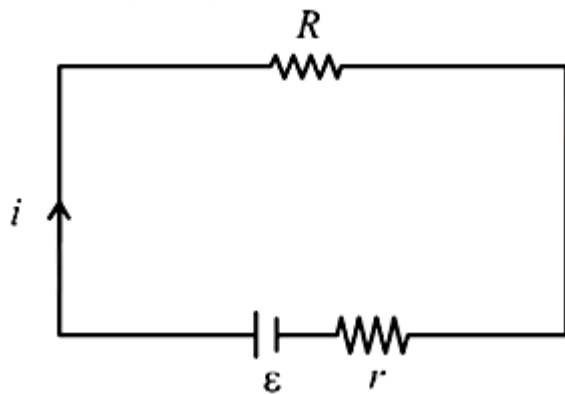
Answer: D

Solution:

$$i = \left(\frac{\varepsilon}{R + r} \right)$$

Power delivered to R.

$$P = i^2 R = \left(\frac{\varepsilon}{R + r} \right)^2 R$$



P to be maximum, $\frac{dP}{dR} = 0$

$$\text{or } \frac{d}{dR} \left[\left(\frac{\varepsilon}{R + r} \right)^2 R \right] = 0$$

or $R = r$

Question276

A heating element has a resistance of 100Ω at room temperature. When it is connected to a supply of 220V a steady current of 2 A passes in it and temperature is 500°C more than room temperature. What is the temperature coefficient of resistance of the heating element?

[Online April 16, 2018]

Options:

A. $1 \times 10^{-4}^\circ\text{C}^{-1}$

B. $5 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$

C. $2 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$

D. $0.5 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$

Answer: C

Solution:

Resistance after temperature increases by 500°C i.e., $R_t = \frac{V}{I} = \frac{220}{2} = 110\Omega$

$R_0 = 100$ (given) temperature coefficient of resistance, $\alpha = ?$

using $R_t = R_0(1 + \alpha t)$

$$110 = 100(1 + \alpha 500)$$

$$\alpha = \frac{10}{100 \times 500}$$

$$\text{or, } \alpha = 2 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$$

Question277

A copper rod of cross-sectional area A carries a uniform current I through it. At temperature T , if the volume charge density of the rod is ρ , how long will the charges take to travel a distance d ?

[Online April 15, 2018]

Options:

A. $\frac{2\rho d A}{IT}$

B. $\frac{2\rho d A}{I}$

C. $\frac{\rho d A}{I}$

D. $\frac{\rho d A}{IT}$

Answer: C

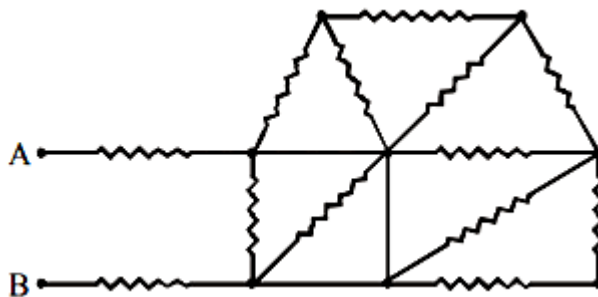
Solution:

$$\text{Charge density } \rho = \frac{\text{charge}}{\text{volume}} = \frac{q}{Ad} \Rightarrow q = \rho Ad$$

$$\text{Also, } q = I T \Rightarrow T = \frac{q}{I} = \frac{\rho Ad}{I}$$

Question 278

In the given circuit all resistances are of value R ohm each. The equivalent resistance between A and B is :



[Online April 15, 2018]

Options:

A. $2R$

B. $\frac{5R}{2}$

C. $\frac{5R}{3}$

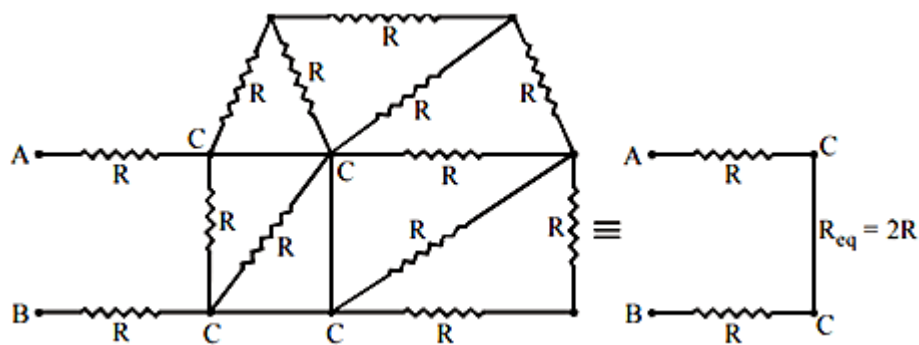
D. $3R$

Answer: A

Solution:

$$R_{\text{series}} = R_1 + R_2 + \dots + R_n$$

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$



Question 279

Two batteries with e.m.f. 12 V and 13 V are connected in parallel across a load resistor of 10Ω . The internal resistances of the two batteries are 1Ω and 2Ω respectively. The voltage across the load lies between:

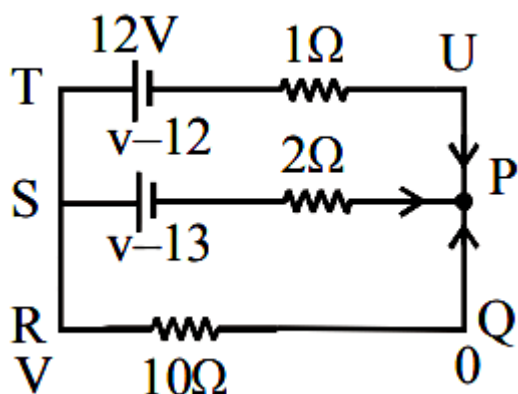
[2018]

Options:

- A. 11.6 V and 11.7 V
- B. 11.5 V and 11.6 V
- C. 11.4 V and 11.5 V
- D. 11.7 V and 11.8 V

Answer: B

Solution:



Using Kirchhoff's law at P we get

$$\frac{V - 12}{1} + \frac{V - 13}{2} + \frac{V - 0}{10} = 0 \quad [\text{Let potential at P, Q, U} = 0 \text{ and at R} = V]$$

$$\Rightarrow \frac{V}{1} + \frac{V}{2} + \frac{V}{10} = \frac{12}{1} + \frac{13}{2} + \frac{0}{10}$$

$$\Rightarrow \frac{10+5+1}{10}V = \frac{24+13}{2} \Rightarrow V \left(\frac{16}{10} \right) = \frac{37}{2}$$

$$\Rightarrow V = \frac{37 \times 10}{16 \times 2} = \frac{370}{32} = 11.56 \text{ volt}$$

Question280

**A constant voltage is applied between two ends of a metallic wire. If the length is halved and the radius of the wire is doubled, the rate of heat developed in the wire will be:
[Online April 15, 2018]**

Options:

- A. Increased 8 times
- B. Doubled
- C. Halved
- D. Unchanged

Answer: A

Solution:

Rate of heat i.e., Power developed in the wire = $P = \frac{V^2}{R}$

Resistance of the wire of length, L $R_1 = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2}$

$\therefore$ Power, $P_1 = \frac{V^2}{R_1}$

Resistance of the wire when length is halved i.e., $L/2$

$R_2 = \frac{\rho \frac{L}{2}}{\pi (2r)^2} = \frac{\rho L}{\pi 8r^2} = \frac{R_1}{8}$

$\therefore$ Power, $P_2 = \frac{V^2}{\frac{R_1}{8}} = \frac{8V^2}{R_1}$

or, $P_2 = 8P_1$ i.e., power increased 8 times of previous or original wire.

Question281

. In a potentiometer experiment, it is found that no current passes through the galvanometer when the terminals of the cell are connected across 52 cm of the potentiometer wire. If the cell is shunted by a resistance of $5\ \Omega$, a balance is found when the cell is connected across 40 cm of the wire. Find the internal resistance of the cell.

[2018]

Options:

- A. $1\ \Omega$
- B. $1.5\ \Omega$
- C. $2\ \Omega$
- D. $2.5\ \Omega$

Answer: B

Solution:

Using formula, internal resistance,

$$r = \left(\frac{l_1 - l_2}{l_2} \right) s = \left(\frac{52 - 40}{40} \right) \times 5 = 1.5\ \Omega$$

Question282

On interchanging the resistances, the balance point of a meter bridge shifts to the left by 10 cm. The resistance of their series combination is $1\text{ k}\Omega$. How much was the resistance on the left slot before interchanging the resistances?

[2018]

Options:

- A. $990\ \Omega$
- B. $505\ \Omega$

C. $550\ \Omega$

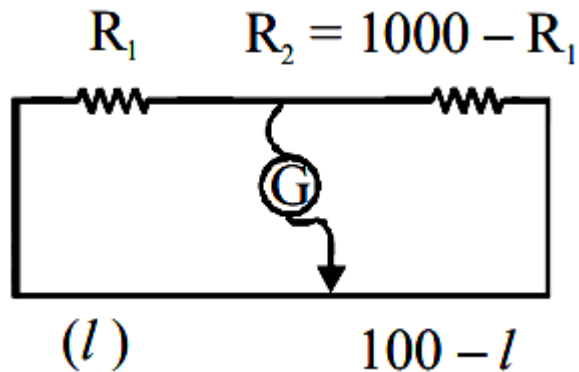
D. $910\ \Omega$

Answer: C

Solution:

$$R_1 + R_2 = 1000$$

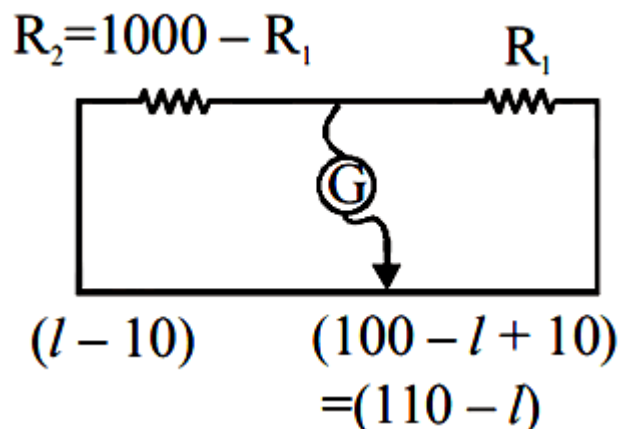
$$\Rightarrow R_2 = 1000 - R_1$$



On balancing condition

$$R_1(100 - l) = (1000 - R_1)l \dots\dots(i)$$

On Interchanging resistance balance point shifts left by 10cm



On balancing condition

$$(1000 - R_1)(110 - l) = R_1(l - 10)$$

$$\text{or, } R_1(l - 10) = (1000 - R_1)(110 - l) \dots\dots(ii)$$

Dividing eqn (i) by (ii)

$$\frac{100 - l}{l - 10} = \frac{1}{110 - l}$$

$$\Rightarrow (100 - l)(110 - l) = 1(l - 10)$$

$$\Rightarrow 11000 - 100l - 110l + l^2 = l^2 - 10l$$

$$\Rightarrow 11000 = 200l$$

$$\text{or, } l = 55$$

Putting the value of 'l' in eqn (i)

$$R_1(100 - 55) = (1000 - R_1)55$$

$$\Rightarrow R_1(45) = (1000 - R_1)55$$

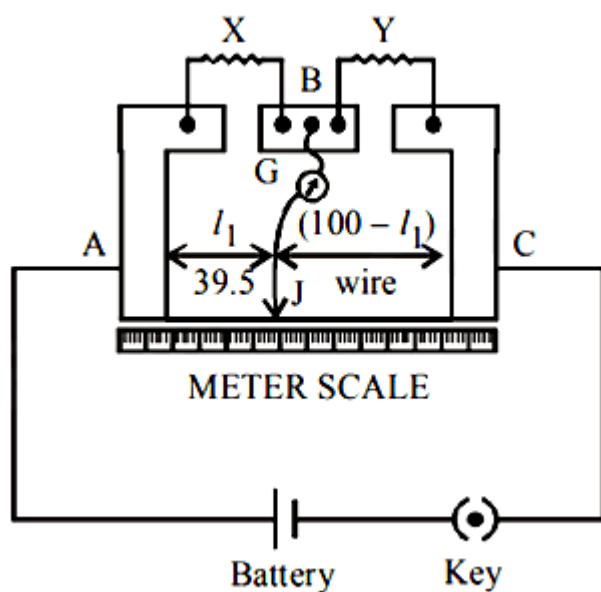
$$\Rightarrow R_1(9) = (1000 - R_1)11$$

$$\Rightarrow 20R_1 = 11000$$

$$\therefore R_1 = 550 \text{ K } \Omega$$

Question 283

In a meter bridge, as shown in the figure, it is given that resistance $Y = 12.5 \Omega$ and that the balance is obtained at a distance 39.5 cm from end A (by jockey J). After interchanging the resistances X and Y, a new balance point is found at a distance l_2 from end A. What are the values of X and l_2 ?



[Online April 15, 2018]

Options:

- A. 19.15Ω and 39.5 cm
- B. 8.16Ω and 60.5 cm
- C. 19.15Ω and 60.5 cm
- D. 8.16Ω and 39.5 cm

Answer: B

Solution:

For a balanced meter bridge,

$$\frac{X}{39.5} = \frac{Y}{(100 - 39.5)} \Rightarrow Y = 39.5 = X \times (100 - 39.5)$$

$$\text{or, } X = \frac{12.5 \times 39.5}{60.5} = 8.16\Omega$$

When X and Y are interchanged l_1 and $(100 - l_1)$ will also interchange so, $l_2 = 60.5\text{cm}$

Question284

A uniform wire of length l and radius r has a resistance of 100Ω . It is recast into a wire of radius $\frac{r}{2}$. The resistance of new wire will be:

[Online April 9, 2017]

Options:

A. $1600\ \Omega$

B. $400\ \Omega$

C. $200\ \Omega$

D. $100\ \Omega$

Answer: A

Solution:

Given, $R_1 = 100\Omega$, $r' = r/2$, $R_2 = ?$

Resistivity of wire, $R = \frac{\rho l}{A} \because \text{Area} \times \text{length} = \text{volume}$

$$\text{Hence, } R = \frac{\rho V}{A^2}$$

Since, $\rho \rightarrow \text{constant}$, $V \rightarrow \text{constant}$

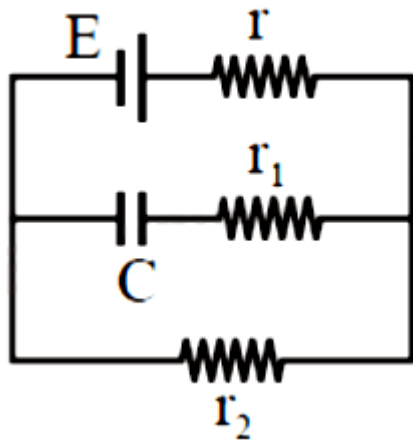
$$R \propto \frac{1}{A^2}$$

$$\text{or } R \propto \frac{1}{r^4} \because A = \pi r^2$$

$$\frac{R_2}{R_1} = 16 \Rightarrow R_2 = 16 \times 100 = 1600\Omega, \text{ Resistance of new wire}$$

Question285

In the given circuit diagram when the current reaches steady state in the circuit, the charge on the capacitor of capacitance C will be :



[2017]

Options:

A. $CE \frac{r_2}{(r+r_2)}$

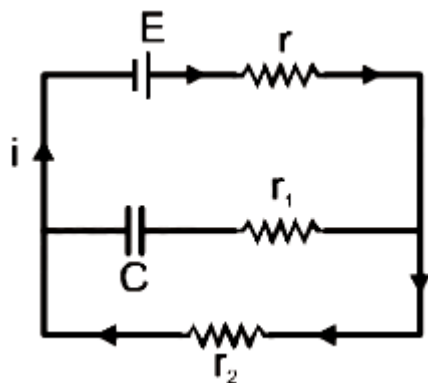
B. $CE \frac{r_1}{(r_1+r)}$

C. CE

D. $CE \frac{r_1}{(r_2+r)}$

Answer: A

Solution:



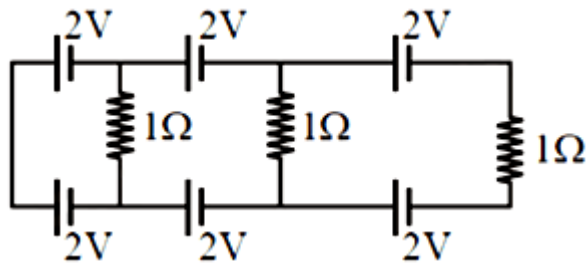
In steady state, flow of current through capacitor will be zero. Current through the circuit, $i = \frac{E}{r+r_2}$

Potential difference through capacitor $V_c = \frac{Q}{C} = E - ir = E - \left(\frac{E}{r + r_2} \right) r$

$$\therefore Q = CE \frac{r_2}{r + r_2}$$

Question286

In the above circuit the current in each resistance is



[2017]

Options:

- A. 0.5A
- B. 0 A
- C. 1 A
- D. 0.25 A

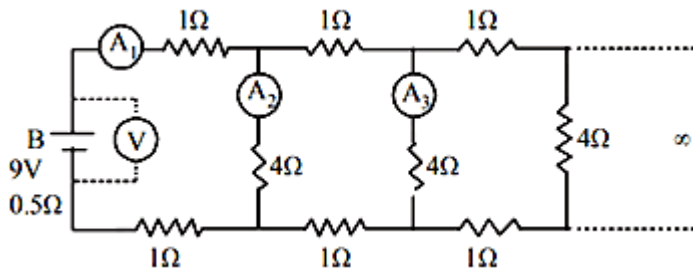
Answer: B

Solution:

The potential difference in each loop is zero.

$\therefore$ No current will flow or current in each resistance is Zero.

Question287



A 9 V battery with internal resistance of 0.5Ω is connected across an infinite network as shown in the figure. All ammeters A_1 , A_2 , A_3 and voltmeter V are ideal.

Choose correct statement.

[Online April 8, 2017]

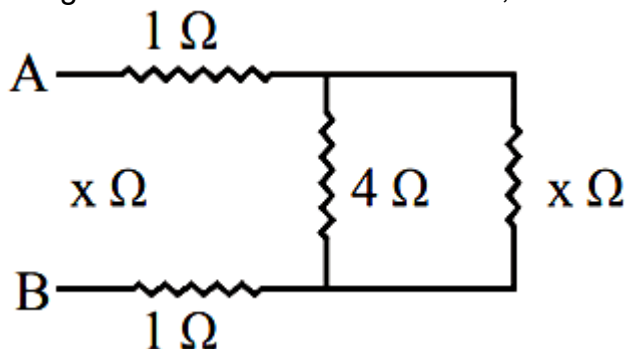
Options:

- A. Reading of A_1 is 2 A
- B. Reading of A_1 is 18 A
- C. Reading of V is 9 V
- D. Reading of V is 7 V

Answer: A

Solution:

The given circuit can be redrawn as,



as 4Ω and $x\Omega$ are parallel $x' = \frac{1}{4} + \frac{1}{x} = \frac{(4+x)}{4x}$

$$x' = \frac{4x}{4+x}$$

& 1Ω and 1Ω are also parallel $x'' = 2\Omega$

Now equivalent resistance of circuit

$$x = \frac{4x}{4+x} + 2 = \frac{8+6x}{4+x}$$

$$4x + x^2 = 8 + 6x$$

$$x^2 - 2x - 8 = 0$$

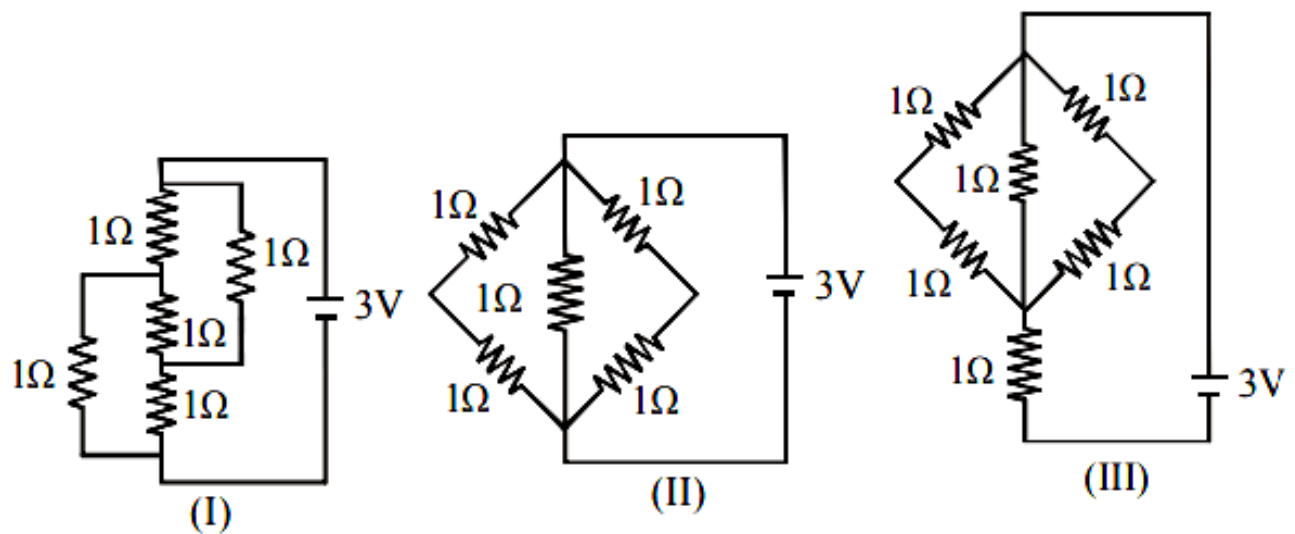
$$x = \frac{2 \pm \sqrt{4 - 4(1)(-8)}}{2} = \frac{2 \pm \sqrt{36}}{2} = \frac{2 \pm 6}{2} = 4\Omega$$

$$\text{Reading of Ammeter } A_1 = \frac{V}{(R+r)}$$

$$A_1 = \frac{9}{4+0.5} = 2 \text{ Ampere}$$

Question 288

The figure shows three circuits I, II and III which are connected to a 3V battery. If the powers dissipated by the configurations I, II and III are P_1 , P_2 and P_3 respectively, then:



[Online April 9, 2017]

Options:

A. $P_1 > P_2 > P_3$

B. $P_1 > P_3 > P_2$

C. $P_2 > P_1 > P_3$

D. $P_3 > P_2 > P_1$

Answer: C

Solution:

From the given circuit, net resistances

$$R_I = 1\Omega, R_{II} = 1/2\Omega, R_{III} = \frac{3}{2}\Omega$$

It is clear that $R_3 > R_1 > R_2$

Hence, $P_3 < P_1 < P_2$

$$\text{As Power (P)} = \frac{V^2}{R} \Rightarrow P \propto \frac{1}{R}$$

Question289

Which of the following statements is false ?
[2017]

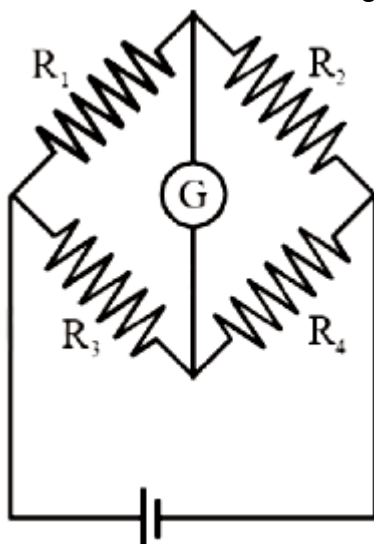
Options:

- A. A rheostat can be used as a potential divider
- B. Kirchhoff's second law represents energy conservation
- C. Wheatstone bridge is the most sensitive when all the four resistances are of the same order of magnitude
- D. In a balanced wheatstone bridge if the cell and the galvanometer are exchanged, the null point is disturbed

Answer: D

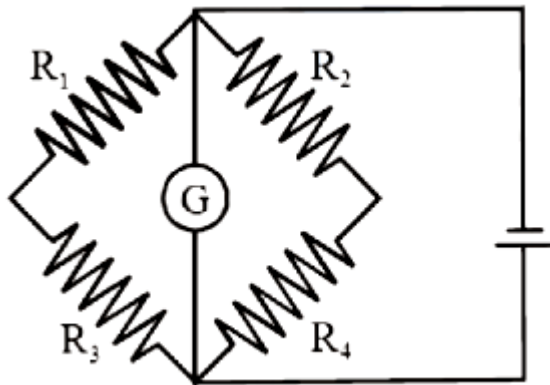
Solution:

There is no change in null point, if the cell and the galvanometer are exchanged in a balanced wheatstone bridge.



On balancing condition $\frac{R_1}{R_3} = \frac{R_2}{R_4}$

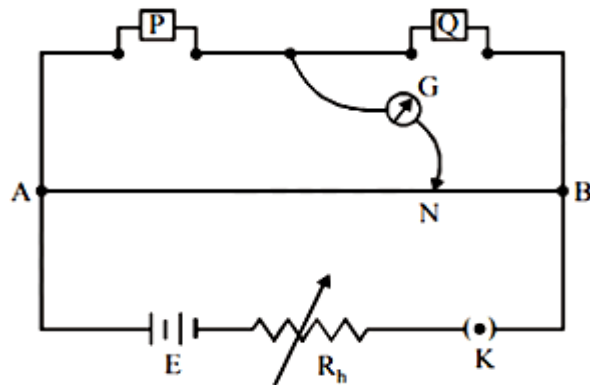
After exchange



On balancing condition $\frac{R_1}{R_2} = \frac{R_3}{R_4}$

Question290

In a meter bridge experiment resistances are connected as shown in the figure. Initially resistance $P = 4 \Omega$ and the neutral point N is at 60 cm from A. Now an unknown resistance R is connected in series to P and the new position of the neutral point is at 80 cm from A. The value of unknown resistance R is :



[Online April 9, 2017]

Options:

A. $\frac{33}{5} \Omega$

B. 6Ω

C. 7Ω

D. $\frac{20}{3}\Omega$

Answer: D

Solution:

In balance position of bridge, $\frac{P}{Q} = \frac{l}{(100-l)}$

Initially neutral position is 60cm. from A, so

$$\frac{4}{60} = \frac{Q}{40} \Rightarrow Q = \frac{16}{6} = \frac{8}{3}\Omega$$

Now, when unknown resistance R is connected in series to P, neutral point is 80cm from A then,

$$\frac{4+R}{80} = \frac{Q}{20}$$

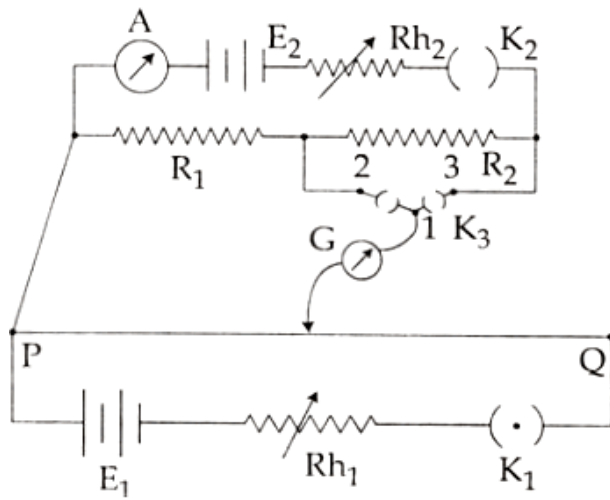
$$\frac{4+R}{80} = \frac{8}{60}$$

$$R = \frac{64}{6} - 4 = \frac{64-24}{6} = \frac{40}{6}\Omega$$

Hence, the value of unknown resistance R is $= \frac{20}{3}\Omega$

Question291

A potentiometer PQ is set up to compare two resistances as shown in the figure. The ammeter A in the circuit reads 1.0A when two way key K_3 is open. The balance point is at a length l_1 cm from P when two way key K_3 is plugged in between 2 and 1, while the balance point is at a length l_2 cm from P when key K_3 is plugged in between 3 and 1. The ratio of two resistances $\frac{R_1}{R_2}$, is found to be:



[Online April 8, 2017]

Options:

A. $\frac{l_1}{l_1 + l_2}$

B. $\frac{l_2}{l_2 - l_1}$

C. $\frac{l_1}{l_1 - l_2}$

D. $\frac{l_1}{l_2 - l_1}$

Answer: D

Solution:

When key is at point (1)

$$V_1 = iR_1 = xl_1$$

When key is at (3)

$$V_2 = i(R_1 + R_2) = xl_2$$

$$R_1 R_1 + R_2 = \frac{l_1}{l_2} \Rightarrow \frac{R_1}{R_2} = \frac{l_1}{l_2 - l_1}$$

Question292

The resistance of an electrical toaster has a temperature dependence given by $R(T) = R_0[1 + \alpha(T - T_0)]$ in its range of operation. At $T_0 = 300\text{K}$, $R = 100\Omega$ and at $T = 500\text{K}$, $R = 120\Omega$. The toaster is connected to a voltage source at 200 V and its temperature is raised at a constant rate from 300 to 500K in 30s . The total work done in raising the temperature is :
[Online April 10, 2016]

Options:

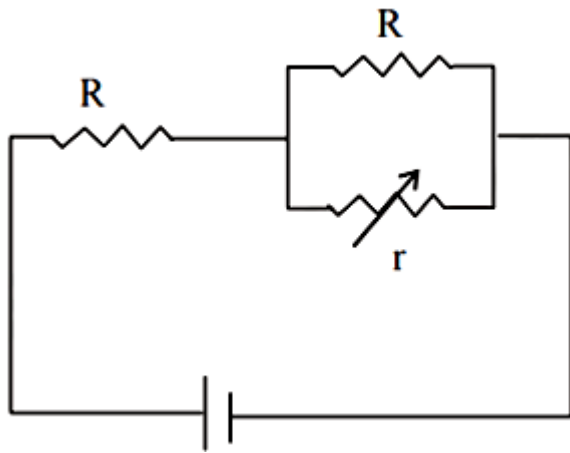
- A. $400 \ln \frac{5}{6} \text{ J}$
- B. $200 \ln \frac{2}{3} \text{ J}$
- C. 300J
- D. $400 \ln \frac{1.5}{1.3} \text{ J}$
- E. None

Answer: E

Solution:

$$\begin{aligned} \text{Work done in } 30\text{s, } W &= \int_0^{30} \frac{V^2}{R} dt \\ \text{or, } W &= \int_0^{30} \frac{(200)^2}{100 \left(1 + \alpha \frac{20t}{3} \right)} dt = \frac{(200)^2}{100} \int_0^{30} \frac{dt}{1 + \frac{20\alpha}{3}t} \\ &= \frac{400 \times 3}{20\alpha} \ln \left(\frac{1 + 20\alpha}{3} \times 301 \right) = 60,000 \ln \left(\frac{6}{5} \right) \\ \because 120 &= 100[1 + \alpha(200)] \\ \therefore \alpha &= \frac{1}{1000} \end{aligned}$$

Question293



In the circuit shown, the resistance r is a variable resistance. If for $r = f R$, the heat generation in r is maximum then the value of f is:

[Online April 9, 2016]

Options:

A. $\frac{1}{2}$

B. 1

C. $\frac{1}{4}$

D. $\frac{3}{4}$

Answer: C

Solution:

Heat energy will be maximum when resistance will be minimum

Question294

When 5V potential difference is applied across a wire of length 0.1m, the drift speed of electrons is $2.5 \times 10^{-4} \text{ ms}^{-1}$. If the electron density in the wire is $8 \times 10^{28} \text{ m}^{-3}$, the resistivity of the material is close to :

[2015]

Options:

A. $1.6 \times 10^{-6} \Omega\text{m}$

B. $1.6 \times 10^{-5} \Omega\text{m}$

C. $1.6 \times 10^{-8} \Omega\text{m}$

D. $1.6 \times 10^{-7} \Omega\text{m}$

Answer: B

Solution:

$$V = IR = (neAv_d)\rho \frac{l}{A}$$

$$\therefore \rho = \frac{V}{V_d \ln e}$$

Here V = potential difference

l = length of wire

n = no. of electrons per unit volume of conductor.

e = no. of electrons

Placing the value of above parameters we get resistivity

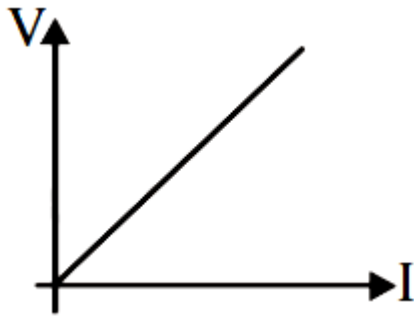
$$\rho = \frac{5}{8 \times 10^{28} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-4} \times 0.1}$$
$$= 1.6 \times 10^{-5} \Omega\text{m}$$

Question295

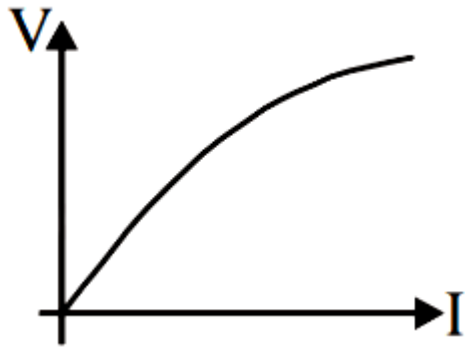
Suppose the drift velocity v_d in a material varied with the applied electric field E as $v_d \propto \sqrt{E}$. Then $V - I$ graph for a wire made of such a material is best given by:
[Online April 10, 2015]

Options:

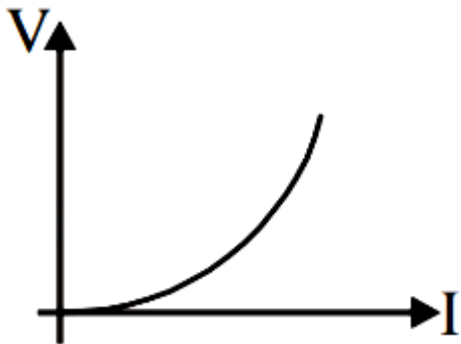
A.



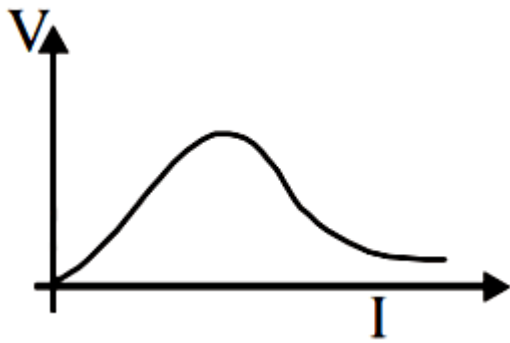
B.



C.



D.



Answer: C

Solution:

$$i = neAV_d \text{ and } V_d \propto \sqrt{E} \text{ (Given)}$$

$$\text{or, } i \propto \sqrt{E}$$

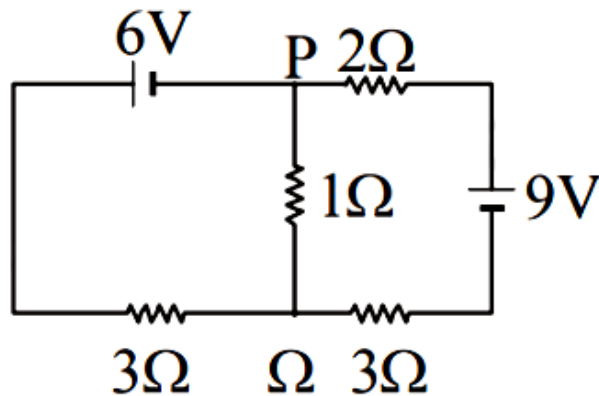
$$i^2 \propto E$$

$$i^2 \propto V$$

Hence graph (c) correctly depicts the $V - I$ graph for a wire made of such type of material.

Question 296

In the circuit shown, the current in the 1Ω resistor is:



[2015]

Options:

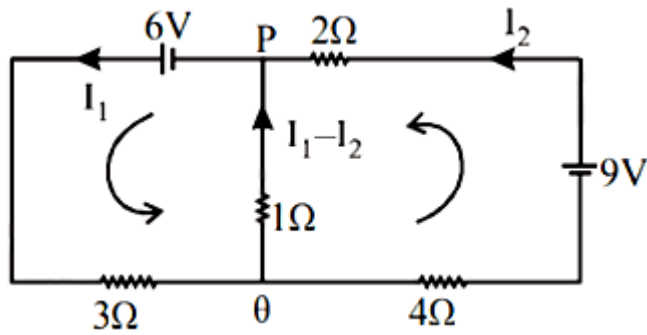
- A. 0.13 A, from Q to P
- B. 0.13 A, from P to Q
- C. 1.3A from P to Q
- D. 0A

Answer: A

Solution:

From KVL

$$-6 + 3I_1 + 1(I_1 - I_2) = 0$$



$$6 = 3I_1 + I_1 - I_2; 4I_1 - I_2 = 6 \dots\dots(1)$$

$$-9 + 2I_2 - (I_1 - I_2) + 3I_2 = 0$$

$$-I_1 + 6I_2 = 9 \dots\dots(2)$$

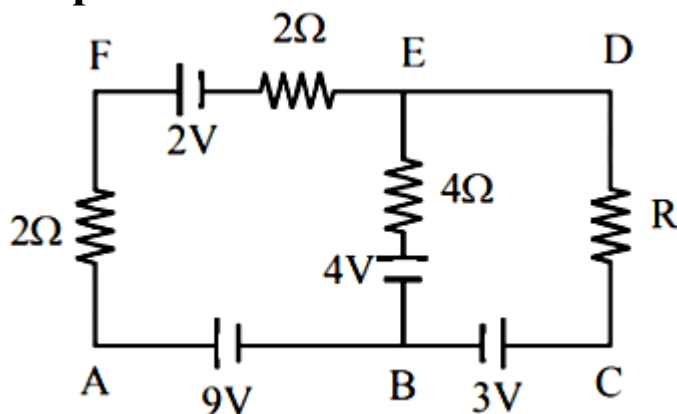
On solving(1) and (2)

$$I_1 = 0.13\text{A}$$

Direction Q to P, since $I_1 > I_2$

Question297

In the electric network shown, when no current flows through the 4Ω resistor in the arm EB, the potential difference between the points A and D will be :



[Online April 11, 2015]

Options:

A. 6 V

B. 3 V

C. 5 V

D. 4 V

Answer: C

Solution:

As no current flows through arm E B then

$$V_n = 0V$$

$$V_F^D = 0V$$

$$V_B^L = -4V$$

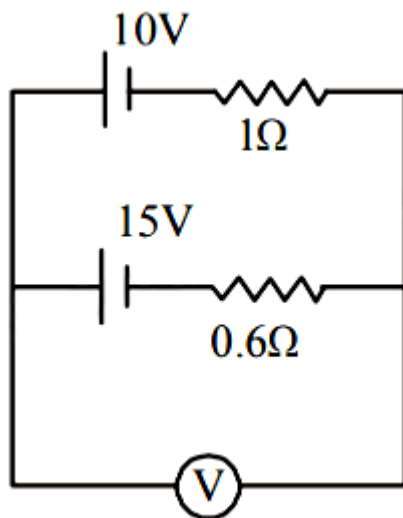
$$V_A^D = 5V$$

So, potential difference between the points A and D

$$V_A - V_D = 5V$$

Question298

A 10V battery with internal resistance 1Ω and a 15V battery with internal resistance 0.6Ω are connected in parallel to a voltmeter (see figure). The reading in the voltmeter will be close to :



[Online April 10, 2015]

Options:

A. 12.5 V

B. 24.5 V

C. 13.1 V

D. 11.9 V

Answer: C

Solution:

As the two cells oppose each other hence, the effective emf in closed circuit is $15 - 10 = 5\text{V}$ and net resistance is $1 + 0.6 = 1.6\Omega$ (because in the closed circuit the internal resistance of two cells are in series).

Current in the circuit,

$$I = \frac{\text{effective emf}}{\text{total resistance}} = \frac{5}{1.6}\text{A}$$

The potential difference across voltmeter will be same as the terminal voltage of either cell.

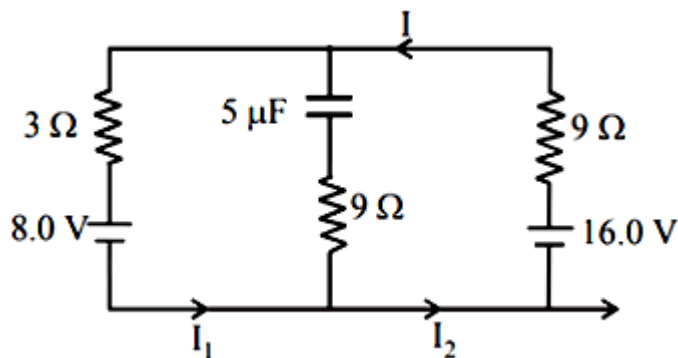
Since the current is drawn from the cell of 15V

$$\therefore V_1 = E_1 - I r_1$$

$$= 15 - \frac{5}{1.6} \times 0.6 = 13.1\text{V}$$

Question299

The circuit shown here has two batteries of 8.0 V and 16.0 V and three resistors $3\ \Omega$, $9\ \Omega$ and $9\ \Omega$ and a capacitor of $5.0\ \mu\text{F}$.



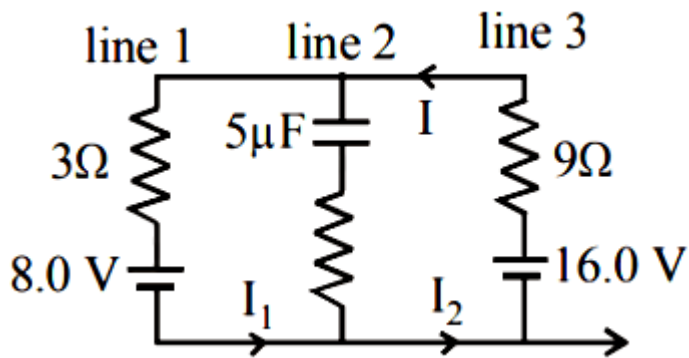
How much is the current I in the circuit in steady state?
[Online April 12, 2014]

Options:

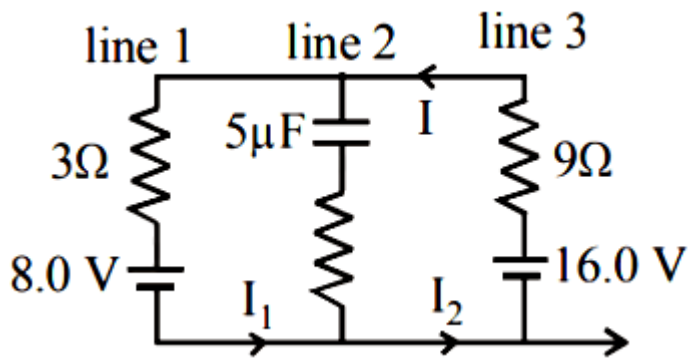
- A. 1.6 A
- B. 0.67 A
- C. 2.5 A
- D. 0.25 A

Answer: B

Solution:



In steady state capacitor is fully charged hence no current will flow through line 2 .
By simplifying the circuit



Hence resultant potential difference across resistances will be 8.0V

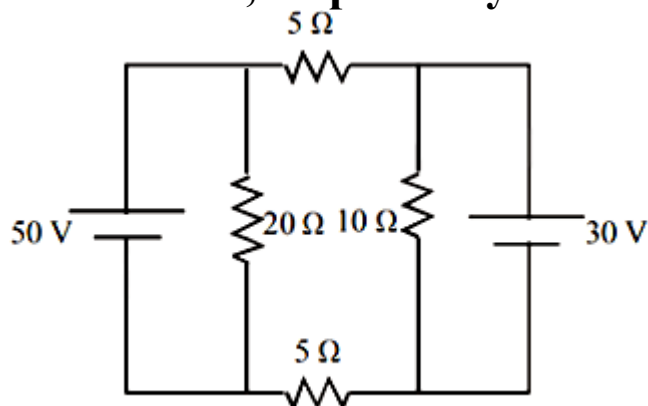
Thus current $I = \frac{V}{R}$

$$= \frac{8.0}{3+9} = \frac{8}{12}$$

$$\text{or, } I = \frac{2}{3} = 0.67\text{A}$$

Question300

In the circuit shown, current (in A) through 50 V and 30 V batteries are, respectively.



[Online April 11, 2014]

Options:

A. 2.5 and 3

B. 3.5 and 2

C. 4.5 and 1

D. 3 and 2.5

Answer: A

Solution:

Current through 50V and 30V batteries are respectively 2.5A and 3A

Question301

A d.c. main supply of e.m.f. 220 V is connected across a storage battery of e.m.f. 200 V through a resistance of 1Ω . The battery terminals are connected to an external resistance 'R'. The minimum value of 'R', so that a current passes through the battery to charge it is:

[Online April 9, 2014]

Options:

A. 7Ω

B. 9Ω

C. 11Ω

D. Zero

Answer: C

Solution:

Given, emf of cell $E = 200V$

Internal resistance of cells $= 1\Omega$

D. C. main supply voltage $V = 220V$

External resistance $R = ?$

$$r = \left(\frac{E - V}{V} \right) R$$

$$1 = \left(\frac{20}{220} \right) \times R \therefore R = 11\Omega$$

Question302

In a large building, there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 heater of 1 kW. The voltage of electric mains is 220 V. The minimum capacity of the main fuse of the building will be:
[2014]

Options:

- A. 8 A
- B. 10 A
- C. 12 A
- D. 14 A

Answer: C

Solution:

Total power consumed by electrical appliances in the building, $P_{\text{total}} = 2500\text{W}$

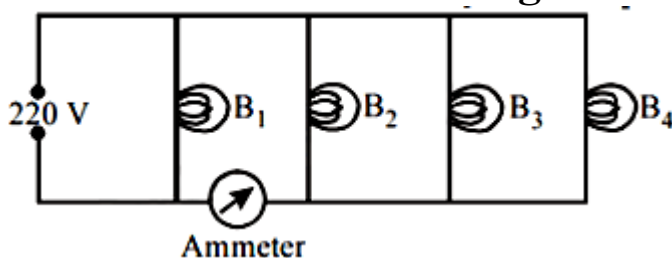
Watt = Volt $\times$ ampere

$$\Rightarrow 2500 = V \times I \Rightarrow 2500 = 220I$$

$$\Rightarrow I = \frac{2500}{220} = 11.36 \approx 12\text{A (Minimum capacity of main fuse)}$$

Question303

Four bulbs B_1 , B_2 , B_3 and B_4 of 100 W each are connected to 220 V main as shown in the figure.



The reading in an ideal ammeter will be:
[Online April 19, 2014]

Options:

A. 0.45 A

B. 0.90 A

C. 1.35 A

D. 1.80 A

Answer: C

Solution:

$$\text{Current in each bulb} = \frac{\text{Power}}{\text{Voltage}}$$

$$= \frac{100}{220} = 0.45\text{A}$$

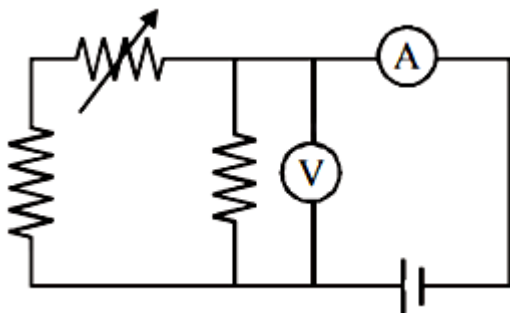
$$\text{Current through ammeter} = 0.45 \times 3 = 1.35\text{A}$$

Question304

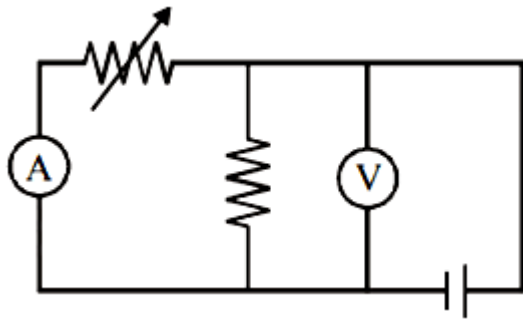
**Correct set up to verify Ohm's law is :
[Online April 23, 2013]**

Options:

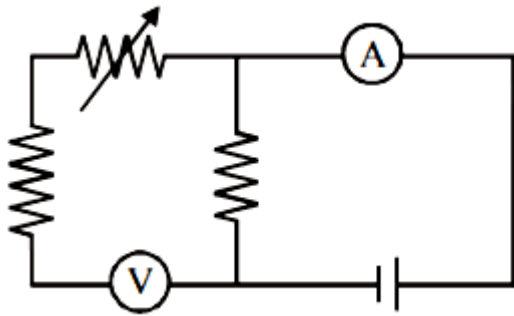
A.



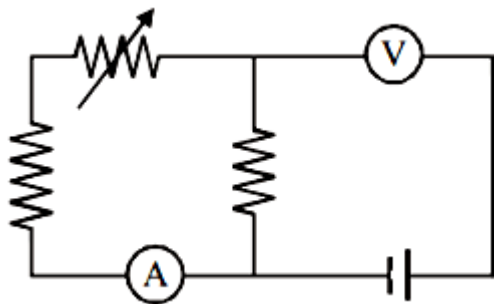
B.



C.



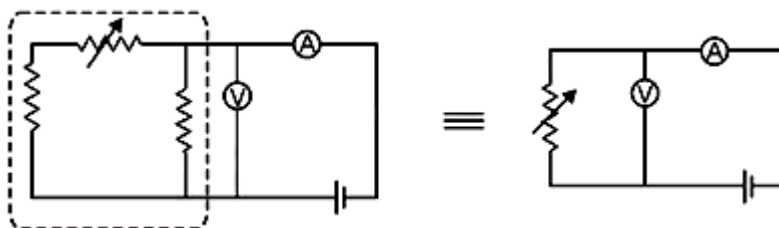
D.



Answer: A

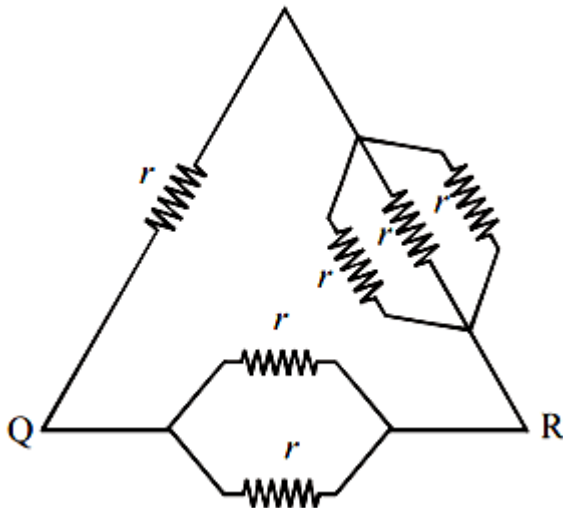
Solution:

In ohm's law, we check $V = IR$ where I is the current flowing through a resistor and V is the potential difference across that resistor. Only option (a) fits the above criteria. Remember that ammeter is connected in series with resistance and voltmeter parallel with the resistance.



Question305

Six equal resistances are connected between points P, Q and R as shown in figure. Then net resistance will be maximum between :



[Online April 25, 2013]

Options:

- A. P and R
- B. P and Q
- C. Q and R
- D. Any two points

Answer: B

Solution:

Resistance between P and Q

$$r_{PQ} = r \parallel \left(\frac{r}{3} + \frac{r}{2} \right) = \frac{r \times \frac{5}{6}r}{r + \frac{5}{6}r} = \frac{5}{11}r$$

Resistance between Q and R

$$r_{QR} = \frac{r}{2} \parallel \left(r + \frac{r}{3} \right) = \frac{\frac{r}{2} \times \frac{4}{3}r}{\frac{r}{2} + \frac{4}{3}r} = \frac{4}{11}r$$

Resistance between P and R

$$r_{PR} = \frac{r}{3} \parallel \left(\frac{r}{2} + r \right) = \frac{\frac{r}{3} \times \frac{3}{2}r}{\frac{r}{3} + \frac{3}{2}r} = \frac{3}{11}r$$

Hence, it is clear that r_{PQ} is maximum

Question306

A letter 'A' is constructed of a uniform wire with resistance 1.0Ω per cm, The sides of the letter are 20 cm and the cross piece in the middle is 10 cm long. The apex angle is 60° . The resistance between the ends of the legs is close to:

[Online April 9, 2013]

Options:

A. 50.0Ω

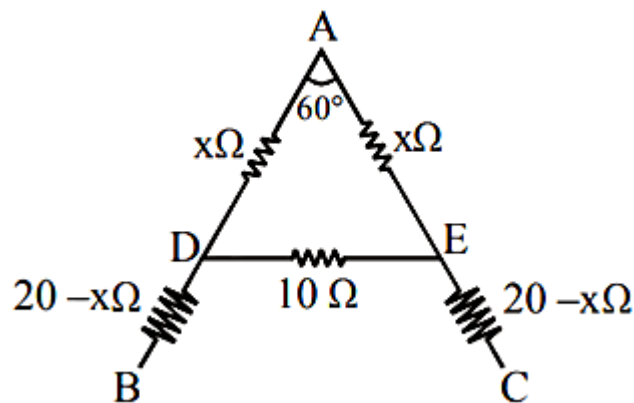
B. 10Ω

C. 36.7Ω

D. 26.7Ω

Answer: D

Solution:



$$\text{For ADE } \frac{1}{R'} = \frac{1}{2x} + \frac{1}{10}$$

$$\text{or } R' = \frac{20x}{10 + 2x}$$

$$R_{BC} = \frac{20x}{10 + 2x} + 20 - x + 20 - x \dots (i)$$

$$\text{or } \frac{20x}{10 + 2x} + 40 = 2x$$

Solving we get

$$x = 10 \Omega$$

Putting the value of $x = 10 \Omega$ in equation (i)

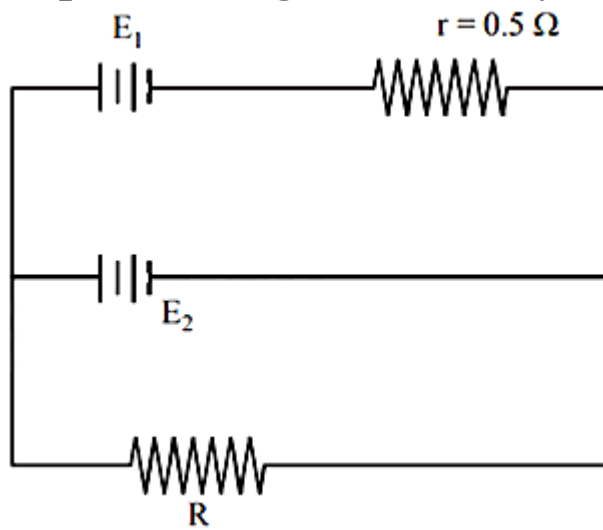
We get

$$R_{BC} = \frac{20 \times 10}{10 + 2 \times 10} + 20 - 10 + 20 - 10$$

$$= \frac{80}{3} = 26.7 \Omega$$

Question307

A dc source of emf $E_1 = 100V$ and internal resistance $r = 0.5 \Omega$, a storage battery of emf $E_2 = 90V$ and an external resistance R are connected as shown in figure. For what value of R no current will pass through the battery?



[Online April 22, 2013]

Options:

- A. 5.5Ω
- B. 3.5Ω
- C. 4.5Ω
- D. 2.5Ω

Answer: C

Solution:

$$\frac{100}{R+r} = \frac{90}{R} \Rightarrow \frac{R+r}{R} = \frac{10}{9} \Rightarrow 1 + \frac{0.5}{R} = \frac{10}{9}$$

$$\Rightarrow \frac{0.5}{R} = \frac{1}{9} \therefore R = 4.5 \Omega$$

Question308

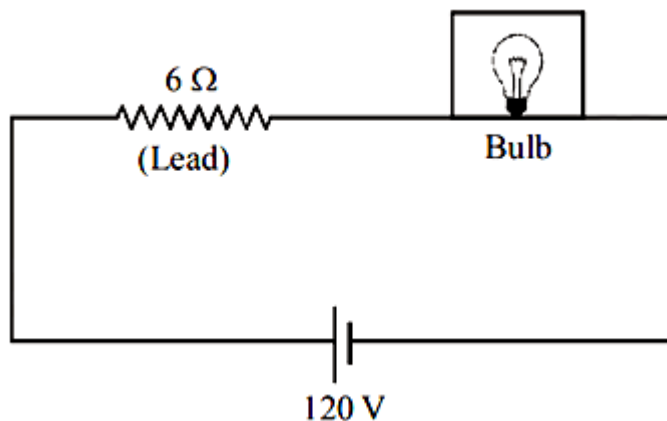
The supply voltage to room is 120V. The resistance of the lead wires is 6W. A 60Ω bulb is already switched on. What is the decrease of voltage across the bulb, when a 240 W heater is switched on in parallel to the bulb?
[2013]

Options:

- A. zero
- B. 2.9 Volt
- C. 13.3 Volt
- D. 10.04 Volt

Answer: D

Solution:



Power of bulb = 60W (given)

$$\text{Resistance of bulb} = \frac{120 \times 120}{60} = 240\Omega \left[\because P = \frac{V^2}{R} \right]$$

Power of heater = 240W (given)

$$\text{Resistance of heater} = \frac{120 \times 120}{240} = 60\Omega$$

Voltage across bulb before heater is switched on,

$$V_1 = \frac{240}{246} \times 120 = 117.73 \text{ volt}$$

Voltage across bulb after heater is switched on,

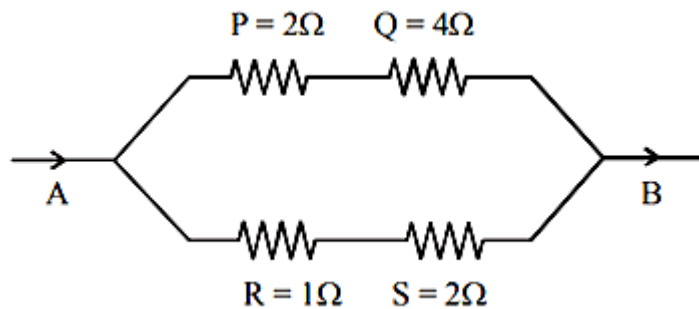
$$V_2 = \frac{48}{54} \times 120 = 106.66 \text{ volt}$$

Hence decrease in voltage

$$V_1 - V_2 = 117.073 - 106.66 = 10.04 \text{ Volt (approximately)}$$

Question309

Which of the four resistances P, Q, R and S generate the greatest amount of heat when a current flows from A to B?



[Online April 23, 2013]

Options:

A. Q

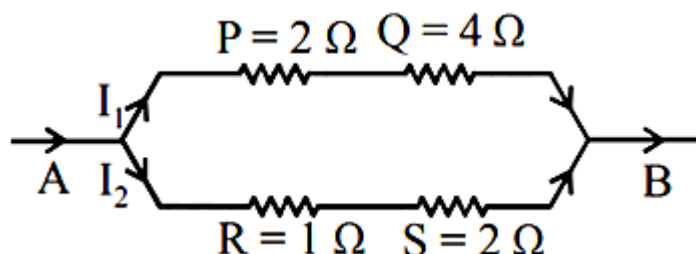
B. S

C. P

D. R

Answer: B

Solution:



$$R_1 = P + Q = 2\Omega + 4\Omega = 6\Omega$$

$$R_2 = R + S = 1\Omega + 2\Omega = 3\Omega$$

$$I_1 R_1 = I_2 R_2$$

$$I_1 = \frac{R_2}{R_1} I_2 = \frac{3}{6} I_2 = \frac{I_2}{2}$$

$$\text{or } I_2 = 2I_1$$

$$\text{Heat flow } H = I^2 R t$$

$$\text{For Q, } H_Q = I_1^2 Q t = \frac{I_2^2}{4} \times 4t = I_2^2 t$$

$$\text{For S, } H_S = I_2^2 S t = I_2^2 \cdot 2t = 2I_2^2 t$$

∴ Greatest amount of heat generated by S.

Question310

In a metre bridge experiment null point is obtained at 40 cm from one end of the wire when resistance X is balanced against another resistance Y. If $X < Y$, then the new position of the null point from the same end, if one decides to balance a resistance of $3X$ against Y , will be close to :

[Online April 9, 2013]

Options:

A. 80 cm

B. 75 cm

C. 67 cm

D. 50 cm

Answer: C

Solution:

$$\text{From question, } \frac{x}{y} = \frac{40}{100 - 40} = \frac{2}{3}$$

$$\Rightarrow x = \frac{2}{3}y$$

$$\text{Again, } \frac{3x}{y} = \frac{Z}{100 - Z}$$

$$\text{or } \frac{3 \times \frac{2y}{3}}{y} = \frac{Z}{100 - Z}$$

Solving we get $Z = 67\text{cm}$

Therefore new position of null point $\cong 67\text{cm}$

Question311

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement 1: The possibility of an electric bulb fusing is higher at the time of switching ON.

Statement 2: Resistance of an electric bulb when it is not lit up is much smaller than when it is lit up.

[Online May 7, 2012]

Options:

A. Statement 1 is true, Statement 2 is false

B. Statement 1 is false, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

C. Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation of Statement 1.

D. Statement 1 is false, Statement 2 is true

Answer: C

Question312

The resistance of a wire is R . It is bent at the middle by 180° and both the ends are twisted together to make a shorter wire. The resistance of the new wire is

[Online May 26, 2012]

Options:

A. $2R$

B. $R/2$

C. $R/4$

D. $R/8$

Answer: C

Solution:

Resistance of wire (R) = $\rho \frac{l}{A}$

If wire is bent in the middle then

$$l' = \frac{l}{2}, A' = 2A$$

∴ New resistance, $R' = \rho \frac{l'}{A'}$

$$= \frac{\rho \frac{l}{2}}{2A} = \frac{\rho l}{4A} = \frac{R}{4}.$$

Question 313

Two electric bulbs rated 25W – 220 V and 100W – 220V are connected in series to a 440 V supply. Which of the bulbs will fuse?

[2012]

Options:

- A. Both
- B. 100 W
- C. 25 W
- D. Neither

Answer: C

Solution:

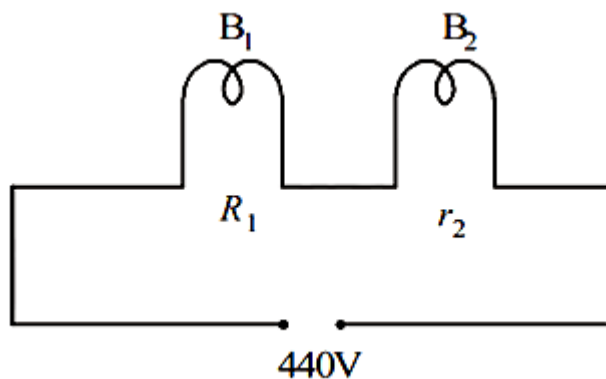
Current capacity of 25W bulb

$$I_1 = \frac{W_1}{V_1} = \frac{25}{220} \text{ Amp}$$

Current capacity of 100W bulb

$$I_2 = \frac{W_2}{V_2} = \frac{100}{220} \text{ Amp}$$

The current flowing through the circuit



Resistance of 25W bulb,

$$R_1 = \frac{V_1^2}{P_1} = \frac{(220)^2}{25}$$

Resistance of 100W bulb

$$R_2 = \frac{V_2^2}{P} = \frac{(220)^2}{100}$$

$$R_{\text{eff}} = R_1 + R_2$$

Current flowing through circuit

$$I = \frac{440}{R_{\text{eff}}}$$

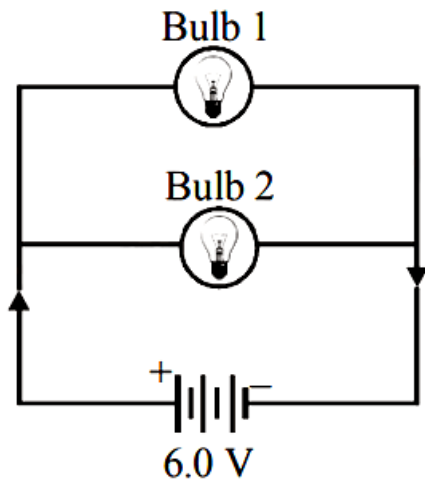
$$I = \frac{440}{\frac{(220)^2}{25} + \frac{(220)^2}{100}}$$

$$= \frac{440}{(220)^2 \left[\frac{1}{25} + \frac{1}{100} \right]}; I = \frac{40}{220} \text{ Amp}$$

$$\therefore I_1 \left(= \frac{25}{220} \text{ A} \right) < I \left(= \frac{40}{220} \text{ A} \right) \text{ \$ Thus the bulb rated 25W – 220 will fuse}$$

Question314

A 6.0 volt battery is connected to two light bulbs as shown in figure. Light bulb 1 has resistance 3 ohm while light bulb 2 has resistance 6 ohm. Battery has negligible internal resistance. Which bulb will glow brighter?



[Online May 19, 2012]

Options:

- A. Bulb 1 will glow more first and then its brightness will become less than bulb 2
- B. Bulb 1
- C. Bulb 2
- D. Both glow equally

Answer: B

Solution:

$$\text{Total resistance} = \frac{6 \times 3}{6 + 3} = 2\Omega$$

$$\text{Current in circuit} = \frac{6}{2} = 3\text{A}$$

Therefore current through bulb 1 is 2A and bulb 2 is 1A.
So bulb 1 will glow more

Question315

Three resistors of 4 Ω , 6 Ω and 12 Ω are connected in parallel and the combination is connected in series with a 1.5 V battery of 1 Ω internal resistance. The rate of Joule heating in the 4 Ω resistor is

[Online May 12, 2012]

Options:

A. 0.55 W

B. 0.33 W

C. 0.25 W

D. 0.86 W

Answer: C

Solution:

Resistors 4Ω , 6Ω and 12Ω are connected in parallel, its equivalent resistance (R) is given by

$$\frac{1}{R} = \frac{1}{4} + \frac{1}{6} + \frac{1}{12} \Rightarrow R = \frac{12}{6} = 2\Omega$$

Again R is connected to $1.5V$ battery whose internal resistance $r = 1\Omega$

Equivalent resistance now,

$$R' = 2\Omega + 1\Omega = 3\Omega$$

$$\text{Current, } I_{\text{total}} = \frac{V}{R'} = \frac{1.5}{3} = \frac{1}{2}A$$

$$I_{\text{total}} = \frac{1}{2} = 3x + 2x + x = 6x$$

$$\Rightarrow x = \frac{1}{12}$$

$$\therefore \text{Current through } 4\Omega \text{ resistor} = 3x$$

$$= 3 \times \frac{1}{12} = \frac{1}{4}A$$

Therefore, rate of Joule heating in the 4Ω resistor

$$= I^2 R = \left(\frac{1}{4}\right)^2 \times 4 = \frac{1}{4} = 0.25W$$

Question316

In an experiment of potentiometer for measuring the internal resistance of primary cell a balancing length l is obtained on the potentiometer wire when the cell is open circuit. Now the cell is short circuited by a resistance R . If R is to be equal to the internal resistance of the cell the balancing length on the potentiometer wire will be

[Online May 26, 2012]

Options:

- A. 1
- B. 2l
- C. $l/2$
- D. $l/4$

Answer: C

Solution:

Balancing length l will give emf of cell

$$\therefore E = K l$$

Here K is potential gradient.

If the cell is short circuited by resistance ' R '

Let balancing length obtained be l' then

$$V = k l'$$

$$r = \left(\frac{E - V}{V} \right) R$$

$$\Rightarrow V = E - V [\because r = R \text{ given}]$$

$$\Rightarrow 2V = E$$

$$\text{or, } 2K l' = K l$$

$$\therefore l' = \frac{l}{2}$$

Question317

It is preferable to measure the e.m.f. of a cell by potentiometer than by a voltmeter because of the following possible reasons.

- (i) In case of potentiometer, no current flows through the cell.**
- (ii) The length of the potentiometer allows greater precision.**
- (iii) Measurement by the potentiometer is quicker.**
- (iv) The sensitivity of the galvanometer, when using a potentiometer is not relevant.**

Which of these reasons are correct?

[Online May 12, 2012]

Options:

- A. (i), (iii), (iv)
- B. (i), (iii), (iv)

C. (i), (ii)

D. (i), (ii), (iii), (iv)

Answer: C

Solution:

To measure the emf of a cell we prefer potentiometer rather than voltmeter because
(i) the length of potentiometer which allows greater precision.
(ii) in case of potentiometer, no current flows through the cell.
(iii) of high sensitivity.

Question318

In a sensitive meter bridge apparatus the bridge wire should possess

[Online May 12, 2012]

Options:

A. high resistivity and low temperature coefficient.

B. low resistivity and high temperature coefficient.

C. low resistivity and low temperature coefficient.

D. high resistivity and high temperature coefficient.

Answer: A

Solution:

Bridge wire in a sensitive meter bridge wire should be of high resistivity and low temperature coefficient.

Question319

**If a wire is stretched to make it 0.1% longer, its resistance will :
[2011]**

Options:

- A. increase by 0.2%
- B. decrease by 0.2%
- C. decrease by 0.05%
- D. increase by 0.05%

Answer: A

Solution:

Resistance of wire $R = \frac{\rho l}{A} = \frac{\rho l^2}{V}$ ($\because V = Al$)

Hence, $R = \rho \frac{l^2}{V} = \text{constant} \times l^2$

$\therefore$ Fractional change in resistance

$$\frac{\Delta R}{R} = 2 \frac{\Delta l}{l}$$

$$100 \times \frac{\Delta R}{R} = 200 \times \left(\frac{\Delta l}{l} \right)$$

$$\therefore \Delta l/l = 0.1\%$$

$$\therefore \% \text{ change in } R = \left[200 \times \left\{ \frac{0.1}{100} \right\} \right] = 0.2\%$$

$\therefore$ Resistance will increase by 0.2%.

Question320

The current in the primary circuit of a potentiometer is 0.2 A. The specific resistance and cross-section of the potentiometer wire are 4×10^{-7} ohm metre and $8 \times 10^{-7} \text{ m}^2$, respectively. The potential gradient will be equal to [2011 RS]

Options:

- A. 1 V /m
- B. 0.5V/m
- C. 0.1 V/m

D. 0.2 V/m

Answer: C

Solution:

Potential gradient

$$\Rightarrow k = \frac{V}{l} = \frac{IR}{l} = \frac{I}{l} \left(\frac{\rho l}{A} \right) = \frac{I\rho}{A}$$

$$k = \frac{0.2 \times 4 \times 10^{-7}}{8 \times 10^{-7}} = \frac{0.8}{8} = 0.1 \text{ V/m}$$

Question 321

Two conductors have the same resistance at 0°C but their temperature coefficients of resistance are α_1 and α_2 . The respective temperature coefficients of their series and parallel combinations are nearly
[2010]

Options:

A. $\frac{\alpha_1 + \alpha_2}{2}, \alpha_1 + \alpha_2$

B. $\alpha_1 + \alpha_2, \frac{\alpha_1 + \alpha_2}{2}$

C. $\alpha_1 + \alpha_2, \frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$

D. $\frac{\alpha_1 + \alpha_2}{2}, \frac{\alpha_1 + \alpha_2}{2}$

Answer: D

Solution:

Let R_1 and R_2 be the resistances of two conductors, then

$$R_1 = R_0[1 + \alpha_1 \Delta t]$$

$$R_2 = R_0[1 + \alpha_2 \Delta t]$$

Here, R_0 is the resistance of conductor at 0°C

$$\begin{aligned}
 \text{In Series, } R &= R_1 + R_2 \\
 &= R_0[2 + (\alpha_1 + \alpha_2)\Delta t] \\
 &= 2R_0 \left[1 + \left(\frac{\alpha_1 + \alpha_2}{2} \right) \Delta t \right] \\
 \therefore \alpha_{\text{eq}} &= \frac{\alpha_1 + \alpha_2}{2} \\
 \text{In Parallel, } \frac{1}{R} &= \frac{1}{R_1} + \frac{1}{R_2} \\
 &= \frac{1}{R_0[1 + \alpha_1\Delta t]} + \frac{1}{R_0[1 + \alpha_2\Delta t]} \\
 &\Rightarrow \frac{1}{\frac{R_0}{2}(1 + \alpha_{\text{eq}}\Delta t)} \\
 &= \frac{1}{R_0(1 + \alpha_1\Delta t)} + \frac{1}{R_0(1 + \alpha_2\Delta t)} \\
 2(1 - \alpha_{\text{eq}}\Delta t) &= (1 - \alpha_1\Delta t)(1 - \alpha_2\Delta t) \\
 \therefore \alpha_{\text{eq}} &= \frac{\alpha_1 + \alpha_2}{2}
 \end{aligned}$$

Question 322

ΔV measured between B and C is
[2008]

Options:

- A. $\frac{\rho I}{\pi a} - \frac{\rho I}{\pi(a+b)}$
- B. $\frac{\rho I}{a} - \frac{\rho I}{(a+b)}$
- C. $\frac{\rho I}{2\pi a} - \frac{\rho I}{2\pi(a+b)}$
- D. $\frac{\rho I}{2\pi(a-b)}$

Answer: A

Solution:

Let j be the current density.

$$\text{Then } j \times 2\pi r^2 = I \Rightarrow j = \frac{I}{2\pi r^2}$$

$$\therefore E = \rho j = \frac{\rho I}{2\pi r^2}$$

Now, $V_B - V_C$

$$= - \int_{a+b}^a \vec{E} \cdot d\vec{r} = - \int_{a+b}^a \frac{\rho I}{2\pi r^2} dr$$

$$= - \frac{\rho I}{2\pi} \left[-\frac{1}{r} \right]_{a+b}^a = \frac{\rho I}{2\pi a} - \frac{\rho I}{2\pi(a+b)}$$

On applying superposition as mentioned we get

$$\Delta V_{BC} = 2 \times \Delta V_{BC}' = \frac{\rho I}{\pi a} - \frac{\rho I}{\pi(a+b)}$$

Question323

For current entering at A, the electric field at a distance 'r' from A is

[2008]

Options:

A. $\frac{\rho I}{8\pi r^2}$

B. $\frac{\rho I}{r^2}$

C. $\frac{\rho I}{2\pi r^2}$

D. $\frac{\rho I}{4\pi r^2}$

Answer: C

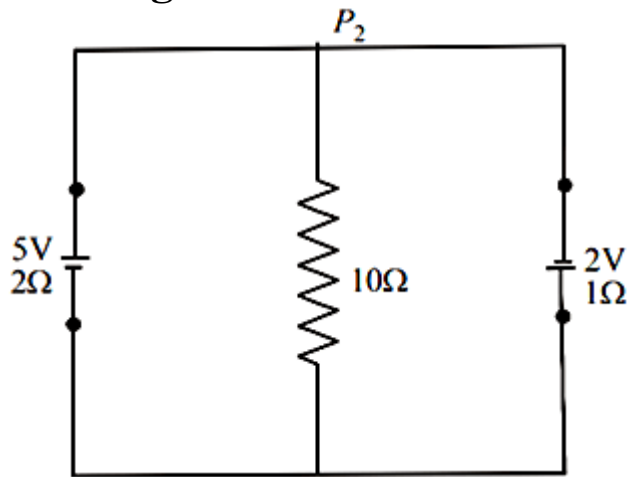
Solution:

As shown in Answer (a) $E = \frac{\rho I}{2\pi r^2}$

Question324

A 5V battery with internal resistance 2Ω and a 2V battery with internal resistance 1Ω are connected to a 10Ω resistor as shown

in the figure.



The current in the 10Ω resistor is
[2008]

Options:

- A. $0.27A$ P_2 to P_1
- B. $0.03A$ P_1 to P_2
- C. $0.03A$ P_2 to P_1
- D. $0.27A$ P_1 to P_2

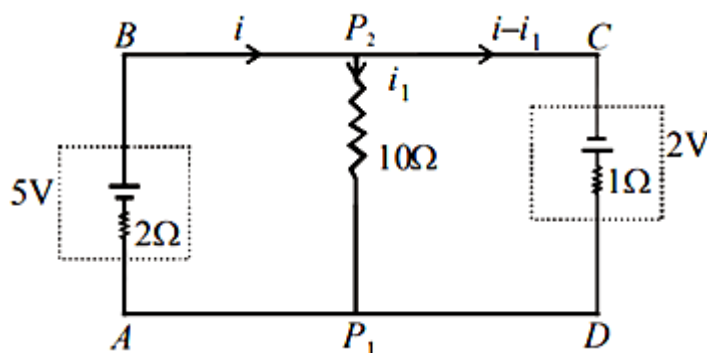
Answer: C

Solution:

Applying Kirchoff's second law in ABP_2P_1A , we get

$$-2i + 5 - 10i_1 = 0$$

$$2i + 10i_1 = 5 \dots\dots(i)$$



Again applying Kirchoff's second law in $P_2CDP_1P_2$ we get,

$$10i_1 + 2 - i + i_1 = 0$$

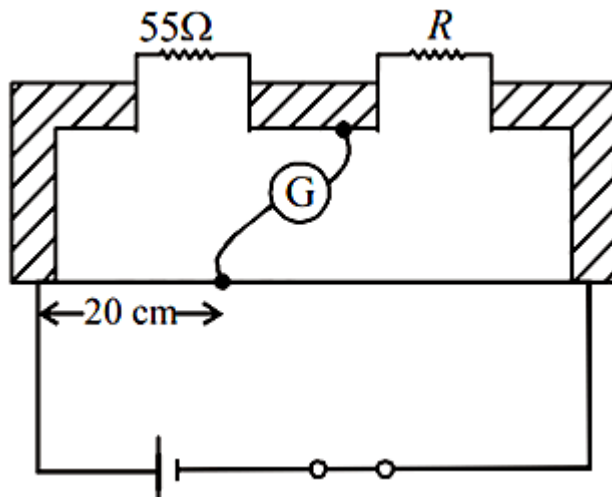
$$2i - 22i_1 = 4 \dots\dots(ii)$$

From (i) and (ii) $32i_1 = 1$

$\Rightarrow i_1 = \frac{1}{32}$ A from P_2 to P_1

Question 325

Shown in the figure below is a meter-bridge set up with null deflection in the galvanometer.



The value of the unknown resistor R is
[2008]

Options:

- A. 13.75Ω
- B. 220Ω
- C. 110Ω
- D. 55Ω

Answer: B

Solution:

Given,

Balance point from one end, $l_1 = 20\text{cm}$

From the condition for balance of metre bridge, we have

$$\frac{55}{R} = \frac{l_1}{100 - l_1}$$

$$\frac{55}{R} = \frac{20}{80}$$

$$\Rightarrow R = 220\Omega$$

Question326

The resistance of a wire is 5 ohm at 50°C and 6 ohm at 100°C. The resistance of the wire at 0°C will be [2007]

Options:

- A. 3 ohm
- B. 2 ohm
- C. 1 ohm
- D. 4 ohm

Answer: D

Solution:

Resistance of a metal conductor at temperature $t^\circ\text{C}$ is given by

$$R_t = R_0(1 + \alpha t)$$

R_0 is the resistance of the wire at 0°C

and α is the temperature coefficient of resistance.

Resistance at 50°C , $R_{50} = R_0(1 + 50\alpha)$ (i)

Resistance at 100°C , $R_{100} = R_0(1 + 100\alpha)$ (ii)

From (i), $R_{50} - R_0 = 50\alpha R_0$ (iii)

From (ii), $R_{100} - R_0 = 100\alpha R_0$ (iv)

Dividing (iii) by (iv), we get

$$\frac{R_{50} - R_0}{R_{100} - R_0} = \frac{1}{2}$$

Here, $R_{50} = 5\Omega$ and $R_{100} = 6\Omega$

$$\therefore \frac{5 - R_0}{6 - R_0} = \frac{1}{2}$$

or, $6 - R_0 = 10 - 2R_0$ or, $R_0 = 4\Omega$

Question327

A battery is used to charge a parallel plate capacitor till the potential difference between the plates becomes equal to the

electromotive force of the battery. The ratio of the energy stored in the capacitor and the work done by the battery will be [2007]

Options:

A. $1/2$

B. 1

C. 2

D. $1/4$

Answer: A

Solution:

Energy in capacitor $= \frac{1}{2}CV^2$

Work done by battery $= QV = CV^2$

where C = Capacitance of capacitor

V = Potential difference,

e = emf of battery

Required ratio $= \frac{\frac{1}{2}CV^2}{CV^2} = \frac{1}{2} (\because V = e)$

Question328

A material ' B ' has twice the specific resistance of 'A'. A circular wire made of ' B ' has twice the diameter of a wire made of 'A'. then for the two wires to have the same resistance, the ratio l_B/l_A of their respective lengths must be [2006]

Options:

A. 1

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. 2

Answer: D

Solution:

Let d_A and d_B are the diameter of wire A and B respectively. Let ρ_B and ρ_A be the resistivity of wire A and B. We have given

$$\rho_B = 2\rho_A$$

$$d_B = 2d_A$$

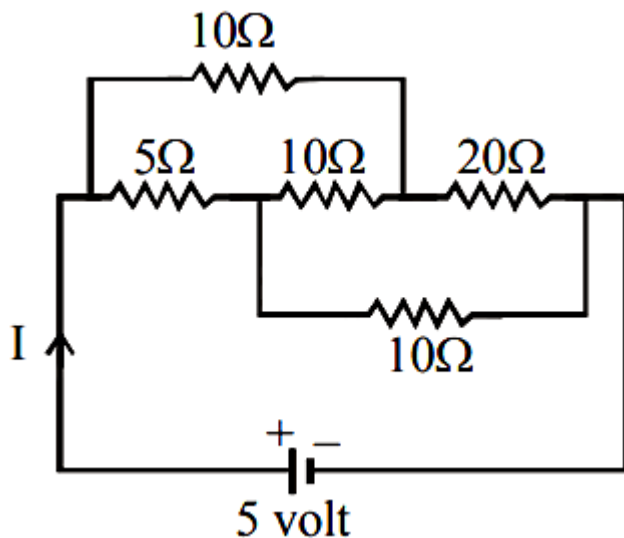
If both resistances are equal $R_B = R_A$

$$\Rightarrow \frac{\rho_B l_B}{A_B} = \frac{\rho_A l_A}{A_A}$$

$$\therefore \frac{l_B}{l_A} = \frac{\rho_A}{\rho_B} \times \frac{d_B^2}{d_A^2} = \frac{\rho_A}{2\rho_A} \times \frac{4d_A^2}{d_A^2} = 2$$

Question329

The current I drawn from the 5 volt source will be



[2006]

Options:

A. 0.33 A

B. 0.5 A

C. 0.67 A

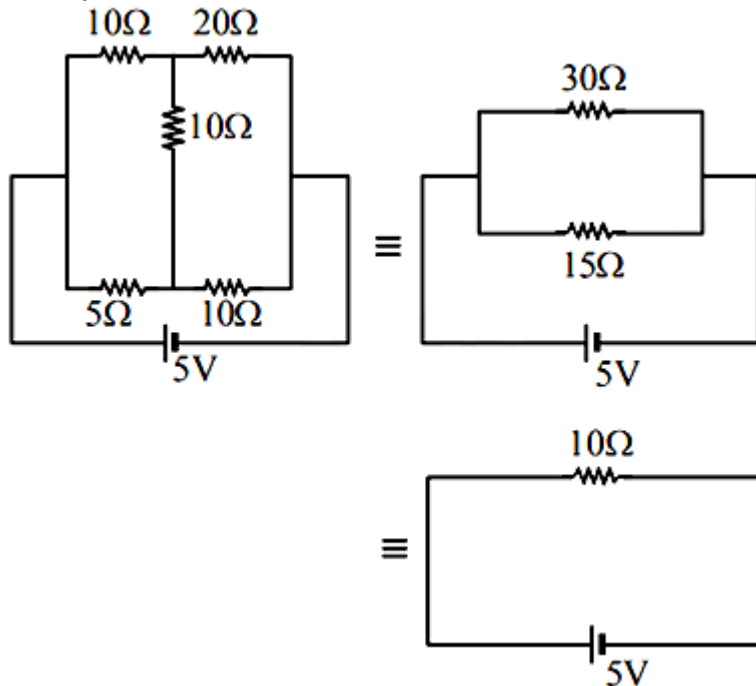
D. 0.17 A

Answer: B

Solution:

The network of resistors is a balanced wheatstone bridge. Hence, no current will flow through centre resistor.

The equivalent circuit is



$$R_{eq} = \frac{15 \times 30}{15 + 30} = 10\Omega \Rightarrow I = \frac{V}{R} = \frac{5}{10} = 0.5A$$

Question330

The Kirchhoff's first law ($\sum i = 0$) and second law ($\sum iR = \sum E$), where the symbols have their usual meanings, are respectively based on
[2006]

Options:

- A. conservation of charge, conservation of momentum
- B. conservation of energy, conservation of charge
- C. conservation of momentum, conservation of charge
- D. conservation of charge, conservatrion of energy

Answer: D

Solution:

Note : Kirchhoff's first law is based on conservation of charge and Kirchhoff's second law is based on conservation of energy.

Question331

A thermocouple is made from two metals, Antimony and Bismuth. If one junction of the couple is kept hot and the other is kept cold, then, an electric current will [2006]

Options:

- A. flow from Antimony to Bismuth at the hot junction
- B. flow from Bismuth to Antimony at the cold junction
- C. now flow through the thermocouple
- D. flow from Antimony to Bismuth at the cold junction

Answer: D

Solution:

At cold junction, current flows from Antimony to Bismuth because current flows from metal occurring later in the series to metal occurring earlier in the thermoelectric series. In thermoelectric series, Bismuth comes earlier than Antimony so at cold junction, current. Flow from Antimony to Bismuth.

Question332

The resistance of a bulb filament is 100Ω at a temperature of 100°C . If its temperature coefficient of resistance be 0.005 per $^\circ\text{C}$, its resistance will become 200Ω at a temperature of [2006]

Options:

- A. 300°C
- B. 400°C
- C. 500°C
- D. 200°C

Answer: B

Solution:

Let resistance of bulb filament be R_0 at 0°C using $R = R_0(1 + \alpha\Delta t)$ we have

$$R_1 = R_0[1 + \alpha \times 100] = 100 \dots (1)$$

$$R_2 = R_0[1 + \alpha \times T] = 200 \dots (2)$$

On dividing we get

$$\frac{200}{100} = \frac{1 + \alpha T}{1 + 100\alpha} \Rightarrow 2 = \frac{1 + 0.005T}{1 + 100 \times 0.005}$$

$$\Rightarrow T = 400^\circ\text{C}$$

Note : We may use this expression as an approximation because the difference in the answers is appreciable. For accurate results one should use $R = R_0 e^{\alpha\Delta T}$

Question333

An electric bulb is rated 220 volt - 100 watt. The power consumed by it when operated on 110 volt will be [2006]

Options:

- A. 75 watt
- B. 40 watt
- C. 25 watt
- D. 50 watt

Answer: C

Solution:

The resistance of the electric bulb is

$$R = \frac{V^2}{P} = \frac{(220)^2}{100}$$

The power consumed when operated at 110V is

$$P' = \frac{V^2}{R}$$

$$\Rightarrow P = \frac{(110)^2}{(220)^2/100} = \frac{100}{4} = 25W$$

Question334

In a Wheatstone's bridge, three resistances P, Q and R connected in the three arms and the fourth arm is formed by two resistances S_1 and S_2 connected in parallel. The condition for the bridge to be balanced will be [2006]

Options:

A. $\frac{P}{Q} = \frac{2R}{S_1 + S_2}$

B. $\frac{P}{Q} = \frac{R(S_1 + S_2)}{S_1 S_2}$

C. $\frac{P}{Q} = \frac{R(S_1 + S_2)}{2S_1 S_2}$

D. $\frac{P}{Q} = \frac{R}{S_1 + S_2}$

Answer: B

Solution:

From balanced wheat stone bridge $\frac{P}{Q} = \frac{R}{S}$ where $S = \frac{S_1 S_2}{S_1 + S_2}$

Question335

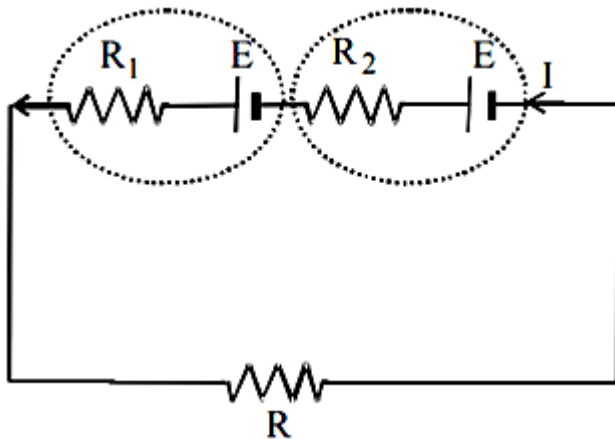
Two sources of equal emf are connected to an external resistance R . The internal resistance of the two sources are R_1 and R_2 ($R_1 > R_2$). If the potential difference across the source having internal resistance R_2 is zero, then
[2005]

Options:

- A. $R = R_2 - R_1$
- B. $R = R_2 \times (R_1 + R_2) / (R_2 - R_1)$
- C. $R = R_1 R_2 / (R_2 - R_1)$
- D. $R = R_1 R_2 / (R_1 - R_2)$

Answer: A

Solution:



Let E be the emf of each source of current

$$\text{Current in the circuit } I = \frac{2E}{R + R_1 + R_2}$$

Potential difference across cell having internal resistance R_2

$$V = E - iR_2 = 0$$

$$E - \frac{2E}{R + R_1 + R_2} \cdot R_2 = 0$$

$$\Rightarrow R + R_1 + R_2 - 2R_2 = 0$$

$$\Rightarrow R + R_1 - R_2 = 0$$

$$\Rightarrow R = R_2 - R_1$$

Question336

Two voltmeters, one of copper and another of silver, are joined in parallel. When a total charge q flows through the voltmeters, equal amount of metals are deposited. If the electrochemical equivalents of copper and silver are Z_1 and Z_2 respectively the charge which flows through the silver voltmeter is [2005]

Options:

A. $\frac{q}{1 + \frac{Z_2}{Z_1}}$

B. $\frac{q}{1 + \frac{Z_1}{Z_2}}$

C. $q \frac{Z_2}{Z_1}$

D. $q \frac{Z_1}{Z_2}$

Answer: A

Solution:

From Faraday's first law of electrolysis, mass deposited

$$m = Z q$$

$$\Rightarrow Z \propto \frac{1}{q} \Rightarrow \frac{Z_1}{Z_2} = \frac{q_2}{q_1} \dots\dots(1)$$

$$\text{Also } q = q_1 + q_2 \dots\dots(ii)$$

$$\Rightarrow \frac{q}{q_2} = \frac{q_1}{q_2} + 1 \quad (\text{Dividing (ii) by } q_2)$$

$$\Rightarrow q_2 = \frac{q}{1 + \frac{q_1}{q_2}} \dots\dots(iii)$$

From equation (i) and (iii),

$$q_2 = \frac{q}{1 + \frac{Z_2}{Z_1}}$$

Question337

An energy source will supply a constant current into the load if its internal resistance is [2005]

Options:

- A. very large as compared to the load resistance
- B. equal to the resistance of the load
- C. non-zero but less than the resistance of the load
- D. zero

Answer: D

Solution:

Current is given by

$$I = \frac{E}{R + r}$$

If internal resistance (r) is zero,

$$I = \frac{E}{R} = \text{constant}.$$

Thus, energy source will supply a constant current if its internal resistance is zero.

Question338

A heater coil is cut into two equal parts and only one part is now used in the heater. The heat generated will now be [2005]

Options:

- A. four times
- B. doubled
- C. halved
- D. one fourth

Answer: B

Solution:

Heat generated,

$$H = \frac{V^2 t}{R}$$

After cutting equal length of heater coil will become half.

As $R \propto l$

$$\text{Resistance of half the coil} = \frac{R}{2}$$

$$H' = \frac{V^2 t}{\frac{R}{2}} = 2H$$

$\therefore$ As R reduces to half, ' H ' will be doubled.

Question339

The resistance of hot tungsten filament is about 10 times the cold resistance. What will be the resistance of 100 W and 200 V lamp when not in use ?

[2005]

Options:

A. 20Ω

B. 40Ω

C. 200Ω

D. 400Ω

Answer: B

Solution:

$$\text{Power, } P = V i = \frac{V^2}{R}$$

$\therefore$ Resistance of tungsten filament when in use

$$R_{\text{hot}} = \frac{V^2}{P} = \frac{200 \times 200}{100} = 400\Omega$$

Resistance when not in use i.e., cold resistance

$$R_{\text{cold}} = \frac{400}{10} = 40\Omega$$

Question340

In a potentiometer experiment the balancing with a cell is at length 240 cm. On shunting the cell with a resistance of 2Ω , the balancing length becomes 120 cm. The internal resistance of the cell is

[2005]

Options:

A. 5.0Ω

B. 1Ω

C. 2Ω

D. 4Ω

Answer: C

Solution:

Initial balancing length, $l_1 = 240\text{cm}$ New balancing length, $l_2 = 120\text{cm}$.

The internal resistance of the cell,

$$r = \left(\frac{l_1 - l_2}{l_2} \right) \times R = \frac{240 - 120}{120} \times 2 = 2\Omega$$

Question341

An electric current is passed through a circuit containing two wires of the same material, connected in parallel. If the lengths and radii are in the ratio of $\frac{4}{3}$ and $\frac{2}{3}$, then the ratio of the current passing through the wires will be

[2004]

Options:

A. $\frac{8}{9}$

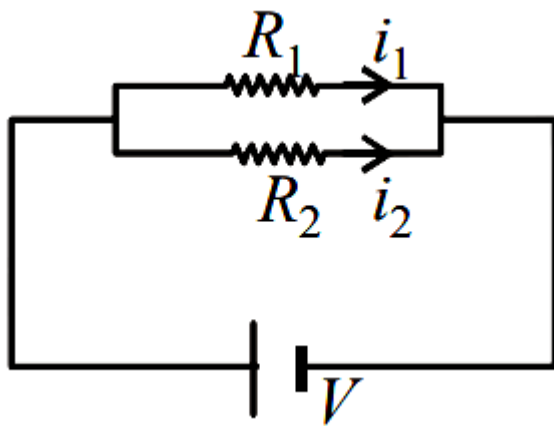
B. $\frac{1}{3}$

C. 3

D. 2

Answer: B

Solution:



Given,

$$\frac{l_1}{l_2} = \frac{4}{3} \text{ and } \frac{r_1}{r_2} = \frac{2}{3}$$

$$R_1 = \frac{\rho l_1}{\pi r_1^2}; R_2 = \frac{\rho l_2}{\pi r_2^2}$$

When wires are in parallel to the circuit potential difference across each wire is same

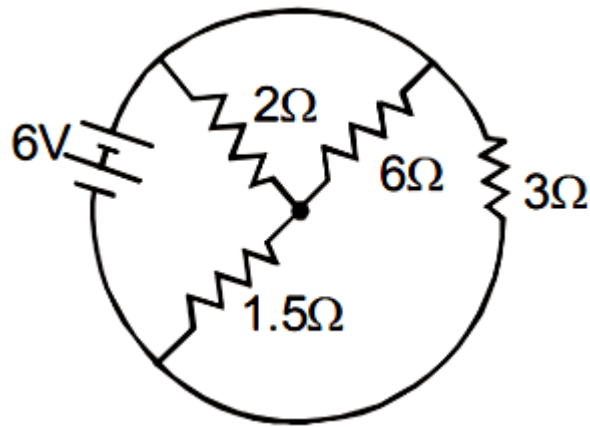
$$i_1 R_1 = i_2 R_2$$

$$\therefore \frac{i_1}{i_2} = \frac{R_2}{R_1} = \frac{\rho l_2}{\pi r_2^2} \times \frac{\pi r_1^2}{\rho l_1} = \frac{l_2}{l_1} \times \frac{r_1^2}{r_2^2}$$

$$= \frac{3}{4} \times \frac{4}{9} = \frac{1}{3}$$

Question342

The total current supplied to the circuit by the battery is



[2004]

Options:

A. 4 A

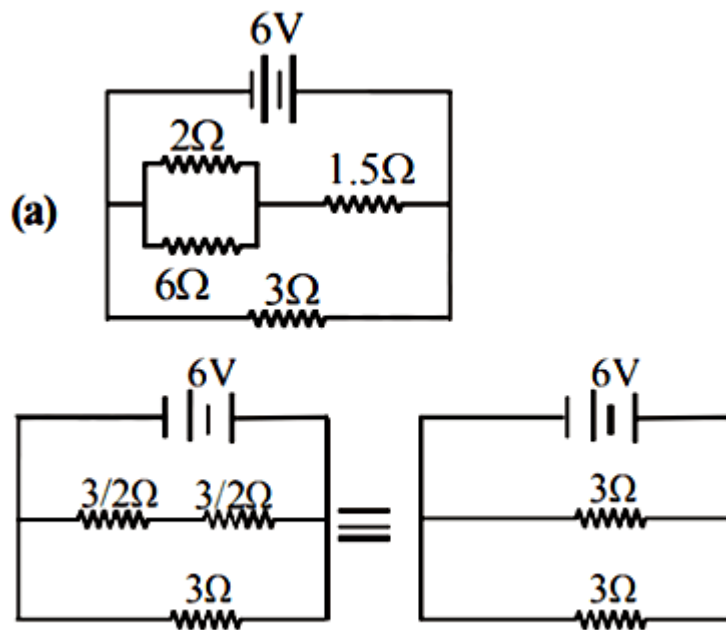
B. 2 A

C. 1 A

D. 6 A

Answer: A

Solution:



hence $R_{eq} = 3/2$; $\therefore I = \frac{6}{3/2} = 4A$

Question343

The resistance of the series combination of two resistances is S. when they are joined in parallel the total resistance is P. If $S = nP$ then the minimum possible value of n is [2004]

Options:

- A. 2
- B. 3
- C. 4
- D. 1

Answer: C

Solution:

Let R_1 and R_2 be the two given resistances

Resistance of the series combination,

$$S = R_1 + R_2$$

Resistance of the parallel combination,

$$P = \frac{R_1 R_2}{R_1 + R_2}$$

As per question $S = nP$

$$\Rightarrow R_1 + R_2 = \frac{n(R_1 R_2)}{(R_1 + R_2)}$$

$$\Rightarrow (R_1 + R_2)^2 = nR_1 R_2$$

Minimum value of n is 4 for that

$$(R_1 + R_2)^2 = 4R_1 R_2$$

$$\Rightarrow (R_1 - R_2)^2 = 0$$

Question344

The thermo emf of a thermocouple varies with the temperature θ of the hot junction as $E = a\theta + b\theta^2$ in volts where the ratio a/b is 700°C . If the cold junction is kept at 0°C , then the neutral

**temperature is
[2004]**

Options:

A. 1400°C

B. 350°C

C. 700°C

D. No neutral temperature is possible for this termocouple

Answer: D

Solution:

Given $E = a\theta + b\theta^2$

$$\Rightarrow \frac{dE}{d\theta} = a + 2b\theta$$

At neutral temperature

$$\theta = \theta_n : \frac{dE}{d\theta} = 0$$

$$\Rightarrow \theta_n = \frac{-a}{2b} = -350 \Rightarrow \frac{d^2E}{d\theta^2} = 2b$$

hence no θ is possible for E to be maximum no neutral temperature is possible.

Question345

**The electrochemical equivalent of a metal is 3.35×10^{-7} kg per Coulomb. The mass of the metal liberated at the cathode when a 3A current is passed for 2 seconds will be
[2004]**

Options:

A. 6.6×10^{57} kg

B. 9.9×10^{-7} kg

C. 19.8×10^{-7} kg

D. 1.1×10^{-7} kg

Answer: C

Solution:

From the Faraday's first law of electrolysis,

$$m = Z it$$

$$\Rightarrow m = 3.3 \times 10^{-7} \times 3 \times 2$$

$$= 19.8 \times 10^{-7} \text{ kg}$$

Question 346

**The thermistors are usually made of
[2004]**

Options:

- A. metal oxides with high temperature coefficient of resistivity
- B. metals with high temperature coefficient of resistivity
- C. metals with low temperature coefficient of resistivity
- D. semiconducting materials having low temperature coefficient of resistivity

Answer: A

Solution:

Thermistors are usually made of metal oxides with high temperature coefficient of resistivity.

Question 347

**Time taken by a 836 W heater to heat one litre of water from
10°C to 40°C is
[2004]**

Options:

- A. 150 s

B. 100 s

C. 50 s

D. 200 s

Answer: A

Solution:

Heat supplied in time t for heating 1L water from 10°C to 40°C

$$\Delta Q = mC_p \times \Delta T$$

$$= 1 \times 4180 \times (40 - 10) = 4180 \times 30$$

$$\text{But } \Delta Q = P \times t = 836 \times t$$

$$\Rightarrow t = \frac{4180 \times 30}{836} = 150\text{s}$$

Question348

In a meter bridge experiment null point is obtained at 20 cm. from one end of the wire when resistance X is balanced against another resistance Y. If $X < Y$, then where will be the new position of the null point from the same end, if one decides to balance a resistance of 4 X against Y [2004]

Options:

A. 40 cm

B. 80 cm

C. 50 cm

D. 70 cm

Answer: C

Solution:

From the balanced wheat stone bridge

$$\frac{R_1}{R_2} = \frac{l_1}{l_2}$$

$$\text{where } l_2 = 100 - l_1$$

$$\text{In the first case } \frac{X}{Y} = \frac{20}{80}$$

$$Y = 4X$$

In the second case

$$\frac{4X}{Y} = \frac{1}{100 - l}$$

$$\Rightarrow \frac{4X}{4X} = \frac{1}{100 - l}$$

$$\Rightarrow l = 50$$

Question 349

The length of a given cylindrical wire is increased by 100%. Due to the consequent decrease in diameter the change in the resistance of the wire will be [2003]

Options:

- A. 200%
- B. 100%
- C. 50%
- D. 300%

Answer: D

Solution:

Since volume of wire remains unchanged on increasing length, hence $A \times l = A' \times l'$
 $\Rightarrow el' = 2l$

$$\therefore A' = A \times \frac{l'}{l} = \frac{A \times l}{2l} = \frac{A}{2}$$

$$\text{Percentage change in resistance} = \frac{R_f - R_i}{R_i} \times 100 = \frac{\rho \frac{l'}{A'} - \rho \frac{l}{A}}{\rho \frac{l}{A}} \times 100$$

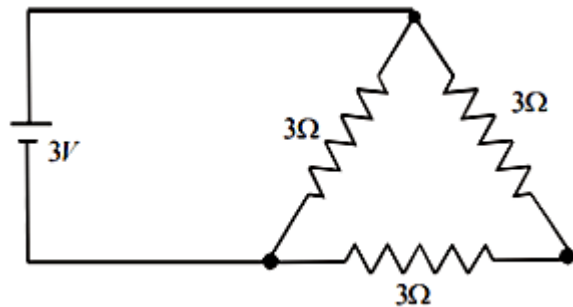
$$= \left[\left(\frac{l'}{A'} \times \frac{A}{l} \right) - 1 \right] \times 100$$

$$= \left[\left(\frac{21}{A/2} \times \frac{A}{1} \right) - 1 \right] \times 100 = (4 - 1) \times 100$$

$$= 300\%$$

Question 350

A 3 volt battery with negligible internal resistance is connected in a circuit as shown in the figure. The current I, in the circuit will be



[2003]

Options:

- A. 1 A
- B. 1.5 A
- C. 2 A
- D. $1/3$ A

Answer: B

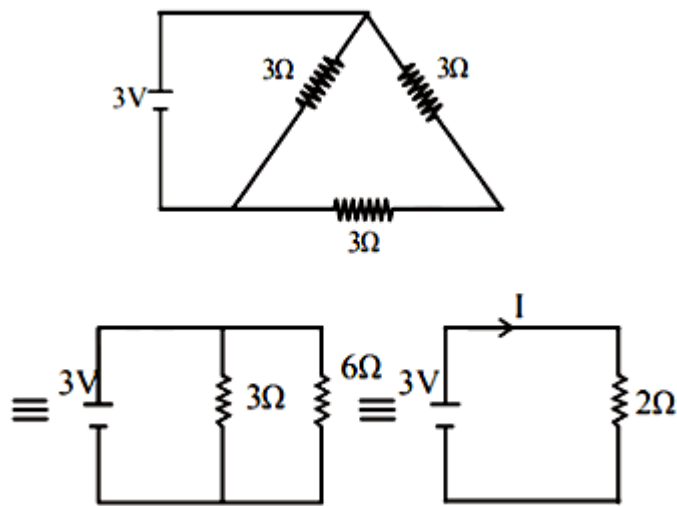
Solution:

In the given circuit, resistance of 3Ω is in parallel with series combination of two 3Ω resistance.

$$R_p = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2\Omega$$

Using ohm's law $V = IR$

$$\Rightarrow I = \frac{V}{R} = \frac{3}{2} = 1.5A$$



Question351

The thermo e.m.f. of a thermo-couple is $25 \mu\text{V}/^\circ\text{C}$ at room temperature. A galvanometer of 40 ohm resistance, capable of detecting current as low as 10^{-5} A , is connected with the thermo couple. The smallest temperature difference that can be detected by this system is
[2003]

Options:

- A. 16°C
- B. 12°C
- C. 8°C
- D. 20°C

Answer: A

Solution:

Let the smallest temperature difference be $\theta^\circ\text{C}$ that can be detected by the thermocouple, then

$$\text{Thermo emf} = (25 \times 10^{-6})\theta$$

Let I is the smallest current which can be detected by the galvanometer of resistance R .
Potential difference across galvanometer

$$IR = 10^{-5} \times 40$$

$$\therefore 10^{-5} \times 40 = 25 \times 10^{-6} \times \theta$$

$$\Rightarrow \theta = 16^\circ\text{C}$$

Question352

The negative Zn pole of a Daniell cell, sending a constant current through a circuit, decreases in mass by 0.13g in 30 minutes. If the electrochemical equivalent of Zn and Cu are 32.5 and 31.5 respectively, the increase in the mass of the positive Cu pole in this time is
[2003]

Options:

- A. 0.180 g
- B. 0.141g
- C. 0.126 g
- D. 0.242 g

Answer: C

Solution:

According to Faraday's first law of electrolysis

$$m = Z \times I \times t$$

When I and t is same, $m \propto Z$

$$\therefore \frac{m_{\text{Cu}}}{m_{\text{Zn}}} = \frac{Z_{\text{Cu}}}{Z_{\text{Zn}}} \Rightarrow m_{\text{Cu}} = \frac{Z_{\text{Cu}}}{Z_{\text{Zn}}} \times m_{\text{Zn}}$$

$$\Rightarrow m_{\text{Cu}} = \frac{31.5}{32.5} \times 0.13 = 0.126\text{g}$$

Question353

A 220 volt, 1000 watt bulb is connected across a 110 volt mains supply. The power consumed will be
[2003]

Options:

- A. 750 watt
- B. 500 watt
- C. 250 watt
- D. 1000 watt

Answer: C

Solution:

We know that resistance,

$$R = \frac{V_{\text{rated}}^2}{P_{\text{rated}}} = \frac{(220)^2}{1000} = 48.4\Omega$$

When this bulb is connected to 110 volt mains supply we get

$$P = \frac{V^2}{R} = \frac{(110)^2}{48.4} = 250W$$

Question354

. The length of a wire of a potentiometer is 100 cm, and the e.m.f. of its standard cell is E volt. It is employed to measure the e.m.f. of a battery whose internal resistance is 0.5Ω . If the balance point is obtained at $l = 30$ cm from the positive end, the e.m.f. of the battery is where i is the current in the potentiometer wire [2003]

Options:

- A. $\frac{30E}{100.5}$
- B. $\frac{30E}{(100 - 0.5)}$
- C. $\frac{30(E - 0.5i)}{100}$
- D. $\frac{30E}{100}$

Answer: D

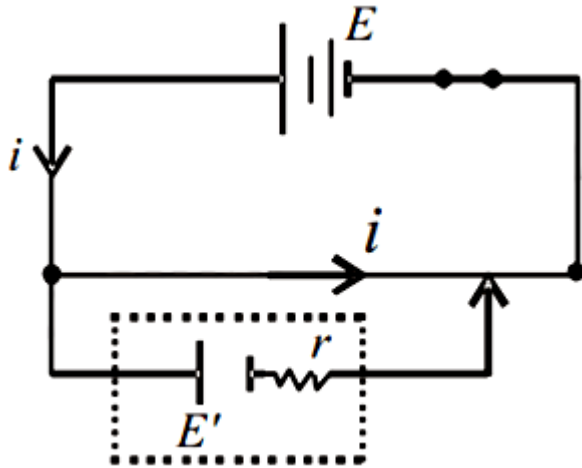
Solution:

From the principle of potentiometer, $V \propto l$

If a cell of emf E is employed in the circuit between the ends of potentiometer wire of length L , then

$$\frac{V}{E} = \frac{l}{L}$$

$$\Rightarrow V = \frac{El}{L} = \frac{30E}{100}$$



Note : In this arrangement, the internal resistance of the battery E does not play any role as current is not passing through the battery.

Question355

An ammeter reads upto 1 ampere. Its internal resistance is 0.81ohm. To increase the range to 10 A the value of the required shunt is
[2003]

Options:

- A. 0.03Ω
- B. 0.3Ω
- C. 0.9Ω
- D. 0.09Ω

Answer: D

Solution:

$$i_g \times G = (i - i_g)S$$

$$\therefore S = i_g \times G \cdot i - i_g = \frac{1 \times 0.81}{10 - 1} = 0.09\Omega$$

Question356

The mass of product liberated on anode in an electrochemical cell depends on

(where t is the time period for which the current is passed).

[2002]

Options:

A. $(It)^{1/2}$

B. It

C. I/t

D. I^2t

Answer: B

Solution:

From the Faraday's first law of electrolysis

$$m = ZIt \Rightarrow m \propto It$$

Question357

A wire when connected to 220 V mains supply has power dissipation P_1 . Now the wire is cut into two equal pieces which are connected in parallel to the same supply. Power dissipation in this case is P_2 . Then $P_2 : P_1$ is

[2002]

Options:

A. 1

B. 4

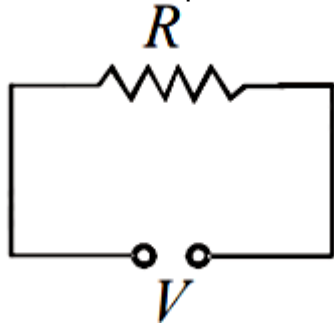
C. 2

D. 3

Answer: B

Solution:

Case 1 Initial power dissipation,

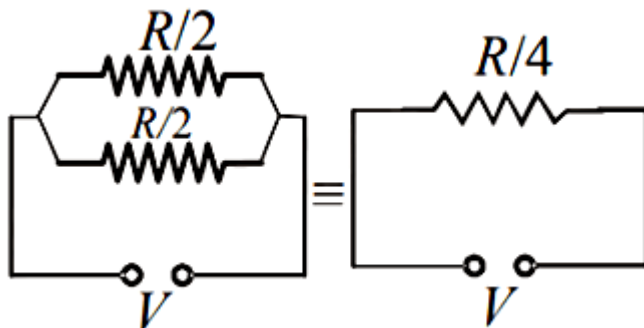


$$P_1 = \frac{V^2}{R}$$

Case 2

When wire is cut into two equal pieces, the resistance of each piece is $\frac{R}{2}$. When they are connected in parallel

$$R_{eq} = \frac{R/2}{2} = \frac{R}{4}$$

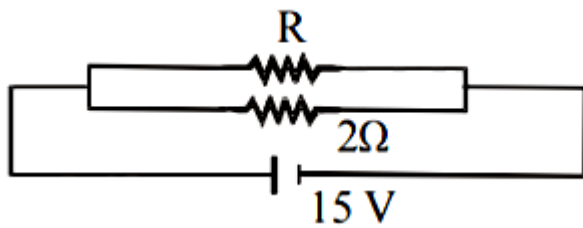


Power dissipated,

$$P_2 = \frac{V^2}{R/4} = 4 \left(\frac{V^2}{R} \right) = 4P_1$$

Question358

If in the circuit, power dissipation is 150 W, then R is



[2002]

Options:

A. 2Ω

B. 6Ω

C. 5Ω

D. 4Ω

Answer: B

Solution:

The equivalent resistance of parallel combination of 2Ω and R is

$$R_{eq} = \frac{2 \times R}{2 + R}$$

$$\therefore \text{Power dissipation } P = \frac{V^2}{R_{eq}} \therefore 150 = \frac{(15)^2}{R_{eq}}$$

$$\Rightarrow 150 = \frac{225 \times (R + 2)}{2R} \Rightarrow \frac{2R}{2 + R} = \frac{3}{2}$$

$$\Rightarrow 4R = 6 + 3R \Rightarrow R = 6\Omega$$

Question 359

If an ammeter is to be used in place of a voltmeter, then we must connect with the ammeter a [2002]

Options:

- A. low resistance in parallel
- B. high resistance in parallel
- C. high resistance in series
- D. low resistance in series.

Answer: C

Solution:

To use an ammeter in place of voltmeter, we must connect a high resistance in series with the ammeter.

Connecting high resistance in series makes its resistance much higher.
